

This book has been dedicated to the life's work of

ARTHUR WILLIAM BELL

1945 — 2018



Broadcasting through the night, from the high desert.

"Come take a ride on the wild side!"

THE SCIENCE OF
UFOLOGY

I . □

*A Skeptic's Textbook for the
Serious Study of the Unidentified*

E V A N S V E N D S E N

First Edition

P R E F A C E

Begun 13 April 2018.

I started writing this book on the night of April 13th, 2018, because that was the night Art Bell died.

The timing is not a coincidence and I am not going to dress it up as one. His death is the reason this book exists. I found out the way most people found out, which is to say secondhand, late, and from a screen. And then I sat there for a long while in a very quiet room, thinking about the fact that the voice that had kept me company through a decade of insomnia had stopped, and that nobody was going to pick up the phone at three in the morning and let a truck driver in Nevada describe what he had just seen without laughing at him. By the time I got up from that chair I had decided that somebody had better write the rest of it down. I opened a blank document before the night was over.

That was the thing about Bell. He let people talk. He did not endorse them, and he did not humiliate them, and if you listen to enough of those tapes you realise how genuinely rare that combination is. Somewhere between the evangelists and the debunkers there was a man with a microphone in the high desert who had figured out that the honest position is usually “well, tell me more.” That is what I started typing towards that night, and somewhere in the first hour of it, it occurred to me quite suddenly that the thing I actually wanted to write about was not flying saucers at all.

This is not a UFO book. I understand that it looks like one. It has the word *Ufology* on the cover in large letters, it has chapters on Roswell and

Rendlesham and Area 51, and if you shelve it anywhere it will be shelved next to books with considerably worse research and considerably better sales. But the subject of this book is not unidentified flying objects. The subject of this book is our spectacular, ongoing, almost artistic failure to communicate.

Consider what actually happens in a UFO case. A person sees something. That something is, by definition, a thing they cannot identify — so already the signal has arrived damaged. They then have to convert it into words, and the words available to them are terrible: “saucer,” “craft,” “triangle,” “it moved like nothing I’ve ever seen.” Those words go to an investigator, who has his own vocabulary and his own hopes. The investigator’s report goes to an agency, which has an interest in the answer. The agency’s summary goes to a journalist, who has a deadline. The journalist’s article goes to the public, which has already decided. By the time the story reaches you it has been translated six times by six parties with six different agendas, and we then act surprised that it does not survive scrutiny.

At no point in that chain does anyone lie, necessarily. That is what I find so interesting. You do not need liars to produce a lie; you only need enough people sincerely talking past one another. Wrap that around a subject with a stigma attached, so that half the witnesses will not speak and half the scientists will not listen, and you have built a machine that reliably converts real events into unusable noise. Seventy-five years of that machine running at full capacity is roughly what we mean when we say “the UFO phenomenon.”

So the questions that organise this book are communication questions. How is the question worded, and what does the wording smuggle in? Who is required to prove what, and who quietly got excused? What did the witness actually say, before it was tidied up? What would count as evidence here, and has anyone bothered to say so out loud? These are unglamorous questions. They are also the only ones that have ever moved a case forward.

And there is a larger version of the problem waiting at the end of the book, which is the part that genuinely keeps me up — Bell’s hours, appropriately. If something out there ever does try to talk to us, the encounter will not fail because of physics. It will fail because we are bad at listening, bad at saying what we mean, and worse at admitting when we have understood nothing at all. We cannot reliably transmit the contents of one human night sky across a single dinner table without it turning into an argument. That

is the skill we are short on. That is what I think is worth practising, and a disreputable subject with a seventy-five-year paper trail turns out to be an excellent place to practise it.

A note on how to use the thing. Chapter 1 is about skepticism, not about UFOs, and it comes first for the reason given above: the whole book is about signal integrity, and Chapter 1 is where we install the instruments. Skip it and the rest is just stories. Where I do not know something, I have said so. Where I lack the equipment or the clearance or the money to settle a question, I have said that too. Those admissions are not gaps in the argument; they are the argument, and they are the one habit I would most like you to steal from me.

I never called in to the show. I meant to. Ten years of meaning to, and then one night in April the line went dead for good, and what I had left was a blank document and about two hundred and fifty pages of things I wanted to say to a man who was very good at letting people say them.

So, in his words, and with his sense of timing:

“Come take a ride on the wild side!”

Evan Svendsen

I N T R O D U C T I O N

“The absence of evidence is not the evidence of absence.”

— COMMONLY ATTRIBUTED TO CARL SAGAN

I am a self-made professor of Ufology. There is no such degree. No university on this planet will hand you a piece of paper that says you have mastered the study of unidentified flying objects, and if one ever does, I would advise you to read the fine print very carefully before you sign the tuition cheque. What I have instead is a decade of case files, a stack of declassified documents that cost me more in printer toner than I care to admit, a standing invitation to lecture at the University of Lethbridge, and a stubborn refusal to accept “swamp gas” as a final answer to anything.

This book began as those lectures. Each chapter that follows corresponds roughly to a week of material, delivered to a room of students who arrived expecting little green men and left, I hope, with something considerably more useful: a working method for thinking about extraordinary claims without either swallowing them whole or spitting them out on reflex.

Which brings me to the single most important structural decision in this volume, and the one that confuses people most. Chapter 1 is not about UFOs. Chapter 1 is about skepticism. The introduction to Ufology proper does not arrive until Chapter 2. This ordering is deliberate and it is not negotiable. Skepticism is a hard dependency. You cannot meaningfully evaluate a single photograph, a single radar return, or a single witness statement until you understand how language distorts questions before they

are even asked, how the burden of proof actually works, why correlation is not causation, and how your own brain will cheerfully lie to you in service of what you already believe. Attempting Chapter 4 without Chapter 1 is like attempting calculus without algebra. You will produce answers. They will be wrong, and worse, you will not know why.

A word on what this book is not. It is not a catalogue of sightings; there are better catalogues, maintained by people with more filing cabinets than I have. It is not an argument that aliens are visiting Earth. I do not know that, you do not know that, and anyone who tells you they know that is selling something. Nor is it a debunking, because debunking is just believing with the sign flipped, and I find it every bit as lazy.

What this book is, is an attempt to treat a disreputable subject with reputable methods. To ask, of each case and each claim: what would count as evidence here? What test could be run? What would have to be true for this to be false? Those are scientific questions, and they can be asked about anything, including things that make you look ridiculous at dinner parties. I have been looking ridiculous at dinner parties for years. You get used to it.

You will find, scattered through these pages, moments where I tell you plainly that I do not know something, or that I lack the resources to test it myself, or that a piece of celebrated evidence collapses the moment you touch it. Those admissions are not weaknesses in the argument. They are the argument. A field that cannot say “I was wrong” is not a field, it is a congregation.

So read Chapter 1 first. Read it twice if you are the sort of person who already knows what you think about all this. Then we will get to the good stuff: Area 51, Roswell, Rendlesham, the physics of time travel, the Drake equation, cattle mutilations, crystal skulls, and the long strange question of what happens on the day we finally answer the phone.

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THE IMPORTANCE OF SKEPTICISM

"I like to regard myself as someone who's capable of critical thought, that is to say, who can evaluate claims."

— BILL NYE

1.1 THE FALLACY OF LANGUAGE

For all the amazing, wonderful things that human language has made possible. Its ability to convey meaning and emotion. Its capacity to transmit culture and history through writing and literature. Human language is fundamentally flawed. Flawed in its ability/hindrane to inject implicit meaning into otherwise explicit questions. For example, in a 1995 survey of unknown sample size conducted by Scripps Howard News Service and Ohio State University;¹ the populous surveyed was presented with the following question:

"Do you believe in UFOs?"
Yes No

The results suggested an even split of 50% with 34% of Seniors (Age 65+),

¹See the References section, entry 1.

and 45% of Church Attendees answering in the affirmative. Such a question when posted in the following format creates many problems. Most often the phrase “Do you believe in UFOs?” is interpreted to read “Do you believe in UFOs AND that UFOs are alien spacecraft that have come to planet earth from a distant star system in effort to take over the world?” In which case the majority would answer HELL NO! Such a fallacy of language is a common problem when conducting public surveys on any number of topics. Much in the way that questions of ghosts or Bigfoot will automatically conjure explanations for such phenomena as the restless undead or hairy woodland beasts who make their living by ripping their prey in half limb by limb. A better written, and more informative question for example should have read:

“Do you believe UFOs are ground in reality?”

Yes No

In which case anyone proclaiming that UFOs are nothing more than trickery of the mind or pranks conducted by irresponsible hoaxers would still be obliged to answer “Yes” to the question posed, because both explanations are ground in one’s rational understanding of reality. It’s only when the idea that UFOs are spaceships with little green men from some distant star system do people tend to lean in the side of non-belief. When in fact explanations for the existence of UFOs are the least interesting things that can be learned from such questions. It doesn’t matter. It does not matter whether UFOs are result of government funding and top-secret black budget projects, or aliens from a distant galaxy, time travelers from the year 2939, or simply hallucinations of the mind. Because in each case regardless of whether the explanation is the one true and genuine reason for the existence of UFOs; all will open new areas of study and investigation regardless. If it’s the government and men in black, then ok what sort of social or political implications do such an apophony pose? If UFOs are REALLY time travelers from the future, then what can be learned about the physics of today that will lead us into the time travel of tomorrow? If UFOs are nothing more than flawed human perception, what does that say about ourselves and who we are as a species? These are the types of questions that the field of Ufology is truly interested in.

Another problem with the question posed from the survey is the constraint of binary answers. Forcing a person to answer in the affirmative

100% Yes or 100% No without allowing any grey area in between. These grey areas that are so important for such provocative questions. Like asking for instance: Are You Religious? Are You Liberal? or Are You Gay? Quick! Answer now yes or no. It's complicated. And complicated questions require careful consideration, thought, and further reasoning. Difficult questions which allow only binary answers are fundamentally flawed; in the way human language is fundamentally flawed and should not be taken as anything more than mere polls for the public enjoyment. Any quoted "scientific" poll pertaining to society's belief in the paranormal should contain prior planning that would consider such limitations as the fallacy of language to induce otherwise unstated meaning. And to consider other aspects like demographics that could potentially play with the results and induce a bias into said conducted survey.

I hesitate in making assertive assumptions or to claim that any of the surveys found in the sourced reference material are in some way flawed. The referenced material which was published in January 2000 by The Mutual UFO Network examines the previous 50+ years of publicly conducted polls and surveys on the topic of UFOs. Polls which were conducted both by universities and by private corporations such as journals and magazines. I am hesitant to dismiss any or all these surveys as quote "scientific" without doing further research into each one. I am unable and without the resources required to personally conduct any survey of my own. Unable to test whether the wording of questions or the number of available options for response in fact play significant role in the information obtained. But I strongly encourage the consideration of such arguments I have made when overseeing the formation of any publicly conducted survey, whether done for scientific research or mere fun and games.

The fallacy of language plays a huge role in all of science and how we go about inferring results, and interpreting data from studies, and should always be taken into careful consideration.

1.2 THE BURDEN OF PROOF

When it comes to science; theories are the gold standard in evidence proof, and the extent of human knowledge. Yes, I know what you are thinking, and I completely agree. They could not have chosen a more misleading term

to represent such a thing. Through the course of everyday conversation, the average individual will use and interpret “theory” to mean something that a person only thinks is true. Or maybe there is evidence it may be true, but no actual proof that it is. In other words: mere speculation. In science, this is what we call a hypothesis. Scientific hypotheses are founded on the notion that the scientist has reason to assume or believe that certain outcomes will result from the conduction of his experiment. But until they actually do the experiment, their hypothesis will remain unproven.

In the field of Ufology there are NO theories. None. There are only rumors, speculation, and any number of explanations for the phenomena. These are the UFO hypotheses; the ancient alien hypothesis, the abduction hypothesis, and the government classified top secret technology hypothesis to name a few. **This is because of one simple fact that the entire foundation of science rests upon: Science MUST be falsifiable.** Any experiment in which the scientist proposes one outcome will happen, must allow for the chance and possibility (even if very slim) that the entire opposite will instead. This is the scientific premise Ufology lacks, and the one thing preventing it from become an officially recognized field of inquiry. The ability to test and form falsifiable experiments on the hypotheses proposed as possible explanations for the subject matter. That is why the burden of proof lies on the Ufologist who studies this subject. Much in the manner that Hitchens’s Razor has inspired many variations on the following quote: “Extraordinary claims require extraordinary evidence.”

FUN FACT

A philosophical razor is not in any way sharp or pointy but is simply an idea or rule of thumb to abide by when drawing conclusions using a skeptical mind. Occam’s Razor for instance is likely the best-known example of such. Occam’s razor states that in the case of 2 possible solutions or answers to a question; chances are the simpler of the two is more likely to be right. Occam’s razor however is often used as a strawman argument to simply debunk UFOs by asserting that conventional wisdom can explain it away. Swamp gas, weather balloons, the planet Venus, hallucinations, mass hallucinations, hysteria, hoaxes, airplanes, ball lightning, frisbees and flares you name it; it’s all rubbish.

Take the case of the hypothetical husband for instance whose fingerprints are found on the weapon used to kill his wife. Well according to Occam’s

Razor, that man is guilty of murder! How else could one explain the existence of his fingerprints on the weapon OTHER than the simplest answer: which is he killed his wife. Preposterous you say! That man loved his wife. Well if you assert that he must have been framed by the true murderer, because what other explanation would there be for such an egregious act from a loving husband? The argument you now present for the husband's innocence entirely contradicts the notion first proposed under Occam's razor, and yet how easy it would be to now assert that this is the new simplest answer. In fact, regardless of what explanation any investigation into the wife's murder may uncover, you will always be able to assert that explanation is the simplest, not because it is simplistic, but because it is true. The investigation into the murder may in fact discover a plot that spans 10 years and includes 3 assassins, 2 lovers, 12 kitchen knives, and a rowboat. Just because the answer is complex does not mean it is false.

However, that does not allow for legitimate argument into the completely opposite and equally absurd explanation for UFO sightings. Which states that Zoltar from the planet Koolon has come 14 million miles to Earth to enslave the human race but has chosen to do so by giving birth to a race of reptilian shapeshifters who have now infiltrated the highest ranks of the US government and plot our very demise through the use of unaccountable tax returns, burning more fossil fuels, and . . . pee-pee tapes. No. If we are to find reasonable legitimacy for the further scientific investigation of UFOs then there must be some common ground that lies both within the realm of possibility and within the validity of such explanations not to assert that UFO eye witnesses are crazy drunks.

It should be understood that the default position of any claim is to reject it until sufficient proof has been presented for acceptance. This is known as the **null hypothesis** — or, in less formal language, the default skeptical position. This position has no bearing on whether the claim is of positive or negative declaration. If the claim is that UFOs exist, then the default position is to reject this belief. This does not mean however you assert that UFOs do NOT exist. Conversely if the claim is that UFOs are total bogus, then the strategy holds; you should reject this claim. Similarly, that does not mean you now assert the opposite is true. The rejection of a claim is not coequal to the assertion of its counterpart.

That is why the burden of proof lies on those who investigate the subject matter. Because nothing should be assumed or believed, and even more

so acted on or regulated without substantial reasons for doing so. This is the plight of the Ufologist and we must agree to take it as such. It should also be noted that when investigation into the subject is conducted it should be done without bias or motive to either prove or disprove such things. Richard Dawkins has said—whether he was the first to say or not—“By all means let’s be open-minded, but not so open-minded that our brains drop out.”

I . 3 A L L T H E E V I D E N C E B U T N O P R O O F

It should also be noted, that scientific theories; although the gold standard in evidence: are NOT 100% bulletproof. That is because any scientific theory that has gone through thousands of experiments, PhD level research, and any number of well documented case studies is fundamentally flawed. Much in the same way language is fundamentally flawed, and for all the same reasons. Scientific theories are based on human perception and understanding of the world, and human perception is often misleading. We aren’t perfect beings, and neither are our theories and ideas perfect philosophies.

Even the greatest thoughts from the greatest minds, are merely good-enough-for-now explanations of the world. For example: Einstein’s work is incomplete. Let me say that a bit louder for the people in the back: Einstein’s work is incomplete! No, I did not say Einstein was wrong. I merely observe that Einstein was flawed because Einstein was human, and humans are flawed. This is the very reason there exists 2 distinct and disjoint theories for relativity. They are: general and special relativity. Both equally as true, and yet only applicable to the situations each describes best. Depending on what you believe; Einstein likely went to his grave unable to solve his greatest challenge: The Theory of Everything. One singular equation that will bind the whole of existence together. My father for instance does not hold such notions. My father believes that Einstein fully comprehended within the realm of his consciousness the key to the universe. But was unable to formulate any equation or combination of language to describe the things he understood; once again resulting from the imperfections of these human inventions.

What then can be known if anything at all? That is the question explored by 17th century philosopher René Descartes. At a time when so many other philosophers were making appeals to God, Descartes took an alternative route by appealing to rationality. He imagined a world fashioned by an evil madman whose sole purpose for existence was to deceive the inhabitants of his false reality. This thinking of false existence has been used many times throughout some of the greatest mind-bending films preying on our fear of the unknown. There is no way however to either prove or disprove such a theory: it's unfalsifiable.

This thought experiment although raises an even bigger question. If everything we know is fake from our past, to our loved ones, to our income and fears; then does any of it even matter? This thought troubled Descartes as he was unable to find a solution. Eventually he would come to realize that the only thing he can know for certain in taking doubt of the world around him was that he knew it. In other words: "I think therefore I am"—René Descartes (1637).²

This method of thinking would eventually lead to the 21st century version of the "evil genius". The brain in a vat connected to a supercomputer. Many great scientists even think it is possible that we are living in a computer simulation. Statistically speaking, if there is 1 true genuine universe with intelligent thinking beings. Then it's possible these beings have reached a stage of technological development in which they can run simulations of their own reality. Hundreds maybe even thousands of reality simulations could be running all at once or in parallel. The chances of any being within each simulation being a resident of the real genuine universe is simply 1 over the number of total simulations being run. And so statistically speaking we are far more likely to be a resident of the false simulated reality than someone who resides in the real world. This is an idea that will be explored further in Chapter 11 when I talk about the science of alternate dimensions.

Again, this idea will raise the question: Does reality even matter if we can't tell it's fake? The answer to this is surprisingly more simplistic than you may initially think, and although the meaning of one's own reality is entirely a subjective predisposition; attempts towards a logically correct answer have led to new insight. The argument runs like this. If the simulation is perfect, then every test you could perform on it returns the same answer it would

²See the References section, entry 2.

return in a base reality, which means the distinction has no observable consequences — and a distinction with no observable consequences is not a scientific claim, it is a mood. If on the other hand the simulation is imperfect, then the imperfections are physical, they are measurable, and hunting for them is ordinary science rather than philosophy. Either the question is untestable and therefore idle, or it is testable and therefore no longer a question about simulation at all but about the behaviour of matter. Notice that this is precisely the structure we will be applying to UFO claims for the next four hundred pages. The gravity you feel, the burn you get from a hot stove, and the grief you feel at a funeral are identical under either hypothesis. All that matters is consistency.

I would like to point out that this is also the number one problem that arises when taking a skeptical approach to thinking. The fact that skepticism is limitless. If you can doubt your very existence, then most certainly the doubt of having seen a UFO is far more within the bounds of held belief. So therefore, it becomes easier to simply assert that the experience was a trick of one's own senses and move on with your life. It takes real tenacity to stand up against the in-crowd and proclaim there is more to the situation than meets the eye. And so, these silent heroes and seekers of the truth should NOT be met with jeering laughter at the gates of discovery, but instead with praises for the courage to speak their mind and demand the truth!

What then can be made of the truth in a world distraught by political scandal, coverup and fake news? All factors are bound tightly together and play significant roles in the field of science, regardless of the specifics on that field of inquiry. Whether you choose to believe it or not, the world is round. That is what scientists call an objective fact. It's true whether you choose to hold that belief or not. We all know however that there are people on this big blue sphere who would claim otherwise. The motivation behind every flat earther is unique to each and should be respected on the grounds of freedom of speech. For it is freedom of speech that leads to discovery. If citizens were not free to express new ideas regardless of how detestable they are, then we may all still think the earth is the center of the universe.

Consider for instance the case of the hypothetical scientist who honestly thinks the earth is flat. While there is plenty of proof that it is not, the amount of given data does not satisfy this individual's threshold of proof. If this individual holds a higher standard of proof than the rest of the

population; there is no logical reason to be angry at them for holding such beliefs. ASSUMING however, and I want to make this very, very clear! They are willing to undergo experiments and test their belief even knowing there is a chance they could be wrong. That is the divide between the hypothetical scientist and real-world examples of flat-earthers. The scientist will undergo experimentation, while the whole of flat earthers resort to the side of cult like assertion of false facts. They are not willing to experiment, and devise means to test their ideas, they simply want you to agree with them because.

Now in defense of the hypothetical flat-earth scientist, and anyone else for that matter holding wildly absurd ideas and a means of testing them, consider the following:

The scientist believes the earth is flat and has a means of testing it. His hypothesis is the experiment will result in data to support the flat earth theory. His experiment has never been done before and is so outside the box it has the scientific community reeling in anticipation. For if his experiment was inside the box or somehow conventional, then it would have been done already in which case this scientist would have the data they need to believe the earth is round. Currently, they do not.

And so, they undergo said experiment, and after all the data is gathered and the numbers are crunched they come to the stunning realization they were wrong. “Ahhha”! The scientist shouts in avid proclamation! “I was wrong after all”. “But wait?” “What is that??” The scientist then notices something in the charts he had not before. Some of data ENTIRELY unrelated to the shape of earth, and yet it was gathered and clearly visible in the chart. And so, they investigate the abnormality further and spend the next 10 years of their life founding a never before seen branch of science based on an abnormality that was only seen as result of their experiment being so unconventional.

In fact, many of the greatest scientific discoveries of all time resulted as complete accident, or because the scientist trying to prove one thing discovers something entirely unrelated. That is the conjecture I put forth. That it is important for people such as the flat-earth scientist to have a platform on which they can present new ideas. Because even though they began with a false belief, they have now found evidence in the contrary THAT EXCEEDS their own standard of believing. In addition, the rest of the scientific community will benefit from new insight into totally unexplored territory.

How much evidence though is required to prove things? What does proof even mean, and how do the beliefs we hold as true impact our behavior? These are all interesting questions and dive much deeper into the heart of reality when explored to their fullest potential. If you have ever had the burden of learning math proofs in post-secondary, then you will understand exactly what is meant by this. For everyone else, all that is needed to be known is that math proofs are hard as (really bad word). For the simple reason that in order to prove anything at all you must first agree on what you are permitted to assume, and mathematics is the only field honest enough to write those permissions down at the top of the page. Consider π . Every schoolchild is told it is the ratio of a circle's circumference to its diameter, which is true and also not a proof of anything. To show that this ratio is the same for every circle, that it is irrational, and that it is transcendental — meaning it is not the root of any polynomial with integer coefficients — took mathematicians until 1882, and Lindemann's proof of the last of those does not involve circles in any recognisable way. It proceeds instead through the exponential function and a result about algebraic numbers, and when you finish reading it you have been given no picture in your head whatsoever. You have been compelled. That is what a proof is: a chain of steps each of which you cannot refuse, arriving somewhere your intuition never went. The proof for π is entirely logical yet makes no sense.³ Hold on to that discomfort, because it is the standard the word "proof" carries, and in this field the word gets used to mean a blurry photograph and a sincere man.

We must accept this simple fact if we are to move forward in our field of discovery. The fact that to date there is NO proof for the existence of UFOs or extraterrestrial beings visiting the planet, but that DOES NOT in the slightest mean there is not evidence. That is the goal of further chapters within the book. To explore what evidence has been presented and in an unbiased fashion examine the possible ways in which we can test such evidence. For then, and only then we may come to a better understanding of the mysterious world around us.

³See the References section, entry 3.

1.4 CLAIMS MADE FROM A SLIPPERY SLOPE

The concept of a slippery slope essentially conditions that once endeavor down a certain path of thought or logical reasoning has begun, it's only a matter of time before things get carried away. My case in point is the infamous History Channel TV show: *Ancient Aliens*. Well written and visually stunning; the show offers little to no actual digestible information. It is done this way on purpose. From a marketing perspective it does its job, and it does this very well for that matter, by capturing and captivating audiences with its wildly absurd claims and ideas. Unfortunately, very little to no actual rigorous scientific experimentation is done throughout the now 13 seasons totaling over 140 episodes.

Functionally speaking, the show suffers from drastic over simplification, and the incessant need to blame every unknown mystery of the world on extraterrestrial intervention. This web of connections the show suggests regarding all the earthly mysteries is both its greatest success and the worst possible thing for genuine researchers. It is successful because it offers something for everyone. Maybe you are into native culture and legends of sky people. They got that. Or perhaps stories of Atlantis deeply affect your existence. Or maybe you yourself are subject to modern day UFO sightings. Yes, yes, yes and yes, they got all that covered plus much more and perfectly wrapped into a nicely edited and dramatically narrated 45 minutes of verbal diarrhea.

This tactic of filming has led to its popular success and obvious financial gain evident from the fact the show is now 9 years running strong. It offers little however for actual discovery aside from the possibility that someone watching from the fence-line who has never formed an opinion on the subject matter may be inclined to think "hey this is interesting, I never even considered that." Aside from that possibility however, not much can be gained by watching the show. Worse is the likelihood that given the sheer amount of information overload, and the fact the show regards aliens as the solution to every problem; may turn away some people who would otherwise be inclined to believe, or at least willing to investigate such unknowns.

This interconnected web of "discovery" only serves to dilute the overall

content and cause sheer rejection in the face of laughter by the masses. There is a reason “Alien Guy” became a meme easily found on any search engine and providing answers to all life’s questions. It leads to the assertion that once any serious question of existence is posed, it’s only a matter of time before we start asking ourselves: “Why is the sky blue?” “Why did she dump me?” “Why was that calculus test so damn hard??” And the answer is always undeniably:

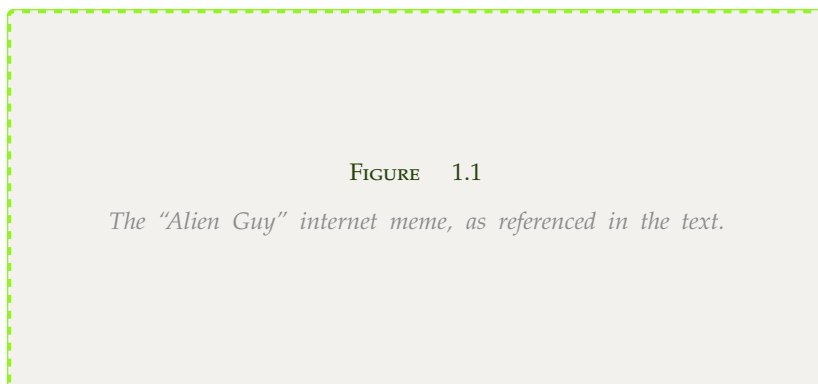


Figure 1.1. The “Alien Guy” internet meme, as referenced in the text.

Image sourced from: <http://i.imgur.com/1LxCoWM.jpg>

This web of interconnected ideas has other unfortunate downsides. Consider the possibility that 1, or 2, maybe even 10 things that *Ancient Aliens* presents as PROOF for alien visitation in the past are ACTUALLY CORRECT. There is no way the average viewer of the show will be able to discern between these factually correct findings and the other 100+ things that have been thrown into this concocted mess of conjecture. And so, the solution is to simply reject the whole of the story as being nothing more than an entertaining idea, and not to be taken seriously. Therefore, any potential insight into the world is simply lost as no investigator, reporter, or investigative reporter will touch certain topics with a 50 ft pole.

Which is why skepticism and scientific investigation are the engines that need to revolutionize this way of thought. To the show’s credit, there have been several scientific tests conducted on certain episodes, but not nearly enough. One such test that comes to mind is an artifact that was discovered in an Egyptian tomb. Archeologists claimed that this artifact modeled the ancient’s representation of birds. Lead show investigator Giorgio A.

Tsoukalos however notes that not a single bird like feature is displayed on this small metallic trinket, and thus reasons that this artifact must instead be a model of modern-day aircraft. (Somehow). A replica of this trinket was produced at a larger scale and tested for flight characteristics.⁴ To basically no one's surprise, it was deemed aerodynamic. I'm not saying that the test was not done scientifically. I merely observe that with modern day understanding of flight, one can very accurately predict whether the model will confine to the structure of something that is aerodynamic even before the test begins. While testing this hypothesis is still critically important, it would be very easy to stage such a "scientific" test and embed the results within the show in effort to earn brownie points from the community of skeptics.

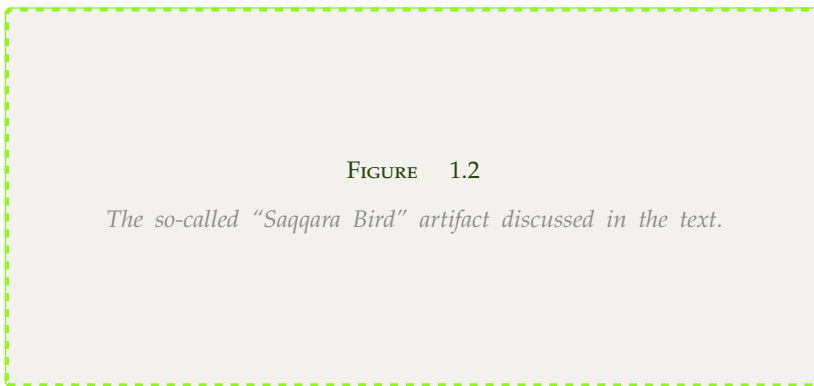


Figure 1.2. The so-called "Saqqara Bird" artifact discussed in the text.

Image sourced from:

https://www.bibliotecapleyades.net/imagenes_ciencia/avionescol02_09.jpg

Equally suspicious is the clear lack of any scientific tests conducted on the show that prove without a doubt something could have been possible WITHOUT alien intervention. I don't find the so-called airplane model found in the tomb to be particularly convincing evidence to support the hypothesis, but I must admit the show tries really hard to persuade otherwise. I think *Ancient Aliens* the show could otherwise benefit overall if they investigated and experimented with ideas that were perhaps not so easily predictable. And furthermore, if any tests were conducted on the show that clearly present flaws to their ideas, these tests should not be edited out. As even

⁴See the References section, entry 4.

the best ideas when investigated will be found to have many flaws. I will be making a more in-depth analysis of this show, and rather the ancient alien hypothesis itself, in Section 9.2 of this book, but for now I think I've said enough.

As for the methodology of skeptical thinking however, I believe that a balanced approach is needed. An approach that does not immediately reject certain ideas because they came from a crazy man or a crazy show. An approach however that does not automatically assume that solutions to all problems are as easy as a singular blanket statement. Only through unbiased investigation and unfiltered scientific study can we reach a potential conclusion of the mysterious. And remember the next time you encounter an episode or idea of the show *Ancient Aliens*: even a broken clock is correct twice a day.

1.5 CORRELATION VS CAUSATION

In the field of statistics there exists a concept often taught in any or all introductory classes to the subject. **The basis of this concept states that evidence for correlation is not the same as evidence for causation.** Take 2 or more things that can be linearly graphed or somehow strung together in a way that may appear scientific and unbiased, this however does not PROVE that these things are related. To illustrate this point, consider the graphs included below.

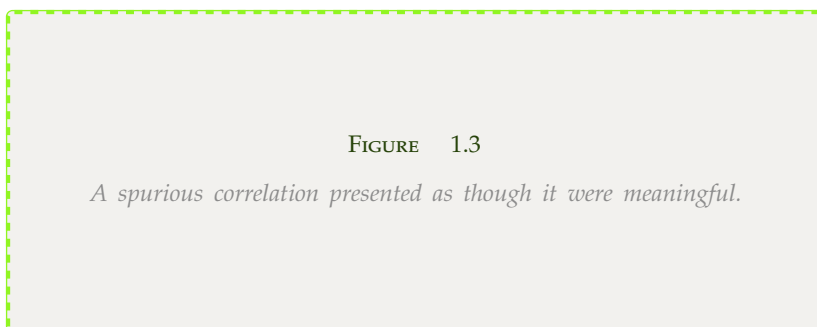


Figure 1.3. A spurious correlation presented as though it were meaningful.

Image sourced from: <http://www.daviddrury.com/wp-content/uploads/2013/01/Screen-Shot-2013-01-02-at-7.07.04-AM.png>

FIGURE 1.4

A second illustration of correlation mistaken for causation.

Figure 1.4. A second illustration of correlation mistaken for causation.

Image sourced from:

<https://wp-assets.futurism.com/2013/11/Correlation-versus-causation-1.png>

FIGURE 1.5

The number of pirates plotted against global average temperature.

Figure 1.5. The number of pirates plotted against global average temperature.

Image sourced from:

<http://blog.sciencegeekgirl.com/wp-content/uploads/2009/03/pirates.jpg>

Look at what those three graphs are actually doing. Each one plots two entirely unrelated quantities, scales the axes so that the wiggles line up, and lets your visual cortex do the rest. The third is my favourite, because it is a joke that was designed to be a joke: as the number of seafaring pirates declined over two centuries, global average temperature rose, and the fit is frankly better than a good deal of published social science. Nobody thinks pirates cooled the planet. The reason the graph is funny is that the graph is convincing, and those two facts sitting side by side should make you permanently suspicious of your own eyes. There are three ordinary explanations for any correlation before you reach for the interesting one — coincidence, in which two trends drift together by luck and the coincidence is guaranteed to occur if you test enough pairs; reversed direction, in which

B causes A rather than A causing B; and the lurking third variable, in which some unmeasured factor drives both. Industrialisation drives the pirates down and the temperature up, and it does not need to know that either graph exists. **Correlation tells you where to look. It never tells you what you have found.** In this field the lurking variable is usually media attention: sightings spike after a film, a broadcast, or a wave of coverage, so any graph of reports against anything else is really a graph of how much people were thinking about the subject that year.

The Face on Mars which was first photographed in 1976 by the Viking 1 orbiter was later proven to be false through the advent of technological development that allowed for improved photographs of the Cydonia region of Mars (see Figure 1.6). These improved photographs clearly showed that the face was indeed nothing more than a small mountain range when under the right conditions of shadows and lighting will appear as the face of man carved into the surface of the rock. Anyone who may have otherwise been on the fence regarding opinion on the subject, and those reporting that such a thing was real and proof of alien life in our solar system were then forced into the objective fact that it is not.

Consider however the implications that it was. A detailed face complete with moles, freckles and even acne scars so perfectly carved into the mountain that it would have had to been placed there by our space brothers and sisters for us to find. Would that really be the implication? Or would there still be a conventional way of explaining the feat? The perfect *correlation* between this hypothetical carving into the surface of a planet on average some 140 million miles away, does not lead to the conclusion that the *causation* is aliens. Similar to how 1000 monkeys typing at random on a keyboard for the duration of eternity will eventually write the complete works of Shakespeare. No logical person would reason that these monkeys have the literary capacity and creativity of grasping the profound influence of William Shakespeare. Neither should it be concluded upon immediate

FIGURE 1.6

The Viking 1 frame of 1976 beside the Mars Global Surveyor image of 2001 — the same landform, better resolution.

Figure 1.6. The Viking 1 frame of 1976 beside the Mars Global Surveyor image of 2001 — the same landform, better resolution.

Image sourced from: https://commons.wikimedia.org/wiki/File:Martian_face_viking_cropped.jpg

discovery that we are no longer alone in this universe.

However small and seemingly impossible the chance that random natural events can cause the illusion of perceived intelligence, it should not be completely ruled out as potential possibility. The possibility that asteroids impacting the surface, geological activity, or volcanic eruptions could have produced the image of a face on the surface of Mars is low; astronomically low, yet what are the chances that life in this one corner of the cosmos would have evolved into the very thing depicted on this rock, and yet it has.

Yet even the existence of our own intelligence starts to look less remarkable — less like the signature of a designer and more like what sufficiently many iterations of a simple rule will produce given time — when you consider just what is capable without even a brain. Conway's Game of Life is a perfect computer science example of what I mean. The rules of the game are simple. In a grid of arbitrary size there exists several pixels where any pixel that is on signifies being "alive" and all the pixels that are turned off represent "death". Rule 1: any live cell with fewer than 2 live neighbors will die in the next state of the simulation as result of "loneliness". Rule 2: any live cell with greater than 3 neighbors will die from "overpopulation". Rule 3: any dead cell with 3 alive neighbors will be birthed and thus become alive, and finally all other cells to which none of the mentioned rules apply will retain their current state.

Although the simulation was intended to be nothing more than a fun thought experiment, with no practical or philosophical intention as admitted by the creator himself: John Conway.⁵ This simulation proves to be quite interesting to watch. Over time the development of certain organisms can be observed. These organisms can be attributed basic characteristics and behaviors if you will. These behaviors can be used to further classify the predicted outcomes of the environment, and fans have even gone as far of naming and categorizing different pixel creatures of the simulation, and their perceived behaviors. These creatures have no brain they have no desires or purpose for life, they are nothing more than points of light on an illuminated computer screen. And yet there is something deeply intrinsic about their movement and characteristics that causes us to personify them and find relations to the real world for such beings.

This personification of computer code is what I find especially fascinating

⁵See the References section, entry 5.

since from a philosophical standpoint there is not much that distinguishes these pixels on a screen from something like a small bug for instance. We know from the scientists and people that study these bugs that they do have brains. These brains are very instinctual and limited only allowing such bugs to do basically just 2 things. Avoid death or being eaten and find food. From our view of the world we assume that these bugs are thinking, but what are the chances that the only thing existing in their tiny bug brain is a few lines of genetic code that causes us to perceive intelligence in their behavior? How big a leap would it be to next assume that we humans are following a larger set of preprogrammed rules that causes the illusion of behavioral patterns such as speech, music and literature? Well that's essentially what DNA is. The genetic code for everything that has ever lived on this planet.

I will be exploring in the next chapter in my subsection about AI the idea that illusions of intelligence can be caused by complete accident, or at the very least not explicitly outlined by the creator of such AI. All these are perfect examples of the concept that correlation does not equal causation. A bug can appear to do intelligent things even if the rules governing its existence have nothing explicitly to do with its behavior at all. Or in the case of computer code and AI deep neural networks, the same principal applies. When it comes to UFOs I will do my best through the rest of this book to ration out the cases where causation arguments could lead to false belief on the premises or causes to such incidents.

1.6 CONFIRMATION BIAS

Confirmation bias is the tendency to notice, remember, and give weight to the evidence that agrees with what you already believe, while quietly discarding everything that doesn't. It is not lying. That is the part people get wrong. The person doing it is entirely sincere, which is precisely what makes it the most dangerous failure mode in this entire field.

Here is how it works in practice. A man believes he saw a craft over his house. He goes looking, and he finds forty-one other reports from the same county that year. Forty-one! He does not go looking for the two hundred reports from that county that were resolved as aircraft, satellites, or Venus, because why would he? He already knows what he saw. The forty-one are

evidence. The two hundred are noise. He has not falsified a thing; he has simply built a case file out of the half of the data that agreed with him, and the case file looks devastating.

Now flip it, because this is the part the skeptic crowd never wants to hear. The debunker does exactly the same thing in the opposite direction. He collects the hoaxes, the drunks, the lens flares, the weather balloons, and he assembles them into an equally devastating file proving the whole subject is rubbish. He ignores the radar-visual cases with multiple independent witnesses and instrument corroboration, because those are inconvenient, and inconvenient data has a way of becoming invisible. Both men have confirmation bias. Only one of them gets invited onto television to explain how rational he is.

The mechanism has a few reliable symptoms, and you should learn to feel them in yourself:

- **Asymmetric scrutiny.** You interrogate the evidence against your position for hours and accept the evidence for it in seconds.
- **Moving the goalposts.** A test comes back against you, so you discover a flaw in the test that you did not mention before you knew the result.
- **Counting hits, forgetting misses.** The psychic got three things right. She got fifty things wrong. Nobody wrote those down.
- **Source laundering.** You trust a claim because it appeared in three places, without checking whether all three copied the same original.

That last one deserves a moment, because it is endemic in Ufology. A single unsourced sentence written in a paperback in 1978 gets quoted in a magazine in 1985, which gets cited by a website in 2001, which gets referenced in a documentary in 2014, and by the time it reaches you it has four independent-looking sources and zero actual evidence behind it. This is how the “fact” that the Roswell debris sprang back to shape, or that Betty Hill’s star map matched Zeta Reticuli, hardened into received wisdom. Trace every claim back to its first appearance. Every single time. Most of the time the trail ends somewhere embarrassing.

FUN FACT

Confirmation bias was demonstrated experimentally by Peter Wason in the 1960s using a task so simple it is almost insulting. Subjects were shown the number sequence 2–4–6 and asked to work out the rule behind it by proposing further sequences. The overwhelming majority proposed 8–10–12, then 20–22–24, then 100–102–104, confirming their guess of “add two” over and over. Hardly anyone tried 3–5–9, which is the only sort of move that could have shown them they were wrong. The actual rule was simply “any three increasing numbers”. People do not naturally test their beliefs. They audition for them.

The cure is not to have no beliefs, which is impossible, and anyone claiming to be free of bias is telling you something useful about their self-awareness rather than their methodology. The cure is procedural. Decide what would count as disconfirming evidence *before* you collect any. Write it down. Write down what result would make you abandon the hypothesis, and then go looking specifically for that. **If you cannot name anything that would change your mind, you are not investigating, you are worshipping.**

One more thing, and it applies to me as much as anyone reading this. The most comfortable position in Ufology is the middle. “Maybe some of it is real.” It feels balanced and reasonable, and it is also perfectly designed to survive any possible evidence, which should worry you. A position that cannot lose is a position that cannot learn. I hold mine loosely and I invite you to hold yours the same way.

1.7 DOUBT DRIVEN SKEPTICISM

There is a variety of skepticism that has nothing to do with inquiry. I call it doubt driven skepticism, and it is the counterfeit currency of this field. Real skepticism is a method: you doubt in order to find out. Doubt driven skepticism is a posture: you doubt in order to be finished. The two look identical from the outside and they produce opposite results, which is why so many people who consider themselves scientific are in fact doing nothing of the kind.

The tell is what happens after the doubt is expressed. A genuine skeptic says “I doubt that, and here is the test that would settle it.” The doubt driven skeptic says “I doubt that,” and then goes to lunch. One of these is

the opening move of an investigation. The other is a conclusion wearing an investigation's clothes.

Recall from Section 1.2 that the rejection of a claim is not coequal to the assertion of its counterpart. Doubt driven skepticism is what happens when a person forgets this. They begin by properly withholding assent from the claim that UFOs are real, which is correct procedure, and then somewhere along the way they quietly upgrade that withholding into the positive claim that UFOs are *not* real. That is a different claim. It has its own burden of proof, and nobody has met it, and almost nobody notices they have taken on the debt.

Here is the deeper problem, and it is the one I raised in Section 1.3: skepticism has no natural floor. There is no point at which doubt exhausts itself and says "well, I suppose that's settled." You can doubt the photograph. Failing that, doubt the photographer. Failing that, doubt the film stock, the developer, the chain of custody, the witness's eyesight, the witness's sobriety, the witness's motives, and failing all of that you can doubt your own existence and go home undefeated. Unlimited doubt is not rigour. It is a machine for never being wrong, and it costs nothing to operate.

FUN FACT

The original Greek *skeptikos* meant simply "inquirer" — from *skeptesthai*, to examine or consider. The word did not mean "one who disbelieves". It meant "one who looks". Somewhere in the last two thousand years we managed to invert the meaning of the word entirely, which given the contents of Section 1.1 should surprise nobody at all.

So how do you tell the difference in the wild? I use four questions, and they work on believers and debunkers alike:

1. **Is the doubt specific?** "That radar return is unreliable" is useful. "Radar can be unreliable" is a fortune cookie.
2. **Does the doubt cost anything?** A doubt that requires no work and risks no reputation is not a scientific contribution.
3. **Is it reciprocal?** Apply the same standard to the mundane explanation. If "weather balloon" is offered with no balloon, no launch record and no recovery, that is an unevidenced claim and it does not get a free pass for being boring.

4. **What would end the doubt?** If nothing would, the doubt is not about the evidence. It is about the doubter.

That third question is where most of the professional debunking of the last seventy years falls apart. Project Blue Book, which we will get to in Section 8.3, closed cases as “probable aircraft” and “possible balloon” with a frequency that would be laughed out of any other branch of government science. A probable is not a finding. It is a hypothesis that was never tested, filed as though it had been, and the sheer administrative confidence of the label did more to end inquiry than any amount of evidence ever could have.

None of which means the mundane explanation is usually wrong. It is usually right. That is the whole point of Occam’s razor and I stand by it, with the caveats from Section 1.2 firmly attached. The overwhelming majority of what gets reported is aircraft, satellites, balloons, Venus, and honest misperception by decent people who were not lying about anything. I would put the number well above ninety per cent and I would not fight you if you said ninety-five.

But ninety-five per cent is not one hundred per cent, and the entire subject lives in the remainder. The residue. The cases that survive competent investigation by people who were actively trying to explain them away and could not. That residue is small, unglamorous, and it is the only thing in Ufology worth a scientist’s afternoon. Doubt driven skepticism cannot see the residue, because it stops working the moment a plausible-sounding explanation appears, whether or not that explanation was ever checked.

So doubt everything, including this book. Especially this book. But doubt the way a mechanic doubts a strange noise — by opening the hood — and not the way a man doubts a knock at the door he would rather not answer.

I . B D E C O N S T R U C T I N G T H E P E E R

R E V I E W

This short final footnote of this chapter is inspired from a single episode of the podcast known as the Joe Rogan Experience and will deal entirely with the research of Dr. Peter Boghossian, Dr. James Lindsay, and Helen Pluckrose. Any information that is presented is sourced from the podcast

episode itself⁶ or from their website.⁷

While their research deals primarily with political science, better known as poli-sci and the newly formed area of gender studies, I draw many parallels with what they say in relation to the field of Ufology. Gender studies, like Ufology for instance is a subject that did not exist a mere 20 years ago. This is not entirely accurate as Ufology has its roots back to the early 50's but you get my point that both are fairly new as field of inquiry and if studied in the traditional mannerisms of science, it could be stated that gender is the most recent example of newly formed science. I know what you are thinking, and I hear those in the back. How could gender studies be a science, when its primary goal is pushing leftist ideology??

Well it would be wrong to outright dismiss an entire field of study based on initial impressions, and nothing more than the political atmosphere at the time. Gender studies for example deals a lot with actual science; mostly biology. It also utilizes aspects of psychology, and even a bit of neuroscience when it comes to the study of the human brain and social constructs. All these things are real science. However the field of gender studies has gained a certain reputation especially among other fields of study for being well how to phrase it . . . politically motivated.

But even politically motivated ventures of “science” have a certain degree of merit, if one can simply take 3 steps backwards to view the larger picture as a whole. The larger picture in this instance is not about gender at all. It is about what happens to any discipline whose conclusions are agreed upon before the research begins, and the reason I raise it here — at some risk of being misread — is that it produced the cleanest experimental demonstration of institutional failure we have. In 2018 three academics, Peter Boghossian, James Lindsay and Helen Pluckrose, ran what they called the grievance studies affair, and I want to be precise about what they proved and what they did not, because both halves matter to us.

What matters for our purposes is the mechanism the three of them exposed, because it is a mechanism and not a scandal. Between 2017 and 2018 they wrote twenty deliberately absurd papers and submitted them to respected journals in the field. Seven were accepted before the project was exposed. Not seven sloppy near-misses — seven formally peer reviewed, published contributions to the scholarly record, including one that was

⁶See the References section, entry 6.

⁷See the References section, entry 7.

recognised for excellence. The papers were nonsense by construction. They passed because they were the right kind of nonsense: correct vocabulary, correct citations, correct politics, correct posture. Peer review checked whether the papers sounded like the field. It never checked whether they were true.

And that is the whole lesson, so let me put it plainly. Peer review is not a truth detector. It never was. It is a small number of busy people, usually unpaid, usually anonymous, reading a manuscript for a few hours and deciding whether it looks like competent work of the sort their field produces. It catches obvious errors and missing controls. It does not replicate your experiment, audit your data, or visit your lab. It is a plausibility filter, and plausibility is exactly the thing a motivated author can manufacture.

Which cuts two ways for us, and I want to be very careful here, because this section is the easiest one in the book to misuse.

The first way is a warning against a bad habit in my own community. “Peer review is broken” has become the universal solvent of the fringe — a reason to dismiss any inconvenient finding while accepting anything published on a blog. That is doubt driven skepticism from Section 1.7 with a lab coat on. Boghossian, Lindsay and Pluckrose did not show that peer review is worthless. They showed it has a specific, diagnosable failure mode: it fails where a field’s conclusions are settled in advance and the review process is checking for conformity rather than for error. A journal of solid-state physics is not vulnerable in the same way, because the referee can check your numbers against a lattice constant that does not care about anyone’s politics.

The second way is the one that concerns Ufology directly, and it is the harder lesson. If a field can be captured by the assumptions of its practitioners, then this field is captured to the bone. Consider what our review process actually looks like. A case is investigated by an organisation whose members joined because they find the subject compelling. It is written up for a publication read almost exclusively by people who already accept the premise. It is refereed, when it is refereed at all, by colleagues who share the founding assumption. Then it is cited by the next investigator as established. That is not peer review. That is a field checking whether new work sounds like the work it already produced — which is precisely the failure the grievance studies papers walked through.

We do it in the other direction too. The debunking literature has its

own orthodoxies, its own approved vocabulary, its own conclusions known before the file is opened. A paper concluding “inconclusive, requires further instrumentation” is unpublishable in either camp. Both audiences want a verdict.

So what do we do about it? Not much, honestly, until the field grows up — but here is what I try to hold myself to, and what I would ask of you.

- **Prefer review from outside the tent.** An anomalous metal sample should go to a metallurgist who has never heard of Roswell and has no stake in the answer. If the finding is real it will survive a hostile lab. If it needs a friendly one, it is not a finding.
- **Publish the failures.** Section 1.4 faulted *Ancient Aliens* for editing out the tests that went badly. We do the same thing when we quietly drop the cases that resolved. The resolved cases are how you calibrate. They are the denominator, and a field with no denominator can prove anything.
- **Cite first appearances, not repetitions.** Section 1.6 covered this. Peer review does not catch source laundering; only you can.
- **State your priors in writing.** If a reader knows what you believed before you began, they can weight your conclusion appropriately. This is not a confession. It is data about the instrument.

I am aware of the irony of this section. Nothing in this book has been peer reviewed. I am a self-appointed professor of a field that does not grant degrees, publishing my lecture notes and asking you to take them seriously. So do not take them seriously — take them provisionally. Check the sources at the back. Trace the claims. Where I have reasoned past the edge of the evidence, say so out loud, because a visible hole in an argument is worth more than a seamless one you cannot inspect.

That is the end of the toolkit. Language distorts the question before you ask it. The burden of proof sits with whoever makes the claim, including me. Evidence is not proof, and proof may never come. One good idea does not license a hundred bad ones. Correlation is not causation. Your brain will curate the evidence on your behalf without asking permission. Doubt is a method and not a destination. And the institutions we trust to check all of this are checking something rather less than we assume.

Keep all seven of those within reach. Starting in the next chapter, we are going to need every one of them.

INTRODUCTION TO UFOLOGY

“Ufology is more than just flying saucers, and men who wear tin foil hats. It’s the study of claims and how one goes about discerning fact from fiction.”

— EVAN SVENDSEN

2.1 REDEFINING THE UNDEFINABLE

Ufology is the study of unidentified flying objects. That sentence takes about two seconds to say and roughly seventy years to argue about, because every single word in it is doing more work than it appears to.

Start with *unidentified*. This is a statement about the observer, not about the object. When I say a thing is unidentified, I am telling you something about the limits of our identification, and nothing whatsoever about what the thing is. This is the single most abused word in the field. The moment somebody says “it was a UFO,” and means by that “it was a spacecraft,” they have smuggled a conclusion into a description. Section 1.1 warned you about exactly this trick. The word describes a failure of our records, and failures of our records are extremely common and not especially exciting.

Then *flying*. Half the interesting reports involve objects that were not obviously flying in any aerodynamic sense — hovering with no visible lift mechanism, accelerating without a sonic boom, or entering water. “Flying” presumes a mode of travel we recognise, which rather undercuts the “unidentified” part.

And *object*. This is the boldest assumption of the three. It asserts that there is a material thing out there rather than an optical event, a plasma, a reflection, a radar artifact, or a perceptual phenomenon happening inside a skull. A great deal of what gets reported is real without being an object at all.

Which is why the terminology keeps mutating. The US government now prefers **UAP** — Unidentified Anomalous Phenomena, having earlier stood for Unidentified Aerial Phenomena. The change from “aerial” to “anomalous” was made specifically to accommodate transmedium and submerged reports. I have mixed feelings about the relabelling. On the one hand, “phenomena” is more honest than “object” and drops the assumption of a craft. On the other hand, “anomalous” quietly reintroduces a different assumption — that the thing is strange — when most reports resolve to something entirely ordinary.

FUN FACT

The term “flying saucer” is a journalistic accident. In June 1947 pilot Kenneth Arnold described nine objects near Mount Rainier as moving “like a saucer if you skip it across water”. He was describing the *motion*, not the shape — he said the objects themselves were crescent-shaped. A reporter compressed this into “flying saucers,” and within weeks the entire world began seeing discs. It is the cleanest demonstration in the field’s history that people frequently see what they have been told to expect.

So here is the working definition I use, and it is deliberately unglamorous. Ufology is the systematic study of reports of aerial and transmedium phenomena that competent investigation has failed to identify, together with the physical, historical, and psychological questions those reports raise. Note what that definition does not contain. It does not contain aliens. Extraterrestrial visitation is one hypothesis within Ufology, and by no means the most defensible one. A field defined by its preferred answer is not a field; it is an advocacy group with footnotes.

One last piece of vocabulary before we go on, because you will meet it constantly. An **IFO** is an identified flying object — a report that was investigated and resolved. IFOs are not failures. IFOs are the denominator. Any honest study of this subject spends the overwhelming majority of its time producing IFOs, and the small residue that resists explanation is the

only thing that has any claim on a scientist's attention.

2.2 THE ADVENT OF AI

I promised in Section 1.5 that I would come back to artificial intelligence, and here we are. The connection is not that computers will solve Ufology, though they may help. The connection is philosophical: AI is our best available laboratory for studying how intelligence can appear where nobody put it.

A deep neural network is, stripped of mystique, a very large pile of numbers that get adjusted until the output stops being wrong. Nobody programs the behaviour. Nobody writes a rule that says “cats have pointed ears”. You show the system several million examples, the weights settle, and something emerges that behaves as though it understands cats. Ask it why it decided a particular photograph contained a cat and it cannot tell you, because there is no reason in there — only a landscape of adjusted numbers that happens to slope in the right direction.

This matters to us for three reasons.

First, it destroys the intuition that intelligent-looking behaviour requires an intelligent designer. Recall the bugs and Conway's cells from Section 1.5. When we encounter a phenomenon that behaves purposefully — an object that appears to react to an approaching aircraft, say, or a light that seems to respond to a flashlight signal — our instinct is to infer a pilot. AI is a standing reminder that apparent purpose is cheap. Thermostats appear to want things. So do slime moulds, and immune systems, and markets.

Second, if we ever do detect an intelligence out there, there is no strong reason to expect it to be biological. A civilisation capable of crossing interstellar distances has almost certainly built machines better than itself, and machines do not require atmospheres, food, sleep, or a return trip. Which means the whole “occupant” folklore of this field — the greys, the reptilians, the tall Nordics — may be an artifact of our inability to imagine a probe. I will return to this in Chapter 6.

Third, and most practically: AI has become an evidentiary catastrophe. As I write this, any competent user can generate a photorealistic video of a metallic disc over their own neighbourhood in an afternoon, with correct lighting, correct motion blur, and correct camera shake. The entire

photographic corpus of Ufology — seventy years of contested stills and grainy footage — has just been retroactively devalued, because nothing that arrives from now on can be authenticated by inspection alone.

FUN FACT

This cuts the other way too, and almost nobody mentions it. If a genuine image were captured tomorrow, it would be indistinguishable from the fakes and therefore worthless as evidence. AI has not merely made faking easier. It has made proof by photograph structurally impossible. Whatever this field becomes, it will have to become instrument-based — radar, spectrometry, calibrated multi-sensor arrays — or it will become nothing at all.

2.3 THE MOST IMPORTANT IMAGE EVER TAKEN BY MANKIND

On Valentine's Day 1990, at the insistence of Carl Sagan and against the better judgement of several people at NASA, the Voyager 1 spacecraft was turned around to photograph the planet it had left thirteen years earlier. It was six billion kilometres away. The engineering team objected, reasonably, that pointing the camera back towards the Sun risked the instrument, and that there was no scientific value in the image. They were right about the second part. There was no scientific value in it at all.

The Earth occupies less than a single pixel. It sits in a band of scattered sunlight, an artifact of the optics, which is the only reason the image has any drama to it. You would not find our planet in that frame if you were not told where to look.

I call it the most important image ever taken because of what it does to the person looking at it, which is the opposite of what most people assume. It is usually invoked to induce humility, and it does, but the humility is not the point. The point is *scale calibration*, and this field desperately needs it.

Here is the calibration. Everything any human being has ever done happened inside that pixel. Every war, every religion, every extinction, every argument about Roswell. Voyager was, at that moment, still inside our own solar system — nowhere near the nearest star. The distance it had travelled in thirteen years is a rounding error against the distance to Proxima

Centauri. Whatever you believe about visitation, you have to believe it against that backdrop.

This works as a corrective in both directions, which is why I like it so much. For the believer, it is a splash of cold water on the assumption that crossing interstellar space is a matter of sufficiently advanced engineering. The distances are not large in the way a continent is large. They are large in a way the human nervous system has no equipment for. For the skeptic, it is an equally cold reminder that the volume of space we have meaningfully searched for other intelligence is, on this scale, roughly equivalent to sampling one glass of water from the Pacific and concluding there are no fish.

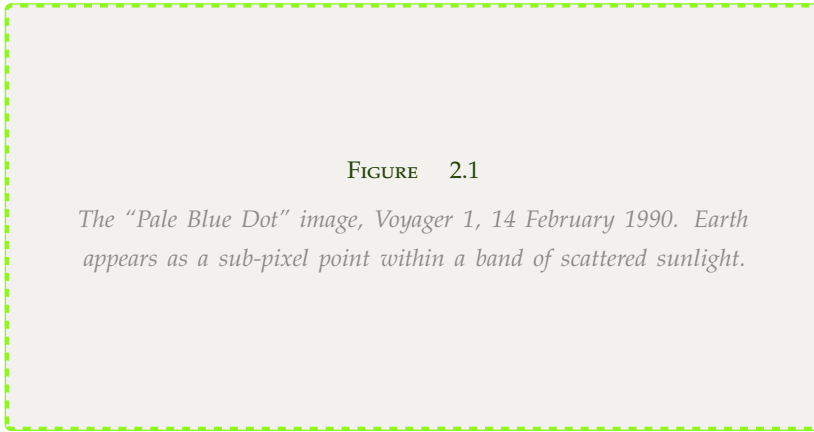


Figure 2.1. The “Pale Blue Dot” image, Voyager 1, 14 February 1990. Earth appears as a sub-pixel point within a band of scattered sunlight.

Image sourced from: <https://photojournal.jpl.nasa.gov/jpeg/PIA00452.jpg>

Sagan’s own commentary on the image is quoted so frequently that it has worn smooth, so I will not reproduce it here. Go and read it properly, in full, from the book it appeared in rather than from a poster. Then come back and read the rest of this book, and notice how often the arguments you encounter — mine included — quietly assume that our little pixel is the interesting part of the picture.

2.4 SETI

The Search for Extraterrestrial Intelligence is the respectable cousin of Ufology, and the relationship between the two is best described as mutually embarrassed. SETI researchers publish in refereed journals, hold university appointments, and would rather you did not mention flying saucers in their presence. Ufologists, in turn, regard SETI as a well-funded exercise in listening to the wrong thing in the wrong place with the wrong assumptions. Both positions contain some truth.

FIGURE 2.2

The 26-metre dish at Green Bank used for Project Ozma, the first modern SETI search.

Figure 2.2. The 26-metre dish at Green Bank used for Project Ozma, the first modern SETI search.

Image sourced from: https://commons.wikimedia.org/wiki/File:Tatel_Telescope.jpg

The basic method has barely changed since Frank Drake pointed a radio telescope at Tau Ceti and Epsilon Eridani in 1960 and called it Project Ozma (see Figure 2.2). You select target stars, you listen in the radio spectrum — traditionally near the 1420 MHz hydrogen line, on the reasoning that any physicist anywhere would recognise it as a natural landmark — and you look for a narrowband signal that nature does not produce. Narrowband is the key word. Natural sources are broad and messy. A signal squeezed into a few hertz is the signature of a transmitter.

Sixty-five years of this have produced exactly one candidate anybody still talks about. On 15 August 1977 the Big Ear telescope at Ohio State recorded a 72-second narrowband signal near the hydrogen line, close to the expected profile in every respect. Astronomer Jerry Ehman circled the printout and wrote “Wow!” beside it. It has never recurred, despite extensive re-observation of that patch of sky. One event, unrepeated, is not a detection. It is a story.

FUN FACT

SETI’s real problem is not sensitivity, it is arithmetic. Earth’s own radio leakage — the unintentional stuff, television and radar — is detectable at interstellar distances only with instruments far better than any we possess, and our leakage has been *decreasing* for decades as we move to fibre, satellites and low-power digital transmission. If our own civilisation is any guide, a

technological species is a bright radio source for perhaps a century and then goes quiet. We may be listening for a phase almost nobody spends any time in.

The deeper criticism, and the one I think lands, is that SETI encodes a specific and questionable hypothesis: that contact will arrive as a deliberate radio broadcast from a civilisation that wants to be found. That is a narrow bet. It excludes probes, it excludes optical and infrared signalling, it excludes technosignatures such as atmospheric industrial pollutants or waste heat, and it excludes the possibility that anybody nearby has already been here — which is precisely the possibility this book exists to examine.

To be fair, the field has broadened considerably. Optical SETI looks for nanosecond laser pulses. Breakthrough Listen, funded privately from 2015, has surveyed far more targets than all previous efforts combined. There is serious work on spectroscopic biosignatures in exoplanet atmospheres, which sidesteps intelligence altogether and simply asks whether anything is alive. That is a better question, and we will get to it.

I raise SETI here, this early, for a methodological reason rather than an astronomical one. SETI is what Ufology would look like if it had grown up properly. It states a hypothesis, specifies in advance what would count as a detection, publishes its null results, and reports its candidate events with the caveats attached. It has found nothing in sixty-five years and it says so plainly. Compare that to a field where a single unrepeatable event becomes a permanent article of faith. The problems with SETI are real, and I will take them apart properly in Section 18.7. Its methods, though, are not the problem. Its methods are the standard we have failed to meet.

2.5 THE UFOLOGIST

What, then, does a Ufologist actually do? Not what television suggests. There is very little night-vision footage and a great deal of paperwork.

The honest answer is that a Ufologist is an archivist who occasionally gets to do fieldwork. The bulk of the job is reading: case files, weather records, air traffic logs, satellite pass predictions, astronomical almanacs, and the previous forty years of contradictory literature about whichever

case you happen to be looking at. When a report comes in, the work is not to determine whether it was a spacecraft. The work is to establish, in order: what the witness actually experienced, what the sky actually contained at that time and place, and whether the first can be accounted for by the second.

That sequence matters and it is usually done backwards. Most investigators start with a hypothesis and go looking. The correct order is to reconstruct the observing conditions before you form any opinion at all — because once you have an opinion, Section 1.6 tells you exactly what will happen to your subsequent reading of the evidence.

There are, broadly, three species of person in this field, and you should learn to tell them apart at a distance.

The **advocate** has already reached a conclusion and is now assembling support for it. He is often knowledgeable, frequently charming, and completely useless as a source, because you cannot tell which of his facts survived scrutiny and which merely survived selection. The **debunker** is the same animal facing the other way, and is treated with more respect because his conclusion is the socially safe one. Then there is the **investigator**, who is boring. She writes up cases that resolved. She says “I don’t know” in print. She publishes the observing conditions even when they undermine her own earlier work. There are perhaps a few dozen of these people alive and most of you have never heard of any of them.

FUN FACT

The most consequential Ufologist in history did not believe in UFOs when he took the job. Dr. J. Allen Hynek was hired by the US Air Force in 1948 as an astronomical consultant specifically to explain sightings away, and he did so cheerfully for years — he is the man who coined the “swamp gas” explanation that got him mocked for the rest of his life. It was the sheer volume of residual cases crossing his desk that eventually turned him, not any single sighting. He went on to devise the close-encounter classification system we still use. A man who changed his mind after twenty years of evidence is worth a thousand who were certain from the start.

Which brings me to credentials, and to my own. There is no degree in Ufology, as I said in the Introduction. This is a genuine structural problem and not merely a personal inconvenience, because it means the field has no mechanism for excluding incompetence and no mechanism for accrediting

competence. Anybody can print a business card. I did. What substitutes for accreditation is your paper trail: the cases you have written up, the sources you have traced, and — most of all — the number of times you have publicly concluded that something exciting was in fact Venus.

2.6 UFOS AROUND THE WORLD

One of the more useful things you can do with a global dataset is look for the places where the reports *differ*, because that tells you which features belong to the phenomenon and which belong to us.

The American pattern is the one everybody knows: discs, then triangles, then a heavy emphasis on government concealment and on abduction. The abduction narrative in particular is startlingly American. It arrives in the early 1960s, it involves medical examination and missing time, and it maps almost too neatly onto the anxieties of a society with a new relationship to hospitals and to surveillance. We will take that apart properly in Chapter 7.

Brazil has the highest concentration of physical-effect and injury cases anywhere in the world. The 1977 events around Colares in Pará state involved dozens of villagers reporting burns and puncture wounds from beams of light, sufficiently many that the Brazilian Air Force ran a formal investigation — Operação Prato — whose files were later partly released. Whatever was happening at Colares, it left marks on people, which makes it a different category of evidence from a light in the sky.

Belgium produced the best-documented mass event in European history. Between November 1989 and April 1990, thousands of witnesses across Wallonia reported large silent triangular craft, and the Belgian Air Force cooperated openly with civilian investigators — an almost unique instance of a military organisation treating the subject as a legitimate question rather than a security embarrassment. The famous radar locks from the F-16 intercept of 30–31 March 1990 have since been substantially reinterpreted, and the single best-known photograph from the wave was admitted as a hoax by its author in 2011. That admission is why the Belgian wave is worth studying rather than why it should be dismissed: the field corrected itself in public.

France is the only nation that has maintained a continuous, publicly funded, publicly accessible investigation. GEIPAN, which sits inside the

national space agency CNES, has catalogued reports since 1977 and publishes its case files online with classifications from A (identified) through D (unexplained after full investigation). Roughly three per cent of its cases end in category D. That number — from a government agency with no stake in the answer and every institutional reason to prefer a tidy file — is the single most quotable statistic in this entire field.

FUN FACT

The cultural variation runs deeper than shapes. Japanese reports feature far fewer occupants and far more object-only sightings. Scandinavian cases skew heavily towards lights and “ghost rockets”, a wave of which crossed Sweden in 1946 — a full year before Kenneth Arnold. Chinese reports cluster around airports and are frequently resolved as drones or launch debris, partly because reporting channels are different. The phenomenon, whatever it is, appears to speak with a local accent. That is either telling you something about the phenomenon or something about the witnesses, and deciding which is most of the work.

What should you take from the international picture? Mainly this: the raw fact of sightings is global and continuous, going back long before 1947, and no serious person disputes it. What varies by culture is the interpretation — the shapes, the occupants, the motives, the cover-up. Section 1.1 told you that language shapes the question. The global record demonstrates that it also shapes the answer, which is why the only figures I trust in this field are the ones produced by people who published their category A cases with the same enthusiasm as their category D.

THE HISTORY OF AREA 51

“Three may keep a secret, if two of them are dead.”

— BENJAMIN FRANKLIN

3.1 SECURITY AND FREEDOM

The most remarkable thing about Area 51 is not that it kept secrets. It is that until 2013 the United States government would not concede that the place existed, while simultaneously prosecuting people for photographing it. You cannot trespass on nothing.

The facility sits on the southern shore of Groom Lake, Nevada, inside the vast Nevada Test and Training Range (see Figure 3.1). Its formal designation has changed repeatedly — Watertown, Paradise Ranch, Homey Airport, Detachment 3 of the Air Force Flight Test Center — and the name “Area 51” appears to derive from Atomic Energy Commission grid numbering. The existence of the base was officially acknowledged in June 2013 with the release of a redacted CIA history of the U-2 programme, and the acknowledgement was almost an afterthought: the document

FIGURE 3.1

Commercial satellite imagery of the Groom Lake facility: runways, hangars, and the dry lakebed.

Figure 3.1. Commercial satellite imagery of the Groom Lake facility: runways, hangars, and the dry lakebed. Image sourced from: https://commons.wikimedia.org/wiki/File:Area51_ikonos.jpg

simply used the name in passing, as though it had never been controversial.

Here is the part that should interest a skeptic more than any rumour of alien hardware. The legal machinery built to protect this base is genuinely extraordinary and entirely documented. The airspace above it, R-4808N, is closed absolutely — known to pilots as “the Box”. The perimeter is patrolled by contractors, universally called “camo dudes”, who are authorised to use deadly force and who are not federal officers. In 1984 the government seized an additional 89000 acres of public land, including Whitesides and Freedom Ridge, specifically to eliminate the last public vantage points — without the congressional hearing normally required. In the 1990s, when former workers sued over exposure to burning toxic waste, the government invoked the state secrets privilege to have the case dismissed, and President Clinton signed an annual exemption from environmental disclosure law that has been renewed by every president since.

FUN FACT

The workforce commutes daily by air. A fleet of unmarked Boeing 737s operates out of a private terminal at Las Vegas McCarran under the callsign “Janet” — origin of the name unknown, though “Joint Air Network for Employee Transportation” is a popular backformation. You can watch them take off from the airport’s public parking garage. A secret you can photograph on a schedule is a strange sort of secret.

I want to be precise about what this proves and what it does not. It proves the state will lie, at scale, for decades, about the mere existence of a place, and will restructure land law and environmental law to do it. That is established fact, not conjecture. It does not prove what the lie was concealing. The documented answer — reconnaissance and stealth aircraft — is entirely sufficient to explain the secrecy, and Section 1.2’s razor applies with full force here.

But notice the cost. Every genuine denial the government issued about Groom Lake was false, and each false denial purchased a little more public willingness to believe the next extraordinary claim. Institutional secrecy manufactures conspiracy theories as a by-product. You cannot spend fifty years teaching a population that official denials mean nothing and then act surprised when they stop believing the true ones.

3.2 THE HISTORY OF AVIATION

Section 12.1 gives you the timeline of flight — balloons to Blackbirds, and how brutally fast it happened. I am not going to repeat it here. What this section is about is a different history that runs alongside it and explains why a place like Groom Lake exists at all: the history of *where* aviation got developed, and how flight testing went from a public spectacle to a state secret in about forty years.

Because it started completely open. Early aviation was a performance industry. The Wrights demonstrated at Fort Myer and at Le Mans in front of crowds and newspapermen. Between 1909 and 1914 the fastest aircraft in the world were flown at ticketed air meets — Reims, Belmont Park, the Gordon Bennett Cup — and the Schneider Trophy seaplane races of the 1920s and 30s drove real engineering progress in engines and airframes while spectators watched from the beach. If you wanted to know the state of the art in 1925, you bought a programme. The Supermarine floatplanes that won the Schneider led more or less directly to the Spitfire, and everybody could see them coming.

Two things ended that. The first was the aircraft becoming a strategic weapon, which happened between the world wars, and the second was the aircraft becoming an *intelligence* weapon, which happened after 1945. Strategic weapons need secrecy about numbers and capability. Intelligence platforms need secrecy about existence.

The American institutional answer developed in stages. McCook Field in Dayton, Ohio, became the Army's engineering centre in 1917; it moved to Wright Field in 1927 and became Wright-Patterson, which is why Section 3.4 keeps coming up in this book — for decades, if the United States acquired a piece of foreign or unexplained hardware, Dayton is where it went to be taken apart. But Ohio has weather, terrain, population, and a fence a determined person can stand outside of. Flight test needs something else: enormous unobstructed emergency landing area, 300-plus clear-air days a year, no overflight of towns, and nobody within camera range.

So in 1942 the Army put its experimental flight-test operation at Muroc Dry Lake in the Mojave, on a playa so flat and hard that a stricken aircraft could put down anywhere for miles. Muroc became Edwards Air Force Base, and the X-plane era happened there: the X-1 breaking Mach 1 in 1947, the D-558, the X-15 to the edge of space in the early 1960s. Notice the

pattern that establishes — the desert dry lake as the natural habitat of the impossible aircraft. Groom Lake, when Kelly Johnson and Richard Bissell went looking for a U-2 site in April 1955, was chosen on exactly the same criteria as Muroc, plus one more: it already sat inside the Nevada Proving Grounds, which meant the airspace was closed and the security apparatus was already built and paid for by the nuclear programme. They did not have to invent secrecy. They inherited it.

And here is the consequence that matters for the rest of this book. From roughly 1955 onward there are two histories of aviation running in parallel. There is the public one in Section 12.1, where the fastest and highest thing flying is a known aircraft with a published service ceiling. And there is the classified one, in which the actual state of the art is fifteen to twenty-five years ahead of the published state of the art and is being flown at night over the American Southwest by people who will deny it on the record. The U-2 was operational three years before Gary Powers made it public by being shot down. The SR-71 was flying in 1964 and its true performance figures stayed classified for decades. The F-117 flew combat missions in Panama before the public knew its shape.

FUN FACT

The gap is not a conspiracy-theory estimate; you can measure it from declassification dates. HAVE BLUE first flew in December 1977 and was declassified in 1988 — eleven years. Tacit Blue flew 135 sorties between 1982 and 1985 and was not revealed until 1996 — fourteen years. The Boeing Bird of Prey flew from 1996 and surfaced in 2002. The RQ-170 Sentinel was photographed in Afghanistan years before it was admitted, and only because Iran captured one. So a reasonable working figure for the classified-to-public lag in American military aviation is somewhere between ten and twenty years. Whatever is flying out of Nevada tonight, you will read about it around 2045.

Which is why I insist that students of this subject learn aviation history properly, and not just the famous version. A person who knows what a Skyhook balloon looks like at 90,000 feet at sunset, or how a lenticular contrail forms, or what a Blackbird's plume looks like from directly beneath, has removed most of the mystery from their own sky. A person who does not know these things is reporting genuine observations with an empty reference library, and will reach for the only explanation they have ever

been given a picture of.

3.3 STEALTH AIRCRAFT

If you want to understand why the skies over Nevada were full of impossible objects in the 1950s, 60s, 70s and 80s, the answer is not complicated and it is not extraterrestrial. It is that the United States was flight-testing aircraft two full generations ahead of anything the public knew existed, in the same airspace, at night.

The sequence is worth learning because it is the single most powerful conventional explanation in this entire field. The U-2, first flown at Groom Lake in August 1955, cruised above 70,000 feet at a time when commercial airliners flew at 20,000 and military aircraft rarely exceeded 40,000. Its silver wings caught sunlight at altitudes where the ground was already dark. The CIA's own declassified history states plainly that U-2 flights accounted for a substantial share of UFO reports in the late 1950s, and that Blue Book investigators were fed deliberately false explanations — natural phenomena, weather inversions — to close the files.

Then came the A-12 and SR-71 Blackbird: titanium, Mach 3.2, 85,000 feet, an airframe that leaked fuel on the ground because it was designed to seal only once heated by friction. Then HAVE BLUE and the F-117 Nighthawk, first flown in 1981 and unacknowledged until 1988, whose faceted geometry produced a radar return the size of a ball bearing and whose silhouette against a night sky is a flat black triangle with no visible engines. Then Tacit Blue, an object so structurally absurd it was nicknamed the Whale.

FIGURE 3.2

The Lockheed F-117 Nighthawk. Flat, faceted, matte black, and unacknowledged for seven years after entering service.

Figure 3.2. The Lockheed F-117 Nighthawk. Flat, faceted, matte black, and

unacknowledged for seven years after entering service.

Image sourced from: <https://www.nationalmuseum.af.mil/Portals/7/images/photos/f-117.jpg>

Hold that F-117 silhouette in mind, because it matters enormously for Chapter 13 and for the Belgian and Phoenix triangle waves. A large silent flat-bottomed triangle with lights at the corners is not evidence of anything exotic. It is a description of hardware that demonstrably existed and was demonstrably being flown over populated areas by governments that denied its existence.

FUN FACT

The F-117's development was concealed by an elegant piece of misdirection: the Air Force did not deny that something was being flown at Groom Lake. It allowed the rumour to circulate that the secret project was a captured or reverse-engineered alien craft. A witness who reports a spacecraft is a witness nobody in Congress will take seriously. UFO folklore has, on at least one documented occasion, functioned as active counter intelligence.

Which forces an uncomfortable question on anyone honest. If the historical record shows that the strangest objects in the sky were ours, and that the secrecy surrounding them was total, then the null hypothesis for any given unexplained aerial object over a testing range is: it is a black programme. Not a probe. Not a visitor. A budget line nobody will admit to. That should be your starting position, and I hold it myself.

3.4 WRIGHT PATTERSON AIR FORCE BASE

Every mythology needs a warehouse, and for American Ufology the warehouse is Hangar 18 at Wright-Patterson Air Force Base near Dayton, Ohio.

The base is real and important, which is exactly why the story attached itself there. Wright Field was the technical intelligence centre of the US Air Force; the Air Technical Intelligence Center at Wright-Patterson ran Project Sign, Project Grudge and Project Blue Book in succession from 1948 to 1969. If any physical material from any American UFO case was ever analysed by the government, Wright-Patterson is the correct place to look. It was

the institutional destination for captured and unusual foreign hardware — German, Japanese, later Soviet. That is documented, mundane, and entirely sufficient to explain why debris of any kind would be shipped there.

The claim, which is not documented, is that a hangar at the base holds recovered extraterrestrial craft and bodies. The story attaches most persistently to Roswell material and to the alleged testimony of Senator Barry Goldwater, who stated in correspondence that he had asked General Curtis LeMay for permission to enter a room at Wright-Patterson known as the “Blue Room” and was refused in extremely forceful terms. Goldwater’s account is genuine — the letters exist and he repeated the story publicly — but note carefully what it establishes: that a US senator was denied access to a secure room. That is not evidence about the room’s contents. It is evidence about compartmented clearance, which functions exactly that way by design.

As for “Hangar 18” specifically: the base has no building with that designation, and the number appears to originate in 1970s paperback literature and a 1980 film of that title. This is the Section 1.6 laundering problem in its purest form. Trace the term and the trail ends at a novel.

What *is* solidly documented at Wright-Patterson is duller and more damning. The Blue Book files were kept there. Their organisation was poor, their staffing was minimal — for long stretches a single officer and a secretary — and case dispositions were frequently recorded with no investigation whatsoever. If you want to be angry about Wright-Patterson, be angry about that. The scandal is not a hidden hangar. The scandal is an open filing cabinet full of unexamined reports.

3.5 S-4 AND BOB LAZAR

FIGURE 3.3

*The “sport model” disc as described
and sketched by Bob Lazar.*

In May 1989, a man appeared in silhouette on Las Vegas television station KLAS under the pseudonym “Dennis” and told reporter George Knapp that he had been employed as a physicist at a facility called S-4, at Papoose Lake south of Groom Lake, reverse engineering the propulsion system of an extraterrestrial craft. He went

Figure 3.3. The “sport model” disc as described and sketched by Bob Lazar.
Image sourced from: https://commons.wikimedia.org/wiki/File:Lazar_sport_model_diagram.svg

public under his own name later that year. His name is Robert Lazar, and the modern Area 51 mythology is very largely his creation.

His claims are specific, which is to his credit — specificity is falsifiable. He described nine craft (see Figure 3.3) in hangars built into the base of a mountain with sloped doors. He described a reactor running on a stable superheavy element he called Element 115, which produced antimatter on proton bombardment and generated a gravitational field that could be amplified and directed to warp spacetime around the craft. He described being briefed with documents concerning long-term extraterrestrial involvement with Earth. He described seeing a craft flight-tested.

Now let us do the work, because this case is an excellent exercise.

What holds up: Lazar's knowledge of the Nevada terrain and of the general geography of test operations was good. He led Knapp and others to a location from which they observed unusual aerial activity on Wednesday evenings, which he had predicted in advance. And element 115 — which did not exist in 1989 — was subsequently synthesised in 2003 and named moscovium in 2016. Lazar's supporters treat this as a bullseye.

What does not hold up is more substantial. Moscovium is not a stable superheavy element; its longest-lived known isotope has a half-life measured in tenths of a second, and it cannot be a fuel for anything. Predicting that element 115 would eventually be made was not prescience — the periodic table has numbered blanks, and physicists had been discussing an "island of stability" in the popular press for years. More seriously, his claimed education has never been substantiated. He states he holds degrees from MIT and Caltech; neither institution has any record of him, no classmate or instructor has ever come forward, and the employment history he has offered is documented only in fragments that do not require any physics credential. He has a 1990 conviction connected to a prostitution business, which is not relevant to physics but is relevant to the claim that he was a cleared contractor.

FUN FACT

The most durable part of Lazar's account is the one nobody argues about: he put the phrase "Area 51" into general public circulation. Before the

1989 KLAS broadcasts the name was known almost exclusively to aviation specialists. Whatever else Bob Lazar did or did not do, he named the most famous secret in the world.

My own assessment, offered as an assessment and not a finding: the claims are unsupported, the credentials appear to be fabricated, and the physics is wrong in a specific and checkable way. That is enough for me to set the case aside. I say this while noting, as Section 1.7 requires, that “he cannot document his degrees” does not prove he was never at Papoose Lake, and that a government which lied about the existence of Groom Lake for fifty years is not a reliable witness against him. The case is not closed by official denial. It is closed by moscovium.

3.6 CULTURAL IMPACT OF AREA 51

Area 51 has become the standard-issue backdrop for secrecy itself, and the trajectory is worth tracing because it shows how a place becomes a symbol and then stops being a place.

The 1989 Lazar broadcasts gave it a name. *Independence Day* in 1996 gave it a location in the popular imagination, complete with an underground hangar — a set, not a photograph, though a startling number of people cannot now tell you which. The Nevada state legislature renamed State Route 375 the “Extraterrestrial Highway” in 1996 as a tourism measure, and the Little A’Le’Inn at Rachel became a genuine economic beneficiary of a rumour. The base now appears in dozens of video games as a level.

Then, in June 2019, a college student created a Facebook event titled “Storm Area 51, They Can’t Stop All of Us” as a joke. Two million people clicked attending. The Air Force issued a formal statement warning that the range was a live training area and that attempts to enter would be met with force. Two small festivals were hastily organised in Rachel and Hiko; roughly 150 people turned up at the gates, six were arrested, and a remote Nevada county came genuinely close to a humanitarian problem because a meme outgrew its punchline.

Around the base a whole pilgrimage geography grew up, and I want to name its stations because they are the physical residue of the cultural

story. Nevada State Route 375 carries roughly two hundred vehicles a day, which makes it one of the least-used numbered highways in the country and an improbable candidate for a tourist attraction. The Little A'Le'Inn opened at Rachel in 1989 — the same year as the Lazar broadcasts, which is not a coincidence — and later turned up in the film *Paul*. Twelve miles from the Groom Lake road turnoff stood the Black Mailbox, the property marker of a rancher named Steve Medlin and the only landmark of any kind within fifty miles. It became a gathering point for night-time skywatchers, and Medlin eventually replaced it with a white one in the hope that people would stop finding it; they did not, and he ultimately listed the property for sale. Before the 1984 and 1995 land seizures, an amateur group known as the Skywatchers Association camped on Whitesides and Freedom Ridge, watched closely by base security from the other side of the line. Since those vantage points were taken, the only remaining spot from which the base is visible to the naked eye is Tikaboo Peak, twenty-six miles out — roughly the distance from Lethbridge to Fort Macleod, if that helps you picture it — and reaching it is a serious hike with a telescope on your back. In 2012 the National Atomic Testing Museum in Las Vegas, a Smithsonian affiliate, opened an Area 51 exhibit, which is a small institutional milestone worth noting: the artefacts of the rumour are now curated inside a federally-affiliated museum, a year before the CIA admitted the base existed.

I find the whole arc instructive rather than annoying. Area 51 in 2026 is not primarily a Ufological subject. It is a case study in how official silence becomes a canvas. The government's refusal to acknowledge the base did not protect the secret — the aircraft were all eventually revealed, on the government's own schedule, with fanfare. What the silence produced was a vacuum, and vacuums get filled by whoever is willing to speak. In this instance that was a man with unverifiable degrees and a television reporter with an hour to fill, and their version has outlasted every declassification since.

There is a lesson in there for anyone in a position of institutional authority who imagines that saying nothing is the same as saying nothing happened. It is not. **Saying nothing is an invitation, and somebody always accepts.**

UFO CSI

“Unlike scientology, science in the true sense of the word is open to unbiased investigation of any existing phenomena.”

— STANISLAV GROF

4.1 MUFON

The Mutual UFO Network was founded in 1969 — as the Midwest UFO Network, in Illinois — in direct response to the closure of Project Blue Book. The logic was straightforward: if the government has stopped collecting reports, somebody had better keep collecting them. More than fifty years later MUFON operates the largest civilian UFO case database in the world, with a public reporting form, a network of trained volunteer field investigators, a monthly journal, and state and national directors.

I cite MUFON material in this book, including in Chapter 1, and I want to be clear about why and with what reservations.

What MUFON does well is scale and continuity. Roughly 500 to 1,000 reports arrive per month. The organisation runs a Field Investigator certification with an actual manual and an actual examination, which is more procedural rigour than any other civilian body in the field has managed to sustain. Cases are assigned dispositions, and — this is the part that matters — a large majority are closed as identified or as insufficient data. That denominator is precisely what Section 1.7 says a serious field must

publish.

What MUFON does badly is govern itself. It has been through repeated internal crises: resignations of senior figures over the leadership's handling of allegations, controversy over funding relationships with wealthy private donors, a 2013 episode in which a state director's public statements caused a mass resignation of investigators, and persistent friction between the organisation's scientific ambitions and its reliance on paid conference appearances by people whose claims its own investigators would fail. The journal's editorial standard varies wildly from issue to issue. Data quality is self-reported and unaudited.

FUN FACT

MUFON's greatest scientific value may be entirely accidental. Because it has collected reports continuously since 1969 using a broadly stable form, its database is one of the few long-baseline datasets in existence — which makes it useful for studying *reporting behaviour* even if every individual case were worthless. You can see the Belgian and Phoenix triangle waves in it. You can see the arrival of camera phones. You can see Chinese lanterns appear as a report category around 2005 and then fade. That is real data about real human beings, whatever else is going on.

The practical advice is this. Use MUFON's database as an index, not as a source. A case number tells you where to start looking; it does not tell you that anybody competent has looked already. And apply the same standard to the two other bodies you will encounter: NUFORC, the National UFO Reporting Center, which is essentially a raw unfiltered feed and should be treated as one, and the now-defunct CUFOs founded by Hynek, whose case work was the best the field has ever produced and whose archive is criminally underused.

4.2 CLASSIFICATION OF COMMON UFO SHAPES

Before you can analyse reports you need a vocabulary that does not smuggle in conclusions. The standard shape taxonomy is imperfect but it is what everybody uses, so learn it.

Discs. The classic saucer. Dominant from 1947 until roughly 1980 and now relatively rare in raw reports, which is itself informative — if the phenomenon were independent of culture you would not expect its shape to go out of fashion.

Triangles. Large, dark, silent, lights at the corners or vertices, slow. Dominant from the mid-1980s onward. See Chapter 3 on the F-117 and Tacit Blue, and note the timing.

Spheres and orbs. Now the most commonly reported category by a wide margin, and the least useful, because a distant unstructured light is compatible with almost everything: aircraft landing lights head-on, Venus, Starlink trains, drones, lanterns, balloons, reflections.

Cigars and cylinders. Elongated, no wings, historically often reported as “airships”. Prominent in the 1896–97 American airship wave, decades before powered flight was public.

Tic-tacs. A recent addition, from the 2004 Nimitz encounter footage: a smooth white oblong with no visible control surfaces, wings, or exhaust.

Chevrons, boomerangs, and V-shapes. The Phoenix Lights of 13 March 1997 are the canonical case, and they are also two separate events conflated into one — an early formation of lights, plausibly aircraft, and a later stationary arc of lights over the Estrella range which the Maryland Air National Guard identified as A-10 flare drops. The second identification is solid. The first is not settled.

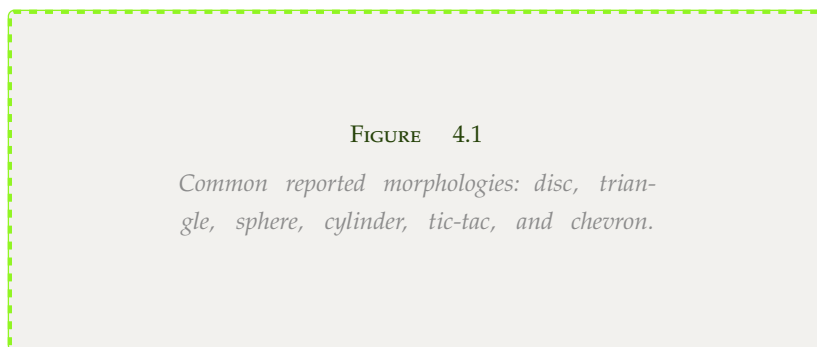


Figure 4.1. Common reported morphologies: disc, triangle, sphere, cylinder, tic-tac, and chevron.

Image sourced from: <https://www.nicap.org/images/shapes.jpg>

Alongside shape you should record the observational parameters that

actually constrain anything: angular size (measured against a thumbnail at arm's length, not guessed in feet), duration, bearing and elevation, apparent motion relative to fixed references, sound, and whether the object was ever occluded by cloud or terrain. **Distance and size estimates from witnesses are worthless in isolation, because the human visual system cannot judge either without a known reference.** A witness who says "it was the size of a football field" is telling you about their impression, not about the object. A witness who says "it was about three finger-widths across and passed behind that water tower" has handed you geometry.

FUN FACT

Hynek's close-encounter scale, which everyone half-remembers from the Spielberg title, is about *proximity and effect*, not shape. CE1: observed within about 500 feet with no physical interaction. CE2: physical traces — scorching, vehicle interference, animal reaction. CE3: occupants observed. CE4 (added later, not by Hynek) : abduction. CE5 (added later still, and not accepted by serious investigators): deliberate human-initiated contact. The scale degrades in reliability at exactly the rate it gains in excitement.

4.3 PHOTOGRAPHIC ANALYSIS

I told you in Section 2.2 that photographic evidence is finished as a category. It is, for anything captured from now on. But the historical corpus still needs analysing, and the techniques are worth knowing because they generalise to any image claim.

Start with the metadata, and start there because it is where most hoaxes die. EXIF data gives you camera model, timestamp, focal length, exposure, and often GPS. Cross-check the claimed time and place against sun and moon position, weather records, and aircraft tracking archives. A great many cases resolve in ten minutes at this stage, and the most common resolution is that the timestamp is wrong and the object is the Moon or Venus.

Then the physics of the image itself. Four questions do most of the work.

Is the object in focus, and does its focus match the background? A nearby small object photographed with a background-focused lens will show characteristic blur. An object sharper than the landscape behind it was not

at the distance claimed.

Does the lighting match? This is the single most reliable discriminator for both physical models and digital composites. The direction, colour temperature, and hardness of light on the object must match the scene. Shadow direction on the object must be consistent with shadows on the ground.

Is there motion blur, and is it consistent? An object claimed to be moving at high speed in a 1/60-second exposure should be smeared. A crisp object in a frame where the tree line shows camera shake was moving with the camera — which means it was attached to it, or very close.

What is the angular size and what does it imply? With focal length, sensor size, and pixel measurement you can compute the object's angular size precisely. Angular size plus any distance assumption gives physical size. This is how the Trent/McMinnville photographs of 1950 (see Figure 4.2) — still among the best-regarded in the field — have been argued over for seventy years: the debate turns entirely on whether brightness measurements of the underside indicate a distant large object or a nearby small one hung from the overhead wires that were visible in the original negatives and cropped out of most reproductions.

FIGURE 4.2

One of the two Trent photographs, McMinnville, Oregon, 11 May 1950. Note the overhead wires at the top of the uncropped frame.

Figure 4.2. One of the two Trent photographs, McMinnville, Oregon, 11 May 1950. Note the overhead wires at the top of the uncropped frame.

Image sourced from: <https://commons.wikimedia.org/wiki/File:McMinnvilleUFO.jpg>

FUN FACT

The most famous UFO photographs almost all have a cropping problem. The McMinnville images originally showed power lines. The 1967 Alex Birch photograph was admitted a hoax by its teenage author. The 1990 Petit-Rechain Belgian triangle photograph — reproduced in hundreds of books as the best triangle image in existence — was confessed as a fake in 2011 by the man who made it, using polystyrene and a lamp. In each instance the field had spent decades analysing the pixels of an image whose provenance nobody had audited. Chain of custody beats image processing every time.

For video, add frame-by-frame examination for compression artifacts, check whether apparent object motion correlates with camera motion, and be extremely careful with footage from infrared and gimballed targeting systems. The 2004 and 2015 US Navy videos are genuine recordings of genuine sensor events, and much of what looks anomalous in them — the sudden “rotation” of the Gimbal object, the apparent zip of the GoFast object across the water — has plausible explanations in the behaviour of the gimbal mechanism and in parallax. “Plausible” is not “established”. Neither is “anomalous”. The honest reading of those videos is that the sensor recorded something the operators could not identify, and that the recording alone cannot resolve size, distance, or speed. Everything beyond that is inference, including the exciting parts.

4.4 INVESTIGATIVE TECHNIQUES

Here is the procedure, in order, and the order is the whole point. Do not skip forward because a case looks interesting.

1. Secure the account before you contaminate it. Get the witness to write or record their narrative before you ask a single specific question, and before they have read anything about similar cases. Every question you ask plants something. “Did you see any windows?” produces windows. This is not a hypothetical risk; it is the documented mechanism behind most of the abduction literature, which we will get to in Chapter 7. Use open prompts only: “What happened next?”

2. Reconstruct the sky. Before forming any opinion, establish what was actually up there. Astronomical software for planetary and stellar positions, ISS and satellite pass predictions, Starlink train schedules, aircraft tracking archives, airport and military notices to airmen, weather records including cloud base and inversion layers, and the location of any nearby launch or test range. Do this for the exact stated time, and then do it again for an hour either side, because witnesses are unreliable about time.

3. Interview separately. Multiple witnesses must be interviewed apart and before they confer. Convergence between independently obtained accounts is meaningful. **Convergence after they have talked to each other is nothing at all.**

4. Go to the site. Photograph the witness’s sightline from their stated

position with a known focal length so angular sizes can be recovered later. Measure bearings with a compass, correcting for magnetic declination. Note every light source, tower, reflective surface, and flight path. An astonishing number of cases resolve when you stand where the witness stood and look where they looked.

5. Collect physical evidence properly or not at all. Soil samples from claimed landing sites need controls — same depth, same soil type, from outside the affected area, collected at the same time. **Without controls a laboratory result means nothing whatsoever.** Metal fragments go to an independent laboratory with no knowledge of the case's origin, as Section 1.8 argued. Photograph everything in place, with scale, before touching it.

6. Write the negative result. If the case resolves, publish it as resolved, with the same care you would have given a positive. This is the step everybody skips and it is the step that makes the difference between a field and a hobby.

One technique deserves an explicit warning: do not use hypnosis, and do not accept evidence derived from it. I will make the full case in Section 7.1, but the summary is that hypnosis does not recover memory, it manufactures confident memory, and once a witness has been hypnotised their original account is permanently compromised. An investigator who hypnotises a witness has destroyed their own evidence.

Finally, a note on temperament. Most of the people who report these things are decent, sober, and frequently reluctant — many are afraid of being mocked, and quite a few have something professional to lose. Treating a witness with respect is not a courtesy, it is methodology: a witness who feels judged will either shut down or start performing, and neither gives you usable data. You can tell someone their sighting was Venus without telling them they are stupid. I have had to do it more times than I can count, and the ones who thank you for it are the reason this is worth doing.

THE PHYSICS OF TIME TRAVEL

“Originally, the burden of proof was on physicists to prove that time travel was possible. Now the burden of proof is on them to prove there must be a law forbidding time travel.”

— DR. MICHIO KAKU

5.1 THE PHYSICS OF TIME TRAVEL

Time travel into the future is not controversial. You are doing it now, at one second per second, and if you would like to do it faster you need only move quickly or sit somewhere with a stronger gravitational field. This is not speculation. It is measured daily, and your satellite navigation would drift by several kilometres a day if the correction were not applied.

Special relativity gives you the first mechanism. Time passes more slowly for a moving observer relative to a stationary one; at 87% of light speed, clocks run at half rate. A crew travelling at that velocity for what they experience as ten years returns to find twenty years elapsed at home. That is forward time travel, it requires no exotic physics, and its only obstacle is energy. General relativity gives you the second: clocks run slower in stronger gravity. Cosmonaut Sergei Krikalev, having accumulated 803 days in orbit, is the most time-displaced human being alive — by roughly 0.02

seconds. Do not spend it all at once.

The past is a completely different problem, and here is the crucial distinction that most popular accounts blur: travel to the past is not obviously forbidden by the equations, and that is precisely what worries physicists.

General relativity permits solutions containing **closedtimelikecurves** — paths through spacetime that return to their own starting point in time as well as space. Kurt Gödel found one in 1949, in a rotating universe, and presented it to Einstein as a birthday present, which tells you something about mathematicians. Others followed: the interior of a rotating black hole; the infinite rotating cylinder of van Stockum and Tipler; the two cosmic strings of Gott; and the wormhole of Morris, Thorne and Yurtsever, which remains the most-discussed because it is the least absurd.

FIGURE 5.1

Embedding diagram of a traversable wormhole: two mouths joined by a throat that exotic matter would have to hold open.

Figure 5.1. Embedding diagram of a traversable wormhole: two mouths joined by a throat that exotic matter would have to hold open.

Image sourced from: <https://commons.wikimedia.org/wiki/File:LorentzianWormhole.jpg>

A traversable wormhole is a topological shortcut — two mouths connected by a throat (see Figure 5.1). Move one mouth at relativistic speed and it becomes desynchronised from the other by ordinary time dilation. Step in one end and out the other and you have moved in time. The difficulty is that keeping such a throat open requires **exotic matter** with negative energy density, in quantities that make “vast” an understatement, and while negative energy density does exist in tiny amounts as a quantum effect — the Casimir effect is measured — nobody has any idea how to accumulate it.

FUN FACT

The Alcubierre drive, endlessly cited in this field as “NASA’s warp drive”, is a genuine 1994 paper by a genuine physicist, and its point was arguably the opposite of what enthusiasts take from it. Miguel Alcubierre showed you can write down a metric that contracts space ahead of a ship and expands it behind, moving the ship without local acceleration. He also noted that it requires negative energy, that the original formulation needed a mass-energy budget comparable to the visible universe, and that the interior would be causally disconnected from the exterior — meaning you could not steer. Subsequent work has reduced the energy requirement substantially, from

“impossible” to “still impossible”. The paper is a demonstration that general relativity is permissive, not a blueprint.

Then there are the objections, and they are serious. The **grandfather paradox** is the famous one, and physics has produced two respectable answers. The Novikov self-consistency principle holds that only self-consistent histories have non-zero probability — you can go back, but you will find that you always were part of the past, and every attempt to create a contradiction fails for boring local reasons. The alternative, which we will meet in Chapter 11, is that the timeline branches, in which case you have not changed your own past at all; you have merely visited someone else’s.

Hawking’s chronology protection conjecture is the sharper objection. It proposes that quantum effects always intervene to destroy a closed timelike curve at the moment of its formation — vacuum fluctuations circulate around the loop, amplifying without limit, and the energy density blows up and destroys the throat. It is a conjecture, not a theorem, because proving it requires a quantum theory of gravity that we do not have. Hawking’s own empirical argument was characteristically brisk: we are not overrun by tourists from the future.

I include this chapter, and put it early, because time travel is the most common exotic explanation offered in Ufology after extraterrestrial visitation — the “ultraterrestrial” or future-human hypothesis. It has one genuine advantage over the interstellar hypothesis: it solves the distance problem, since a time traveller need not cross space at all. It has one crippling disadvantage: every mechanism we can write down requires physics we cannot perform, and several of them may be forbidden outright. Keep both facts in mind as you read the cases that follow. Some of them are interesting. None of them require this chapter to be true.

S . 2 R E N D L E S H A M F O R E S T

Over three nights in late December 1980, in a pine forest in Suffolk, England, between two active NATO air bases holding nuclear weapons, a series of events occurred that produced the best-documented military UFO case in European history. It is frequently called Britain’s Roswell, which is unfair to

Rendlesham — the documentation here is far better.

The core facts are not in dispute. In the early hours of 26 December, security personnel at the east gate of RAF Woodbridge observed lights descending into Rendlesham Forest. Three men — Staff Sergeant Jim Peniston, Airman First Class John Burroughs, and Airman Edward Cabansag — were sent to investigate on foot, initially believing an aircraft had crashed. Two nights later, on 28 December, the deputy base commander of RAF Bentwaters, Lieutenant Colonel Charles Halt, led a team into the forest with a Geiger counter and a cassette recorder. The recording exists. So does the memorandum Halt wrote on 13 January 1981, titled “Unexplained Lights”, which was released under the US Freedom of Information Act in 1983.

The Halt memo is the case’s spine, so read what it actually says. It describes a glowing object, metallic in appearance, triangular, roughly two to three metres across the base, illuminating the forest with a white light and having a red pulsing light on top and a bank of blue lights underneath. It describes the object manoeuvring through the trees and departing. It describes three depressions found the following day in a triangular pattern, and beta/gamma readings at those depressions of 0.07 milliroentgens per hour, noted as ten times background. It describes, on the third night, a red sun-like object which appeared to throw off molten metal and then split into five white objects.

That is an extraordinary document written by a serving lieutenant colonel, and its authenticity is not in question. Now the conventional case, which is also strong.

The forest lies about eight kilometres from Orford Ness lighthouse, whose beam sweeps at a low angle across exactly that terrain (see Figure 5.2), and whose flash rate matches the pulsing witnesses described. Halt’s own recording captures the team discussing the lighthouse and then dismissing it. Astronomer Ian Ridpath established that the brilliant object low in the sky on the third night was the star Sirius, and the “molten metal” and splitting behaviour is consistent with atmospheric scintillation and with the other

FIGURE 5.2

Orford Ness lighthouse, about eight kilometres from Rendlesham Forest, sweeping the treeline at a low angle.

Figure 5.2. Orford Ness lighthouse, about eight kilometres from Rendlesham Forest, sweeping the treeline at a low angle.

Image sourced from: https://commons.wikimedia.org/wiki/File:Orfordness_Lighthouse.jpg

objects reported — which were at bearings matching Deneb, Vega and other bright stars. The radiation readings, on examination of the instrument's scale, fall within the normal background range for that area; the “ten times background” assessment came from an operator unfamiliar with the instrument. The three depressions were identified by a local forester as rabbit diggings, and there was a well-known local phenomenon of a farmer's tractor and of poachers in the forest that night.

FUN FACT

The Ministry of Defence's released files show that the case was assessed as being of no defence significance — and also show that nobody in the MoD conducted anything resembling an investigation before reaching that conclusion. Both the believers and the debunkers of Rendlesham have relied on the same absence: the government neither confirmed nor examined anything. The 2001 release of the “Rendlesham file” was greeted as vindication by both sides simultaneously, which is usually a sign that the file contains very little.

Then there is the part that belongs in this chapter. Jim Penniston has stated — not in 1980, but in accounts developed from the 1990s onward, some under hypnosis — that he touched the craft, that its surface was warm and covered with symbols resembling Egyptian hieroglyphs, and that he received a download of binary code. That claim, and what came of it, is the subject of the next three sections.

My position on Rendlesham: something happened in that forest, several trained men were frightened by it, and Halt's contemporaneous recording is the closest thing this field has to an honest primary document. The lighthouse and stellar explanations account for the majority of the reported observations and cannot be waved away. The residue — and there is a residue, chiefly the close-range 26 December encounter — is contaminated beyond repair by thirty years of hypnosis, book deals, and mutual contamination between witnesses who have been in the same room at the same conferences for decades. This is the tragedy of the case, and Section 4.4 told you exactly how it happened.

5.3 UNDERSTANDING BINARY

To evaluate what comes next you need to understand binary, and it takes about two minutes, so let us do it properly.

Our ordinary numbers are base ten: each position is worth ten times the one to its right, because we have ten fingers and no better reason. Binary is base two: each position is worth twice the one to its right, and only the digits 0 and 1 exist. The place values are 1, 2, 4, 8, 16, 32, 64, 128, and so on. To read a binary number, add up the place values wherever there is a 1.

Binary	Decimal
0000 0001	1
0000 1010	$8 + 2 = 10$
0100 0001	$64 + 1 = 65$
0110 0101	$64 + 32 + 4 + 1 = 101$

Computers use binary because a switch is either on or off, and a wire either carries current or does not. That is the entire reason. There is nothing mystical about it, and nothing universal about it either, which will matter shortly.

To store text, you need a convention mapping numbers to characters. The dominant one is **ASCII**, agreed in 1963 by an American standards committee. In ASCII, capital A is 65, lowercase a is 97, and the digit 1 is 49. Those values are not derived from anything. They are the result of committee decisions about teleprinter compatibility in the early 1960s. Had the committee met on a different day, A might be 71.

FUN FACT

This is the point that demolishes an entire genre of claim. If a message from a non-human intelligence arrived encoded in eight-bit ASCII, the correct conclusion would not be “they can speak English”. It would be “a human being wrote this”. ASCII is a parochial artifact of Cold War American office equipment. Expecting an alien or a future civilisation to use it is like expecting them to file their reports in triplicate. There is more on genuinely universal encoding in Section 18.2, and the answer is mathematics — primes, physical constants, ratios — not character sets.

Binary itself is a reasonable candidate for a shared representation, since base two is the minimal positional system and any species that builds

switches will find it. Binary *plus ASCII* is a fingerprint. Keep that distinction sharp for the next section.

S . 4 T H E I M P O S S I B L E N O T E B O O K

Jim Penniston kept a notebook. He has stated that in the hours after the 26 December encounter he sketched the object he had touched, copied the symbols from its surface, and then — over the following days — compulsively wrote out pages of ones and zeros, without knowing why or what they meant. The notebook pages have been photographed and published, and the sketch of the craft with its glyphs is one of the more striking artifacts in the field.

In 2010, a researcher named Nick Ciske ran the digits through an ASCII decoder. They produced English text. The decoded message reads, in the version most often quoted:

```
EXPLORATION OF HUMANITY 666 8145 52.0942532 N 13.131269 W
CONTINUOUS FOR PLANETARY ADVAN... EYES OF YOUR EYES ORIGIN
52.0942532 N 13.131269 W ORIGIN YEAR 8100
```

The coordinates have been matched to various locations of significance — Hy Brasil, a legendary island west of Ireland; the Great Pyramid; Sedona; the Nazca lines; Portland, England. The year 8100 is taken to indicate that the visitors were future humans, which is why this case sits in a chapter about time travel rather than one about extraterrestrials.

Now. Section 1.2 says the burden of proof falls on the claimant, and Section 1.7 says apply your hardest scrutiny to the evidence you like best. So:

The message is in English. Not merely encoded in an American character set — actually composed in English, with English word order and English spelling. It uses decimal degrees of latitude and longitude, a coordinate system defined by a British prime meridian and a Babylonian sexagesimal circle, and it specifies a year in a calendar reckoned from a Christian event. A message from 8100 AD, or from anywhere else, that arrives in English-language ASCII with WGS-84 coordinates and an Anno Domini date has told you precisely one thing about its origin, and it is not a flattering thing.

The provenance is worse. The notebook was not made public in 1980,

or 1985, or 1990. The binary pages entered the record decades later, after Penniston had undergone hypnotic regression — see Section 7.1 — and after he had become a regular figure on the conference circuit. The decoding was performed by someone who knew what case he was working on and what answer would be interesting. There is no chain of custody, no dated third-party examination of the pages, and no independent verification of when the digits were written. Section 1.8 covered exactly this: an analysis performed by a partisan on an unauthenticated artifact is not evidence, however competent the analyst.

And the decoding itself is not clean. The digit sequences require judgement about where characters begin and end, and different investigators reading the same pages have produced different text. When a decoding requires the decoder to make choices, and the decoder knows what he hopes to find, Section 1.6 tells you what the output will be.

FUN FACT

The coordinates have been “identified” as at least eight different sites of anomalous interest, which is a statistical inevitability rather than a mystery. Pick any latitude and longitude on Earth and something legendary lies within a few hundred kilometres, because legends are dense on the ground. This is the Section 1.5 problem wearing a compass.

I regard the notebook as the single most instructive artifact in this book, and not because it is genuine. It is instructive because it is *almost* good enough to be compelling, and because it fails on a point — the character encoding — that is checkable by anybody who has read the previous section. That is what a real methodology gets you. You do not need access to Suffolk, or to Penniston, or to a laboratory. You need base two and about five minutes.

S . S D I E G L O C K E

Die Glocke — The Bell — is a claimed Nazi secret weapon: a metallic bell-shaped device, roughly three metres tall, containing two counter-rotating cylinders filled with a violet mercury-like substance code-named “Xerum 525”, which when energised produced intense radiation, killed several of

the scientists working on it, and was variously an antigravity device, a time machine, or a means of generating a torsion field.

The entire story traces to a single source: the Polish journalist Igor Witkowski, who stated in his 2000 book *Prawda o Wunderwaffe* that he had been shown transcripts of the interrogation of SS General Jakob Sporrenberg by an unnamed Polish intelligence officer, was permitted to copy them by hand, and was not allowed to retain or photograph the documents. The story reached the English-speaking world through Nick Cook's *The Hunt for Zero Point* in 2001, and from there into hundreds of documentaries.

So the evidentiary chain is: an unnamed officer showed an author documents nobody else has seen, which the author transcribed by hand and cannot produce. That is not a chain. That is a single link with nothing on either end. Sporrenberg was a real person, tried and executed in Poland in 1952 for genuine atrocities, and his actual trial record contains nothing about any such device.

The supporting evidence usually offered is a concrete structure near the Wenceslas Mine at Ludwikowice in Poland — a ring of eleven concrete pillars, presented in documentaries as a test rig for the Bell. Industrial archaeologists identify it as a cooling tower support for the adjacent mine's power plant, of a standard design found at other sites. It looks exotic only if you have never seen a cooling tower footing.

FUN FACT

The Nazi wonder-weapon genre exists because the real programmes were genuinely astonishing. The V-2 was the first object to reach space. The Me 262 was the first operational jet fighter. The Type XXI submarine was a generation ahead of anything afloat. The Horten Ho 229 was a flying wing with a plausibly reduced radar signature, and its shape is the ancestor of the B-2. When a regime demonstrably built five impossible things, a sixth is an easy sell. The difference is that the real programmes left factories, blueprints, prototypes, and thousands of witnesses. Die Glocke left one journalist's handwriting.

There is also a moral point here, and I do not usually make them. The Nazi-antigravity genre frequently shades into admiration — for the engineering, the secrecy, the supposed superiority — and from there into the Vrill and Thule occult material and eventually into the territory I address in

Section 19.7. Be aware of where this road goes. A great deal of “suppressed Nazi technology” literature is doing work that has nothing to do with physics.

S . E . JOHN TITOR

Beginning in November 2000, a poster on internet forums including the Time Travel Institute and Art Bell’s BBS claimed to be a soldier from the year 2036, sent back to 1975 to retrieve an IBM 5100 computer needed to debug legacy UNIX systems, and stopping in 2000–2001 for personal reasons. He posted under the name John Titor, produced diagrams of a “C204 gravity distortion time displacement unit” installed in a Chevrolet, photographs of the device, and a scanned military insignia. He answered questions for several months and then, in March 2001, said he was going home and stopped posting.

The Titor material is the most successful piece of internet folklore the UFO-adjacent world has produced, and it is worth studying as folklore rather than as physics.

His predictions failed, and they failed specifically. He stated that a civil war would begin in the United States around 2004–2005 over a disputed federal authority and escalate into a brief nuclear exchange with Russia in 2015, reducing the country to a rural agrarian society of five regional capitals. He stated that CERN would discover the basis of time travel around 2001 with the production of miniature black holes. He stated the 2004 Olympics would be the last. None of this happened. There was no 2015 nuclear war, the Olympics continue, and CERN’s actual 2012 discovery was the Higgs boson, which is not a portal.

His defenders invoke the branching-worlds escape — he said explicitly that his timeline diverged from ours by a few per cent, meaning no prediction can ever falsify him. Note what that does. **An unfalsifiable claim is not a strong claim, it is an empty one**, and Section 1.4 covered the mechanics of this retreat.

The technical details are the tell. The IBM 5100 detail is the celebrated “hit”: Titor said the machine had an undocumented ability to emulate older IBM mainframe code, which was not public knowledge and was confirmed by an IBM engineer. It is a real feature. It was also discussed in

IBM technical circles and in print, and the person best placed to know it would be a computing hobbyist in 2000 — not a soldier in 2036, for whom a fifty-year-old portable computer would be an artifact of no more personal familiarity than a Victorian loom is to us.

FUN FACT

In 2003 an Italian researcher, working with a lawyer, traced the Titor material to a Florida entertainment company and to two brothers, one of whom had a background in computing and the other in law. A linguistic analysis published in 2009 by John Håkan Larsson found that the writing style matched. No confession has ever been made, and the family has consistently declined to comment — which is, in fairness, exactly what innocent people also do.

What makes Titor interesting is not the claim but the shape of it. He was modest, non-messianic, occasionally rude, refused to profit, made no attempt to be believed, and left on schedule. That is a far more sophisticated performance than the usual prophet routine, and it is why the story has outlived every prediction in it. If you ever wonder why this field is so hard to clean up, consider that a well-constructed fiction with a consistent voice will always beat a badly written truth.

S . 7 THE BERMUDA TRIANGLE

The Bermuda Triangle is the region bounded roughly by Miami, Bermuda, and Puerto Rico, in which ships and aircraft are said to disappear at an anomalous rate under mysterious circumstances. It is the most commercially successful geographic mystery in history and it is also, as an anomaly, entirely manufactured — and we can say that with unusual confidence, because somebody did the work.

The name was coined by Vincent Gaddis in a 1964 *Argosy* magazine article. The concept was made famous by Charles Berlitz's 1974 bestseller *The Bermuda Triangle*, which sold in the millions. And in 1975 a librarian in Arizona named Lawrence David Kusche did something nobody in the genre had bothered to do: he went to the primary sources for every case Berlitz and his predecessors cited — Lloyd's records, US Coast Guard reports, newspaper archives, contemporary weather data.

FIGURE 5.3

The Bermuda Triangle as usually drawn — the boundaries vary by tens of thousands of square miles between sources.

Figure 5.3. The Bermuda Triangle as usually drawn — the boundaries vary by tens of thousands of square miles between sources.

Image sourced from: https://commons.wikimedia.org/wiki/File:Bermuda_Triangle.png

His findings, published as *The Bermuda Triangle Mystery: Solved*, are devastating in their specificity (see Figure 5.3). Some vessels listed as vanishing in calm weather had disappeared during documented severe storms. Some had not disappeared in the Triangle at all — one loss placed there occurred in the Pacific. Some had not disappeared: at least one ship reported as lost with all hands had arrived safely in port. In several cases the “mystery” consisted entirely of details Berlitz had omitted, such as a wreck having been found. Kusche’s summary is the best

sentence written about this subject: the legend was “a manufactured mystery,” perpetuated by writers who used careless research and sometimes deliberate sensationalism.

And the statistics settle it. Lloyd’s of London stated that the Triangle is no more dangerous than any comparable stretch of ocean, and charges no premium for it. The US Coast Guard says the same. The area happens to be one of the most heavily trafficked shipping and aviation regions on Earth, it contains the Gulf Stream, and it sits in the Atlantic hurricane track. A high absolute number of losses in a high-traffic, high-weather corridor is arithmetic, not anomaly — the Section 1.5 lesson again.

FUN FACT

The founding case, Flight 19 — five TBM Avengers lost on a training flight from Fort Lauderdale on 5 December 1945 — is genuinely tragic and not at all mysterious once you read the Navy’s own board of investigation. The flight leader became convinced his compass was faulty and that he was over the Florida Keys when he was in fact over the Bahamas, and led the formation further out to sea in deteriorating weather until they ran out of fuel after dark. Radio transcripts capture the disorientation in real time. The board initially cited pilot error, then amended it to “causes or reasons unknown” after the flight leader’s mother objected — and that amended phrase, a compassionate gesture to a grieving parent, became the evidentiary foundation of a global mystery. The rescue aircraft that also vanished that night was a PBM Mariner, a type nicknamed the “flying gas tank”; a tanker crew reported seeing an

explosion and an oil slick at the appropriate time and place.

I put the Triangle at the end of a chapter on time travel deliberately, because the standard exotic explanations for it — time slips, dimensional portals, electromagnetic vortices — require everything in Section 5.1 to be not just possible but common, in one specific patch of the Atlantic, for no stated reason. Whereas Kusche’s explanation requires only that authors do not check things. Section 1.2 tells you which of those two hypotheses is carrying the lighter load.

L I F E I N T H E U N I V E R S E

A N D E . G . E . S

“This is one corner. . . of one country, in one continent, on one planet that’s a corner of a galaxy that’s a corner of a universe that is forever growing and shrinking and creating and destroying and never remaining the same for a single millisecond. And there is so much, so much to see.”

— THE DOCTOR

E . I T H E D R A K E E Q U A T I O N

In 1961, ahead of a small conference at Green Bank, West Virginia, Frank Drake wrote an equation on a blackboard. He was not attempting to calculate an answer. He was attempting to organise an agenda — to break the question “how many civilisations are there?” into components that could be discussed separately. It has since been printed on more t-shirts than any other equation except Einstein’s, and misunderstood at a comparable rate.

$$N = R_* \cdot f_p \cdot n_e \cdot f_\ell \cdot f_i \cdot f_c \cdot L$$

Read it left to right, and notice the change in the quality of the knowledge as you go.

R_* , the rate of star formation in our galaxy, is measured. Roughly one

and a half to three solar masses per year. This is astronomy and it is solid.

f_p , the fraction of stars with planets, was a wild guess in 1961 and is now measured. Before 1992 we knew of no planets outside our solar system. As I write, the confirmed count exceeds 5800, and the statistical conclusion from the Kepler and TESS surveys is that planets are the rule rather than the exception. Call it close to one. Drake's colleagues guessed conservatively and were wrong in the interesting direction.

n_e , the average number of habitable planets per system, is being narrowed but is still soft — estimates for Earth-size planets in the habitable zones of Sun-like stars range from roughly 2% to 20% of such stars, and the definition of "habitable" is doing an enormous amount of concealed work, as Section 6.3 will argue.

f_ℓ , the fraction of habitable planets where life actually appears, is where the equation stops being science. We have one data point, and we are it. From a sample of one you can derive nothing — the fraction could be one, or it could be one in 10^{40} , and no observation currently distinguishes those. The one relevant fact is that life on Earth appears in the record remarkably early, within a few hundred million years of the surface becoming habitable, which is at least suggestive that abiogenesis is not desperately improbable. It is not proof of anything; a lottery winner also notices that they won quickly.

f_i , the fraction of life that becomes intelligent, is worse. On Earth it took roughly four billion years and required a specific and contingent sequence: oxygenation, eukaryotes, multicellularity, several mass extinctions that cleared incumbents, and a final primate lineage that got lucky. Complex life may be the bottleneck rather than life itself.

f_c , the fraction that develops detectable communication, and L , the average lifetime of such a civilisation, are pure speculation. L is the term everybody skips past and it is arguably the most important in the equation, because it is the only one we might influence. Drake's own 1961 estimate produced N of around 10000. Modern authors have published values from 10^{-10} to millions.

FUN FACT

The Drake Equation's real function is not arithmetic, it is diagnostic — it shows you precisely where the ignorance lives. In 1961 six of the seven terms were unknown. Today two are measured, one is being measured, and four

remain unknown. That is genuine progress on a question that was thought unanswerable, and it happened entirely on the astronomy side. Every term we have nailed down was nailed down by building better telescopes, not by arguing.

The honest use of the equation is as a filter on other people's confidence. Anybody who quotes you a number for N is telling you about their assumptions concerning f_ℓ , f_i and L , and nothing else. This is why Section 1.3 warned about mathematics being used as rhetoric: multiplying four measurements by three guesses does not yield a measurement. It yields a guess with better manners.

6.2 FERMI'S PARADOX

Over lunch at Los Alamos in 1950, in the middle of a conversation about a *New Yorker* cartoon, Enrico Fermi asked: "Where is everybody?" The question has become the central problem of the field, and it is a genuine paradox rather than a rhetorical flourish, because the two halves of it are both well-supported.

The first half: the galaxy is old. It is roughly 13 billion years old (see Figure 6.1), and there are stars with planets that predate our Sun by billions of years. Even at speeds we could achieve with physics we already understand — a few per cent of light speed, using nuclear pulse or beamed propulsion — a civilisation that expanded outward and settled systems which then launched their own probes would colonise the entire galaxy in something between five and fifty million years. That is a geological eyeblink against 13 billion. Self-replicating probes would do it faster and cheaper. So if anybody at all ever started, they should be here already, and we should be surrounded by evidence.

FIGURE 6.1

The Milky Way in plan with the Sun's position marked. Full settlement takes millions of years against an age of billions.

Figure 6.1. The Milky Way in plan with the Sun's position marked. Full settlement takes millions of years against an age of billions.

Image sourced from: https://commons.wikimedia.org/wiki/File:Milky_Way_Arms_ssc2008-10.svg

The second half: we observe nothing. No probes, no megastructures, no radio traffic, no industrial signatures in exoplanet spectra, no artifacts in the solar system. The silence is not merely an absence of contact; it is an absence of the physical debris that any expansive technological civilisation should leave behind.

That is the paradox. The resolutions fall into a few families, and I will give you the ones that are actually defensible.

Nobody is out there. A term in the Drake equation is catastrophically small. The “Rare Earth” argument holds that the specific conditions here — plate tectonics, a large stabilising moon, a Jupiter clearing debris, a long-lived magnetic field — are rare in combination. Alternatively abiogenesis itself is a fluke.

They die first. The Great Filter, in Robin Hanson’s formulation. Somewhere between dead chemistry and galactic expansion there is a step almost nothing survives. The uncomfortable question is whether the filter is behind us or ahead of us. If we ever find independent microbial life nearby, that is genuinely bad news: it moves the filter forward, towards us.

They are quiet. Broadcasting may be understood, everywhere, as a bad idea — the “dark forest” argument, which I will treat properly in Section 18.5. Or technological maturity may mean efficiency, and efficient civilisations are dim.

We cannot see them. Our search has been absurdly shallow. We have examined a vanishing fraction of the sky, in a narrow band of frequencies, for a few decades, for one specific kind of signal. The often-quoted comparison is that all SETI observation to date is equivalent to sampling a hot tub’s worth of the Earth’s oceans. Absence of evidence here is very weak evidence of absence.

They are here and we have not established it. This is the hypothesis that gives this book its subject matter. I list it honestly and rank it low, for the reason Section 1.2 requires: it is the option carrying the heaviest burden, and seventy years of trying has not produced a single artifact that survives independent examination.

FUN FACT

The zoo hypothesis and the simulation hypothesis are both technically resolutions of Fermi's Paradox, and both are unfalsifiable, which puts them in the same category as John Titor's branching timelines. They can be true. They cannot be investigated. A hypothesis that forbids no observation is not competing with the others; it has left the competition.

6.3 THE MISCONCEPTION OF WATER

"Follow the water" has been NASA's search strategy for thirty years, and it is a defensible engineering decision dressed up as a law of nature. The distinction matters.

The genuine case for water is good. It is an excellent solvent, so chemistry happens in it. It is liquid across a wide temperature range. It is unusual in being less dense as a solid, so bodies of it freeze from the top and stay liquid underneath. It moderates temperature. And crucially, hydrogen and oxygen are the first and third most abundant chemically active elements in the universe, so water is everywhere — in comets, in nebulae, in interstellar clouds. If you are designing a search on a budget, water is where you look.

The misconception is the inference that ran the other way: that because our life needs water, life needs water. That is a sample of one, again, and it has quietly narrowed the search in ways worth naming.

FIGURE 6.2

*Cassini radar mosaic of Titan's
northern lakes — liquid methane
and ethane, mapped in detail.*

Figure 6.2. Cassini radar mosaic of Titan's northern lakes — liquid methane and ethane, mapped in detail.

Image sourced from: <https://commons.wikimedia.org/wiki/File:PIA20023-Titan-NorthPole-Lakes.jpg>

Alternative solvents are chemically serious, not science fiction. Liquid ammonia is polar, dissolves many of the same compounds, and stays liquid at temperatures where water is ice. Liquid methane and ethane exist in enormous quantities on Titan, where the lakes are mapped (see Figure 6.2); a 2005 paper by Benner and colleagues argued that hydrocarbon solvents actually have advantages for prebiotic chemistry, since water aggressively hydrolyses many of the molecules life needs to build. Supercritical carbon dioxide is a good solvent and would be stable on

high-pressure worlds. Formamide, hydrogen fluoride, sulphuric acid — all have been argued.

And the “habitable zone” concept, defined as the orbital band where liquid surface water is possible, is narrower still than the water criterion. It excludes the most promising locations in our own solar system. Europa has a subsurface ocean beneath its ice, containing perhaps twice the water of all Earth’s oceans, kept liquid by tidal heating and completely independent of solar distance. So does Enceladus, which is actively venting that ocean into space through its south polar plumes, where Cassini flew through and detected water, salts, silica, methane, molecular hydrogen and complex organics. Ganymede and Callisto probably have oceans too. None of these bodies is in the habitable zone. All of them are better candidates than Mars.

FUN FACT

Earth’s own deep biosphere makes the point. Hydrothermal vent ecosystems run on chemosynthesis, oxidising hydrogen sulphide, with no sunlight at all — an entire food web that does not care whether the star exists. Bacteria have been recovered from kilometres deep in continental rock, metabolising on timescales of centuries. *Deinococcus radiodurans* survives radiation doses thousands of times lethal to us. Tardigrades survive vacuum. If our own single sample of life is this flexible, using our own surface conditions as the template for the search is an odd choice.

Carbon, incidentally, is a much stronger requirement than water, and I want to be fair about which assumptions actually hold up. Carbon forms four stable bonds, catenates into long chains and rings, and builds the enormous molecular diversity that any conceivable metabolism needs. Silicon, its usual proposed substitute, sits below it in the periodic table with the same valence but forms much weaker bonds to itself, and its oxide is a solid rather than a gas — silicon-based respiration would mean exhaling sand. Carbon chauvinism is defensible. Water chauvinism is a budget decision. Learn to tell the difference, because most popular writing on astrobiology does not.

6.4 EXTRATERRESTRIAL BIOLOGICAL ENTITIES

EBE — Extraterrestrial Biological Entity — is a piece of vocabulary that arrived via the alleged MJ-12 documents and associated 1980s testimony, and which has since been adopted uncritically by the field. I use it here under protest, because it is a bureaucratic euphemism that lends institutional weight to claims which have none, and I want to be explicit that nothing in this section is established.

The folklore has produced a fairly stable taxonomy, and its stability is the interesting part.

Greys. Short, grey-skinned, hairless, enormous black almond eyes, no external ears, vestigial nose and mouth, thin limbs, large cranium. The dominant type since roughly 1965.

Nordics. Tall, human-like, blond, benevolent, spiritually inclined. Dominant in the 1950s contactee period, essentially extinct in the modern record.

Reptilians. Scaled, upright, hostile, associated with the underground-base and control-conspiracy material of Chapter 17.

Mantids and insectoids. Tall, thin, described as supervisory, appearing mostly in later abduction accounts.

Here is what should bother you. **Every one of these is a human being with a parameter adjusted.** The reported entity types are shown in Figure 6.3. The Greys are a foetus, or a projection of ageing: enlarged head, large eyes, receding features, small body. The Nordics are an idealised human, and the fact that their heyday was the 1950s, in a culture with very particular ideas about who was ideal, is not a coincidence anybody should have to explain. Reptilians are the reptile-brain fear response wearing clothes. Nothing in this catalogue is genuinely alien —

there is no entity in the corpus with radial symmetry, no distributed nervous system, no colonial organism, no silicon chemistry, nothing that a person could not have imagined by adjusting a face in a mirror. Compare the actual

FIGURE 6.3

The standard entity types of the reported corpus, drawn to scale: Grey, Nordic, Reptilian, hairy dwarf.

Figure 6.3. The standard entity types of the reported corpus, drawn to scale: Grey, Nordic, Reptilian, hairy dwarf. Image sourced from: https://commons.wikimedia.org/wiki/File:Alien_grey_scale_diagram.svg

diversity of life on this planet, where an octopus — a close relative, sharing our entire evolutionary history — has a distributed brain, chromatophore communication, and three hearts. Real biology is stranger than the entire EBE catalogue.

And there is a hard convergent-evolution question underneath the folklore. Some features are plausibly universal: bilateral symmetry is efficient for directed movement, sensory organs cluster near the front and near the brain, and eyes have evolved independently dozens of times on Earth because photons are worth detecting. Manipulative appendages are likely for a tool user. But the specifics — humanoid bipedalism, two arms, a face arranged like ours — are contingent on our particular lineage of tetrapods that crawled out of water with four limbs. Get that far in the argument and you notice that the folklore has assumed away the entire question.

FUN FACT

The most likely form for a visitor is the one that never appears in the reports: a machine, or nothing recognisable as a body at all. Our own first and only interstellar emissaries — Pioneer, Voyager, and the New Horizons bus — are unmanned. As Section 2.2 argued, biology is a poor engineering choice for interstellar travel: it needs atmosphere, food, radiation shielding, and a return trip, and it dies. The occupant literature has spent seventy years describing pilots for craft that would not need any.

What about actual candidate biology? There is a small amount of real science worth knowing about. The Viking labelled-release experiment on Mars in 1976 produced a positive result that has never been fully explained, though the consensus attributes it to perchlorate chemistry in the soil. The 1996 claim of microfossils in Martian meteorite ALH 84001 has not held up — the magnetite chains and carbonate globules are now generally attributed to non-biological processes — but it prompted a generation of work on how you would recognise biology if you found it, and that methodological legacy outlived the claim. The 2020 detection of phosphine in Venus's atmosphere generated enormous excitement and has been substantially contested on re-analysis of the data. Enceladus remains the best target in the solar system and nobody has been back.

That is the state of the evidence for extraterrestrial life: several intriguing

signals, all of them contested, none of them confirmed. It is a considerably weaker position than the confident taxonomies of the abduction literature would suggest, and considerably more interesting, because every item on that short list is being actively worked on by people who publish their errors.

ALIEN ABDUCTION & PARAPSYCHOLOGY

"In order to understand more it is imperative that we improve our knowledge before choosing which side of the fence we feel compelled to belong."

— J. P. ROBINSON

7.1 WHAT IS HYPNOSIS?

Hypnosis is a state of focused attention with heightened suggestibility. That is the whole of it. It is not sleep — EEG readings during hypnosis resemble waking attention, not any sleep stage. It is not unconsciousness, it does not surrender the will, and it cannot make a person do something they find genuinely abhorrent. What it reliably does is lower the threshold at which a suggestion becomes an experience.

It has real clinical uses. There is decent evidence for hypnosis in pain management, in irritable bowel syndrome, in smoking cessation as an adjunct, and in anxiety around dental and surgical procedures. Some people are highly susceptible and some are barely susceptible at all; it appears to be a stable trait, roughly normally distributed.

What hypnosis absolutely does not do — and this is not a controversial position, it is the settled position of the relevant professional bodies — is recover accurate memory.

Understand why, because the mechanism matters. Memory is not storage. **It is not a recording that can be replayed with sufficient effort.** Every act of recall is an act of reconstruction, assembled from fragments, and the reconstruction is rewritten and re-stored each time, incorporating whatever context is present at the moment of recall. Elizabeth Loftus spent a career demonstrating this experimentally. Change one word in a question — “how fast were the cars going when they *smashed*” versus “*hit*” — and you change both the speed estimate and whether subjects later remember broken glass that was never there. Her group implanted a false childhood memory of being lost in a shopping mall in roughly a quarter of subjects, using nothing but a suggestion from a relative.

Now add hypnosis. Under hypnosis the subject is more suggestible, less critical of implausibility, and — this is the destructive part — reports the resulting material with *increased subjective confidence*. Hypnotic recall produces more detail, more vivid detail, and more errors, while making the subject certain. Confabulation under hypnosis is not lying. The person genuinely remembers it afterwards, and cannot distinguish it from a real memory, because neurologically it now is one.

FUN FACT

The legal system worked this out and acted on it. Hypnotically refreshed testimony is inadmissible or severely restricted in most US jurisdictions, and the American Psychological Association, the American Medical Association and the British Psychological Society have all issued statements against its use for memory retrieval. The reason was not theoretical: it was the wave of wrongful convictions and the “satanic ritual abuse” panic of the 1980s and 1990s, in which therapists using regression techniques generated thousands of detailed, confidently-held, entirely false memories, destroyed families, and imprisoned innocent people. The abduction literature was built with the same tools, in the same decade, frequently by people with no clinical training at all.

So here is my position, stated plainly, because it governs the rest of this chapter. **Any account whose content originates under hypnosis is not evidence about the world.** It is evidence about the session. This is not a claim that the witnesses are dishonest — almost none of them are, and many are clearly suffering. It is a claim about what the instrument can measure. Handing a sincere frightened person to a regression practitioner with a

theory does not recover what happened to them. It replaces it.

7.2 BETTY AND BARNEY HILL

On the night of 19 September 1961, driving home to Portsmouth, New Hampshire on Route 3, Betty and Barney Hill saw a light in the sky that appeared to follow their car. Barney stopped, looked at it through binoculars, and became frightened. They drove home in a hurry, arriving about two hours later than they expected. This is the founding case of the entire abduction genre — everything that follows in the literature, from the medical examination to the missing time to the greys themselves, begins here.

What makes the case invaluable is that its development is fully documented, step by step, which is almost unique. We can watch the story being built.

In the days immediately afterwards, the Hills reported a sighting. That is all. They reported it to Pease Air Force Base, and the Blue Book entry describes lights and a possible aircraft. Betty then borrowed a book by Donald Keyhoe from the library. Within about ten days she began having a series of vivid nightmares involving being taken aboard a craft, examined by humanoid beings with large eyes, and having a needle inserted into her navel. She wrote the dreams down. She discussed them with Barney, repeatedly, over the following months.

FIGURE 7.1

Betty and Barney Hill, whose 1961 encounter became the template for the abduction narrative.

Figure 7.1. Betty and Barney Hill, whose 1961 encounter became the template for the abduction narrative. Image sourced from: https://commons.wikimedia.org/wiki/File:Betty_and_Barney_Hill.jpg

Two and a half years later, in early 1964, both Hills underwent hypnotic regression with Dr. Benjamin Simon (see Figure 7.1), a Boston psychiatrist. Under hypnosis they produced detailed abduction accounts. Barney's account included elements matching Betty's dreams, which he had by then heard many times. Simon — and this is the part almost never quoted — concluded that the abduction narrative was not a real memory. His professional assessment was that it was a fantasy, originating with Betty's dreams, transferred to Barney through their conver-

sations, and that the Hills sincerely believed it. He said so publicly and repeatedly for the rest of his life.

The case reached the public through a 1965 newspaper leak and then John Fuller's 1966 bestseller *The Interrupted Journey*, followed by a 1975 television film watched by tens of millions. The greys entered popular culture through that broadcast.

The famous star map deserves specific attention because it is the case's strongest evidentiary claim. Betty, under hypnosis, said she was shown a map of trade routes and later drew it from memory. In 1968 an amateur astronomer, Marjorie Fish, spent years building physical models of nearby star systems and found an orientation matching Betty's sketch, with Zeta Reticuli as the origin. This was published in *Astronomy* magazine in 1974 and caused a sensation.

It does not survive. The match depended on the stellar distance data available in the early 1970s; when the Hipparcos satellite produced accurate parallax measurements in the 1990s, several of the stars in Fish's model moved substantially and the pattern dissolved. More fundamentally, Carl Sagan and Steven Soter pointed out at the time that the exercise was statistically empty: with enough stars to choose from and freedom to select the viewing angle, the scale, and which dots count, you can match almost any arrangement of dots. Take Betty's sketch, they noted, and you can find matches to random dot patterns. This is the Section 9.4 ley-line problem in three dimensions.

FUN FACT

There is a strong candidate for what the Hills actually saw, and it is oddly poignant. On that night Jupiter and Saturn were both prominent low in the southwest along their route — a bright object and a fainter companion, which is exactly the initial description — and a vehicle driving a winding road through the White Mountains will see such objects apparently move, stop, and follow. Barney was an interracial-marriage-era postal worker in 1961 New Hampshire, active in the NAACP, driving at night through rural country, already anxious. The emotional weight of that night was real. What it was about may have had less to do with the sky than with the road.

7.3 CLOSE ENCOUNTERS

Hynek's classification, introduced in *The UFO Experience* in 1972, is the field's one genuinely useful piece of taxonomy, and I gave you the outline in Section 4.2. Here is what each level is actually worth.

Close Encounter of the First Kind. An object observed at close range — Hynek used about 500 feet — with no interaction and no physical trace. Evidentially this is a sighting with better resolution. It rises or falls entirely on witness quality and on what Section 4.4 step two turns up about the sky.

Close Encounter of the Second Kind. An encounter leaving physical effects: scorched or depressed ground, damaged vegetation, vehicle engine or electrical failure, radiation readings, animal distress, burns. This is the only category with real evidentiary potential, because physical traces can be sampled, controlled, and analysed by people who do not know where the sample came from. The Ted Phillips catalogue collected several thousand physical trace cases. The frustrating truth is that the great majority of them were sampled without controls, which as Section 4.4 explained renders the laboratory results meaningless. The best-known CE2 in the field, Cash–Landrum, appears in Section 15.5, and it is the exception that shows what the category could be.

The other case worth putting beside it is Falcon Lake, in Manitoba, on 20 May 1967, because it is the closest thing this field has to a CE2 with a paper trail. Stefan Michalak, an industrial mechanic who was prospecting for quartz, reported two glowing objects, one of which landed. He said he approached it, sketched it, touched its hull with a gloved hand (see Figure 7.2) — the glove came away melted — and was then struck by a blast of hot gas from a grid of holes in the side, which set his shirt alight and knocked him backwards. He walked out of the bush vomiting. What

follows is the part that separates this case from the anecdotes: he presented with nausea, diarrhoea, headaches, weight loss and a geometric pattern of burns on his chest, he was examined at the Mayo Clinic, and the Royal

FIGURE 7.2

Michalak's sketch of the Falcon Lake object and the grid of vents that set his shirt alight.

Figure 7.2. Michalak's sketch of the Falcon Lake object and the grid of vents that set his shirt alight.

Image sourced from: https://commons.wikimedia.org/wiki/File:Falcon_Lake_incident_sketch.jpg

Canadian Mounted Police, the Royal Canadian Air Force and the Canadian Department of National Defence all generated files on him. He never sought money and never changed his account before his death in 1999. The Canadian government, in 2018, put the incident on a legal-tender \$20 silver coin.

I want to be exact about what that file actually contains, because Falcon Lake is routinely overstated in both directions. The burns are real and photographed; the melted glove and the scorched shirt exist; soil samples from the site were radioactive enough that a section of ground was eventually cordoned. Against that: the radiation testing done on Michalak himself came back negative, his blood counts did not show the marrow depression that a dose capable of producing his reported symptoms would cause, the radioactive soil was of a grade and isotope profile consistent with material that could have been introduced, and the site was visited repeatedly by civilians before it was ever controlled — which, as Section 4.4 keeps insisting, is fatal. So the honest verdict is neither “proven landing” nor “hoax”. It is a genuinely unexplained injury with a physical residue, investigated too late and too loosely to settle. That is precisely the shape of the best evidence in ufology, and you should get used to it.

Close Encounter of the Third Kind. Occupants observed. Hynek was reluctant to include this category and said so, on the grounds that it would damage the credibility of the rest of the scale. He included it because the reports existed and he refused to discard data he found embarrassing — which is, for my money, the single most admirable act in the history of this field.

Close Encounter of the Fourth Kind. Abduction. Added later, not by Hynek. Note that the scale’s evidential quality is now inversely proportional to its number: CE2 has physical samples, CE4 has a hypnosis transcript.

Close Encounter of the Fifth Kind. Deliberate human-initiated contact, promoted from the 1990s by advocates of “CE5 protocols” — group meditation to summon craft. There is no controlled evidence for any of it, the reported results are precisely what expectation and group dynamics would produce in a field at night, and the practice has no place on a scale intended to classify observations.

The abduction narrative itself has a remarkably fixed structure, and structure is diagnostic. The sequence is: capture (often from a bedroom, often preceded by paralysis), transit through a beam or corridor, examination

on a table by beings performing medical procedures, focus on reproductive organs, a “conference” or telepathic communication frequently including an environmental warning, a tour of the craft, return, and missing time. Budd Hopkins and David Jacobs treated this consistency as corroboration — how could thousands of unconnected people report the same thing?

The answer is that they were not unconnected. *The Interrupted Journey* was a bestseller in 1966; the television film reached tens of millions in 1975; *Communion* sold in the millions in 1987 with a grey face on the cover. By the time the great mass of cases arrived, the script was in every supermarket in America. And the majority of those cases were developed by a small number of investigators using regression, asking questions shaped by the accounts they had already collected. Section 1.6 and Section 7.1 between them predict exactly this outcome: high consistency, high confidence, no independent corroboration.

7.4 GENETIC MOTIVATION

The standard explanation offered within the abduction literature for the reproductive focus of the examinations is a breeding programme: the visitors are harvesting gametes, producing hybrids, and monitoring the results across generations. Jacobs built an entire late career on it. It deserves examination on the merits, because it is a biological claim and biology can adjudicate it.

It does not work, and it fails on grounds that have nothing to do with skepticism about UFOs.

Interbreeding requires close common ancestry. Genetic compatibility is not a matter of similar appearance — it requires compatible chromosome number and structure, compatible gene regulation, compatible developmental timing, and compatible molecular machinery down to the codon assignments. Humans cannot interbreed with chimpanzees, and we share about 98.8% of our DNA with them and a common ancestor six million years ago. We cannot interbreed with any other species on this planet, all of which descend from the same origin and use the same genetic code. The proposition that an organism from an independent biosphere — with its own chemistry, its own information storage, possibly not using DNA at all — could produce viable offspring with us is not merely improbable. It is a

category error. It is like expecting to run Windows software on a plant.

Second, the technology assumption is incoherent. A civilisation capable of interstellar travel is, necessarily, vastly ahead of us in every technical domain, including molecular biology. Such a civilisation would not need to abduct farmers repeatedly over decades to obtain genetic material. A single tissue sample, obtained once, contains the entire genome. We can already sequence a human genome from a cheek swab for a few hundred dollars, and synthesise arbitrary DNA sequences to order. A species that crosses light years but requires a sixty-year sampling programme with a table and instruments is doing molecular biology worse than we currently do it.

Third, the stated purposes are contradictory across the literature: they need our DNA because they are dying out; they are creating a hybrid species to replace us; they are correcting a genetic defect in themselves; they are preparing a vessel for consciousness transfer. Each is offered with equal confidence by different authors reading the same case files.

FUN FACT

There is a much older version of this story, and it is nearly universal. Incubus and succubus legends across medieval Europe describe nocturnal visitation by entities that paralyse the sleeper and extract or impart seed. The changeling folklore of Northern Europe describes children taken and replaced, and fairy abduction describes missing time — days spent in the fairy mound that were years outside. Jacques Vallée made this argument in *Passport to Magonia* in 1969 and it has never been answered: the abduction narrative is the fairy narrative with the technology updated. Either the phenomenon has changed costume, or something about human beings produces this story reliably in every century.

Which brings me to the explanation I find far more likely, and which is also considerably more respectful of the witnesses than pretending to take the hybrid programme seriously. Sleep paralysis is a documented neurological event experienced by something like 8 to 30 per cent of people at least once. During REM sleep the body's voluntary muscles are inhibited. If consciousness returns before that inhibition lifts, the person is awake, aware, and completely unable to move. The state is frequently accompanied by hypnagogic hallucination — a sensed presence in the room, figures at the bedside, pressure on the chest, buzzing or humming sounds, a sense of being watched, floating, and overwhelming terror. Every element of

the classic bedroom abduction onset is a documented feature of a common sleep disorder. Add a regression session with a practitioner who believes in a breeding programme, and the rest writes itself.

7.5 MISSING 411

Missing 411 is a series of books by David Paulides, a former police detective, cataloguing disappearances in national parks and wilderness areas that he argues share anomalous characteristics. He is careful — deliberately and, I think, shrewdly — not to state a cause. The books present clusters, note recurring details, and leave the reader to supply an explanation.

The recurring profile includes: disappearances near water or boulder fields; victims found with clothing removed or neatly folded; children found improbable distances from where they vanished, sometimes at elevation; dogs unable to pick up a scent; sudden weather at the time of disappearance; German surnames; disappearances clustering geographically; and survivors with no memory of the missing period.

Some of this is genuinely strange, and I do not want to be glib about a subject that involves dead children. But the methodological problems are severe and they are of a specific kind worth learning.

The first is the missing denominator, which is Section 1.7's central complaint. Hundreds of millions of visits are made to US national parks annually. Any set of features you select will occur in some subset of cases. Without knowing the base rate — how often do *found* missing persons have their clothes off, how often is there sudden weather in mountains, how common are German surnames in the American population — a list of coincidences is unanalysable. Paulides does not provide base rates.

The second is that the anomalous details have well-documented mundane causes, and several are signatures of specific known processes. Paradoxical undressing is a recognised feature of the late stages of hypothermia: as the peripheral vasoconstriction fails, the victim feels a rush of heat and removes clothing. It is documented in mountain rescue literature and in forensic pathology, and it accounts for one of the eeriest items on the list. Terminal burrowing — the hypothermic victim's tendency to crawl into a confined space — accounts for bodies found in places that seem impossible to reach and that searchers walked past. Small children can and do cover

remarkable distances and climb, and they hide from searchers calling their names because they have been taught to avoid strangers. Scent tracking fails routinely for entirely understood reasons involving wind, water, temperature gradients and contamination. Weather in mountains changes suddenly as a matter of course, which is why people die in mountains.

The third is a refusal-to-conclude problem. By declining to state a hypothesis, the books become unfalsifiable while retaining all the rhetorical force of a claim. The National Park Service has stated it does not maintain a comprehensive centralised missing-persons database of the kind Paulides describes requesting, and his account of that refusal is presented as suppression rather than as a records-management deficiency, which is what it is.

FUN FACT

A researcher named Kyle Polich attempted the obvious control: he compiled a comparison set of wilderness disappearances that were resolved and checked how many exhibited the “411” profile markers. A great many did. When your anomaly profile is equally present in the solved cases, you have not identified a pattern in disappearances. You have identified a pattern in wilderness.

I include this section because Missing 411 is the most methodologically instructive body of work currently circulating in the paranormal world. Everything in it is a real case. Every detail is documented. And the whole is still worthless as evidence, for the single reason that nobody counted what the ordinary rate would be. That is a more advanced failure than a hoax, and much harder to spot.

7.6 THE EVIDENCE FOR ABDUCTION CASES

I have spent five sections dismantling the abduction literature, so let me now do the thing Section 1.7 demands and lay out the strongest case for it, as fairly as I can.

The consistency argument. Thousands of accounts, from people of different ages, countries, educational levels and cultures, share detailed

features — including details not present in popular media at the time of the report. Some early cases predate the availability of the script.

The witness quality argument. Abduction reporters are not, as a group, mentally ill. This has been tested. A 1983 study by Ted Bloecher, Aphrodite Clamar and Budd Hopkins, and more rigorously the work of psychologist Nicholas Spanos in 1993, found that experiencers score within normal ranges on standard psychopathology measures. They are not psychotic, not unusually fantasy-prone by clinical measures, and frequently reluctant — many have lost relationships and jobs over their accounts, and many kept quiet for years.

The physical marks argument. Some experiencers present scoop marks, straight-line scars, and puncture wounds. A few cases involve claimed implants, and several have been surgically removed and examined.

The multiple-witness argument. A handful of cases involve more than one person reporting a shared abduction, and a very few involve independent observation of an object at the same time and place — the 1975 Travis Walton case, in which six logging crew members gave consistent accounts of Walton being struck by a beam of light and reported his disappearance to police the same night, is the strongest instance in the corpus.

The authority argument. John Mack, a tenured professor of psychiatry at Harvard Medical School and a Pulitzer winner, investigated experiencers and concluded that conventional psychiatric explanations were inadequate. Harvard convened a committee to review his work and, after fourteen months, took no action against him.

Now the counter-argument, item by item, because the whole point of Chapter 1 was that this is the work.

Consistency: the script was culturally available from 1966 and dominant from 1987, and the majority of the corpus was collected after both dates by a handful of investigators using leading methods. The genuinely pre-script cases are few, and the small number that exist are the only ones I regard as evidentially interesting.

Witness quality: agreed, and it is important. But sanity is not accuracy. Perfectly healthy people experience sleep paralysis, confabulate under hypnosis, and hold false memories with complete conviction. Demonstrating that experiencers are not insane refutes a claim nobody serious was making.

Physical marks: the marks are real. Their attribution is not. Human

beings acquire unexplained scars constantly and normally do not notice; once a person is looking for evidence, they find their own body. Of the implants surgically removed and independently examined, none has ever been shown to be of non-terrestrial manufacture — the published analyses have found glass, carbon fibre, and metals with terrestrial isotope ratios, which is decisive. Section 1.8 explains why the isotope point is the one that settles it.

Multiple witnesses: Walton is the best case and it is still not clean. His account developed over subsequent years, he had a documented prior interest in the subject, he failed one polygraph in 1975 — a fact suppressed for two decades — and the crew's reports concern what they saw from a truck, not what happened during the five days. I regard it as genuinely unresolved rather than explained, which is the most I will grant any case in this chapter.

Mack: his integrity is not in question and his clinical observations about the effect of the experience on his patients were careful and valuable. His conclusion, however, rested on regression material, and no amount of professional standing repairs a compromised instrument. Section 1.8 again: credentials are not a method.

FUN FACT

The cleanest experiment ever run on this subject is almost never mentioned. In 1993 Nicholas Spanos and colleagues gave people who had never reported an abduction a hypnotic session with leading suggestions — and a substantial fraction produced detailed abduction narratives, indistinguishable in structure and emotional intensity from those of self-identified experiencers. That is the experiment. The narrative can be manufactured on demand, in a laboratory, in an afternoon.

My conclusion, and I want to state it in a way that does not insult anybody: the abduction experience is real. People are genuinely experiencing something, it frightens them badly, and it is not fraud and not madness. The evidence that anybody has been physically removed from their bedroom by non-human entities is, after sixty years and thousands of cases, nil. Not weak. Nil — there is no artifact, no photograph, no medical finding, no missing-person record, no independent corroboration of a single abduction event. A phenomenon this widespread, if physical, would have left

something behind by now.

7.7 INTRODUCTION TO

PARAPSYCHOLOGY

The chapter title promised parapsychology and here it is, and I owe you an explanation of why it is bolted onto a chapter about abduction rather than given a chapter of its own.

The reason is that the two subjects share a nervous system. Abduction research and psychical research both study a private experience that leaves no residue; both rely on testimony recovered under conditions that can generate it; both have a century or more of accumulated case files and no replicable result; and both, crucially, keep reporting the same secondary phenomena. Read enough abduction literature and you will find telepathic communication, out-of-body episodes, precognitive warnings, poltergeist activity in the household afterwards, and an insistence that the entities operated on consciousness rather than on matter. John Mack noticed that. Jacques Vallée built a whole thesis on it. Whatever you think is happening, the phenomenology of abduction is closer to the phenomenology of the séance room than to the phenomenology of a flight test. So we should look at what happened when serious people spent a hundred and forty years trying to nail the séance room down.

7.7.1 DEFINITIONS, PROPERLY SEPARATED

Parapsychology is the scientific study of claimed phenomena that, if real, cannot be accounted for by currently understood physical mechanisms of information transfer or causation. The standard taxonomy, which is worth learning because the field's precision about its own terms is genuinely admirable:

Psi is the umbrella term. It divides into **ESP** — extrasensory perception, meaning information acquired without a sensory channel — and **psychokinesis**, meaning influence on a physical system without a physical channel. ESP subdivides into **telepathy** (mind to mind), **clairvoyance** (mind to remote object or event), and **precognition** (information from the future). Psychokinesis subdivides into **macro-PK**, the visible kind that Uri Geller

performed and that has never once survived controlled conditions, and **micro-PK**, the statistical kind: tiny deviations in random number generators, which is where the modern field actually lives.

Separately there is **survivalresearch** — mediumship, apparitions, reincarnation claims, near-death experiences — which asks whether consciousness persists after death. Section 7.8 and Section 11.4 handle that.

7.7.2 A VERY SHORT HISTORY

The Society for Psychical Research was founded in London in 1882, and its founding membership is the part that surprises people: Henry Sidgwick, the Cambridge moral philosopher, was its first president. Later presidents and members included the physicists William Barrett, Oliver Lodge and Lord Rayleigh, the psychologist William James, the philosopher Henri Bergson, and the Prime Minister Arthur Balfour. This was not a fringe operation. It was the Victorian intellectual establishment attempting to apply experimental method to spiritualism, and its early achievement was largely negative and largely excellent: it exposed fraudulent mediums in detail, including some its own members badly wanted to be genuine.

J. B. Rhine moved the subject into the laboratory at Duke University in the 1930s, coined the term “extrasensory perception”, and introduced the Zener card deck and statistical evaluation. Rhine’s contribution was methodological — he turned anecdote into a countable experiment — and his results did not replicate outside his own laboratory, and in at least one case involved a collaborator, Walter Levy, caught falsifying data in 1974.

The cast of the government period deserves naming, because their biographies are themselves an argument. Hal Puthoff, a laser physicist with a Stanford doctorate in electrical engineering, built the protocols and ran the paranormal research programme at the Stanford Research Institute through the 1970s. Ingo Swann, an artist and self-described psychic who claimed abilities from early childhood, is generally credited with co-developing the coordinate remote-viewing method. Joseph McMoneagle, who enlisted in 1964 and came to the work through Army intelligence, was designated Remote Viewer No. 1 when the operational unit stood up in 1977 and is the only one of the three who was ever formally decorated for the output. There were roughly three dozen military volunteers through the programme’s life. And here is the detail that nobody in the field’s promotional literature

volunteers: Puthoff and Swann were both, at the relevant period, members of the Church of Scientology, an organisation whose doctrine already asserted that the abilities being tested were real and recoverable by training. Several of the principals met either at SRI or through the church.

I raise it not as a smear but as a methodological fact of exactly the kind Section 1.6 told you to look for. When the people designing the experiment, the people being tested, and the people scoring the transcripts share a prior religious commitment to the phenomenon existing, the resulting protocol will leak. That is not a claim about anyone's honesty; it is a claim about how blinding fails. It also explains the programme's peculiar signature — results that were statistically odd in-house and vanished when outsiders ran the sessions. And it is why the lineage matters: the remote-viewing work sat downstream of MKULTRA, which had already spent a decade demonstrating that the American security establishment would fund almost any hypothesis about the mind, and had already produced an institutional habit of running human research without controls that would survive external review.

Then came the government period, which is the part relevant to the rest of this book. Between 1972 and 1995 the CIA and the Defense Intelligence Agency funded remote-viewing research at the Stanford Research Institute and later at SAIC, under a sequence of code names ending with STARGATE. Real money, roughly \$20 million, real classification, real operational tasking. The programme was terminated in 1995 after an external review commissioned by the CIA — Jessica Utts, a statistician sympathetic to the field, and Ray Hyman, a psychologist who was not — produced a remarkable pair of reports. Utts concluded that a statistically significant effect had been demonstrated. Hyman agreed that the statistics were anomalous but held that methodological flaws and the absence of any mechanism made a paranormal conclusion unwarranted. Both agreed the programme had never produced intelligence of operational use. Read those two reports side by side; they are the single best education available in what a genuinely disputed borderline result looks like, as opposed to the fake disputes that fill most of this field.

FUN FACT

The STARGATE files were declassified and are on the CIA's own public reading room, which means the sentence "the CIA studied psychic phenomena" is entirely true and is also the setup for the most misleading move in the paranormal repertoire. Yes, they studied it. For twenty-three years. Then they read the assessment and shut it down. The funding is evidence that the question was taken seriously; the termination is evidence about the answer. People quote the first half.

7.7.3 WHERE THE EVIDENCE ACTUALLY STANDS

I want to be careful here, because parapsychology is the one place in this book where the honest skeptical position is more interesting than a flat dismissal.

FIGURE 7.3

Ganzfeld apparatus: halved ping-pong balls, red light and white noise, producing uniform sensory input.

Figure 7.3. Ganzfeld apparatus: halved ping-pong balls, red light and white noise, producing uniform sensory input.

Image sourced from: https://commons.wikimedia.org/wiki/File:Ganzfeld_experiment_setup.png

The **ganzfeld** telepathy experiment is the field's best paradigm. A receiver in mild sensory isolation (see Figure 7.3) attempts to identify which of four images a distant sender viewed; chance is 25%. Meta-analyses across hundreds of sessions have repeatedly reported hit rates in the low-to-mid 30s, which is small but not trivial. Charles Honorton and Ray Hyman jointly published a "joint communique" in 1986 agreeing on stricter protocol standards — a genuinely honourable episode in the history of scientific disagreement. Under the tightened protocols the effect persisted in some labs and vanished in

others.

The **random number generator** experiments, particularly the Princeton Engineering Anomalies Research programme run out of Princeton's engineering school from 1979 to 2007, reported deviations from expectation on the order of one part in ten thousand, accumulated across tens of millions of trials. PEAR's own conclusion after twenty-eight years was that the effect was real but too small and too unstable to characterise.

And in 2011 Daryl Bem published nine experiments in the *Journal of Personality and Social Psychology* reporting precognition — that participants'

responses were influenced by stimuli presented afterwards. This is the most consequential paper in the history of parapsychology, and not for the reason Bem intended. It used entirely standard social-psychology methods, passed peer review at a mainstream journal, failed to replicate, and in doing so helped trigger the replication crisis in psychology as a whole. Bem's paper is now taught in methods courses as the *reductio*: if these normal, accepted, everyday research practices can produce a significant result for something impossible, then those practices cannot be trusted to establish anything.

That is the state of play. A century and a half of work, a serious methodological tradition, and a residual effect that is always small, always near the detection threshold, never gets larger with better instruments, and shrinks as controls tighten. Compare that with any real phenomenon: the effect size for aspirin on headache, or the charge on an electron, does not diminish when you improve the apparatus. Real effects get *clearer*. This is the diagnostic test, and psi has been failing it in the same direction for a hundred years.

"The case for psychic functioning seems better than it ever has been . . . I also have to admit that I do not have a ready explanation for these observed effects."

— Ray Hyman

Why include all this in a UFO textbook? Because it is the closest thing we have to a completed experiment on our own discipline. Parapsychology did everything Ufology is constantly told to do. It got into universities. It adopted statistics, controls, blinding, pre-registration, and meta-analysis. It obtained government funding and classified operational trials. It attracted physicists and Nobel laureates. It ran for a hundred and forty years. And it still cannot demonstrate its central claim on demand. That should be sobering for anyone who believes the only thing standing between Ufology and a definitive answer is respectability and a research budget. It is possible to do this properly, at length, with money, and still find nothing — and it is possible that finding nothing is the correct outcome, honestly arrived at.

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7 . B W H A T I S A G H O S T ?

You may reasonably ask what ghosts are doing in a book about UFOs. Two answers. First, because the field itself keeps mixing them — a substantial minority of UFO cases contain poltergeist activity, apparitions and precognition, and Skinwalker Ranch in Section 13.8 is unintelligible without acknowledging it. Second, and more importantly, because ghost hunting is the perfect control group. It is a field with more enthusiasts, more equipment, and better site access than Ufology has ever had, and it has produced nothing in a century and a half. Studying how that happened is cheaper than repeating it.

“Ghost” is at least five distinct claims wearing one word, and separating them is most of the work:

The survival hypothesis. A ghost is the persisting consciousness of a dead person. This requires consciousness to be separable from the brain, which is discussed properly in Section 11.4.

The recording hypothesis. The “stone tape” idea: environments somehow store and replay events. No proposed physical mechanism exists, and stone has no known information storage capacity of the required kind.

The apparitional experience. A perceptual event occurring in the observer. Well-documented, common, and requiring no external entity at all.

Poltergeist activity. Physical disturbance — objects moving, sounds, electrical failures — usually centred on a person rather than a place, and disproportionately on an adolescent, which the older literature noticed and drew exactly the wrong conclusion from.

Interdimensional or non-human entities. Vallée’s ultraterrestrial idea, and Section 11.5’s subject.

The conventional explanations for hauntings are unusually well established, and two are worth knowing in detail.

Carbon monoxide poisoning produces, at sub-lethal chronic doses: headache, nausea, confusion, dread, auditory hallucination and visual apparition. There is a celebrated 1921 case — reported in the *American Journal of Ophthalmology* — of a family who moved into a house and experienced footsteps, voices, and figures at the bedside, all of which resolved when a faulty furnace was found. Any haunting confined to one building with a heating system should be investigated with a gas detector before an EMF meter.

Infrasound — sound below about 20 Hz, inaudible but perceptible — produces anxiety, chills, sensations of presence, and, near 18–19 Hz, visual disturbance, because that frequency is close to the resonant frequency of the human eyeball. Vic Tandy, an engineer, identified this in 1998 after tracing a “haunted” laboratory to a newly installed extractor fan; when the fan was fixed, the ghost left. Old buildings with large fans, plumbing, or wind-driven structural resonance generate infrasound routinely.

Add to those: mould, particularly *Stachybotrys*, associated with cognitive symptoms in some studies; sleep paralysis, as in Section 7.6, which accounts for the enormous class of bedroom apparitions; and pareidolia and expectation, which account for the rest. A person told a house is haunted will experience a haunted house.

FUN FACT

The suggestion effect has been measured directly. Richard Wiseman ran studies at Hampton Court Palace and the Edinburgh Vaults in which visitors were led through locations and asked to report unusual experiences. Reports clustered strongly in areas with prior haunting reputations — and also correlated with measurable environmental variables: air movement, temperature variation, lighting, and ceiling height. People reliably feel watched in places with certain physical properties. The ghost is in the architecture.

7.9 GHOST HUNTING

Modern ghost hunting is a methodological catastrophe conducted with great sincerity, and I want to go through the instruments one at a time, because every criticism here transfers directly to UFO fieldwork.

The EMF meter. Used to detect “spirit energy”. It detects electromagnetic

fields, of which a house is full: wiring, appliances, phones, the meter operator's own equipment, and the investigator's radio. There has never been any demonstrated link between EMF readings and apparitional experience — and note the irony, since if there is a real effect here it runs the other way: strong fields may cause the experience, which would make the meter a detector of causes rather than of ghosts.

The full-spectrum or infrared camera. Infrared photography captures reflected IR. Dust, insects, moisture droplets and lens flare are all rendered as bright “orbs”. The orb phenomenon appeared in the literature at precisely the moment consumer digital cameras with built-in flash became common, for the entirely understood optical reason that a flash close to the lens axis illuminates near-field dust. There is no orb literature from the era of large-format film with off-axis flash.

Thermal imaging. A thermal camera shows surface temperature. It is genuinely useful for finding drafts, insulation gaps, and plumbing — which is to say, for finding the conventional cause. It is routinely used instead to photograph “cold spots” whose measured magnitude is within instrument tolerance.

Audio recording and EVP. Electronic voice phenomena — voices claimed to appear in recordings. The mechanism is auditory pareidolia, and it is one of the strongest perceptual effects known. The human auditory system is aggressively tuned to extract speech from noise; given random audio and a suggestion of what to listen for, most people will hear it clearly, and once heard cannot un-hear it. Broadband noise, radio bleed, and digital compression artifacts supply the raw material. The decisive test — play the clip to listeners with no prior suggestion and no transcript, and see whether they independently agree on the words — is essentially never performed.

The spirit box. A device that sweeps AM/FM frequencies rapidly, producing fragments of broadcast audio. This is a random word generator connected to a suggestible listener. It cannot fail to produce apparent responses, which is exactly why it must not be used.

FUN FACT

Every ghost-hunting technique shares one design flaw: not a single one has a null condition. No instrument in the standard kit can produce a result meaning “nothing is here”. An EMF meter reading zero means the ghost

is inactive; a reading means the ghost is present. Silence on the recorder means try again. This is the Section 1.4 problem in hardware. A detector that cannot register an absence is not a detector, it is a prop — and if you have ever wondered why a hundred and fifty years of enthusiastic effort has produced no result, that is the reason, and it is entirely sufficient.

One last observation, and it applies to this whole chapter. There is a version of this research that is done properly: it is called parapsychology, it has occupied laboratories at Duke, Princeton, Edinburgh and elsewhere for the better part of a century, it uses randomised protocols, blinding, pre-registration and meta-analysis — and its results, after all that rigour, hover indistinguishably close to chance. That is a real answer, arrived at honestly, and it is far more interesting than any orb photograph. If a field with laboratory conditions, funding, and unlimited access to its subject matter found nothing, then a field working with handheld meters in the dark at two in the morning is not going to find it either.

THE GOVERNMENT AND MEN IN BLACK [MIB]

“Leaks are not the problem; they are the solution. They reveal a disconnect between what people want and need to know and what they actually do know. The greater the secrecy, the more likely a leak.”

— HEATHER BROOKE

8.1 PRESIDENTIAL UFOS

Every American president since Truman has been asked about UFOs, and the answers form a surprisingly useful record — not of what is in the vaults, but of what the office knows and how it behaves when it does not know something.

Truman, 1950, in a press conference: “I can assure you that flying saucers, given that they exist, are not constructed by any power on earth.” This is quoted constantly as a confession. Read the sentence again. It is a conditional, and its point is that the United States is not building them — a denial of domestic manufacture, which in 1950 was true of saucers and false of a great deal else.

Eisenhower disappeared from Palm Springs on the evening of 20 February 1954 and reappeared the following morning; the

FIGURE 8.1

Eisenhower delivering the farewell address of 17 January 1961 and its warning about the military-industrial complex.

press was told he had chipped a tooth and been taken to a dentist. The dentist was named, appeared at a function that night, and confirmed it. This is the basis of the persistent claim that Eisenhower met with extraterrestrials at Edwards Air Force Base and signed a treaty. What Eisenhower actually did in office that bears on this book is warn, in his 1961 farewell address (see Figure 8.1), about the “military-industrial complex” — a man who had commanded the largest military operation in history telling his country that its arms industry had acquired unaccountable influence. That is the real disclosure and nobody made a movie about it.

Kennedy is the subject of Section 8.5 and I will not preempt it.

Carter filed a formal sighting report with the International UFO Bureau in 1969, describing an object he and roughly ten others watched in Leary, Georgia in October 1969 — it changed colour and brightness, appeared to approach and recede, and lasted ten to twelve minutes. He campaigned on a promise to release UFO files. In office, he asked NASA to look into it; NASA declined, on the grounds that there was nothing to study without new data. Carter’s case, incidentally, is a textbook Section 4.4 exercise: the sighting was in the west after sunset, and Venus was at greatest brilliance in that position on that date. Carter himself later said he suspected it was Venus. A president reported a UFO honestly and then revised his own conclusion — which is more intellectual integrity than the field usually manages.

Reagan is the most genuinely interesting. He described, on more than one occasion, an incident in 1974 in a Cessna over California in which he and the pilot observed a light that accelerated away at speed. And repeatedly, in front of the UN General Assembly in 1987 and privately to Gorbachev at Geneva in 1985, he raised the idea that an alien threat would unite humanity: “how quickly our differences worldwide would vanish if we were facing an alien threat from outside this world.” Gorbachev confirmed the conversation happened. Reagan was a man who thought in narratives and had watched *The Day the Earth Stood Still*; he was using an invasion as a rhetorical device for arms control. It worked, in the sense that Geneva produced progress.

Clinton asked Webster Hubbell, his associate attorney general, to find out about UFOs and about who killed Kennedy, and was, by Hubbell's account, stonewalled on both. John Podesta, his chief of staff, has publicly and repeatedly advocated declassification.

Obama handled the question with jokes for eight years and then, in 2021, said plainly that there is "footage and records of objects in the skies that we don't know exactly what they are."

FUN FACT

The pattern across seventy years is not one of men guarding a secret. It is one of men discovering that the executive branch is enormous, compartmented, and not obliged to tell the president things he has no operational need to know. Clinton asked and did not get an answer. That is not evidence of aliens. It is evidence about the structure of the American state, which is the far more disturbing finding and the actual subject of this chapter.

B . 2 B L A C K B U D G E T F U N D I N G

The classified portion of the US defence budget is real, publicly acknowledged, and enormous. For recent fiscal years the requested classified figure has run in the vicinity of \$80–100 billion annually, spread across the National Intelligence Program, the Military Intelligence Program, and classified line items inside the services' research and procurement accounts. This is not a conspiracy theory; it is printed in budget documents as aggregate numbers with no detail, and Congress votes on it.

How it works matters more than the number. Funds are appropriated to unnamed programme elements, sometimes with deliberately meaningless nicknames. Oversight is restricted to specific committees, and within those, to cleared members and staff. Special Access Programs add a further layer, and "unacknowledged" SAPs are those whose existence is itself classified — meaning a cleared official may be legally obliged to deny that a programme exists. That is the mechanism people are reaching for when they talk about a cover-up, and it is genuine.

The historical track record proves it works. The U-2 flew in 1955 and was denied. The SR-71's predecessor, the A-12, was built by roughly a few thousand people at Area 51 and stayed secret for years. The F-117 flew

operationally for six years, from 1983, before its existence was acknowledged in 1988 — an aircraft with a genuinely bizarre faceted shape, flying at night over Nevada, in the exact period when Nevada UFO reports surged. The B-2, the Manhattan Project with 130,000 employees, the CORONA satellite programme, the existence of the NSA itself: all concealed successfully for years or decades.

So the claim “the government can keep a secret this big” is not naive. It is demonstrated.

The counter-claim is more specific and it is where I part company with the disclosure movement. Every one of those programmes eventually surfaced — through congressional inquiry, through budget forensics, through defectors, through the simple attrition of retirement and death. Average time to exposure has been on the order of five to twenty-five years. A crash-retrieval and reverse-engineering programme running continuously since 1947 would be, by a wide margin, the longest-held technical secret in human history, held by an institution that could not keep the Pentagon Papers, the Church Committee revelations, MKULTRA, COINTELPRO, Iran-Contra, Abu Ghraib, or a quarter of a million diplomatic cables. The same organisation that leaks constantly on everything else has, on this alone, achieved perfection.

FUN FACT

A specific piece of budget evidence does exist and deserves mention because it is real: the Advanced Aerospace Threat Identification Program was funded at about \$22 million between 2007 and 2012, largely at the initiative of Senator Harry Reid, and administered substantially through Robert Bigelow’s aerospace company. Its existence was revealed in December 2017 by the *New York Times* and *Politico*. So there was a secret government UFO programme, it was concealed, and it came out — in ten years, through journalism, exactly as the pattern predicts. It also produced no craft, no bodies, and a set of videos discussed in Section 12.5.

B . 3 PROJECT BLUE BOOK

The United States Air Force ran three consecutive public UFO investigations, and their progression tells you what the institution was actually doing.

Project Sign, 1948. Established after the Arnold sighting and the pilot

deaths associated with the Mantell crash. Sign's staff produced an "Estimate of the Situation" concluding that the objects were probably interplanetary. General Hoyt Vandenberg, Air Force Chief of Staff, rejected the estimate for insufficient proof and it was destroyed. Note what happened: the investigators concluded aliens, and the leadership said the evidence did not support it. That is the opposite of the cover-up narrative and it is documented.

Project Grudge, 1949–1951. Named with more candour than intended. Grudge operated from the conclusion that reports were misidentification and hoax, and its function was debunking. It was, by the assessment of Air Force personnel at the time, a public relations exercise. Its files were poor.

Project Blue Book, 1952–1969. Seventeen years, 12,618 reported cases. Under Captain Edward Ruppelt, who took over in 1952 and who coined the term "UFO" precisely to escape the connotations of "flying saucer", Blue Book was briefly a serious operation: standardised reporting forms, statistical analysis, a genuine attempt at method. Ruppelt left in 1953 and the project declined into a complaints department, staffed at times by a single officer and a secretary.

The final numbers: 701 cases, roughly 5.5%, classified "unidentified". The remainder were attributed to astronomical objects, aircraft, balloons, weather phenomena, hoaxes, and insufficient data. J. Allen Hynek was the project's scientific consultant for its entire duration, and his trajectory is the most important thing about it: he arrived as a debunker, spent twenty years reading reports, and left convinced the residue was real and had been handled disgracefully. He was also personally responsible for the "swamp gas" explanation of the 1966 Michigan sightings, which he regretted for the rest of his life and which taught him, he said, what happens when you offer a plausible-sounding answer you have not verified.

The 1953 CIA-convened **Robertson Panel** is the single most consequential document here (see Figure 8.2). A group of scientists including Luis Alvarez and Lloyd Berkner met for four days and concluded that UFO reports posed no direct threat — but that the *volume* of reports posed a real threat, by clogging military communications channels during a period

FIGURE 8.2

A page of the declassified Robertson Panel report, January 1953, recommending a public "debunking" programme.

Figure 8.2. A page of the declassified Robertson Panel report, January 1953, recommending a public "debunking" programme.

Image sourced from: https://commons.wikimedia.org/wiki/File:Robertson_Panel_report_page.jpg

when an actual Soviet attack was the concern, and by leaving the public vulnerable to psychological warfare. Their recommendation was a programme of “debunking” to reduce public interest, potentially using mass media and, explicitly, Disney.

Read carefully, that is a genuine government conspiracy to manipulate public opinion about UFOs, and it is fully documented in the panel’s own declassified report. Its motive was not hiding aliens. It was air defence bandwidth and Cold War psychology. The consequence, however, was that official statements from 1953 onward were shaped by a policy of reducing interest rather than a policy of finding out — which poisoned the well permanently, and which is why every denial issued since is reasonably suspected of being institutional rather than factual. The government created the modern UFO conspiracy culture on purpose, by deciding that reassurance mattered more than accuracy.

B . 4 THE AFTERMATH OF BLUE BOOK

Blue Book was closed on 17 December 1969 on the recommendation of the **Condon Report** — formally the *Scientific Study of Unidentified Flying Objects*, produced at the University of Colorado under Edward Condon with Air Force funding, published in 1968 at some 1465 pages.

Its conclusion: further study was unlikely to be productive, nothing had come out of twenty-one years of investigation that advanced science, and there was no national security implication.

The Condon Report is a genuine scandal, and not for the reason usually claimed. The problem is internal: Condon expressed contempt for the subject publicly before the study concluded, and a project coordinator’s memo — the “Low memo”, written by Robert Low before the project began — proposed that the study could be run so that “the trick would be, I think, to describe the project so that, to the public, it would appear a totally objective study but, to the scientific community, would present the image of a group of nonbelievers.” That memo leaked. Two staff members were fired after circulating it. Hynek and James McDonald both attacked the report’s integrity.

And here is the detail that decides it: roughly 30% of the cases examined *in the body of the report* were left unexplained by the report's own investigators, a higher unidentified rate than Blue Book's. Condon's summary conclusions, written by Condon, do not reflect the contents of the chapters they summarise. Anybody can verify this; the report is public. It is the classic institutional manoeuvre — commission a study, and write the executive summary separately.

The American Institute of Aeronautics and Astronautics reviewed the report and disagreed with its conclusion, stating that the study did not support the claim that nothing of scientific value could come from further work. Nobody cared. Condon's summary was what the press printed, Blue Book closed, official study ended, and the field was handed over entirely to amateurs for the next forty-eight years, which is a substantial part of why this book has to spend its first chapter teaching method.

FUN FACT

Official study did not actually end — it just stopped being public. The 1969 termination memo instructed that reports of national security significance should continue to be handled through normal military channels, specifically JANAP 146 and Air Force Regulation 200-2, both of which predated Blue Book and neither of which was rescinded. So the public office closed and the classified reporting channel stayed open. Then in 2017 the Pentagon acknowledged AATIP, and in 2022 Congress established the All-domain Anomaly Resolution Office. Fifty-three years after announcing there was nothing to investigate, the Department of Defense established a permanent office to investigate it.

8.4.1 THE GRUSCH TESTIMONY

Which brings the story to 26 July 2023, and to the single most serious allegation ever made about this subject under oath. David Grusch, then thirty-six, a former Air Force intelligence officer who had served on the National Reconnaissance Office and had been the National Geospatial-Intelligence Agency's representative to the Unidentified Aerial Phenomena Task Force, told a subcommittee of the House Oversight Committee that the United States operates a multi-decade programme to recover and reverse-engineer craft of non-human origin, and that "non-human biologics" had been retrieved from them. He first went public in June of that year through

The Debrief and in an interview with the journalist Ross Coulthart, and he filed a complaint with the Intelligence Community Inspector General alleging that he had been retaliated against for reporting the programme through lawful channels.

I want to be extremely careful here, because both the enthusiastic and the dismissive readings of this are wrong. What is genuinely new is procedural, and it is not small: a serving intelligence officer with verified access and a clean record made these claims in open session, under oath, under penalty of felony, and his Inspector General complaint was found urgent and credible — which is a statutory determination about whether a complaint warrants investigation, not a finding that the underlying claim is true. Nothing remotely like that had happened in the previous seventy-six years.

What is not new is the evidentiary situation. Grusch stated repeatedly that his knowledge was derived from interviews with some forty other individuals whom he considered credible, that he had not personally seen craft or bodies, and that the specifics — names, locations, programme designations — could only be given in classified session. **That is testimony about testimony, which is the oldest structure in this book.** Section 1.5 gave you the rule and I am not going to suspend it for a witness I happen to find interesting. The Department of Defense denied having any verifiable information about the programme he described; AARO's historical review, published in March 2024, reported finding no evidence that any US programme had ever possessed extraterrestrial technology, and attributed a portion of the recovered-craft folklore to genuine but conventional classified projects and to a deliberate practice of using UFO cover stories to protect them.

So where does that leave us? With a named, credentialled, on-the-record witness alleging an extraordinary thing, an official review denying it, and no physical evidence in the public domain from either side. If the allegation is true, the classification structure is the reason we cannot check it. If it is false, the classification structure is also the reason we cannot check it. That symmetry is uncomfortable and I am not going to resolve it for you. What I will say is that the correct posture toward Grusch is neither belief nor mockery but the one this whole course argues for: hold the claim open, insist that the resolution come through documents and materials rather than through further interviews, and notice which side of the argument is asking you to accept a chain of unnamed sources.

8.5 THE JFK ASSASSINATION

The UFO connection to the Kennedy assassination rests on two documents, and I am going to be blunt about them because this is where the field embarrasses itself most reliably.

The first is a memorandum allegedly dated 12 November 1963, from Kennedy to the Director of Central Intelligence, requesting a review of UFO intelligence files and referencing cooperation with the Soviet Union on space activities. The second is a purported CIA document known in the community as the “burned memo”, containing the line about “Lancer” — the Secret Service codename for Kennedy — being a problem.

The November 1963 memo is real, and it says something less interesting than advertised: it is a National Security Action Memorandum directing cooperation with the USSR on space matters, which is consistent with Kennedy’s September 1963 UN speech proposing a joint Moon mission. That proposal is genuinely remarkable and genuinely relevant to Cold War history. It is not about aliens. The associated “UFO files” memo has never been authenticated, exists in photocopy, and comes through the same channels as the material in Section 8.6.

The burned memo is not authenticated by anybody, was supplied by a single anonymous source to one researcher, and shows physical damage consistent with someone deliberately burning a photocopy.

Here is why this matters beyond the specific claim. The Kennedy assassination has genuine unresolved features: the acoustic evidence, the House Select Committee’s 1979 conclusion of probable conspiracy, the documented CIA and FBI withholding of information from the Warren Commission, the Mafia and anti-Castro connections, and the fact that thousands of records remained sealed into the 2020s. There is a serious body of work there. Attaching UFOs to it does not strengthen the UFO case; it weakens the assassination case, by giving anybody who wants to dismiss all of it an easy reason to.

This is the mechanism I called conspiracy accretion in Section 1.6. A theory that explains everything explains nothing, and a movement that adopts every available mystery as a subsidiary of its own is not investigating — it is collecting.

B . E M A J E S T I C 1 2

In December 1984 a roll of undeveloped 35mm film arrived in the mail at the home of Jaime Shandera, a television producer working with researcher William Moore. Developed, it contained photographs of an eight-page briefing document dated 18 November 1952, prepared for President-elect Eisenhower, describing “Operation Majestic 12” — a group of twelve scientists and officials overseeing the recovery of crashed extraterrestrial craft and bodies at Roswell and elsewhere.

The named members are real and impressive: Vannevar Bush, James Forrestal, Roscoe Hillenkoetter, Nathan Twining, Hoyt Vandenberg, Detlev Bronk, Donald Menzel and others — exactly the group you would assemble for such a task, which is either evidence of authenticity or evidence that the forger did their homework.

The documents are forgeries, and the case against them is technical and decisive:

Date format. The documents use “18 November, 1952” — with a comma after the month in a military-style date. This is wrong, and consistently wrong, for the period and the agencies involved. Correct usage is “18 November 1952”. The comma is a distinctively civilian American habit.

Typeface. Analysis identified the type as consistent with a Smith-Corona machine and, more damagingly, the documents’ formatting conventions do not match verified Truman-era Executive Office documents.

The Truman-Forrestal memo. An attached memo bears Truman’s signature, and that signature is a pixel-for-pixel match — identical size and shape — to Truman’s signature on a genuine, publicly available 1 October 1947 letter to Vannevar Bush. Signatures are never identical. It was photocopied and pasted.

Classification markings. The documents use “TOP SECRET / MAJIC EYES ONLY”, a classification and control marking that appears nowhere in the vast declassified record of the period, and their handling instructions are inconsistent with the security manuals in force.

Provenance. Anonymous film in the mail. There is no chain of custody at all.

The FBI investigated and stamped its copy “BOGUS”. The Air Force Office of Special Investigations concluded the same. And in the 1980s AFOSI

is documented — in the Paul Bennewitz affair — to have deliberately fed fabricated UFO material to researchers as a counterintelligence exercise, in Bennewitz’s case with results that destroyed the man’s mental health. Richard Doty, an AFOSI agent, has admitted participating in disinformation. William Moore has admitted cooperating with intelligence contacts.

FUN FACT

The most consequential thing about MJ-12 is not whether it is fake — it is fake — but that it worked. Almost the entire modern architecture of UFO conspiracy belief descends from these eight pages: the crash retrieval programme, the secret oversight committee, the “EBE” vocabulary of Section 6.4, the treaty narrative, the underground facilities of Chapter 17. A document rejected by the FBI in 1988 is still the load-bearing wall of a belief system followed by millions. That is worth more study than the document.

B . 7 THE MEN IN BLACK

The Men in Black arrived in the literature through Albert Bender, who ran the International Flying Saucer Bureau and shut it down abruptly in 1953, telling subscribers he had been visited by three men in dark suits who told him to stop. Gray Barker turned this into *They Knew Too Much About Flying Saucers* in 1956, and John Keel expanded the mythology in the 1970s. Bender himself later published a different account — *Flying Saucers and the Three Men* (1962) — describing telepathic entities in a mountain base in Antarctica, which tends not to get quoted.

The reported profile is consistent: two or three men, black suits of slightly wrong cut, black cars of an old model in impossibly new condition, pale or oddly tinted skin, monotone or stilted speech, unfamiliarity with ordinary objects and idioms, knowledge of things they should not know, and either a warning to stop talking or a strange interrogation. They arrive shortly after a sighting and before it is public.

The mundane explanations account for most of it and they are not boring.

Actual government agents did visit UFO witnesses, routinely, and this is documented. Air Force investigators under AFR 200-2 interviewed witnesses; so did the FBI, and so

FIGURE 8.3

Air Force investigators interviewing a witness under AFR 200-2 — the documented origin of the “men in black” visit.

did the OSI. In the 1950s and 1960s a federal agent's uniform was, functionally, a dark suit, a hat, and a government sedan (see Figure 8.3) — and agents were frequently instructed to discourage witnesses from talking to the press, which was Robertson Panel policy from 1953. An anxious witness visited by two men in dark suits with government credentials who told them not to discuss the matter publicly has not had a paranormal experience. They have had an accurate one.

Add to that: journalists, insurance adjusters, private researchers with self-important manners, Mormon missionaries, and outright pranksters. And the “uncanny” details — stilted speech, wrong idioms, unfamiliarity with a ballpoint pen — are exactly what memory does to a frightening encounter recalled years later after exposure to the folklore, which is Section 7.1 operating without a hypnotist.

The cultural trajectory is the joke. By 1997 the MIB were a Will Smith comedy franchise, and the modern reported cases have been contaminated by it beyond recovery.

B . B THE ILLUMINATI

The Bavarian Illuminati was founded on 1 May 1776 by Adam Weishaupt, a professor of canon law at Ingolstadt. It was a real organisation with real goals: Enlightenment rationalism, opposition to clerical control of education, and reform of the Bavarian state. It recruited within Masonic lodges, reached perhaps 2000 members at its peak, and was suppressed by the Elector of Bavaria's edicts of 1784 and 1785. Weishaupt fled and died in obscurity in 1830. That is the history, and it is not in dispute.

The conspiracy version was manufactured almost immediately, by two authors reacting to the French Revolution: Augustin Barruel in *Mémoires pour servir à l'histoire du jacobinisme* (1797) and John Robison in *Proofs of a Conspiracy* (1798). Both argued the Revolution was engineered by surviving Illuminati. This was politically useful to people who preferred not to consider that the French had revolted over bread and taxes.

The modern version accumulated by the same process I described in Section 1.6: absorb Masonic symbolism, absorb the *Protocols of the Elders of Zion* (a Tsarist secret police forgery of around 1903, demonstrated as such by *The Times* in 1921, and the direct ancestor of most modern global-conspiracy structure), absorb central banking, absorb the eye on the dollar bill — which is the Eye of Providence, a common eighteenth-century Christian motif, on a Great Seal designed in 1782 by people who were not shy about explaining their choices — and finally, in the 1990s, absorb aliens.

I want to name the specific reason I refuse to treat this material as a harmless parlour game. The structure of the Illuminati narrative — a small hidden group of financiers manipulating nations, media, and money — is the structure of antisemitic conspiracy theory, because that is literally where the modern version came from. The *Protocols* were antisemitic propaganda that contributed materially to the Holocaust. When a modern theory says “Illuminati” or “globalist bankers” in the slot where the *Protocols* said something else, the argument has not changed; only the label has. David Icke’s reptilian version, which I will address in Chapter 17, maps onto this lineage in a way its adherents mostly do not realise and should.

FUN FACT

The genuinely useful lesson here is about how power actually operates, because it is not hidden. Lobbying disclosures, campaign finance records, corporate board interlocks, the revolving door between regulators and the industries they regulate, Bilderberg and Davos attendee lists — all public, all documented, all considerably more influential than any secret society. The people who run things do not need to hide. They file paperwork. Any evening spent reading lobbying registrations will teach you more about who controls the world than the entire Illuminati literature, and it has the advantage of being true.

FROM ANCIENT TO MODERN TIMES

"We are made of stardust, our whole body consists of material that has been here before the beginning of time."

— GIORGIO A. TSOUKALOS

9.1 CLASSIFYING IMPOSSIBILITIES

Graham Hancock has spent thirty years making one argument, and it deserves to be stated properly before anybody argues with it. The claim is not that aliens built the pyramids. It is that a sophisticated human civilisation existed before the end of the last Ice Age, that it was destroyed in the cataclysms of the Younger Dryas around 12,900 years ago, and that what survived was carried by refugees into the cultures we call the beginnings of history — as astronomy encoded in myth, as agriculture appearing suspiciously fully formed, as megalithic building traditions separated by oceans. On this reading, the ancient world is not the dawn of human achievement. It is the amnesia after a collapse.

It is a genuinely interesting thesis and it fails on evidence rather than on principle, which is a compliment. A civilisation leaves middens, quarries, harbours, pollen signatures, metallurgical slag, genetic bottlenecks, isotopes in ice. We have found none of that from the right period, and we have looked

at that period intensively for other reasons. Göbekli Tepe — Hancock's best exhibit, and it is a good one — is around 11,500 years old, monumental, sophisticated, and built by people whose toolkit and diet we can characterise in detail. It pushed the date of monumental architecture back by millennia. It did not produce a lost Atlantis; it produced hunter-gatherers doing something nobody expected hunter-gatherers to do. Chapter 19 returns to the submerged-city version of the same idea.

I open with Hancock deliberately, because his structure is the structure of this entire chapter. Something in the ancient record looks impossible. A modern reader supplies the missing cause. And whichever cause you supply — lost civilisation, visiting spacecraft, divine intervention — you are performing the same operation. The useful skill is not picking a favourite. It is sorting the impossibilities before explaining them: which are impossible because the evidence is genuinely anomalous, which are impossible only relative to what the ancient author knew, and which are impossible only because somebody described them badly a very long time ago and the description is all we have left.

Begin with the middle category, because it comes with a ready-made test.

A miracle, in the strict theological sense, is an event caused by direct divine intervention in violation of natural law. That definition is precise and it has a consequence people rarely follow through on: a miracle is only a miracle relative to what the observer knows about natural law. Move the observer, and the category moves with them.

The Ancient Alien Hypothesis supplies spacecraft. Religion supplies God. Hancock supplies a drowned civilisation. Conventional history supplies exaggeration, metaphor, and craftsmanship we have underestimated. All four are the same move — filling an explanatory gap with the most impressive thing available to the person doing the filling. Only one of those four is required to suspend a law of nature, and there is a classical test for that case.

So before we go further, the ground rules from Chapter 1, applied to text:

Ancient documents are not neutral reports. They are written by literate elites, for purposes — legitimising a dynasty, glorifying a deity, explaining a ruin nobody remembered building. They are copied by hand for centuries, with drift. They are translated, often several times, and translation of

technical vocabulary from a dead language is a guessing game.

And the correct null hypothesis for an ancient account of something amazing is not “it didn’t happen”. It is: something happened, and the description we have is the last surviving link in a long chain of retelling. Ezekiel saw something, or believed he did, or wrote a vision for a reason. Determining which is history; determining what he *physically* saw is, in almost all cases, not recoverable.

FUN FACT

David Hume gave the sharpest test in 1748, and it is still the one to use. The question is never “is this event possible?” but “which is more likely — that the laws of nature were suspended, or that the testimony is mistaken?” Human testimony is demonstrably fallible; the laws of nature have never once been observed to fail. This is not an argument that miracles cannot happen. It is an argument about where to place your bet before the evidence arrives.

9.2 THE ANCIENT ALIEN HYPOTHESIS

The Ancient Alien Hypothesis holds that extraterrestrials visited Earth in antiquity, were recorded as gods, and contributed technology or genetic material to early humans. Its modern popular form comes from Erich von Däniken’s *Chariots of the Gods?* (1968), which sold something like 70 million copies, though the idea predates him — Charles Fort gestured at it, and Morris Jessup and the Soviet writer Matest Agrest published versions in the 1950s.

Let me first state the strongest version of the case, because there is one and it is worth engaging with.

Sky gods are close to universal. Cultures with no contact describe beings descending from above, teaching agriculture, writing, metallurgy and law, and departing with a promise to return. Some ancient engineering is genuinely difficult to account for — the largest stones at Baalbek weigh over 800 tonnes and one unmoved block exceeds 1600. Some ancient astronomical knowledge is precise beyond apparent need. And the Dogon of Mali were reported by Marcel Griaule to possess knowledge of Sirius B, a white dwarf invisible without a telescope.

Now the problems, in order of severity.

It is an argument from ignorance, structurally. Every instance takes the form: I cannot see how they did this, therefore someone else did. **That inference is invalid regardless of the candidate you insert.** It also has a poor track record, because the gaps keep closing — and Section 9.5 is about exactly that.

It underestimates ancient people, systematically and in one direction. The hypothesis has never once been applied to European megaliths in a form its adherents find equally exciting, and it is applied overwhelmingly to non-European achievement: Egyptian, Mesoamerican, Andean, Southeast Asian. There is a name for the assumption that brown people could not have built their own monuments, and the fact that its modern proponents do not intend it does not make the pattern less real. Egyptians left us quarry marks, workers' villages, bakeries, medical records of construction injuries, administrative papyri recording the delivery of limestone blocks — the Merer diary, found at Wadi al-Jarf in 2013, is a working logbook of a boat crew hauling casing stone to Giza. We know who built the pyramids. Their names are on the walls, and some of them scratched graffiti about which work gang was better.

The material evidence is absent, and it is the kind of absence that counts. A visiting technological civilisation would leave residue that survives: alloys that do not occur naturally, isotopic anomalies, machined tolerances, plastics, radioactive traces, refined materials in stratified archaeological context. We have excavated an enormous number of sites with increasingly sensitive methods and found none of it. Not one anomalous artifact from a controlled excavation — as Section 1.8 explains, that is where such a find would have to come from to mean anything.

The Dogon case dissolved. Later anthropologists, notably Walter van Beek, worked with the Dogon and could not find the Sirius B tradition Griaule reported. Griaule had visited in the 1930s and 1940s — after Sirius B was well publicised in European popular science, and after European visitors including astronomers had been in the region. The likeliest explanation is feedback from Griaule's own questioning, which is Section 1.6's leading-question problem in fieldwork rather than in a therapist's office.

9.2.1 SITCHIN, THE ANUNNAKI, AND THE TWELFTH PLANET

If von Däniken supplied the hypothesis, Zecharia Sitchin supplied its scripture, and since his names turn up constantly in this field you should know exactly where they come from. Sitchin was a Russian-born American writer who published *The 12th Planet* in 1976 and eventually seven volumes of what he called the Earth Chronicles. His claim was that the Sumerian cuneiform record — the oldest written literature we possess, from the oldest civilisation we can name — is not mythology but history. The Anunnaki, in his reading, were flesh-and-blood visitors from Nibiru, a twelfth planet on a 3600-year elliptical orbit, who came to Earth to mine gold and genetically engineered humanity from local hominids as a labour force. Every element of the modern ancient-astronaut vocabulary — Anunnaki, Nibiru, the gold mining, the engineered slave species — is Sitchin's invention, not Sumer's, and he was a fixture on *Coast to Coast AM* under both Art Bell and George Noory, which is how it reached millions.

FIGURE 9.1

A Sumerian administrative tablet of the kind Sitchin cited. The signs he read as "rocketship" say nothing of the sort.

Figure 9.1. A Sumerian administrative tablet of the kind Sitchin cited. The signs he read as "rocketship" say nothing of the sort.

Image sourced from: https://commons.wikimedia.org/wiki/File:Sumerian_cuneiform_tablet.jpg

The problem is not that Sitchin was a kook, though mainstream Assyriology regarded him as one. The problem is specific and checkable: **the translations are wrong.** Sitchin was self-taught in the cuneiform languages (see Figure 9.1) and worked without the philological apparatus. Assyriologists — Michael Heiser wrote the most detailed rebuttal — have gone through his key terms line by line. The word he renders as "rocketship" does not mean that. *Anunnaki* means, roughly, those of princely seed, and the Sumerian texts place them in the sky and the underworld as deities, not in vehicles. There is no

Sumerian word for Nibiru's supposed orbit, no tablet listing twelve planets, and the Sumerians did not know Neptune or Uranus existed. When a body of work rests on translation and the translation fails, nothing downstream of it survives. That is not a sceptic's opinion; it is the ordinary standard applied to any scholar who works from a text.

As for Nibiru, the ufological literature likes to say NASA supports the

idea of a large gravitational body beyond the Kuiper belt. What NASA-affiliated astronomers have actually proposed — Batygin and Brown, in 2016 — is Planet Nine: a hypothetical body of perhaps five to ten Earth masses, inferred from the clustered orbits of distant trans-Neptunian objects, at several hundred astronomical units, with a period on the order of ten thousand years or more. It has never been observed, its existence is contested, and even if confirmed it would be a frozen world with no possible relation to Sumerian labour disputes. A distant, cold, unconfirmed planet and an inhabited world that swings through the inner solar system every 3600 years are not the same claim wearing different clothes. This is a move you will see again and again, so learn it now: take a real, cautious, technical result, keep the headline, and quietly swap the content.

The same surgery is worth performing on Robert Bauval, whose *Orion Correlation Theory* argues that the three Giza pyramids map the belt stars of Orion. The correlation is loose, requires choosing a particular epoch and ignoring the pyramids that do not fit, and the Egyptians left us their own extensive account of what the alignments were for — the northern circumpolar stars and the king's transformation — which is both better attested and more interesting. Bauval's water erosion argument about the Sphinx enclosure, developed with Robert Schoch, is the most respectable thing in this territory and the one place where a real geological dispute exists. It is also not an argument about aliens, and no Egyptologist has yet been persuaded that it beats the conventional dating.

FUN FACT

The sky-god universality argument has a much better explanation than visitors, and it is the same one that explains why Section 7.4's abduction narrative matches fairy folklore. Every human culture is under the same sky, subject to the same weather, dependent on the same sun, frightened by the same lightning, and possessed of the same brain — one that models the world in terms of agents with intentions, because that heuristic kept our ancestors alive. Universal myths are evidence of universal humans. That is a real discovery about our species, and it is more interesting than a spaceship.

9.3 UFOS IN ANCIENT ARTWORK

The standard gallery is short and every item in it has an art-historical answer. I will go through them because knowing the answers is the only inoculation.

The Madonna with Saint Giovannino (15th century, Palazzo Vecchio). A disc-shaped object with rays in the sky, a man and his dog looking up. It is a standard Renaissance depiction of a cloud with divine radiance — the visual convention for the Annunciation or the glory of God — and comparable clouds appear in dozens of contemporary works nobody has heard of. The shepherd and dog are conventional too.

The Baptism of Christ by Aert de Gelder (1710). A bright disc emitting beams over Christ's head. This is the Holy Spirit descending, which the text of Matthew 3:16 requires the painter to depict. The alternative is that de Gelder painted a UFO into a baptism scene and nobody in 1710 remarked on it.

The Crucifixion (1350, Visoki Dečani monastery, Kosovo). Two objects with figures apparently inside them, flanking the cross. They are personifications of the Sun and Moon — a standard Byzantine convention with a long documented pedigree, present in countless Orthodox crucifixion scenes, representing creation mourning.

The Nazca lines. Enormous geoglyphs in the Peruvian desert, only fully legible from the air, therefore runways or signals. They are not runways: they are shallow surface scrapings a few centimetres deep in loose stone, which would not survive a bicycle let alone a landing. They are constructible from the ground with stakes and cord, which has been demonstrated repeatedly, and are visible from surrounding foothills. They are almost certainly ritual pathways related to water and were walked, not viewed.

The Palenque sarcophagus lid (Maya, 7th century). K'inich Janaab' Pakal reclining, read as an astronaut in a capsule. The full iconography is documented and internally consistent: Pakal falls into the underworld at the moment of death, with the world tree rising above him, the Celestial Bird at its top, and the maw of the underworld below. The glyphs around the border name him and date it. Rotate the image ninety degrees to make it look like a rocket and you have rotated a religious scene out of its own orientation.

Ezekiel's wheel. Ezekiel 1 describes wheels within wheels, full of eyes,

with four living creatures. Josef Blumrich, a NASA engineer, published *The Spaceships of Ezekiel* in 1974 reconstructing a lander from the text. The text is apocalyptic vision literature written in Babylonian exile, deploying Babylonian and Assyrian imagery — the four-faced creatures correspond closely to known Mesopotamian composite guardian figures — and its purpose is theological: God’s throne is mobile, therefore God is present in exile, outside the destroyed Temple. That is the point of the passage and it is a devastating one for its original audience.

The Dendera light and the “helicopter hieroglyphs” at Abydos. The former is a lotus and a serpent, a standard creation motif, in a crypt whose full inscriptions explain it. The latter is a genuine curiosity with a dull answer: it is a palimpsest, where Ramesses II’s titles were carved over Seti I’s and the filling plaster fell out, superimposing two sets of glyphs. This is visible on inspection and documented in the epigraphic record.

FUN FACT

Here is the test that settles the whole category. Ancient artists depicted what they saw with remarkable fidelity when they were depicting objects — Egyptian tomb painting gets bird species right to the feather, and Greek pottery renders ship rigging accurately enough to reconstruct from. If flying craft had been observed, we would have detailed, repeated, unambiguous depictions with consistent features, in secular as well as religious contexts, and they would be described in the accompanying inscriptions. Instead we have a handful of religious symbols that require reinterpretation against their own captions.

S . 4 L E Y L I N E S

In 1921, an English businessman and amateur photographer named Alfred Watkins was looking at a map of Herefordshire when he noticed that certain ancient sites — standing stones, mounds, churches on older sacred sites, hilltop beacons — appeared to fall into straight lines. He published *The Old Straight Track* in 1925, proposing that these alignments were prehistoric trading routes surveyed by sighting between landmarks.

Watkins’ original hypothesis was archaeological and entirely reasonable in outline. People do navigate by sightlines. Some ancient landscapes do contain deliberate alignments — the solstice axis at Stonehenge and

Newgrange are real, measured, and beyond dispute.

The problem is statistical, and it is a lovely example of why Chapter 1 exists. England contains many thousands of ancient sites. If you scatter that many points randomly on a map and permit a line to be a few hundred metres wide, you will find an enormous number of “alignments” by pure chance. This was demonstrated formally — in 1980 the astronomer Robert Forrest ran the numbers and showed that the observed frequency of Watkins-style alignments is what you would expect from random placement, and in 1989 the mathematician Chris Marshall generated convincing ley networks from randomly generated site data. You can do this yourself with a map of pizza restaurants, and several people have, producing spectacular “ley lines” across Manhattan.

By the 1960s the concept had been detached from archaeology entirely and reattached to earth energies, dowsing, and eventually to UFOs — the claim being that sightings cluster along leys, that craft navigate by them, and that they represent lines of geomagnetic force. There is no measured geomagnetic anomaly along any ley line. Ordnance Survey and geological survey data have been available for a century, and the alignments do not correspond to anything in them.

FUN FACT

The UFO-ley connection has an author: Tony Wedd, in 1961, who combined Watkins with the French “orthoténie” work of Aimé Michel, who had claimed in 1958 that sightings during the 1954 French wave fell along straight lines. Michel’s orthoteny was tested properly in the 1960s by Jacques Vallée among others, and it collapsed: the lines were an artifact of selecting which reports to plot. Vallée, to his considerable credit, published the negative result about his own earlier enthusiasm. That is what Section 4.4 step six looks like in practice.

I keep ley lines in the book for one reason: they are the cleanest demonstration in the entire field that pattern-finding is not evidence. **Your brain is extremely good at spotting alignments and extremely bad at estimating how many alignments chance produces.** Every time you see a map in this literature with lines drawn on it — and you will see many, connecting pyramids, connecting crash sites, connecting sacred mountains — ask the only question that matters: how many lines would I have found in random

data? Nobody who draws those maps has ever once told you.

9.5 THE IMPOSSIBLE STONE

The strongest argument the ancient astronaut literature has is not artwork. It is masonry, and I want to give it a fair hearing, because standing in front of some of these walls does something to your confidence that reading about them does not.

The items usually cited: the Great Pyramid, 2.3 million blocks, some casing stones fitted to tolerances measured in fractions of a millimetre. The Trilithon at Baalbek (see Figure 9.2), three blocks around 800 tonnes each, raised into a wall course, with the abandoned Stone of the Pregnant Woman and a larger 2014 find nearby exceeding 1,600 tonnes. Sacsayhuamán, polygonal Inca walls of irregular multi-tonne blocks fitted without mortar so closely that a knife blade will not enter. The Osirion at Abydos. The unfinished obelisk at Aswan, which would have weighed around 1,100 tonnes. Puma Punku's andesite H-blocks, with flat faces and interior right angles.

The conventional answers are unglamorous, well tested, and — this is the part people miss — experimentally demonstrated.

Cutting. Egyptian hard-stone work used copper saws with quartz sand abrasive, and it is the sand doing the cutting, not the metal. Drill cores from Egyptian granite show the characteristic striations. Denys Stocks spent decades replicating Egyptian stoneworking with period tools and successfully cut, drilled and dressed granite. Andesite at Puma Punku was worked with harder hammerstones and abrasive; the “machined” interior angles show tool marks under examination and are not, in fact, perfectly square.

Fitting. Inca polygonal masonry was achieved iteratively — place a block, mark the contact, lift, dress, replace, repeat. It is laborious rather than

FIGURE 9.2

The Trilithon at Baalbek: three courses of roughly 800 tonnes each, moved by documented Roman engineering.

Figure 9.2. The Trilithon at Baalbek: three courses of roughly 800 tonnes each, moved by documented Roman engineering.

Image sourced from: https://commons.wikimedia.org/wiki/File:Baalbek_trilithon.jpg

mysterious, and unfinished Inca walls survive showing the process mid-sequence, with dressing scars and abandoned blocks. The technique also has a functional reason nobody mentions: irregular interlocking mortarless masonry is superbly earthquake resistant, and it is still standing in a seismic zone that has flattened everything the Spanish built.

Moving. This is where the intuition fails hardest, and it fails because of arithmetic. A 60-tonne block on a lubricated wooden sledge on a prepared trackway requires a pulling force well within the reach of a few hundred coordinated men — a wall painting in the tomb of Djehutihotep depicts 172 men hauling an alabaster colossus, with a man pouring liquid in front of the sledge, and a 2014 experiment at Amsterdam confirmed that wetting sand roughly halves the required force. For the Baalbek trilithon, the Roman engineering literature and surviving evidence of capstans, levers, ramps and earth embankments account for it; the blocks were quarried a few hundred metres uphill from their final position, which is a hint the “impossible” framing always omits.

Precision. Ancient surveying was extremely good. The Great Pyramid’s base is level to around two centimetres across 230 metres, which is achievable with water channels, and its sides are aligned to true north within about three arcminutes, achievable by stellar observation. These are not exotic methods. They are patient ones.

FUN FACT

What these sites really demonstrate is organisation, not technology — and organisation is the harder achievement. The Great Pyramid required feeding, housing, sanitising and administering a workforce of tens of thousands for two decades, with supply lines, bakeries, breweries, medical care, and a bureaucracy tracking it all in writing. We found the workers’ town at Giza, with its bakeries and its cattle bones and the fractures set by physicians. The astonishing thing about the pyramids is not that stone can be cut. It is that human beings four and a half thousand years ago could run a project of that scale at all. Handing that to visitors is not generous to the ancients. It is theft.

S . E THE IMPOSSIBLE GLASS

There is a class of claim about desert glass that gets cited as evidence of ancient nuclear warfare, and it is worth taking apart because it involves real

physical evidence with a real explanation — my favourite kind of case.

Libyan Desert Glass. Sheets and fragments of nearly pure silica glass scattered across about 6500 square kilometres of the Great Sand Sea between Libya and Egypt. It is roughly 98% silica, yellow-green, and about 29 million years old. Tutankhamun's pectoral contains a carved scarab of it, which is a genuinely wonderful fact. Its origin is impact-related: it contains iridium enrichment, high-temperature mineral phases including cristobalite, and traces of meteoritic material, indicating formation at temperatures above 1600°C, probably from an airburst or a nearby impact melt. Twenty-nine million years predates humans by a wide margin, which disposes of the ancient war reading without further argument.

Trinitite, by contrast, is genuine nuclear glass — formed at the Trinity test site in July 1945, greenish, and radioactive, containing fission products and traces of the device and its tower. Comparing the two is instructive. Nuclear glass carries an unmistakable signature: caesium-137, strontium-90, plutonium isotopes, europium-152 from neutron activation. It is detectable for millennia and it is not subtle.

So when Mohenjo-daro is claimed as an ancient nuclear site — vitrified surfaces, skeletons in the streets, elevated radioactivity — the test is available and has been applied. There is no elevated radioactivity at Mohenjo-daro. The claim traces to a single unsourced assertion repeated for decades without anybody measuring. The site shows evidence of decline consistent with hydrological change in the Indus system, and the “skeletons in the street” are burials and deposits from different strata that were reported carelessly in early excavation.

Vitrified forts in Scotland and elsewhere are real: Iron Age dry-stone ramparts whose stones have partially melted and fused. Experimental archaeology has reproduced the effect by burning timber-laced ramparts, and the debate in the literature is whether the burning was destruction by enemies, ritual, or deliberate construction technique — a real scholarly disagreement about temperature and duration, being argued with reproducible fires.

Fulgurites — lightning-strike glass — form tubes in sand where a strike has fused silica along its path, and account for a great deal of “anomalous” glass in deserts.

FUN FACT

The Mahabharata's descriptions of the Brahmastra — “a single projectile charged with all the power of the Universe... an incandescent column of smoke and flame as bright as ten thousand suns” — are genuinely striking, and often quoted. The quotation is also, in most circulating versions, not a translation of the Mahabharata. It is a paraphrase that entered the literature through mid-twentieth-century popular writing, after Hiroshima, by authors who had a modern image in their heads. Compare it to a scholarly translation and the resemblance largely evaporates. Oppenheimer did quote the Gita at Trinity — “Now I am become Death” — and the feedback loop starts there.

9.7 THE STRANGE CASE OF BRAWLEY OATES

In the early 1960s, in Elberton, Georgia, a man named Brawley Oates bought the property surrounding a small granite outcrop called Wonder Hill — a spot where, according to local legend and to a good many visitors, things do not behave as they should. Water appears to run uphill. A ball rolls the wrong way. A person standing in the structure leans at an impossible angle. Oates built the attraction that later generations knew as a “gravity hill” house, and the story attached to him afterwards is that he drank the water from the site to cure his arthritis and that it worked.

I include this case for a reason that has nothing to do with whether Oates was cured. Gravity hills — and there are hundreds catalogued worldwide, from the Electric Brae in Ayrshire to Magnetic Hill in New Brunswick to Confusion Hill in California — are the single best available demonstration that direct sensory experience is unreliable in a completely specific and measurable way.

FIGURE 9.3

A gravity hill with the true horizon obscured: the road that looks uphill surveys as downhill.

Figure 9.3. A gravity hill with the true horizon obscured: the road that looks uphill surveys as downhill.

Image sourced from: https://commons.wikimedia.org/wiki/File:Gravity_hill_road.jpg

Here is what is happening. Human beings do not sense the gravitational vertical directly with any precision. We infer level from visual cues: the horizon, the angle of trees, the line of a road, the walls of the room we are standing in. Obstruct the true horizon — with a slope, a ridgeline, a stand of trees growing at an angle on a hillside, or a

deliberately tilted building — and the visual system will confidently construct a false reference frame, and your body will believe it. Surveying instruments at gravity hills (see Figure 9.3) consistently show that the apparently uphill slope is in fact downhill. The water is obeying physics. Your inner ear and your visual cortex are having an argument, and vision wins.

The purpose-built versions — “mystery spots” — amplify this with rooms constructed out of plumb, so that every reference line in the observer’s field of view is tilted consistently. Stand in one and you will experience an overwhelming sensation of leaning, and you will not be able to reason your way out of it. That is the whole point of visiting.

FUN FACT

Elberton, Georgia has a genuinely stranger claim to fame that requires no anomaly at all. In 1980 a man using the name R.C. Christian commissioned, and paid for, an enormous granite monument outside town — the Georgia Guidestones — inscribed in eight languages with ten instructions for rebuilding civilisation after a collapse, including maintaining humanity under 500 million. Nobody ever established who paid for it. It became a fixture of New World Order literature for four decades, and in July 2022 somebody bombed it and the remains were demolished the same day. A real unsolved mystery, in the same small town, with better documentation than any UFO case in this book.

The methodological point stands on its own: if a hillside in Georgia can convince every visitor that water flows uphill, then “I saw it with my own eyes” establishes considerably less than people think. This is Section 1.5’s argument, standing outdoors, available for anybody to test with a spirit level.

B . B A N I G H T F R O M H E L L

On the night of 21 August 1955, at a farmhouse near Kelly, in Hopkinsville, Kentucky, eleven people — the Sutton family and their guests, including five children — reported a siege. Billy Ray Taylor went outside to the well and saw a bright object cross the sky and descend beyond a treeline. Shortly

afterwards a small glowing figure approached the house: roughly three feet tall, with an oversized round head, enormous ears, thin arms ending in claws, and a silvery, apparently metallic sheen.

Elmer “Lucky” Sutton and Taylor fired at it with a shotgun and a .22 rifle. They reported hitting it, hearing a sound like bullets on a metal bucket, and watching it flip backwards and scuttle away unharmed. Over roughly three hours, figures appeared repeatedly at windows and on the roof, and the men fired repeatedly through screens and walls.

At about eleven o’clock the family drove to Hopkinsville police station. Chief Russell Greenwell, who by all accounts was a sober man with no interest in flying saucers, judged them genuinely terrified and not drunk — and noted specifically that this was a family who had nothing to gain. Roughly twenty officers, including state police and military police from Fort Campbell, went to the farm. They found bullet holes, damaged screens, and nothing else. No tracks, no residue, no craft. Officers reported hearing an unusual sound and seeing a luminous area, but nothing conclusive. After the police left, the family reported the figures returned and the shooting resumed until dawn.

The Suttons were mocked relentlessly, and gave up their farm and moved. There is no sign of financial motive; the family consistently refused to profit from it.

The leading conventional explanation is unusually specific and unusually good. Great horned owls — and barn owls — are territorial, will defend a nest aggressively, approach a lit farmhouse at night, and have enormous eyes, prominent ear tufts, a wingspan around 1.5 metres, claws, a hissing threat display, and a habit of walking on the ground when injured or defensive. At night, illuminated from below by lamplight and seen by terrified people, an owl at close range has been proposed — notably by Joe Nickell, who examined the case in detail — as the source, and a shotgun blast that misses will produce exactly the observed result: the thing flips over backwards and is unharmed.

Against that: eleven witnesses, three hours, repeated close approaches at multiple windows, and men who lived on a farm and knew what owls were. I do not think the owl explanation is comfortable, and I am not going to pretend it is. What I will say is that the two hypotheses are not symmetrical — owls in Kentucky are established, and small metallic humanoids are not.

FUN FACT

Kelly–Hopkinsville is one of the few cases in this book that produced something durable and good. The town holds an annual Little Green Men Days festival, the family’s account was never monetised, and — the detail I find most telling — the entities were reported as silver, and the press turned them green. “Little green men” as a phrase owes a great deal to a Kentucky farmhouse where nobody actually saw anything green.

U F O S & R E L I G I O N

“Any sufficiently advanced technology is indistinguishable from magic.”

— SIR ARTHUR C. CLARKE

10.1 IN THE BEGINNING

The Ancient Alien reading of scripture rests almost entirely on one word, and I want to deal with it properly, because it is the most-repeated claim in this entire genre and almost everybody repeating it has never looked at the grammar.

The word is *Elohim*. In Genesis 1:1 — *bereshit bara Elohim* — it is the subject. The claim is that *Elohim* is plural, therefore Genesis describes multiple beings, therefore gods, therefore visitors. And the plural form is real: the *-im* suffix is the standard Hebrew masculine plural.

Here is what the argument omits. In Genesis 1:1 the verb *bara* — “created” — is third person masculine *singular*. Hebrew, like most inflected languages, requires agreement. A plural subject with a singular verb, sustained consistently across the entire text, is not an accident and not a suppressed clue. It is a recognised grammatical construction: linguists call it an honorific or intensive plural, and Hebrew uses the same device elsewhere — *Adonai* (lords) for a singular lord, *Behemoth* as a singular beast. English does something comparable with the royal “we”.

Genesis 1:26 — “Let us make man in our image” — is the other pillar,

and it is a genuine exegetical puzzle that has been argued for two thousand years, with serious candidate readings: the divine council of heavenly beings (which appears explicitly elsewhere in the Hebrew Bible, as in Job 1 and Psalm 82), a plural of deliberation, a Trinitarian reading in Christian tradition, or a residue of the older Canaanite pantheon from which Israelite religion emerged. That last one is the most historically interesting and the one biblical scholars find most productive: there is strong textual and archaeological evidence that Israelite monotheism developed out of a polytheistic West Semitic context, with El and Yahweh originally distinct. That is a real and well-supported finding about religious history, and it explains the plural language completely without any spacecraft.

The methodological lesson is broader than Genesis. **Every argument in this chapter depends on translation, and translation is where the evidence quietly gets manufactured.** Chariot, whirlwind, glory, host, cloud, wheel — each is a rendering choice, made by translators from a language with a limited surviving corpus, and each has been read back as technical vocabulary by people working from an English Bible. If you want to make a textual argument, you need the original language, or at minimum a critical apparatus. Otherwise you are analysing the King James translators' vocabulary of 1611, which is a legitimate subject but not the one you think you are studying.

10.2 THE NEPHILIM

Genesis 6:1–4 is four verses long, obviously fragmentary, and one of the strangest passages in the Hebrew Bible. The sons of God saw that the daughters of men were fair, took wives, and produced offspring; the Nephilim were on the earth in those days, and afterward; these were the mighty men of old, men of renown. Then the flood narrative begins.

The passage is doing something the surrounding text does not explain, and it clearly assumes knowledge the reader was expected to have and we do not. That gap is what makes it a magnet.

Bene ha-Elohim, sons of God, has three traditional readings: divine beings or angels; the righteous line of Seth intermarrying with the line of Cain; or dynastic rulers claiming divine descent, which is how Near Eastern kings routinely described themselves. *Nephilim* is usually derived from a root

meaning “to fall” — hence “fallen ones” — and is elsewhere associated with unusual stature: Numbers 13:33 has the spies reporting giants in Canaan, and Deuteronomy 3:11 gives Og of Bashan’s bed as nine cubits, which is around four metres and is plainly a boast.

The expanded version everybody actually knows comes from the Book of Enoch, and I will treat that in Section 10.4. What matters here is the modern claim: that the Nephilim were human-extraterrestrial hybrids, and that this corroborates the abduction breeding programme of Section 7.4.

It fails on the same biological grounds and one additional textual one. Biologically, Section 7.4 already established that cross-biosphere reproduction is a category error, not a long shot. Textually, the passage’s own framing is theological: it appears immediately before the flood as part of an explanation for why the world had become corrupt. Its narrative function is to establish transgression of the boundary between divine and human — which is precisely the sin the flood punishes. It is a story about categories being violated, told by a culture for whom that boundary was the fundamental one. **Reading it as a laboratory report removes the only thing the author was trying to say.**

FUN FACT

Giants are close to universal in mythology and there is a good non-mythological reason for at least some of it. Europeans and Mediterranean peoples regularly dug up bones far too large for any animal they knew — and dwarf elephant and mammoth skulls, in particular, have a single large central nasal cavity that looks exactly like one enormous eye socket. Adrienne Mayor’s work on this is excellent. The Cyclops may well be a fossil elephant, found by people with no concept of extinction and a perfectly rational willingness to reason from evidence. They were doing palaeontology with the wrong reference set. Give them credit for trying.

10.3 FLOOD MYTHOLOGY

Flood narratives are genuinely widespread, and the standard list is long: the Mesopotamian accounts in Atrahasis and the Epic of Gilgamesh, Genesis, Deucalion in Greek myth, Manu in the Hindu tradition, and stories from China, Australia, the Americas and the Pacific. The parallels between the

Mesopotamian and biblical versions in particular are close enough to be decisive about relationship: a warned man, a built vessel, animals preserved, a bird sent out to test for land, a mountain landing, a sacrifice.

The conventional explanation is not one thing, and its multiplicity is what makes it convincing.

Textual transmission. The Gilgamesh flood is older than Genesis and the two are related by cultural contact — Israel sat within the Mesopotamian sphere and the exile put its scribes in Babylon directly. This explains the close parallels, and only those.

Actual local floods. Mesopotamia is a floodplain between two rivers with a documented history of catastrophic inundation; Leonard Woolley found a thick flood deposit at Ur in the 1920s. For an inhabitant of a river civilisation, a regional flood *is* the destruction of the world, because the world is the valley.

Real post-glacial sea level rise. This is the serious one. Between roughly 20000 and 7000 years ago, global sea level rose about 120 metres as the ice sheets melted, not smoothly but with pulses. Enormous areas of habitable coastal land were drowned — Doggerland between Britain and Denmark, the Sunda Shelf in Southeast Asia, the Persian Gulf basin, and the land bridge at Beringia. Every coastal population on Earth experienced generations of retreat, and oral tradition can carry that. Aboriginal Australian traditions describing specific coastlines now underwater have been argued, in peer-reviewed work by Nunn and Reid, to preserve accurate geographic detail across something like 7000 to 10000 years.

That is a genuinely astonishing finding and it does not require a single global deluge.

Which is fortunate, because a global flood is physically impossible in a way worth stating plainly, since it is the one claim in this chapter that can be checked with arithmetic. Covering Everest requires roughly three times the total water present in the Earth's atmosphere, oceans, ice caps and groundwater combined. There is nowhere for it to come from and nowhere for it to go. The geological column shows no global flood layer — it shows continuous, ordered, regionally varying deposition with intact fossil sequences, ancient soils, footprints, raindrop impressions, and desert dune systems, all of which require dry land. Ice cores from Greenland and Antarctica preserve unbroken annual layering back over

100,000 years and in Antarctica considerably further, which is not compatible with submersion. And the genetic evidence rules out a recent bottleneck of eight humans; human genetic diversity, and its geographic structure, requires a continuously large population.

FUN FACT

The Black Sea deluge hypothesis, proposed by William Ryan and Walter Pitman in 1997, argued that rising Mediterranean water breached the Bosphorus around 5600 BCE and flooded a freshwater lake catastrophically, displacing everyone around it. It is contested — the sedimentary evidence supports a breach but the argued speed and violence are disputed — but it is a real scientific hypothesis about a real event that could have generated a real flood tradition, published in journals, argued with core samples. That is what an interesting answer looks like, and it took some of the fun out of nothing.

10.4 BOOKS OF FORBIDDEN

KNOWLEDGE

“Forbidden” is a marketing term. The texts in question are not forbidden — most are freely available in translation, several are canonical in some traditions, and the scholarship on them is extensive and public. **What actually happened to them is more mundane and more interesting: they lost a vote.**

1 Enoch is the important one. Composed in sections between roughly the third century BCE and the first century CE, it expands Genesis 6 enormously: two hundred Watchers descend under the leader Semjaza, take human wives, and — this is the crucial part — teach humanity forbidden arts. Azazel teaches metalworking and weapons, and cosmetics. Others teach astrology, meteorology, root-cutting and enchantment. The offspring are giants who devour the earth. The archangels intervene, the Watchers are bound, and the flood cleanses.

Enoch is quoted in the New Testament — the Epistle of Jude cites it directly — and was treated as authoritative by several early Church Fathers. It remains canonical in the Ethiopian Orthodox Tewahedo Church, which is how the complete text survived; the Ge’ez version was reintroduced to Europe by James Bruce in 1773, and Aramaic fragments turned up at

Qumran among the Dead Sea Scrolls, confirming its antiquity.

Why it was excluded is not a mystery. It is not by Enoch, it is a composite of clearly different periods, its cosmology conflicted with developing orthodoxy, and its account of the origin of evil competed with the doctrine that located it in human free will. Those are ordinary editorial grounds.

And note what the text is actually about, because the ancient astronaut reading always stops at “beings descended and taught us things” and never reads on. The knowledge the Watchers impart is knowledge of *technology and craft*, and the text’s judgement is that it was catastrophic — weapons, warfare, vanity, manipulation of nature. 1 Enoch is a sustained argument that technical knowledge acquired without wisdom destroys the world, written by someone watching Hellenistic empires make war on each other with excellent equipment. It is, if you like, the first anti-technology treatise, and reading it as a record of beneficent alien tutors inverts its thesis exactly.

The Book of Jubilees, **2 Enoch**, and **the Book of Giants** extend the same material. **The Nag Hammadi codices**, found in Egypt in 1945, are a genuine treasure — fifty-two Gnostic texts including the Gospel of Thomas and the Apocryphon of John, which describe the material world as the flawed work of a lesser creator, the Demiurge, and the Archons who administer it. Modern writers map the Archons onto reptilians. The mapping is superficially tempting and historically backwards: Gnosticism is a philosophical response to the problem of evil, developed in a Platonic and Jewish milieu, and its Archons are cosmic principles, not species.

FUN FACT

The genuinely suppressed texts are the ones nobody talks about because nobody has them. We know of lost works only by citation: the Gospel of the Hebrews, most of Epicurus, the bulk of Sappho, Berossus’s Babylonian history, and Celsus’s anti-Christian polemic, which survives only in the extensive quotations Origen made in order to refute it. That is what suppression actually looks like — silence, not availability. Anything you can buy on Amazon with “forbidden” in the subtitle has, by definition, failed to be forbidden.

10.5 STAR PEOPLE

A great many indigenous traditions include beings associated with the stars, and this material is handled worse than anything else in the field. So let me be careful.

The Lakota tradition includes the Pleiades and an origin narrative connecting people to the stars. Hopi tradition includes the Kachinas, who arrive from the sky, and Hopi ceremonial life is structured around them. Australian Aboriginal astronomy is sophisticated, ancient, and encodes seasonal and navigational information with great precision — the Emu in the Sky, formed from the dark dust lanes of the Milky Way rather than from stars, is a genuinely brilliant piece of observational practice, and its position through the year signals emu breeding and egg collection. Cherokee, Zuni, Dogon, Maori and Polynesian traditions all contain star lore, and Polynesian stellar navigation was accurate enough to settle the Pacific.

The ancient astronaut treatment takes these traditions, extracts the sky elements, discards the context, and presents them as testimony to visitation. There are three things wrong with that, and only one of them is evidential.

First, it is bad method: selecting the fraction of a tradition that fits and ignoring the rest is the confirmation bias of Section 1.6 applied to other people's religion.

Second, it is inconsistent: the same treatment is never applied to, say, Greek constellation myths, which are also sky beings who interact with humans. Zeus is not read as a visitor. Distinguishing between the two is a decision about whose mythology counts as mythology.

Third — and this is the part that should carry weight — living peoples with living traditions have repeatedly and publicly objected to it. Hopi religious authorities have objected to the Kachina material being used this way. These are not dead cultures whose literature is available for reinterpretation. They are ongoing religious practices belonging to specific communities, and taking their sacred narratives, stripping the meaning, and repurposing them as evidence for a Western hypothesis is a continuation of a long pattern that those communities can identify faster than the people doing it.

What these traditions actually demonstrate is that pre-literate societies conducted excellent naked-eye astronomy over enormous timespans and

encoded it in narrative because narrative is the best available storage medium when you have no writing. Aboriginal Australian traditions preserving coastline changes from the end of the last glacial period, as Section 10.3 noted, are the same phenomenon: mythology as archive. That is a serious finding in cognitive and cultural science, it is being published, and it credits the people who did the work.

10.6 VIMANAS

The Sanskrit epics — the Mahabharata and Ramayana, along with the Puranas — describe *vimanas*: flying palaces and chariots, sometimes vast, sometimes armed, used by gods, kings and demons. Ravana's Pushpaka Vimana carries a city. There are aerial battles. There are weapons of appalling power, including the Brahmastra of Section 9.6.

This is real literature and it is magnificent, and it should be read. The question is what it is evidence of.

The key text in the modern claim is the *Vaimanika Shastra*, presented as an ancient treatise on aeronautics (see Figure 10.1) with detailed descriptions of aircraft types, materials, and propulsion. It is not ancient. Its actual provenance is documented: it was produced between 1918 and 1923 by Pandit Subbaraya Shastry, who stated that he received it through psychic channelling, and it was transcribed by G. R. Josyer and published in 1952 with an English translation. A 1974 study by aeronautical engineers at the Indian Institute of Science in Bangalore — Mukunda, Deshpande, Nagendra, Prabhu

and Govindaraju — examined the described craft in detail and concluded that the geometries and materials described were aeronautically unworkable and that the text showed no understanding of flight. It is a twentieth-century document. Nothing about it is in dispute among people who have looked.

The epics themselves are a different matter and deserve better than either treatment they usually get. Their composition spans centuries, with

FIGURE 10.1

A plate from the Vaimanika Shastra: channelled drawings that a 1974 engineering study found unflyable.

Figure 10.1. A plate from the *Vaimanika Shastra*: channelled drawings that a 1974 engineering study found unflyable.

Image sourced from: https://commons.wikimedia.org/wiki/File:Vimana_Shakuna_plate.jpg

the Mahabharata reaching roughly 100,000 verses — the longest poem in existence. They are theological, dynastic and philosophical works in which the flying palaces belong to the same register as the shapeshifting, the divine weapons, the curses, the multi-headed beings, and the incarnations of gods. Extracting the aircraft while leaving the curses is not reading the text; it is mining it.

And there is a real history of Indian technical achievement being ignored while this argument consumes attention: the decimal place-value system and zero, Aryabhata computing π to four places and proposing the Earth's rotation in the fifth century, Brahmagupta's rules for negative numbers, the Kerala school's infinite series work anticipating parts of calculus by centuries, Sushruta's surgical corpus, Panini's grammar — which is a formal generative system of such rigour that twentieth-century linguists were startled by it. That is the genuine record, it is enormous, and it is available. It has the disadvantage of requiring study.

FUN FACT

The Mahabharata's aerial battles are frequently cited alongside the claim that Sanskrit's precision proves it was engineered or transmitted. Sanskrit is indeed precise, and Panini's grammar of roughly the fourth century BCE describes it in about 4000 rules with a formalism that has been compared to Backus–Naur form. It was written by a man. That is the point.

10.7 UFO CULTS

This section exists because the subject has a body count, and anybody writing about UFOs owes it an honest accounting.

FIGURE 10.2

Festinger, Riecken and Schachter's When Prophecy Fails (1956), the study behind cognitive dissonance theory.

Figure 10.2. Festinger, Riecken and Schachter's *When Prophecy Fails* (1956), the study behind cognitive dissonance theory.

Image sourced from: https://commons.wikimedia.org/wiki/File:When_Prophecy_Fails_first_edition.jpg

The Seekers, 1954. Dorothy Martin, a Chicago housewife, received channelled messages predicting a flood on 21 December 1954 and rescue by spacecraft. A group left jobs and marriages. The social psychologist Leon Festinger and colleagues infiltrated the group and published *When Prophecy Fails* in 1956 (see Figure 10.2) — and the finding is one of

the most important in the study of belief. When the flood did not come, the committed members did not abandon the belief. They intensified it, received a new message explaining that their faith had spared the Earth, and began proselytising for the first time. Festinger developed the theory of cognitive dissonance from this. Disconfirmed belief, held publicly and at cost, gets stronger. Remember that when you watch a disclosure deadline pass.

Heaven's Gate. Marshall Applewhite and Bonnie Nettles built a group around the doctrine that human bodies were vehicles to be shed in order to board a craft, which they eventually associated with Comet Hale-Bopp. In March 1997, thirty-nine people including Applewhite killed themselves in a house in Rancho Santa Fe, California, in shifts, having recorded farewell videos. Several men had undergone voluntary castration. They were calm, articulate, and technically competent — the group ran a web design business, and their site is still online.

The Order of the Solar Temple. Between 1994 and 1997, at least seventy-four members died in Switzerland, Quebec and France, by murder and suicide, in a doctrine involving transit to a planet near Sirius. Children were among the dead.

Aum Shinrikyo is not usually classed here but belongs in the discussion: a millenarian group with a syncretic doctrine that released sarin on the Tokyo subway in March 1995, killing thirteen and injuring thousands. It had recruited scientists and was attempting to produce biological and chemical weapons.

The Raëlians, founded by Claude Vorilhon in 1974 on a claim of contact with the Elohim — and note the vocabulary, straight out of Section 10.1 — teach that humanity was created in a laboratory. They are non-violent, and in 2002 claimed the first human clone, a claim never substantiated.

The recurring structure is worth learning, because it is the same every time: a charismatic leader with privileged access to information; a coming transformation with a date; isolation from outside contact; escalating financial and personal commitment; reinterpretation rather than abandonment when predictions fail; and a doctrine in which the body or the world is a thing to be escaped. That last element is the dangerous one. Any belief system that devalues this life in favour of a transit elsewhere has removed

the safety rail.

I do not think believing in UFOs makes anybody a cultist, and the overwhelming majority of people in this field are ordinary, curious, and harmless. But the field's epistemology — trust the insider, distrust the institution, treat disconfirmation as suppression — is the exact epistemology these groups run on, and that is not a coincidence. It is why Chapter 1 exists.

10.8 THE WRATH OF GOD

Sodom and Gomorrah, Genesis 19: fire and sulphur from the sky, cities destroyed, Lot's wife turned to a pillar of salt for looking back. The ancient astronaut reading is a nuclear strike, and the argument runs through the salt, the fire, and the instruction not to look.

The conventional readings are better and there are several, which is honest.

The region — the southern Dead Sea, if that is where the cities were — is geologically active: it sits on the Dead Sea Transform, a major fault system. It has bitumen and natural asphalt seeps, which Genesis 14 itself mentions, and pockets of methane and hydrogen sulphide. An earthquake igniting seeping hydrocarbons produces burning material falling from the sky and a sulphurous stench, and the area has natural salt formations — Mount Sodom is a salt diapir, and pillars of salt are literally standing there for anybody to photograph. A theological story explaining a striking landscape feature is one of the oldest genres there is.

A 2021 *Scientific Reports* paper proposed a cosmic airburst destroying Tall el-Hammam in the Jordan Valley around 1650 BCE, citing shocked quartz, melted pottery and elevated platinum. It attracted immediate and substantial methodological criticism, and the case is not settled — but it is a real hypothesis being tested with real samples, and if it holds up it is a fine example of an ancient catastrophe story preserving an actual event.

The Exodus plagues get similar treatment, and the naturalistic sequence is genuinely elegant: an algal bloom turning water red and toxic, driving frogs ashore, whose deaths produce insects, which transmit livestock disease, and so on — with the Thera eruption around 1600 BCE sometimes invoked for the darkness and the hail. The chronology is a problem and the connection

is speculative, but the internal causal chain is coherent.

And there is one detail in the Exodus material worth pausing on, because it is a real physical phenomenon rather than a stretch. A strong sustained wind over a shallow water body produces wind setdown, exposing a passable bed which returns violently when the wind drops. This is observed, modelled, and has drowned people. Whether it happened at a specific place three thousand years ago is unknowable; that it can happen is measured.

FUN FACT

There is a persistent claim that the Ark of the Covenant was an electrical device — a gold-plated wooden box, two conductive surfaces separated by an insulator, hence a capacitor, hence the deaths of those who touched it. The physics is worth checking rather than repeating. A capacitor of those dimensions would store a trivial charge, it has no mechanism to accumulate or sustain one, and the deaths described in 2 Samuel 6 and Numbers are theological in framing: the holy is dangerous to the unauthorised, which is the entire point of the priesthood the text is establishing. Sometimes a gold box is a gold box.

THE SCIENCE OF ALTERNATIVE DIMENSIONS

“The greatest question of all is whether our experience on this planet is ‘it’ or whether there is something else. Things in the supernatural realm give support, strangely perhaps, to the things we take on faith.”

— ART BELL

11.1.1 DIMENSIONAL SPACETIME

“Another dimension” is the most abused phrase in ufology, and the abuse comes from a genuine ambiguity in the English word. In physics a dimension is a degree of freedom — an independent number you need to specify a position. Our spacetime has four: three of space, one of time. In popular usage a dimension is a *place* — a parallel world you could travel to. **These are not the same concept, and almost every interdimensional UFO claim silently swaps the first for the second.**

The eleven-dimensional figure people cite is real and comes from a specific place. String theory replaces point particles with one-dimensional vibrating strings, and its mathematics only stays consistent in a particular number of dimensions — ten for superstring

FIGURE 11.1

A two-dimensional section of a Calabi–Yau manifold, the geometry proposed for the compactified dimensions.

theories, twenty-six for the original bosonic version. In 1995 Edward Witten argued that the five known superstring theories were limits of a single underlying eleven-dimensional framework, which he called M-theory. The extra seven dimensions are proposed to be compactified: curled up at a scale near the Planck length, around 10^{-35} metres, in configurations called Calabi–Yau manifolds (see Figure 11.1).

The standard analogy is a garden hose seen from a distance. It looks like a line — one dimension — but an ant on its surface has two directions available, one of which wraps around and is invisible from far away. Compactified dimensions are supposed to be like that wrapped direction: everywhere, but too small to notice.

Here is what matters for our purposes, and I am going to be blunt about it. If the extra dimensions are curled up at 10^{-35} metres, then nothing can travel through them, because there is nowhere to go. A nucleus is roughly 10^{20} times larger than that scale. Saying a craft came from the eleventh dimension is like saying a lorry arrived via the gap between two atoms. The claim isn't bold; it's dimensionally illiterate.

And string theory itself has a problem that ufologists ought to find sobering rather than encouraging. It has, after fifty years, produced no confirmed experimental prediction. The number of possible compactifications — the “landscape” — is estimated at something like 10^{500} , which means the framework can accommodate almost any low-energy physics and therefore forbids almost nothing. A theory that cannot be wrong is in exactly the position Section 1.3 described. Some very serious physicists, Lee Smolin and Peter Woit among them, argue it has stopped being physics. Others expect it to be vindicated. The dispute is live, public, conducted in journals, and settled by nobody yet — which is what an honest hard problem looks like.

FUN FACT

Large extra dimensions are actually testable, and were tested. If some extra dimension were millimetre-scale rather than Planck-scale, gravity would deviate from the inverse-square law at short range, and energy would leak into

it in high-energy collisions — producing missing energy, or even microscopic black holes, at the Large Hadron Collider. Physicists looked carefully. Torsion-balance experiments have confirmed inverse-square gravity down to tens of micrometres, and the LHC found no missing-energy signature. That is how you do it: take the exotic idea, derive a consequence, go and look. Nobody found anything, and the search itself was the achievement.

1.1.2 MANY WORLDS HYPOTHESIS

The Many Worlds Interpretation is genuine physics and it is genuinely strange, so it is worth getting right rather than gesturing at.

Quantum mechanics describes a system by a wavefunction which evolves smoothly and deterministically according to the Schrödinger equation, and which generally represents a superposition of possible outcomes. Measurement, however, yields a single definite result, with probabilities given by the Born rule. Reconciling those two facts is the measurement problem, and it is unsolved — not in the sense that nobody has an answer, but in the sense that several answers exist and no experiment currently distinguishes them.

The Copenhagen interpretation says the wavefunction collapses on measurement. This works and is what most physicists use in practice, but it never satisfactorily defines what counts as a measurement.

Hugh Everett, in his 1957 Princeton thesis, proposed removing collapse entirely. If the Schrödinger equation always applies, with no exceptions for observers, then measurement entangles the observer with the system and the superposition persists — the observer simply becomes a superposition of observers, each perceiving one definite outcome and each unaware of the others. Decoherence, worked out later by Zeh, Zurek and others, explains why the branches stop interfering: interaction with the environment scrambles the relative phases irreversibly and fast.

The result is a picture with no collapse, no randomness, and no special role for observers — bought at the price of an unimaginable number of coexisting branches. Everett's work was ignored for a decade, he left academia for defence analysis, and he died at fifty-one without seeing it taken seriously.

Now the part relevant to this chapter. The branches of the MWI are, by the theory's own mechanics, *not accessible*. Decoherence is what makes branching effective, and it is effectively irreversible for any macroscopic system — the phase information is dispersed into astronomically many environmental degrees of freedom. **There is no mechanism in the formalism for travelling between branches, sending a signal between them, or receiving a visitor from one.** Many Worlds is not a multiverse of destinations. It is a claim about why measurement looks probabilistic.

So when an interdimensional hypothesis borrows MWI for credibility, it is borrowing the name of a theory whose central technical feature — decoherence — specifically forbids what is being proposed. That is worse than having no theory. It is having one that says no.

11.2.1 THE MAN FROM TAURED

Which brings me to the story that gets told in every lecture hall where this subject is discussed, usually by someone who heard it from someone. In July 1954, the account goes, a well-dressed European man arrived at Tokyo's Haneda Airport and presented a passport issued by the nation of Taured. He gave his profession as business, said he had flown this route several times, and produced a driving licence, a chequebook and other documents from the same country. Taured does not exist. Shown a map, he indicated the territory of Andorra and became agitated, insisting his country had existed for a thousand years. He was detained in a hotel room on an upper floor with guards outside. The next morning the room was empty — and so, reportedly, was the property room holding his documents.

FIGURE 11.2

Tokyo's Haneda international arrivals hall in the 1950s — the setting of a story with no surviving paperwork.

Figure 11.2. Tokyo's Haneda international arrivals hall in the 1950s — the setting of a story with no surviving paperwork.

Image sourced from: https://commons.wikimedia.org/wiki/File:Haneda_Airport_1955.jpg

It is a magnificent story. Now let me do to it what I have been asking you to do all term. There is no contemporaneous Japanese press report (see Figure 11.2), no immigration record, no named officer, and no airport file. The trail runs back to a 1981 anthology of strange stories by Colin Wilson and John Grant, which retold it without a primary source, and behind that to a French tale of a mysterious traveller circulating in the 1950s. There are two versions of the year, two of

the airport, and two of the country's name. Every element that would make the account checkable is precisely the element that is missing. What we have is not a case. It is a modern legend with an unusually good hook, and its persistence is a fact about narrative appeal rather than about cosmology — which is worth noticing, because if the branches of the wavefunction were sending us visitors, the evidence would look like paperwork, not folklore.

1.1.3 THE MANDELA EFFECT

A large number of people share confident, detailed, identical false memories. That is the actual phenomenon, it is real, it is well documented, and the explanation is more interesting than the parallel-universe reading.

The name comes from Fiona Broome, who around 2009 found many people who remembered Nelson Mandela dying in prison in the 1980s. He was released in 1990 and died in 2013. The standard catalogue: the Berenstain Bears remembered as Berenstein; “Luke, I am your father” (the line is “No, I am your father”); “Mirror, mirror on the wall” (“Magic mirror on the wall”); the Monopoly man’s monocle, which he does not have; Pikachu’s tail tip, which is not black; Curious George’s tail, which he does not have; “Life is like a box of chocolates” (it is “*was*”); and the Fruit of the Loom cornucopia, which never existed.

The mechanisms are established experimental psychology, and there are four of them working together.

Memory is reconstructive. This is the finding of Section 1.6 restated: recall is not retrieval of a stored record but reconstruction from fragments, and each reconstruction can be re-encoded with alterations. Elizabeth Loftus’s work demonstrates this repeatedly and quantitatively.

Schema-driven completion. We fill gaps with what fits. “-stein” is a far commoner English surname ending than “-stain”, so the brain regularises it. A wealthy cartoon capitalist with a top hat and cane should have a monocle — and Mr. Peanut, who does, was in the same visual culture. A cornucopia belongs on a fruit logo.

Source confusion. People remember content and lose its origin. Parodies, imitations, misquotations in later media, and packaging redesigns all leave

traces that get attributed to the original. “Luke, I am your father” has been quoted that way in advertising and comedy for forty years; the misquotation is more available than the line.

Social contagion. Once articulated, a false memory spreads and gains confidence. Asking “do you remember the cornucopia?” plants it, and the internet asks at enormous scale. And this is why the effect *feels* paranormal: the shared quality is the surprising part, but shared cognitive architecture plus shared media environment predicts shared errors. We all run the same hardware on the same inputs. Of course we make the same mistakes.

FUN FACT

The Mandela Effect is the single most useful thing in this book, and I mean that. It is a large-scale, self-administered demonstration that vivid, detailed, confidently held memory can be simply and verifiably wrong — because unlike a UFO sighting, you can check. The logo is available. The film exists. Go and look, watch your own certainty survive the disconfirmation for a second or two, and then remember that feeling next time you read a witness statement in Chapter 5. Including mine.

11.4 LIFE AFTER DEATH

Near-death experiences are reported with striking cross-cultural consistency: a sense of leaving the body, movement through darkness or a tunnel, light, encounters with figures or deceased relatives, life review, a boundary, and reluctance to return. Bruce Greyson developed a standardised scale for them, Raymond Moody popularised the topic in 1975, and the phenomenology is real, well-catalogued, and often profoundly transformative for the person.

The neuroscience accounts for a surprising amount of it, feature by feature.

Tunnel vision is what cerebral hypoxia does to the visual system: peripheral retinal and cortical function degrades before central, producing a constricting field with a bright centre. Fighter pilots in centrifuge training experience it routinely and call it greyout, along with euphoria and dreamlets during G-induced loss of consciousness — James Whinnery’s research on this is directly relevant and largely ignored in NDE literature.

Out-of-body experience can be induced by stimulating the temporopari-

etal junction, which integrates vestibular, proprioceptive and visual information into the sense of embodied self. Olaf Blanke's work does this deliberately in the lab. Disrupt the integration and the felt location of the self detaches. This is also what happens in sleep paralysis — see Section 7.6 — and in ketamine anaesthesia, which produces dissociation and NDE-like features reliably enough to be studied.

Life review plausibly reflects disinhibited temporal lobe and hippocampal activity, and temporal lobe seizures produce vivid autobiographical flooding along with intense religious feeling.

The 2013 Borjigin study at Michigan found a surge of coherent high-frequency gamma activity in rats after cardiac arrest, exceeding waking levels — and comparable surges have since been recorded in dying human patients, including a 2022 case captured incidentally during EEG monitoring. A dying brain is not a quiet brain. It is briefly, intensely active.

The honest limits: the timing of NDE content relative to measurable brain activity is hard to establish, and the AWARE studies led by Sam Parnia — which placed images in resuscitation rooms visible only from the ceiling, precisely to test veridical perception — produced no confirmed hits in the visual test across many years and hundreds of cardiac arrests. That is a genuine negative result from a genuine attempt, and it deserves respect from both sides.

Why is this in a UFO book? Because Section 7.3's abduction narrative and the NDE share their entire structure: paralysis, a floating sensation, a passage, beings of light, a message about humanity's future, a life review, and a return with transformed values. Two literatures describing what is very plausibly one set of neurological events, interpreted through two different cultural vocabularies. Whichever way that resolves, it tells us something.

11.5 SHADOW CREATURES

Shadow people — dark humanoid figures seen peripherally, often at night, often accompanied by dread, sometimes wearing a hat — are reported consistently enough to constitute a real perceptual phenomenon. The hat is the detail that convinces people, because how would strangers independently invent a hat?

Several mechanisms produce this, and they stack.

Hypnagogic and hypnopompic hallucination, at sleep onset and waking, produces vivid figures with full conviction of reality. Combined with sleep paralysis — REM atonia persisting into wakefulness — it produces the complete package: immobility, a felt presence, a dark intruder, crushing chest pressure, and terror. This is the Old Hag, the *mara*, the *kanashibari*, the *jinn* sitting on the chest, the incubus. Every culture on Earth has a name for it, which tells you it is in the brainstem rather than the bedroom.

Peripheral vision is rod-dominated: high sensitivity, poor acuity, essentially no colour. A peripheral figure is therefore reported as dark, featureless and human-shaped by default, because that is the only kind of thing low-resolution monochrome peripheral processing can return. The hat is likely the visual system resolving an ambiguous silhouette using the most available human template.

Pareidolia and face detection run continuously and with a deliberate bias toward false positives, because the cost asymmetry favours it. Mistaking a shadow for a person is cheap; mistaking a person for a shadow was historically fatal.

Felt presence can be induced directly by stimulating the temporoparietal junction, and is reported in high-altitude climbing, sensory deprivation, extreme fatigue, bereavement, and solo endurance situations. Shackleton's fourth man; the "third man factor".

Environmental contributors are real and measurable: infrasound below about 20 Hz has been associated with unease and shivers, and Vic Tandy's classic 1998 investigation traced a "haunted" laboratory to a 19 Hz fan resonance — and to the resulting eyeball resonance producing a grey figure at the edge of vision. Carbon monoxide poisoning produces dread and hallucination and can kill you, which is why every account of a haunted bedroom should be met with the question of whether anybody checked the boiler. Mould exposure and certain medications contribute.

FUN FACT

The Hat Man deserves one more note, because the internet did something to him that is directly relevant to Chapter 7. Before roughly 2000, shadow-figure reports were varied. After the hat became a widely shared motif online, the hat became far more prevalent in new reports. That is cultural transmission

shaping the content of a genuine perceptual experience, in real time, in an archive you can search. It is the same process as Section 7.3's grey alien, and here we got to watch it happen.

11.6 THE PHILADELPHIA EXPERIMENT

The story: in October 1943, at the Philadelphia Naval Shipyard, the destroyer escort USS Eldridge was rendered invisible and teleported to Norfolk, Virginia and back, as part of a project to defeat magnetic detection. Crew members were fused into the deck, driven mad, or vanished. The technology derived from Einstein's unified field work and involved Nikola Tesla. It became a 1984 film, and it is one of the founding legends of the field.

The provenance is fully traceable, and this is one of the rare cases where we can watch a legend assemble from parts.

It begins with Carl M. Allen, writing as Carlos Miguel Allende, who in 1955–56 sent a series of letters to the Office of Naval Research and to the astronomer Morris K. Jessup, who had published a book on UFOs. The letters were handwritten, in multiple ink colours, with erratic capitalisation, and claimed Allende had witnessed the event from the deck of the SS Andrew Furuseth (see Figure 11.3). He cited a Philadelphia newspaper story he could not name and never produced. He later admitted to investigators that parts were invented. The annotated copy of Jessup's book that ONR reproduced — the “Varo edition” — became scripture; ONR's own position, stated repeatedly since, is that it never conducted any invisibility research.

The documentary record contradicts the story directly. The Eldridge's deck log survives. She was commissioned in August 1943 and in October 1943 was conducting shakedown and convoy escort work in the Atlantic;

FIGURE 11.3

USS Eldridge (DE-173). Her deck log places her nowhere near the claimed events.

Figure 11.3. USS Eldridge (DE-173). Her deck log places her nowhere near the claimed events.

Image sourced from: https://commons.wikimedia.org/wiki/File:USS_Eldridge.jpg

her movements are accounted for and she was not in Philadelphia on the claimed date. Surviving crew members held reunions and denied it flatly, with several publicly annoyed about it.

What the Navy *was* doing at Philadelphia is genuinely interesting and is almost certainly the seed. **Degaussing** — wrapping a hull in energised cables to neutralise its magnetic signature and defeat German magnetic mines and torpedoes — was real, widespread, urgent, and classified. It was routinely described in the yard as making a ship “invisible” to mines. That is the whole trick: a real classified programme, a real piece of jargon, and a decade of retelling.

The physics closes the case. Optical invisibility of a macroscopic steel object is not a matter of insufficient power; metamaterial cloaking works only over narrow bandwidths at scales appropriate to the wavelength, and a 100-metre destroyer across the visible spectrum is not a harder version of the same problem. Teleporting 1240 tonnes of steel and crew 200 miles has no physical mechanism whatsoever, and Einstein never completed a unified field theory — he worked on it fruitlessly for thirty years, which is a matter of public record and a rather poignant one.

FUN FACT

Morris Jessup’s involvement ended badly. He was a serious man — he had done real astronomical work on double stars — and he was drawn into the Allende material, became increasingly isolated and distressed, and died by suicide in 1959. That death was immediately absorbed into the legend as evidence of silencing. It is the ugliest habit this field has: a man’s collapse repurposed as corroboration. Whatever else you take from this book, don’t do that.

I T S A B I R D ! I T S A
P L A N E !

"I can assure you that flying saucers, given that they exist, are not constructed by any power on Earth."

— HARRY S. TRUMAN

12.1 THE HISTORY OF AVIATION

A UFO is an *unidentified flying object*, and identifying flying objects requires knowing what is up there. Most people do not, and that single gap in general knowledge is responsible for a large share of the reports in this book. So: a short history of what human beings have put in the air, and when.

Balloons came first, and they matter more than anybody expects. The Montgolfier brothers flew a hot-air balloon at Annonay in June 1783; Jacques Charles flew hydrogen in December of the same year. Balloons were used for military observation at Fleurus in 1794 and extensively in the American Civil War. By the late nineteenth century they were commonplace, which is worth remembering when you read a nineteenth-century airship sighting.

Heavier-than-air flight arrived on 17 December 1903 at Kill Devil Hills, North Carolina — 12 seconds, 120 feet. What the Wrights actually contributed was not the wing but the control problem: three-axis control, coupled roll and yaw, and their own wind tunnel to get usable lift data after

they found the published tables wrong. That is engineering of a high order and it is why they succeeded where better-funded rivals failed.

Then the timeline compres, and the compression is the point. Powered flight to a monoplane fighter is eleven years. To all-metal airliners, twenty-five. Frank Whittle and Hans von Ohain developed turbojets independently in the 1930s; the Heinkel He 178 flew in 1939 and the Me 262 entered service in 1944. Chuck Yeager exceeded Mach 1 in the Bell X-1 in October 1947 — the same year as Roswell and Kenneth Arnold, which is not a coincidence and I will return to it. Sputnik, 1957. Gagarin, 1961. Armstrong, 1969. Sixty-six years from a sand dune in North Carolina to the Sea of Tranquility.

FIGURE 12.1

The SR-71 Blackbird: classified while flying, and unlike anything the public had a category for.

Figure 12.1. The SR-71 Blackbird: classified while flying, and unlike anything the public had a category for.

Image sourced from: https://commons.wikimedia.org/wiki/File:Lockheed_SR-71_Blackbird.jpg

The SR-71 Blackbird first flew in 1964, cruised above Mach 3 at 85000 feet, leaked fuel on the ground (see Figure 12.1) because its panels were built loose to allow for thermal expansion, and looked — from below, at that altitude, in sunlight — like nothing anybody had a category for. The U-2 flew above 70000 feet from 1955, at a time when commercial aviation topped out around 20000 and airline crews were reporting silver objects far above them. The CIA's own declassified history of the U-2 programme states plainly that the aircraft accounted for a substantial fraction of UFO reports in the late 1950s, and that

Blue Book investigators were quietly given enough information to dismiss those reports without explaining why.

That is the single most important fact in this chapter. There is a documented, admitted case of the US government using the UFO phenomenon as cover for a classified programme. It is not a theory. And it cuts both ways: it vindicates the conspiracy-minded about the *behaviour*, and it removes the anomaly from the *sightings*.

FUN FACT

The F-117 Nighthawk flew in 1981 and was not acknowledged until 1988. For seven years there were faceted, matte-black, impossibly angular aircraft operating at night out of Tonopah, and people saw them. The Nevada

triangle-UFO reports of the 1980s look considerably less mysterious once you know the flight schedules — and the B-2, first seen publicly in 1988, is a large silent-seeming flying wing that from directly below is a black triangle. Chapter 4's Phoenix Lights deserve to be read with that in mind.

12.2 THEORY OF FLIGHT

You cannot judge whether an object's flight is impossible without knowing what makes flight possible. This section is the physics, briefly, because it is the tool you need for Chapter 16's video analysis and for every "it performed a 90-degree turn at 7000 miles an hour" claim.

Four forces act on an aircraft: lift, weight, thrust, drag. Lift is generated by the wing turning the airflow downward; by Newton's third law, the reaction is upward. The popular "longer path over the top" explanation is wrong, and demonstrably so — it predicts the wrong magnitude and fails to explain inverted flight. Circulation theory, via Kutta-Joukowski, gets it right.

The number that matters is the one nobody mentions: **g-loading**. A turn is an acceleration, and the acceleration required scales as velocity squared over radius. **This is why the reported manoeuvres are the real problem, not the speeds.**

Run it. An object at 3000 miles per hour — roughly 1340 metres per second — making a turn with a radius of 100 metres experiences an acceleration of v^2/r , which is about 18000000 m/s², or around 1.8 million g. Human tolerance is roughly 9 g sustained with a suit and training; the highest survived brief peak on record is around 46 g (John Stapp, who did it to himself on a rocket sled and detached his retinas). Structural limits for a fighter airframe are in the region of 9 to 12 g. Even an unmanned airframe of any known material fails three or four orders of magnitude below what these reports require.

The consequence is important and it is not the one either camp usually draws. If an object really performed those manoeuvres, the occupants and the structure would need to be shielded from inertia itself — which means the interesting claim is not propulsion but inertial mass manipulation, and

no known physics offers a route to it. *Or* the reported values are wrong, which happens the moment you misjudge range: angular motion across your field of view is unremarkable for a nearby small object and impossible for a distant large one, and observers cannot judge range on an unfamiliar object against an empty sky. That is not a slur on witnesses. **It is a documented limitation of monocular distance estimation beyond a few hundred metres.**

Two more numbers worth carrying. **Sonic boom:** anything exceeding roughly 767 mph at sea level produces a shock wave that is heard for miles. **Silent supersonic flight in atmosphere is not a stealth feature; it is a violation of gas dynamics.** **Heating:** at Mach 3 the SR-71's leading edges reached over 300°C, which is why it was built of titanium. An object doing Mach 20 in the lower atmosphere would be incandescent and would ablate.

FUN FACT

The hardest number for the exotic-craft hypothesis is not any of these. It is the absence of turbulence. Every one of those manoeuvres displaces an enormous mass of air, and displaced air is visible on radar as a disturbance, audible as noise, and detectable by aircraft flying behind — wake turbulence is why air traffic control enforces separation minima. Objects reported to shift from hover to hypersonic leave no wake, no boom, no thermal plume and no downwash on the water beneath them. In physics, if you move something you have to move what it was in.

1 2 . 3 F O O F I G H T E R S

During the Second World War, Allied aircrews — particularly night fighter crews of the US 415th Night Fighter Squadron over the Rhine valley in late 1944 — reported small glowing balls of light, usually orange or red, that paced their aircraft, held formation, appeared to manoeuvre intelligently, and did not attack. The crews named them foo fighters, from the “Smokey Stover” comic strip catchphrase “where there’s foo, there’s fire”. Japanese and German crews reported comparable things in their own theatres, which is the detail that makes the case robust: these were not one air force’s rumour.

The reports were taken seriously precisely because the obvious hypothesis was enemy technology, and each side suspected the other. Neither had

it. That mutual suspicion, incidentally, is a general lesson about secret weapons: the people who would know are usually the ones asking.

The candidate explanations are several, and unusually good.

St. Elmo's fire — corona discharge from a charged airframe — is well known to aircrew, appears at propeller tips, wingtips and windscreen edges, glows blue or orange, and moves. It is common in the electrical conditions of the northern European winter.

Ball lightning is real, now photographed and reproduced under some conditions, and remains partially unexplained — luminous spheres persisting for seconds, associated with thunderstorms, occasionally reported entering aircraft.

Ice crystal and lens artefacts, reflections from a windscreen, and internal reflections in gunsight optics account for a category of paced lights, and the *pacing* is the tell: an object that maintains perfect station with your aircraft regardless of your manoeuvre is very likely attached to your aircraft, either physically or optically.

Autokinesis makes a single point light in a dark field appear to drift and dart. Night fighter crews were staring into blackness for hours, which is the exact configuration in which this occurs.

And some were real hardware. German flak used pyrotechnic tracking rounds and there were experimental programmes; V-1s and V-2s were airborne; and the Me 163 rocket interceptor and Me 262 left unfamiliar signatures at night.

What I find valuable about the foo fighters is not the identification. It is the timing. These reports predate Kenneth Arnold in 1947, and they are noticeably *different* in content: lights that pace, not discs that fly. Once the flying saucer template existed, the reports changed shape. That is a cultural fingerprint on raw perception, visible in the archive, and it is Section 4.1's argument with a date stamp.

12.4 THE CREDIBILITY OF PILOTS

The most common appeal in this field is to the authority of the pilot: trained observer, thousands of hours, no reason to lie, therefore his report should be weighted heavily. The appeal is reasonable, and it is also substantially

wrong, and explaining why requires distinguishing two things that get conflated.

Pilots are highly reliable about *what they did, what their instruments read, and what procedures they followed*. That is what their training is for and their record is excellent.

Pilots are *not* reliable at judging the size, range, or speed of an unfamiliar object against an empty sky, and there is no training that makes anybody reliable at it, because the necessary visual cues are absent. This is not my opinion; it is the consistent finding of the aviation human-factors literature, and it is why the industry built an entire safety culture around not trusting the eyeball.

The specific failure modes:

The empty-field problem. With no reference objects, the eye has no basis for range, and the brain therefore assigns one from assumed size — circularly. A weather balloon at 20 miles and a metallic disc at 2 miles produce identical retinal images.

Relative motion and parallax. A distant object appears to move with your aircraft; a nearby one appears to shoot past. This is why Venus has been chased by fighters — repeatedly and embarrassingly — and why a celestial body appears to pace, then dart away when the pilot turns.

Vection and false horizons. At night, over water, or in cloud, the vestibular system misreports attitude confidently, which is what killed John Kennedy Jr. Pilots are trained explicitly to disbelieve their own sensations and fly the instruments, which is an institutional admission of exactly the point I am making.

Canopy and HUD artefacts. Reflections, scratches, coatings, and symbology produce apparent objects with apparent motion.

Workload and attention. A crew running an intercept has extremely high cognitive load, and load degrades peripheral perception and memory encoding.

The key illustration is Kenneth Arnold, the founding case. On 24 June 1947 he was a private pilot with a legitimate reputation, flying near Mount Rainier (see Figure 12.2), and he reported nine objects moving in a way he described as “like a saucer if you

FIGURE 12.2

Kenneth Arnold with his own drawing of the objects seen near Mount Rainier — crescents, not saucers.

skipped it across water”. His *motion* simile became “flying saucer” in the press — he did not describe the objects as saucer-shaped, and he complained about the misquotation for the rest of his life. His speed estimate of over 1200 mph was derived by assuming a size he had no way to know. The most likely candidates are a formation of pelicans catching sunlight, or lenticular cloud fragments. And the entire modern era of this subject grew from a misreported simile.

FUN FACT

The airlines eventually solved the pilot-credibility problem in a way ufology never has, and it is the best argument in this book for how to handle testimony. After a series of accidents caused by confident, senior, experienced men being wrong, aviation adopted Crew Resource Management, mandatory cross-checking, instrument primacy, and the non-punitive Aviation Safety Reporting System run by NASA specifically so that reports could be filed without career risk. Commercial aviation became the safest form of travel ever devised by systematically distrusting individual perception. Ufology’s response to the same problem was to add “he was a very experienced pilot” to the beginning of the sentence.

12.5 UFOS & UAP

Somewhere around 2017 the vocabulary changed, and the change matters more than it looks. **UFO** — unidentified flying object, coined by Captain Edward Ruppelt of Project Blue Book in 1952 precisely because “flying saucer” had become unusable — had by the 1990s acquired the same problem. Say “UFO” in a briefing room and everyone hears *alien*, which is exactly the inference the term was invented to avoid. So the U.S. Navy adopted **UAP**: originally Unidentified Aerial Phenomena, broadened in 2022 to Unidentified *Anomalous* Phenomena when the Pentagon’s remit was extended to objects in space and underwater as well as in the air. The official body is now AARO, the All-domain Anomaly Resolution Office, which succeeded the UAP Task Force, which succeeded AATIP, the Advanced Aerospace Threat Identification Program.

I am mildly cynical about the rebrand and I will say why: it is a laundering operation for respectability, and it worked. Identical claims that were career-ending in 1995 became briefable to the Senate Intelligence Committee in 2019 without any new evidence being produced — only a new noun. That is worth noticing as a lesson about institutions rather than about physics. But I also think the term is genuinely better, because it drops the two assumptions smuggled inside “flying object”: that the thing is flying, and that it is an object. A sensor artifact is neither.

12.5.1 THE NIMITZ INCIDENT

On 14 November 2004, aircraft from Carrier Strike Group Eleven, operating with USS *Nimitz* about a hundred miles southwest of San Diego, were vectored onto an unknown contact. The cruiser USS *Princeton*, running the then-new SPY-1B radar upgrade, had been tracking objects intermittently for several days: appearing at around 80,000 feet, descending to roughly 20,000 feet, and holding.

Commander David Fravor, then commanding officer of the Black Aces — Strike Fighter Squadron 41 — was airborne on a training sortie with his wingman when *Princeton* tasked him with a real-world intercept. What he reported seeing, and has reported consistently for two decades in terms that have not inflated, was a disturbance in the sea surface roughly the size of a Boeing 737, and above it a white object about forty feet long, shaped like an elongated Tic Tac, with no wings, no rotors, no exhaust plume and no visible means of propulsion, moving erratically over the water. As he descended to engage it, he says it rose to meet him, then accelerated away in a manner he could not match. *Princeton* then reported reacquiring a contact at the CAP point sixty miles away — the squadron’s pre-briefed rendezvous, which is the detail that makes the case unsettling rather than merely odd.

A later flight that day captured the FLIR footage subsequently released as “FLIR1” or the “Tic Tac” video. Fravor was not the pilot who filmed it. In 2017 the video appeared in *The New York Times* alongside the disclosure of AATIP, and in April 2020 the Department of Defense formally released it and two other clips, confirming they were genuine Navy recordings and that the objects remained unidentified.

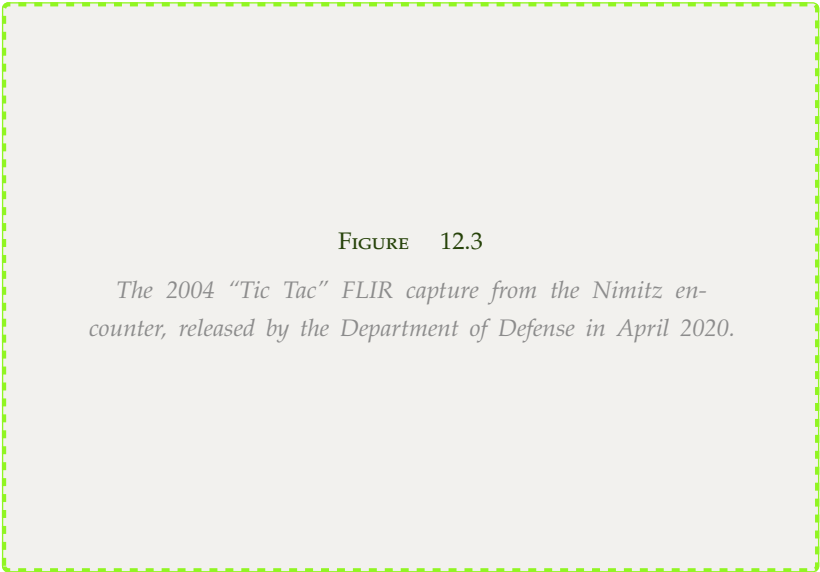


FIGURE 12.3

The 2004 “Tic Tac” FLIR capture from the Nimitz encounter, released by the Department of Defense in April 2020.

Figure 12.3. The 2004 “Tic Tac” FLIR capture from the Nimitz encounter, released by the Department of Defense in April 2020.

Image sourced from: <https://example.org/placeholder-flir1.jpg>

12.5.2 WHY THIS CASE IS DIFFERENT, AND WHERE IT IS WEAK

Different, first. Fravor is about as good a witness as this subject will ever get: a Navy commander with roughly 3,000 flight hours, TOPGUN-qualified, who reported through the chain of command at the time, gained nothing and risked a good deal, and whose account has not grown in the telling. There were multiple independent sensor systems — shipboard radar, airborne radar, infrared — and multiple aircrew across two flights. Section 12.4 argued that pilot credibility is oversold in general; this is the case that survives that argument best, because the claim does not rest on his eyes alone.

Now the weaknesses, because they are real and the enthusiast literature omits them. The *Princeton’s* radar had just been upgraded and was reportedly generating spurious tracks, which is exactly the profile of a new-software artifact. The video itself has been analysed in detail by Mick West and others, and the “impossible” motion in it is largely explicable: the apparent rotation is the gimbal of the ATFLIR pod rolling as it tracks, and the apparent high-

speed departure at the end coincides with the operator changing zoom level, whereupon a distant object at the edge of tracking lock simply leaves frame. Those are not hand-waves; they are demonstrated with the instrument's own documented behaviour. And critically, the compelling parts of Fravor's account — the water disturbance, the ascent, the acceleration — are *not* in the video at all. The footage everyone argues about is not the encounter everyone finds impressive.

So my position: the Nimitz case is the strongest single item in the modern corpus and it still gives us radar returns from an instrument with a known fault, an infrared clip with a plausible mundane reading, and unverifiable testimony — excellent testimony, but testimony. AARO's 2024 historical review found no evidence of extraterrestrial technology in any case it examined, including this one, while leaving a residue of genuinely unresolved reports. That residue is where this subject actually lives, and Section 1.8 is why I will not fill it in with a guess.

FUN FACT

The 2021 Office of the Director of National Intelligence preliminary assessment examined 144 incidents from 2004–2021 and managed to explain exactly one — a deflating balloon. That statistic is quoted constantly as “143 unexplained”. What the report actually said is that most cases lacked sufficient data to reach *any* conclusion. “We could not determine what it was” and “it was something extraordinary” are different sentences, and the entire modern disclosure debate runs on blurring them.

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12.6 METEOROLOGICAL PHENOMENA

A large fraction of what people see is weather, and weather is stranger than most people's mental catalogue allows for. Here is the working list I would want anybody to have memorised before reporting anything.

Lenticular clouds form in standing waves downwind of mountains: smooth, layered, lens-shaped, stationary while the wind passes through them, often with a hard-edged disc appearance and stacked “pile of plates” structure. They are the single best natural explanation for a hovering metallic disc, and they hover for hours.

Noctilucent clouds, at around 80 km in the mesosphere, glow electric blue after sunset long after the sky is dark. **Nacreous or polar stratospheric clouds** produce iridescent sheets at 15–25 km.

Sun dogs, sun pillars, 22° haloes, circumzenithal arcs and parhelic circles are all ice-crystal optics, and they produce geometrically precise bright spots and lines in the sky that look manufactured because they are governed by crystal geometry.

Temperature inversions bend light, producing superior mirages, looming, and the Fata Morgana — which can lift a distant ship, light, or coastline into the sky, stretch it, invert it, and make it shimmer and appear to move. This is also the principal cause of *radar anomalies*: ducting in an inversion layer returns ground clutter at apparent altitude and apparent speed. Section 12.5's radar-visual cases need to be read against it.

Sprites, blue jets and ELVES are genuine upper-atmosphere electrical discharges above thunderstorms, confirmed only in 1989 despite decades of pilot reports that were dismissed — which is a case where the aircrew were right and the scientists were wrong, and I include it because intellectual honesty requires it.

Green flashes, mirages of Venus, and atmospheric scintillation make stars and planets change colour rapidly and appear to jump. Venus at maximum brightness, low on the horizon, through turbulent air, is the single most reported UFO in history and has been chased by military aircraft on multiple continents.

Meteors and re-entries deserve their own note. A bolide can be brighter than the full moon, fragment visibly, appear to travel slowly and horizontally when seen end-on, and be reported as a formation of lit windows in a

structured craft — the 1968 Zond IV re-entry over the eastern United States produced exactly that report from multiple witnesses, described as a cigar-shaped craft with square windows. It was a rocket body breaking up.

FUN FACT

And then there are the ones that are not weather at all. Starlink satellite trains produce a perfect line of moving lights and generated a global spike in reports from 2019. Chinese lanterns produce orange orbs in silent formation that drift with the wind and wink out — the 2007 Alderney and countless UK “fleet” sightings are lanterns, and the correlation with wedding season is not subtle. High-altitude research balloons at 100,000 feet catch sunlight for an hour after dark. Google’s Project Loon put balloons in the stratosphere for years. If a thing in the sky is silent, slow, and drifting downwind, the wind is the identification.

THE EARTHFILES

“Hey, what if those crop circles are just ads for Target?”

— DENNIS MILLER

The premise of this chapter derives itself from a weekly podcast of the same name. Edited and produced by documentary filmmaker and investigative journalist Linda Moulton Howe, Howe’s weekly show deals primarily with — for lack of better terms — what can only be described as real-life X-Files cases.

Like many of the invented cases and stories of that 1990s television show, the real-life counterparts are often dark and mysterious, and otherwise have no scientific, rational, or sound justification for their existence. For the completeness of information, I have chosen to include various details and photographs of these cases. Many of which are not pretty to look at, and very upsetting to read. Due to the nature of this information, I feel the need to warn you, the student, that some of what you are about to read is extremely disturbing.

I also want to set out how this chapter is going to work, because it is not the same as the chapters before it. Every case here is associated, in the public mind and usually in the archival record as well, with a single investigator. That is not an accident of my sourcing. It is a fact about the field. When a rancher in Colorado finds something he cannot explain, he does not call a university pathology department. He calls the person who has been on the radio talking about it. That pattern — the private archive, the trusted intermediary, the case file that exists in one place only — is itself one of the

most important things a student of this subject needs to understand, and Section 13.1 is where we look at it directly.

13.1 CATTLE MUTILATIONS

For anyone not already aware of what exactly this is, or why it is so significant, the story often goes something like this. Farmer Joe wakes up bright eyed and early in the morning to tend to his farm. Upon arriving inside the barn, he is immediately aware that one of his prize steers is missing. Nowhere to be found. He takes the tractor out to drive around his 300-acre property and see if he can find this missing cow. Somewhere along the outskirts of his fence, he finds his missing cow, but is immediately shocked and horrified by the sight in front of him.

Laying before him, his cow is dead. The cavity of the chest is ripped open with clean cuts, as though surgical tools were used to commit such a crime. The random gnawing and biting of the skin from predators such as wolves does not account for the perfectly clean cuts on the belly of the beast. Inside the cow, Farmer Joe notices that all internal organs are missing. The heart, the liver, the lungs — there is nothing but empty space inside.

The face of the cow is rather disfigured, as though all bones had been broken right before it met its untimely demise. In the sockets where eyes should be, there is nothing; only the frightening stare into oblivion from this now dead animal remains. The cow has had all fluid from inside removed. No blood, no water, no urine is to be found. And the grass surrounding the beast is not touched. Not one drop of blood is to be found, suggesting this final resting location of Farmer Joe's cow is likely not where the actual killing took place.

The problem is that no sign of dragging the animal to this location can be found. The surrounding grass does not have any traces of blood, nor has it been flattened in the manner one would expect with the dragging of a several-hundred-pound animal overtop. It is as though this cow was dropped there from above, or was magically teleported to that very location.

FIGURE 13.1

A classic mutilation photograph of the kind circulated from the mid-1970s onward: excised soft tissue at the head, an apparently clean wound margin, and no visible blood on the surrounding ground.

Figure 13.1. A classic mutilation photograph of the kind circulated from the mid-1970s onward: excised soft tissue at the head, an apparently clean wound margin, and no visible blood on the surrounding ground.

Image sourced from:

<http://www.ufonut.com/wp-content/uploads/2017/10/Animamute1342699.jpg>

FIGURE 13.2

A second widely reproduced example. Note what the photograph cannot tell you: how long the animal had been dead, what the ambient temperature was, whether a necropsy was performed, or who took the picture.

Figure 13.2. A second widely reproduced example. Note what the photograph cannot tell you: how long the animal had been dead, what the ambient temperature was, whether a necropsy was performed, or who took the picture.

Image sourced from:

https://pm1.narvii.com/6488/90fe827fca358e76bc0e962346cc52d87a845848_hq.jpg

The local authorities are notified but, upon further investigation, are unable to make heads or tails of what could possibly account for this vicious and brutal murder. So, in the absence of answers, Farmer Joe then contacted Linda Moulton Howe — investigative reporter — to come take a look.

13.1.1 LINDA MOULTON HOWE AND THE MAKING OF A
MYSTERY

That last sentence is the one to slow down on, because it is the hinge of the entire subject. The modern cattle mutilation phenomenon is not primarily a veterinary story or a law-enforcement story. It is a broadcasting story. And the single person most responsible for its shape, its vocabulary, and its persistence across five decades is Linda Moulton Howe.

Howe was born in Boise, Idaho, in 1942, and received a master's degree in communication from Stanford University in 1968. Her early career was conventional, successful, and had nothing to do with any of this: she worked in local television as a documentary producer, winning regional Emmy awards and a Peabody for environmental and public-health work, and from 1978 to 1983 she was director of special projects at KMGH-TV, the CBS affiliate in Denver. Her focus in that period was primarily environmental — pollution, water, public health in the Mountain West. She was, in other words, a working investigative journalist with a portfolio of documented, awarded reporting before she ever looked at a dead cow. Students of this field should hold on to that, because the lazy version of the skeptical position is to imply that everyone who investigates anomalies is a crank who could not get hired elsewhere. Howe could get hired anywhere.

In 1979 and 1980, working out of Denver, she produced and directed *A Strange Harvest*, a one-hour documentary on the livestock deaths then being reported across eastern Colorado and New Mexico. It aired in 1980, won a regional Emmy, and is the founding text of the subject. Almost everything the general public believes about cattle mutilations — the surgical precision, the bloodlessness, the cored orifices, the absent tracks, the unmarked helicopters, the reluctance of officials to engage — reaches them through that film or through material downstream of it. Howe interviewed ranchers, sheriff's deputies, and veterinarians on camera, and she reached a conclusion that she has held ever since: that the excisions were performed with a technology not available to the ranchers or to conventional predators, and that the most

probable explanation involved a non-human intelligence conducting some form of biological harvest.

She did not stop there, and this is what distinguishes her from the many people who produced one documentary and moved on. She kept the case files. She published *An Alien Harvest* in 1989 and *Glimpses of Other Realities* in two volumes through the 1990s. She became one of the most frequent guests on *Coast to Coast AM*, appearing regularly from around 1991 under Art Bell and continuing under George Noory from 2003 until her run as a regular ended in 2019 — Section 20.4 covers what that platform meant for the field. In 1999 she launched Earthfiles as a website, and from 2007 onward as the podcast and video channel that gives this chapter its name. Across that span she has produced something genuinely rare: a continuous, single-authored, five-decade archive of anomaly reports, with names, dates, locations, and photographs, maintained by one person who answered the phone when ranchers called.

WHY THIS MATTERS METHODOLOGICALLY

Howe's archive is simultaneously the best and the worst thing that happened to cattle mutilation research. Best, because without it we would have almost no documentation of the 1970s wave at all — the sheriff's reports are thin, the necropsies were rarely performed, and the carcasses are long gone. Worst, because a body of evidence curated by a single investigator who holds a specific hypothesis is a body of evidence with a selection filter attached, and there is no way to unpick the filter after the fact. If the interesting cases are the ones that got recorded, and the interesting cases are the ones that fit, the archive will confirm the hypothesis no matter what was actually happening in the field. This is not a charge of dishonesty. It is a structural problem, and it applies to every private archive in this book, including the ones assembled by people I like.

The critical case against Howe's interpretation is not that she invented anything. **It is that her diagnostic standards were those of a documentary producer rather than a pathologist.** Joe Nickell and other investigators have described her as credulous — specifically, as accepting witness characterisation of wounds as “surgical” without independent forensic confirmation, and as treating the absence of an official explanation as evidence for an extraordinary one. In *A Strange Harvest* the veterinary voices that support anomaly are foregrounded and the ones that support scavenging are not.

Later Earthfiles material has extended the same method to claims — underground bases, government insiders, retrieved biological entities — where the sourcing is a single anonymous informant, and where a journalist's normal corroboration requirements appear to have been suspended.

13.1.2 WHAT THE INVESTIGATIONS ACTUALLY FOUND

Beginning in earnest in the mid-1970s and centred on Colorado, New Mexico, Kansas, Nebraska and Montana, ranchers found dead cattle with what appeared to be surgically precise excisions: an eye, an ear, the tongue removed at the root, the jaw stripped to clean bone, the rectum and genitals cored out, and the udder taken. The wounds appeared bloodless. Predators reportedly avoided the carcasses. There were no tracks. Sometimes there were reports of unmarked black helicopters.

The scale was real enough to matter: Colorado had hundreds of reports by 1975, the state governor requested federal help, and in 1979 the FBI opened an investigation of cases in New Mexico under Senator Harrison Schmitt's pressure. Kenneth Rommel, a retired FBI agent, was hired to run it on a \$44,170 grant from the Law Enforcement Assistance Administration. He spent a year on it, examined cases as they came in rather than after the trail had gone cold, and in June 1980 — the same year *A Strange Harvest* aired — published *Operation Animal Mutilation*, 297 pages, which remains the most thorough field investigation the subject has ever received.

The conclusion of that investigation, and of every veterinary pathologist who has examined fresh cases, is normal death followed by scavenging — and the mechanism explains every one of the strange features, which is what a good explanation does.

The precision. Blowfly larvae, ants, and small scavengers begin at the moist, soft-tissue orifices: eyes, mouth, nostrils, anus, vulva, udder. That is not selective interest in reproductive organs; it is where the openings are. Larval feeding produces a smooth, scalloped margin with no tearing, which looks under casual inspection like a cut, and looks under a microscope like exactly what it is — pathologists distinguish them without difficulty.

The bloodlessness. Blood settles by gravity and clots within hours; a carcass drained by livor mortis into the down-side tissues has no free blood at an upper-surface wound. Nothing was exsanguinated. This is basic post-mortem physiology and it is the single detail most responsible for the

mystery, because ranchers know living animals and pathologists know dead ones.

The stripped jaw. Soft tissue is consumed first; hide contracts and splits along the line of least resistance as it dries, exposing bone cleanly.

The absent tracks. Hard ground, wind, and the interval between death and discovery. Rommel documented cases where reported “no tracks” sites had tracks visible on examination.

The predator avoidance. Not consistently true, and coyotes are selective about putrefying meat.

Rommel also found that the reported rate did not exceed baseline cattle mortality. Cattle die constantly — of blackleg, anthrax, lightning, bloat, plant toxicity, calving complications and being struck by lightning while standing under the only tree. What changed in 1975 was not the death rate. It was that ranchers started looking at carcasses with a hypothesis in mind, and a media narrative told them what to look for.

FUN FACT

The black helicopters were probably real, and that is the most interesting part. The mid-1970s saw genuine unmarked helicopter activity across the American West, some of it associated with mineral survey, some with law enforcement narcotics work, and some never accounted for. Also real: a 1975 wave of livestock deaths coinciding with a documented outbreak of clostridial disease. Also real: at least some cases were deliberate mutilation by humans, including cult activity and insurance fraud, and several people were prosecuted. Three real phenomena, one imagined connection.

13.1.3 TWO DOCUMENTS, ONE YEAR

So we have a clean natural experiment, and it is worth stating plainly because it is the sort of thing that only becomes visible when you line the dates up. In 1980, two major works on cattle mutilations were released. One was a broadcast documentary made by an award-winning journalist who interviewed the ranchers, filmed the carcasses as she found them, took the witnesses at their word about what a surgical cut looks like, and concluded that something not of this Earth was harvesting tissue. The other was a 297-page federally funded field report by a retired FBI agent who attended fresh scenes, commissioned necropsies, checked the reported “no

tracks” sites himself, compared the reported death rate against baseline herd mortality, and concluded predation.

One of those documents won an Emmy, spawned a genre, and is still cited constantly. The other sits in a New Mexico state archive. That asymmetry has nothing to do with which one was more careful, and everything to do with which one was more watchable. If there is a single lesson in this chapter, it is that the survival of a claim in public memory is determined by its narrative fitness and not by its evidential quality — and that a field which gets its case files from broadcasters rather than from laboratories will inherit the broadcaster’s incentives whether anyone intends it to or not.

I want to be fair to Howe in closing, because it would be easy not to be. She asked questions that the relevant authorities were, in the 1970s, genuinely not asking. She recorded testimony that would otherwise have vanished. She has been consistent, public, and reachable for fifty years, which is more than can be said for most of the anonymous sources in this book. Her interpretation of the evidence is, I think, wrong, and wrong in a way that a working pathologist could have corrected in an afternoon. Those two judgements are compatible, and holding both at once is roughly the whole skill this textbook is trying to teach.

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10.2 ANIMAL-HUMAN HYBRIDIZATION

Chimeras and hybrids are a real, active, and heavily regulated area of biology, and knowing what is actually possible is the only way to evaluate the claims in Section 7.4 and Section 10.2.

The distinctions matter. A **hybrid** results from fertilisation between two species and has mixed DNA in every cell — a mule, a liger. A **chimera** contains cells from two organisms in separate lineages, mixed but not

blended. A **transgenic** organism carries specific inserted genes.

What has actually been done: human–animal chimeras exist in research and are strictly limited. Human cells have been introduced into pig, sheep and monkey embryos, with human contribution remaining very low — typically a tiny fraction of cells — and the embryos destroyed within days under protocol. Human–mouse neural chimeras have been made to study brain development. Human genes are routinely expressed in animals for research and pharmaceutical production, and human insulin has been made in bacteria since 1978. Pig heart valves have been used in humans for decades, and in 2022 a genetically modified pig heart was transplanted into a human patient who survived two months.

What has *not* been done, and cannot be: viable hybridisation between species separated by any substantial evolutionary distance. The barriers are hard. Chromosome number and structure must be compatible enough for meiosis — humans have 46, chimpanzees 48, and even that gap makes viable fertile offspring effectively impossible. Gamete recognition proteins are species-specific. Developmental gene regulation is timed differently. Mitochondrial and nuclear genomes must be co-adapted. Hybrids between close species are usually sterile precisely because of this: mules cannot breed.

And notice what that implies for the alien hybrid claim. Horse and donkey diverged around four million years ago, share nearly all their biochemistry, and produce a sterile offspring. An organism from another biosphere would not share our chirality conventions, our genetic code, our amino acid set, our nucleotides, or our replication machinery — if it used nucleic acids at all. It is not a distant relative. It is not a relative. Section 7.4 makes the point at length; here I will simply say that the abduction breeding programme is not improbable in the way a long shot is improbable. It is a category error, like proposing to cross a fern with a filing cabinet.

FUN FACT

Humans are already chimeras, routinely and without technology. Foetal cells cross the placenta and persist in the mother for decades — microchimerism — and have been found in maternal brain tissue. Cells can pass between twins in utero. And roughly 8% of the human genome is endogenous retroviral sequence: ancient viral DNA integrated into our germ line, some of it now

essential, including a gene co-opted for placental function. We are a mosaic of other things already. Nature did the interesting version first.

13.3 CROP CIRCLES

This is the cleanest case in the entire book, because we have confessions, we have a reconstruction, and we can watch the phenomenon respond to human incentives in real time.

Crop formations — flattened geometric patterns in standing cereal crops — appeared in numbers in southern England from the late 1970s, concentrated in Wiltshire and Hampshire around Avebury, Silbury Hill and the Pewsey Vale. They began as simple circles and became, over two decades, extraordinarily elaborate: fractals, three-dimensional illusions, and in 1996 the Julia Set beside Stonehenge, and in 2001 a 400-metre pattern at Milk Hill with 409 circles.

The claimed anomalies were: bent-not-broken stem nodes, altered germination rates, magnetic particles in soil, electronic equipment failure, dowsing responses, and the sheer complexity.

FIGURE 13.3

Bower and Chorley's plank-and-rope method, and a formation of the kind declared unfakeable.

Figure 13.3. Bower and Chorley's plank-and-rope method, and a formation of the kind declared unfakeable.

Image sourced from: https://commons.wikimedia.org/wiki/File:Crop_circle_Switzerland.jpg

In September 1991, Doug Bower and Dave Chorley — two men in their sixties from Southampton — announced that they had made the original formations starting in 1978, and demonstrated the method for the press (see Figure 13.3): a length of rope, a plank with a loop for the foot, and a baseball cap with a wire sight for keeping lines straight. They then made a formation and invited a prominent cereologist to authenticate it, which he did, on camera, declaring that no human could have made it. That footage exists. Bower and Chorley said they had started in a pub

after discussing Tully, and that they kept going because the reaction was funny.

Since then: circle-making has become an art form with named practitioners, competitions, commissions from advertising agencies — Nike, Mitsubishi, Weetabix and others have paid for them — and documented overnight construction of complex formations under observation. John Lundberg’s Circlemakers group has built formations for television with cameras running. Physicist Richard Taylor published in *Nature* in 2011 on the increasing technological sophistication of circle-making, including the use of GPS and laser levels.

The alleged physical anomalies did not survive testing. Node bending occurs naturally through phototropic and gravitropic response in flattened stems — lay a plant down and it bends at the nodes as it tries to grow upright, within days. Germination differences track mechanical damage. The magnetic particles are meteoritic dust, which falls everywhere. And the diagnostic test is decisive: formations appear overnight, in crops, in the specific regions where circle-makers live, in the months when crops are standing, and they became more complex in step with the interest and the competition. Crop circles have never appeared in a field under continuous surveillance in a way that excluded human access, and they follow the aesthetics of human mathematical fashion — fractals arrived after fractals were in popular culture.

FUN FACT

My favourite detail is the economics. Crop formations cluster in Wiltshire not because of ley lines but because Wiltshire has tourists, farmers who charge admission, and a concentration of circle enthusiasts. Some farmers made more from gate money than from the crop. Where there is a market, there is supply. And the phenomenon has declined markedly since the 2000s — not because the visitors left, but because the makers got older and the novelty passed. Anomalies do not usually retire.

13.4 THE TRUMPETS OF JERICO

Since roughly 2008, and heavily from 2011, people worldwide have recorded low, groaning, trumpet-like or metallic-grinding sounds apparently coming from the sky, lasting minutes, sometimes with no visible source. Recordings came from Kiev, Florida, British Columbia, Germany, Costa Rica, and dozens

of other places. They were widely framed as apocalyptic, the trumpets of Revelation, or the sound of a cloaked craft.

Several distinct real phenomena are in the mix, and the strongest single point about this case is that the recordings are not all recordings of the same thing.

Industrial noise carried by inversion. A temperature inversion refracts sound downward and can carry industrial noise — pile drivers, mining, rail yards, gas flares, ventilation fans, ice-road construction — many kilometres, arriving with no apparent direction and apparently from above. This accounts for a large share, and in several cases the specific source was identified: a nearby plant, a rail marshalling yard, a snow grooming operation.

Ice quakes and glacial calving produce extraordinary groaning and booming. So does lake ice — and the sounds are eerie enough that recordings of them are frequently mislabelled as sky trumpets.

Aurora-related and geomagnetic acoustic effects are documented but poorly understood.

Skyquakes are a genuine long-standing unexplained category: the Barisal Guns in Bengal, the Seneca Guns on Seneca Lake and the Carolina coast, *mistpouffers* in Belgium, and *brontidi* in Italy. These have been reported for centuries and are variously attributed to supersonic aircraft, distant thunder from over-the-horizon storms, submarine slope failure, methane release, and minor seismicity. Some remain unexplained and the literature is honest about it.

And an enormous fraction are hoaxes. This is not speculation. Multiple viral “sky trumpet” videos were traced to the same sound library, and specifically to sound design assets — several were identified as an audio track from a film or from the popular “Red Weather” stock sample, layered over unrelated footage. Video forensics on the most-shared examples found identical waveforms across supposedly independent recordings on different continents. **Identical waveforms are not a coincidence; they are a file.**

FUN FACT

The honest position on skyquakes is that a residue of them is unexplained, and I want to put that plainly because this chapter has been unsympathetic. The Seneca Guns were recorded by James Fenimore Cooper. Nobody has

produced a complete account. But note the shape of the residue: it is a *sound with no identified source*, not a sound with an identified extraterrestrial source. Unexplained means unexplained. It does not mean explained by the most exciting hypothesis available, and if you take one habit from this book, take that one.

13.5 PARANORMAL SKIES

Beyond craft, the sky produces a catalogue of genuinely odd reports, and I want to run through them because the pattern across the category is instructive.

Falls of fish and frogs are real and documented since antiquity, with modern cases verified by authorities — Honduras has an annual event, and there are well-attested cases from England, Japan and Australia. The mechanism is waterspouts and tornadic updraught lifting the contents of a shallow pond and depositing them downwind, which explains why the animals are typically all one species and one size class: the pond sorted them.

Coloured rain — red rain in Kerala in 2001 — was analysed and found to contain spores of a local algae, *Trentepohlia*. It had been proposed as extraterrestrial by Chandra Wickramasinghe, who has proposed that about a great many things.

Angel hair is spider silk, from mass ballooning dispersal events, which coat landscapes in filament and drift down from great height.

Star jelly is variously slime mould, cyanobacteria, or the oviducts of frogs deposited by predators, all of which have been identified in specific cases.

Ice falls — megacryometeors — are real, up to several kilograms, and mostly but not entirely attributable to aircraft; some fall from clear skies and remain argued over.

Airborne debris accounts for much more than people expect: pieces of aircraft, cargo, balloon payloads, and space junk. Around 100 tonnes of meteoritic material arrives daily, mostly as dust.

The Hessdalen lights in Norway deserve better than a paragraph.

Recurring luminous phenomena in a specific valley, under continuous instrumented observation since 1984 with cameras, spectrometers, radar and magnetometers — an ongoing scientific field study of an anomalous light phenomenon. Results suggest a plasma or combustion process possibly involving valley geology and dust, with some events correlating with radar returns. It is unresolved. *This* is how you study an anomaly: pick a place where it recurs, put instruments there, and wait for years. Compare it to the field's usual method of interviewing people about something that happened once, decades ago.

Earthquake lights were dismissed for centuries and are now accepted, with a proposed mechanism involving stress-induced charge separation in certain rock types. Another case where the witnesses were right.

Tunguska is the one that gets folded into ufology, and it belongs here because of how it is misused rather than because of what it was. On the morning of 30 June 1908 something detonated several kilometres above the Podkamennaya Tunguska river in Siberia. The airburst flattened roughly 770 square miles of forest and some eighty million trees, registered around 5.0 on seismographs, and sent atmospheric pressure waves that were recorded on barometers in London thousands of kilometres away. Estimates of the yield run in the range of ten to fifteen megatons, which is to say something on the order of a thousand times the Hiroshima device. Crucially, there is no impact crater and no large recovered meteorite, because the body broke up under aerodynamic stress before reaching the ground.

That absence of a crater is the entire foothold for the exotic readings — a crashing alien craft, a natural nuclear explosion, an antimatter fragment, a small black hole. None of them is needed. An airburst is the expected behaviour of a stony asteroid or comet fragment of that size entering at that speed, and it is exactly what happened on a smaller scale over Chelyabinsk on 15 February 2013, an event captured by hundreds of dashboard cameras and analysed to death, which produced the same signature: bright bolide, high-altitude detonation, blast damage over a wide area, no crater, and only small recovered fragments. Tunguska is not a UFO case. It is the best available demonstration that the sky is genuinely dangerous without any help from anybody, and the reason I include it is that the field's instinct on encountering an unexplained gap — reach for the most exotic filler — is precisely the instinct this course exists to break.

13.6 DYATLOV PASS

In early February 1959, nine experienced ski hikers from the Ural Polytechnic Institute, led by Igor Dyatlov, died on the eastern slope of Kholat Syakhl in the northern Urals. The details are why the case has never gone away.

Their tent was found cut open from the inside. All nine had left it in severe cold — something like -25 to -30°C — inadequately dressed, several barefoot or in socks. Five died of hypothermia. Four, found later in a ravine under deep snow, had severe internal injuries: Thibeaux-Brignolle had a major skull fracture, Dubinina and Zolotaryov had multiple crushed ribs, with no corresponding external wounds — the investigator's phrase was that the force resembled a car crash. Dubinina was missing her tongue and eyes. Some clothing showed elevated radioactivity. Some bodies had an orange-brown discolouration. Other hikers in the region reported orange spheres in the sky that month. The Soviet inquiry closed with the formulation that a “compelling natural force” had caused the deaths, and the area was closed for three years.

The explanations, in ascending order of plausibility:

Avalanche was long dismissed because the slope angle seemed too shallow and there was no obvious debris. In 2019 the Russian authorities reopened and closed the case with a slab avalanche finding, and in 2021 Puzrin and Gaume published a modelling paper in *Communications Earth & Environment* showing that cutting into the slope to pitch the tent, combined with persistent katy winds loading the slab, could produce a delayed, small, localised slab release capable of causing exactly those crush injuries in people lying in a tent. The injuries match the biomechanics of compression under a slab against a hard surface, which also explains the absence of external wounds.

Paradoxical undressing and **terminal burrowing** are established hypothermia behaviours: as core temperature falls, peripheral vasodilation produces a false sensation of heat and victims remove clothing, then seek enclosed spaces. This accounts for the state of the bodies and for the ravine.

The soft-tissue loss is scavenging and decomposition in running water — the ravine bodies were found in a stream bed after months, and eyes, tongue and lips are the first tissues lost, as Section 13.1 explained.

The radioactivity was on the clothing of members who had worked with

radioactive materials at the institute, which the file records.

The orange spheres are almost certainly R-7 rocket launches from Baikonur or Plesetsk — the timing and geography fit, and rocket plumes in the upper atmosphere produce exactly that.

FUN FACT

I find Dyatlov Pass the most sobering case in this book, and not because of the mystery. Nine young people died in the cold, badly, and their families spent sixty years being told the file was closed by a state that closed files as a matter of habit. The secrecy produced the legend — not because there was a cover-up of aliens, but because a government that will not explain anything teaches its citizens that nothing is explained. Chapter 8 is about the American version of the same lesson.

13.7 CRYPTOZOOLOGY

Cryptozoology is the search for animals whose existence is unconfirmed, and it belongs in this book for one reason: it is the closest available control group. It uses the same evidence types as ufology, faces the same problems, and — crucially — it occasionally *wins*, which lets us see what a win looks like.

The wins are real and worth knowing. The okapi was confirmed in 1901, the mountain gorilla in 1902, the Komodo dragon in 1912, the coelacanth in 1938 after being presumed extinct for 66 million years, the saola in 1992, and the giant squid was photographed alive only in 2004 and filmed in 2012. New species are described continuously — thousands a year — including large vertebrates. Anybody who tells you biology is finished is wrong.

Notice, though, what confirmation required in every case: a body, a specimen, a type description, and a place in the taxonomic literature. Not footage. Not testimony. A carcass.

Which is what the famous cryptids lack.

Bigfoot. The Patterson–Gimlin film of 1967 (see Figure 13.4) remains the centre-piece, and after nearly sixty years of analysis it is unresolved in the sense that neither side can force the other's hand. Bob Heironimus

FIGURE 13.4

Frame 352 of the Patterson–Gimlin film: the most analysed and least conclusive footage in cryptozoology.

claimed to have worn the suit. What is not unresolved: no body, no bones, no teeth, no scat with novel DNA, no roadkill, and — this is the killer — no fossil record of any great ape in the Americas, ever. A breeding population of large primates requires thousands of individuals over thousands of years, and thousands of individuals die. A 2014 study by Sykes and colleagues sequenced 30 alleged samples: every one was a known animal, mostly bear, and two supposed Yeti samples were Himalayan brown bear.

Loch Ness. Sonar surveys, including comprehensive sweeps, found nothing large. A 2018 environmental DNA survey sampled the entire loch and found no reptilian DNA at all — and a great deal of eel DNA, which produced the current best hypothesis and a rather charming one. The 1934 “surgeon’s photograph” was admitted in 1994 to be a toy submarine with a model head. And the loch is oligotrophic: its food web cannot support a large predator population.

Thylacine, by contrast, is the interesting one. It genuinely existed, the last confirmed individual died in Hobart in 1936, and sighting reports continue. That is a case where the animal is real, extinction is recent, habitat exists, and the search is legitimate — and it still has not produced a specimen in ninety years, which tells you how hard producing a specimen is even when the animal was definitely there.

FUN FACT

The lesson cryptozoology teaches ufology is about *trail cameras*. There are now millions of motion-triggered cameras in North American forests, operating continuously for two decades, and they have produced an extraordinary photographic record of wildlife — including rare, shy, nocturnal animals in enormous numbers. Not one unambiguous Sasquatch. Meanwhile the same interval put a high-resolution camera in every pocket on Earth, and the quality of UFO photographs has not improved. When the observing capacity increases by orders of magnitude and the evidence does not, that is data.

13.8 SKINWALKER RANCH

A roughly 500-acre property in the Uinta Basin, Utah, near Fort Duchesne, has been the site of reported anomalies since at least the 1970s: cattle mutilations, orbs, poltergeist activity, large unidentified animals, disappearing livestock, magnetic and radiation anomalies, and injuries to investigators.

The history is well documented. The Sherman family bought the ranch in 1994 and reported extensive phenomena. In 1996 Robert Bigelow, the aerospace entrepreneur, bought the property and ran the National Institute for Discovery Science on it, staffing it with scientists including Colm Kelleher and involving journalist George Knapp; their book *Hunt for the Skinwalker* appeared in 2005. NIDS closed in 2004. Bigelow later held the AAWSAP contract discussed in Chapter 8 — and this connection is genuinely important, because the Defense Intelligence Agency programme that produced part of the 2017 story was administered by the man who owned the ranch, and some of AAWSAP's work involved it. Brandon Fugal bought the property in 2016 and the History channel series began in 2020.

My assessment, and I will be direct.

The research problem is structural. Nearly all reported phenomena come from residents and investigators who were financially, professionally or emotionally invested. NIDS operated for eight years with equipment and staff and published essentially nothing in peer-reviewed venues. The television series generates anomalies weekly on a production schedule, which is the least plausible feature of the entire story — genuine anomalous phenomena do not respect a filming calendar. Access is controlled, and independent researchers have been unable to conduct unsupervised work.

The environment supplies a lot. The Uinta Basin is a major oil and gas region with substantial methane, hydrogen sulphide and natural seepage; it hosts high electromagnetic activity from industrial operations; it is under military airspace, with the Utah Test and Training Range not far away; and the geology includes uranium-bearing formations, which accounts for background radiation readings that startle people who have not measured background elsewhere. **Every radiation or magnetic “anomaly” reported without a control measurement is not a measurement.**

And the cultural dimension deserves respect. *Skinwalker* — *yee naaldlooshi* — is a specific and deeply serious element of Navajo belief, referring to

a witch who takes animal form, and it is not discussed casually within the culture. The ranch's name attaches that tradition to a commercial entertainment property, on Ute land rather than Navajo, and Navajo voices have objected. Section 10.5's point applies again.

What would change my mind is not complicated: continuous instrumented monitoring, with published calibration and control data, conducted by researchers with no stake in the outcome, with raw data released. The Hessdalen project in Section 13.5 has done exactly that in Norway for forty years on a fraction of the budget. The difference between the two is the difference between research and content.

THE ROSWELL INCIDENT

“The Roswell incident, for instance, had over three hundred witnesses — some describing the bodies, some the craft, some the military procedures. Were they all perpetuating their own lives in a myth?”

— DWIGHT SCHULTZ

14.1 SPACESHIP DOWN!

Roswell is the founding myth of American ufology, and it is also the best-documented case in the field — which is an awkward combination, because the documentation is what dismantles it. I want to lay out the sequence carefully, in order, with the dates, because almost nobody who has an opinion about Roswell knows the chronology.

Before the chronology, one piece of context that I think is indispensable and almost always left out. Five years earlier, at 3:16 on the morning of 24 February 1942 — less than three months after Pearl Harbor — air raid sirens sounded over Los Angeles and the 37th Coast Artillery Brigade opened fire on something in the sky over the city (see Figure 14.1) with .50-calibre machine guns and 13-pound anti-aircraft shells. The barrage

FIGURE 14.1

The Los Angeles Times photograph of 26 February 1942: searchlights converging during the barrage.

Figure 14.1. The *Los Angeles Times* photograph of 26 February 1942: searchlights converging during the barrage.

Image sourced from: <https://commons.wikimedia.org/wiki/File:BattleOfLosAngeles.jpg>

ran for the better part of four hours and expended roughly 1400 rounds of anti-aircraft ammunition. Shrapnel fell on the city for over an hour, several houses and vehicles were wrecked with property damage in the hundreds of thousands of dollars, and five people died — three in traffic accidents during the blackout and two of heart attacks. After the war, Japanese records confirmed what the Navy had said within a day: there were no Japanese aircraft over Los Angeles that night, or any other night. The most likely trigger was a stray weather balloon, amplified by jumpy gunners, searchlights and smoke from the guns themselves.

I raise the Battle of Los Angeles because of what it tells you about 1947. If any physical object had actually been brought down over an American city in 1942, the United States would have already had five years of institutional practice at crash recovery, materials analysis and news management by the time Mac Brazel walked into a sheriff's office. It did not. What it had instead was a well-rehearsed habit of shooting first and explaining later, and a public that had watched its own artillery lose an argument with a balloon. **Both the credulity and the official clumsiness that define Roswell were already in place before Roswell.**

In late June or early July 1947, ranch foreman William "Mac" Brazel found a scatter of debris on the Foster ranch, roughly 75 miles northwest of Roswell, New Mexico. He described rubbery strips, tinfoil, tough paper, sticks, and tape with flowers or figures printed on it. He collected some, showed it to neighbours, and on 7 July took it to Sheriff George Wilcox in Roswell, who contacted Roswell Army Air Field.

RAAF was not an ordinary base. The 509th Bombardment Group stationed there was the only nuclear-capable bomb group in the world — the unit that had dropped the atomic bombs two years earlier. Its personnel and its security posture were exceptional, which matters for what happened next.

Major Jesse Marcel, the group intelligence officer, went out with counter-intelligence officer Sheridan Cavitt and recovered material. And on 8 July 1947 the base public information officer, Walter Haut, issued a press release, approved by base commander Colonel William Blanchard, stating that the 509th had come into possession of a "flying disc". The Roswell Daily Record ran it that afternoon under the headline "RAAF Captures Flying Saucer On

Ranch in Roswell Region”.

Within hours the story was reversed. General Roger Ramey at Eighth Air Force headquarters in Fort Worth stated it was a weather balloon and a radar reflector, and photographs were taken in Ramey’s office of Marcel with foil, sticks and rubber laid out on the floor. On 9 July the papers ran the correction.

And then — this is the part that decides everything — **nothing happened for thirty years**. No follow-up. Brazel gave interviews and went back to ranching. Marcel served out his career. The story appears in none of the major UFO books of the 1950s and 1960s; Donald Keyhoe, who was hunting exactly this kind of case and had good military contacts, never mentioned it. Blue Book, running through 1969 and reviewing thousands of cases, did not treat it as significant.

In 1978 nuclear physicist and lecturer Stanton Friedman met Jesse Marcel, and in 1980 Charles Berlitz and William Moore published *The Roswell Incident*. Everything else — the craft, the bodies, the autopsies, the second crash site, the threats — enters the record after 1978, in testimony given thirty to fifty years after the event, much of it by people who were not present.

FUN FACT

The 1947 press release is genuinely puzzling and I do not want to wave it away. A public information officer at the world’s only nuclear bomb group announced the capture of a flying disc, with his commander’s approval, and the announcement was retracted within hours. Something embarrassing happened. But note the shape of the embarrassment: an officer at a base full of officers with the highest clearances in the American military looked at the debris and thought “flying disc” was a sensible thing to put on paper. Whatever was on that ranch, the people nearest to it did not recognise it and did not treat it as the most important object in human history. They put it on the floor of a general’s office and photographed it.

14.2 A SIMPLE MISTAKE

The conventional explanation — and it is not the weather balloon of 1947, which is important — is Project Mogul, and I will treat Mogul itself in Section 14.4. Here I want to address the question of *how the mistake happened*, because the field’s standard objection is that Jesse Marcel was an intelligence

officer who could recognise a balloon.

He could, and that objection is weaker than it sounds, for three reasons.

First, Marcel identified it as a balloon at the time — or at least he participated in the recovery, filed his report, and did not spend the next three decades insisting otherwise until he was asked in 1978. His 1978–79 account differs substantially from what he said in 1947 and from the recollections of Cavitt, who was with him and who stated flatly and repeatedly that it was a balloon and that he recognised it as such.

Second, Marcel’s own account contains verifiable errors about his own biography — he claimed a degree and a pilot’s rating that his service record does not support — which is not an accusation of lying so much as evidence of the ordinary self-enhancement that thirty years of retelling produces in anybody.

Third, and most importantly, the debris description *matches* the object. Mogul balloon trains used neoprene rubber balloons, aluminium foil radar reflectors, balsa wood struts, and — this is the detail that closes it — the reflectors were assembled by a New York toy manufacturer using tape with pinkish-purple flower and geometric patterns on it, because that was the tape stock available. Brazel’s “flowers” on the tape is not a strange detail. It is the strongest corroboration in the entire case, and it points at a toy factory.

The “memory metal” claim — foil that unfolded itself when crumpled — entered the record later, and foil-laminated paper does spring back somewhat. Nobody kept a piece, which is the other notable fact: on a ranch in New Mexico in 1947, with debris scattered over a field, with neighbours handling it and children playing with it, not one fragment survived in private hands. The base’s own recovery could not have been that thorough; the material simply was not remarkable enough for anybody to keep.

14.3 CONTROVERSY BEGINS

The post-1978 Roswell literature accumulated a set of claims that are worth listing individually, because they were added by different people at different times and several have been retracted.

Bodies. Glenn Dennis, a mortician, described a nurse and a field autopsy; the nurse he named could not be located and the details did not check out. The most-repeated element of his account is a telephone call from the

base mortuary officer asking the funeral home how small a hermetically sealed coffin they could supply, which in retelling became a request for three child-sized coffins — Dennis had worked at Ballard's since he was fifteen and took the call himself, he said, thirty-two years before he first told anyone about it. It is the single most evocative detail in the whole case and there is no independent record of the call. Frank Kaufmann described an intact craft and bodies; after his death, his papers revealed forged documents and a fabricated service record. Gerald Anderson's account was undermined when a document he supplied was shown to contain a fax timestamp predating its supposed date.

The alien autopsy film, released in 1995 by Ray Santilli to enormous publicity, was admitted by Santilli and by sculptor John Humphreys in 2006 to be a staged reconstruction filmed in a London flat, with organs bought from a butcher.

The MJ-12 documents, addressed in Section 8.6, contain typographic and formatting anachronisms, an incorrect date format, and a signature that appears to be lifted from another Truman document.

Walter Haut's 2002 affidavit, released after his death, describes a craft and bodies. Haut was 80 when it was drafted, in poor health, and the document is inconsistent with his numerous earlier interviews across four decades in which he said he never saw any debris and only wrote what he was told.

The nurse, the second site, the train of aircraft, the sealed hangar — each traces to one or two witnesses, first speaking decades later, frequently in contact with the researchers who were developing the story.

I want to be careful about the accusation implied here, because most of these people were not liars. The contamination mechanism is Section 1.6's, and it is well understood: witnesses were interviewed repeatedly, by enthusiasts, using leading questions, over years, with each interview re-encoding the memory. Testimony converged because the interviewers converged, and everybody read the same books. Roswell is a case study in how a shared narrative is constructed out of genuine but malleable recollection, and if you want to understand how any oral tradition forms, this one happened inside the era of tape recorders.

14.4 PROJECT MOGUL

Project Mogul was real, classified Top Secret at the time, and its purpose explains everything.

In 1947 the United States had no way of knowing when the Soviet Union tested an atomic bomb. Seismic detection was inadequate and the airborne sampling programme was immature. Maurice Ewing of Columbia had proposed that the atmosphere contains a sound channel — an altitude band where low-frequency acoustic energy propagates over vast distances with little loss, analogous to the SOFAR channel in the ocean. Mogul was the attempt to put microphones in that channel using constant-altitude balloon trains and listen for Soviet nuclear tests and missile launches.

The hardware: long trains, sometimes over 200 metres, of multiple neoprene balloons carrying microphones, radiosondes, ballast systems, and multiple ML-307 radar target reflectors made of foil-laminated paper on balsa frames. The launches were from Alamogordo Army Air Field, roughly 100 miles west of the Foster ranch. Flight 4, launched 4 June 1947, was tracked and then lost, on a trajectory toward the Foster ranch area. The project's own records, including the diary of Charles Moore, one of the engineers, survive and were examined.

Mogul was declassified and the Air Force published two reports: *The Roswell Report: Fact Versus Fiction in the New Mexico Desert* in 1994, and *The Roswell Report: Case Closed* in 1997 (see Figure 14.2). These followed a GAO audit requested by Congressman Steven Schiff. The 1997 report addressed the bodies by pointing to Project High Dive and Excelsior — high-altitude anthropomorphic dummy drops conducted in New Mexico to test parachute systems — and this is the weakest part of the Air Force case, because those drops occurred in the 1950s, which the report acknowledges while suggesting time compression in witness memory. I think that is a genuine stretch, and I say so.

But the debris explanation does not depend on the dummies. It depends

FIGURE 14.2

The Air Force's two Roswell reports, 1994 and 1997, produced after a GAO audit.

Figure 14.2. The Air Force's two Roswell reports, 1994 and 1997, produced after a GAO audit.

Image sourced from: https://commons.wikimedia.org/wiki/File:Roswell_Report_Case_Closed_cover.jpg

on: a classified programme, using exactly the materials described, launched from exactly the right direction, with a lost flight on the right trajectory, at the right time, whose components were unfamiliar to base personnel precisely because the programme was classified above them. That last point is the resolution of the whole affair. Marcel had a high clearance and still had no need to know about Mogul, which is how compartmentalisation works and why Section 8.2 matters.

FUN FACT

Mogul worked, incidentally, though not the way intended — the acoustic detection approach was superseded, and the Soviet test of August 1949 was detected by airborne radiological sampling from a WB-29. But the balloon technology developed for Mogul fed directly into Skyhook and into the high-altitude reconnaissance balloon programmes of the 1950s, which produced their own generation of UFO reports. Classified balloons have accounted for more sightings than any other single human artefact, and they are still doing it: February 2023 saw the US shoot down a Chinese surveillance balloon and then three unidentified objects, at least some of which were probably amateur radio pico-balloons. Seventy-six years on, the same object still confuses the same air force.

14.5 ART'S PARTS

In 1996, the broadcaster Art Bell — whose role in all of this gets a chapter of its own, Section 20.4 — received an anonymous package containing metal fragments, accompanied by a letter claiming they came from the Roswell debris and had been kept by a family member in the military. The material became known as “Art’s Parts”, and Bell, to his credit, made them available for analysis rather than simply talking about them.

What happened next is a decent illustration of how physical analysis actually works.

The samples were examined by several parties over years, including at Los Alamos-adjacent facilities and by materials scientists working with UFO researchers. The most-discussed piece was a multi-layered specimen described as bismuth and magnesium in alternating thin layers — bismuth layers on the order of a micron, magnesium-zinc layers thicker. The claim was that such a laminate had no known terrestrial application and could

not have been manufactured in 1947.

The deflating findings came in stages. The material's composition was consistent with samples from an industrial process — multilayer bismuth-magnesium structures had been produced experimentally in the semiconductor and thermoelectric research literature, and similar laminates turned up in the waste stream of a specific manufacturing process. Isotopic analysis showed ordinary terrestrial ratios, which is the single most important test and the one that would have settled things in the other direction: extraterrestrial material from another stellar environment would show anomalous isotopic abundances, as meteorites do. It did not.

And the provenance was worthless from the start. **An anonymous package with no chain of custody cannot establish anything about origin, no matter what the metallurgy shows.** This is Section 1.8's point: an artefact without documented context is a curiosity, not evidence. Even if the laminate had been genuinely inexplicable, there would be no way to connect it to a ranch in New Mexico other than an unsigned letter.

The same deflation happened to the other famous Roswell artefact story. In a 2008 episode of *UFO Hunters*, material from Stanton Friedman's collection — descended from the "memory metal" that Marcel's son described playing with as a boy — was submitted for compositional analysis. It came back closest to acetate, chemically unremarkable and used today for spectacle frames and office stationery, and never in aerospace. That result is worth holding beside the memory-metal testimony itself, which is vivid, consistent and sincerely given: Jesse Marcel Jr. described foil that could not be torn and that unfolded itself when crumpled. Both things are true at once. The recollection is real and the sample is acetate, and the honest conclusion is that the sample is not the thing being recollected — which is exactly the chain-of-custody problem again, arriving from the other direction.

Linda Moulton Howe pursued the analysis for years and reported the results as they came, and I will say clearly that this is more than most claims in the field receive. The samples were tested. The tests came back terrestrial. **That is a functioning process, and the answer being boring is not a failure of the process.**

14.6 TECHNOLOGICAL BOOM

The reverse-engineering claim is that the post-war explosion in American technology — the transistor, integrated circuits, lasers, fibre optics, night vision, Kevlar, stealth — derived from recovered alien hardware. Philip Corso's *The Day After Roswell* (1997) is the primary source, claiming he distributed Roswell material to defence contractors from a Pentagon desk.

The claim fails because the actual history of each invention is documented in detail, with predecessors, false starts, competing labs, patent trails, and named individuals who spent years stuck.

The transistor was demonstrated at Bell Labs in December 1947, by Bardeen, Brattain and Shockley — five months after Roswell, which is the timeline the claim requires. Semiconductor rectification had been understood since the 1870s; cat's whisker crystal detectors were in widespread use; Julius Lilienfeld patented a field-effect concept in 1926; Bell Labs had a dedicated solid-state programme from 1945 with a written research plan; and German and British researchers were pursuing the same thing. Bardeen's contribution was the theory of surface states, which solved a problem the team had been failing at for two years. The laboratory notebooks survive, dated and witnessed.

The laser descends from Einstein's 1917 treatment of stimulated emission, through Townes's maser in 1953, to Maiman's ruby laser in 1960, with a documented patent dispute involving Gordon Gould that lasted decades. That dispute is itself the disproof: people fought bitterly over credit for an idea, in public, in court.

Fibre optics runs from Daniel Colladon's light guiding in 1841 through Kao and Hockham's 1966 paper identifying material purity as the limiting factor, to Corning's low-loss fibre in 1970. Kao got a Nobel Prize for it in 2009.

Integrated circuits — Kilby at Texas Instruments and Noyce at Fairchild in 1958–59, independently, with a patent war.

Stealth came from a Soviet paper. Pyotr Ufimtsev's work on the physical theory of diffraction, published openly in the USSR in 1962 and translated by the US Air Force in 1971, gave Lockheed's Denys Overholser the mathematics to compute radar cross-section from faceted surfaces. The Soviets did not classify it because they saw no application. That is how the F-117 was born,

and it is a much better story than a crashed saucer.

FUN FACT

The deeper problem with the reverse-engineering claim is that it gets technology backwards. You cannot reverse-engineer an artefact from a civilisation whose physics you do not have. Hand a smartphone to Isaac Newton — the finest mind of his century, with a well-equipped workshop — and he cannot get anything from it. Not the semiconductors, not the lithography, not the radio protocol, not the software. He does not have electromagnetism. Technology is not an object; it is an enormous stack of prior understanding and industrial capacity, and objects are the visible top of it. If we had a genuine alien device we would have learned nothing from it for a century, and the absence of an inexplicable unexplained gap in our own record is the most telling evidence there is.

14.7 THE CULTURAL IMPACT OF ROSWELL

Whatever fell on the Foster ranch, what grew from it is real, large, and measurable — and as a phenomenon it is more interesting than the debris.

Roswell, New Mexico now has the International UFO Museum and Research Center, founded in 1991 by Walter Haut and Glenn Dennis among others, drawing on the order of 200,000 visitors a year. The city holds an annual festival over the July anniversary. The streetlights have alien eyes. The local economy has a UFO line item, and the museum maintains a research library that is genuinely useful — I have used it.

The cultural output is enormous: *Independence Day* (1996), *The X-Files* (1993–2018), *Roswell* (1999–2002), *Men in Black* (1997), *Dark Skies*, *Stargate*, something over a hundred books, and a permanent place in the vocabulary. “Area 51” and “Roswell” function as shorthand in languages that have no other American place names in common use.

FUN FACT

A small piece of calendar trivia that I have never seen printed anywhere, and that I checked myself because nobody else seemed to have bothered. Walter

Haut's press release went out on Tuesday, 8 July 1947. The fiftieth anniversary of that release — 8 July 1997, the peak of the enormous golden-jubilee festival that drew tens of thousands of people to a town of fifty thousand — also fell on a Tuesday. The Gregorian calendar repeats on a fifty-year interval only when the leap-year pattern cooperates, which it does here, so the coincidence is arithmetic rather than mystical. But it is a nice detail for a field that spends so much of its energy looking for meaning in patterns: here is a genuine pattern, exactly and verifiably true, and it means absolutely nothing. Keep that feeling. You will need it in Chapter 19.

The political impact is real too. Roswell is why the GAO conducted an audit and why the Air Force published two reports; it is a background condition for the 2017 disclosures of Chapter 8 and for the 2021 and 2023 Congressional hearings. A rancher's fence line generated federal paperwork for fifty years.

And the intellectual impact is the one I care about. Roswell established the template for every subsequent case: initial official statement, rapid retraction, decades of silence, delayed testimony, documentary battles, and a permanent unresolvable residue. It taught a generation that the correct response to an official explanation is suspicion — which, given Chapter 8's actual record of official lying, is not an unreasonable lesson to have learned. Roswell is a story about trust, and the reason it will not die is that the institutions involved spent the same fifty years earning the distrust that keeps it alive.

FUN FACT

Here is my honest position after all of it. I think Mogul explains the debris, comfortably. I think the bodies were added later and I can trace who added them. And I still find the 8 July press release strange, and I still cannot fully account for why a base commander approved it. That residue is about two percent of the case and it is not big enough to hold a spacecraft. But it is not zero, and a professor of Ufology who tells you a case is one hundred percent closed is selling you the same certainty as the man with the autopsy film.

THE NUCLEAR CONNECTION

"I am become death, the destroyer of worlds."

— J. ROBERT OPPENHEIMER

15.1 THE MANHATTAN PROJECT

The correlation at the heart of this chapter is not disputed by anybody, and it is the single strongest pattern in the sighting record: modern UFO reports begin in the 1940s, in the American southwest, in the airspace around nuclear weapons development, and they cluster around nuclear facilities for eighty years afterwards.

FIGURE 15.1

*The Trinity fireball at 16 milliseconds,
16 July 1945 — about 100
miles from the Foster ranch.*

Figure 15.1. The Trinity fireball at 16 milliseconds, 16 July 1945 — about 100 miles from the Foster ranch.

Image sourced from: https://commons.wikimedia.org/wiki/File:Trinity_Test_Fireball_16ms.jpg

The Manhattan Project ran from 1942 to 1946 under Leslie Groves, with Oppenheimer directing the Los Alamos laboratory. Trinity detonated on 16 July 1945 at Alamogordo (see Figure 15.1) — about 100 miles from the Foster ranch of Chapter 14 — and Hiroshima and Nagasaki followed in August. The programme employed some 125,000 people at its peak across Oak Ridge, Hanford, Los Alamos and dozens of other sites, at a cost around \$2 billion in 1945 dollars, and it was the most tightly compartmentalised effort in Ameri-

can history. Most participants did not know what they were building.

Then came the sightings. Kenneth Arnold, 24 June 1947, near Mount Rainier. Roswell, July 1947. The 1948 Thomas Mantell crash while pursuing what was almost certainly a Skyhook balloon. Green fireballs over New Mexico from December 1948, concentrated over Los Alamos, Sandia and Kirtland, investigated by Lincoln LaPaz of the University of New Mexico under Project Twinkle. The 1952 Washington DC radar events. Sightings at every major weapons site.

So: correlation, undisputed. The entire question is which way the causal arrow points, and I want to set out the two readings honestly because this chapter is where I am least dismissive.

The extraterrestrial reading: fission and fusion detonations are detectable at interstellar distances by gamma and neutron signature; a monitoring intelligence would react to a species acquiring the ability to destroy itself; therefore the arrival correlates with Trinity. Section 15.6 supplies the ongoing pattern.

The mundane reading: nuclear facilities are where the most restricted airspace, the most advanced classified aircraft, the most sensitive radar, the most rigorous reporting requirements, and the most highly trained observers all coincide. You get more reports there because reporting is mandatory, observation is continuous, and there is far more unusual hardware in the sky. The same period saw the U-2, the balloon programmes, sounding rockets from White Sands, and the first jets — all tested in the same desert for the same reasons of isolation and clear weather.

Both readings predict the same correlation, which is why eighty years of argument has not settled it. What would distinguish them is whether the reports at nuclear sites are qualitatively different from reports elsewhere, and Section 15.6 is my attempt at that question.

FUN FACT

The green fireballs are the part of this I cannot dismiss. LaPaz was a competent meteoriticist who specifically concluded these were not meteors: wrong colour for the spectrum of ablating iron or nickel, wrong trajectory — horizontal rather than descending — no sound, no recovered material, and they recurred over a specific and highly sensitive area. Project Twinkle

set up instrumented watches and largely failed through underfunding and poor coverage, which is a very familiar story. The modern explanation is copper-bearing meteoritic material or an unusual atmospheric electrical phenomenon. LaPaz never accepted it, and he was the man who had actually looked.

15.2 THE LAW OF ATTRACTION

I have used a slightly mischievous section title here, and I mean two things by it.

The first is the serious version: what, physically, would attract attention to Earth? This is a real and answerable question about our detectability.

Our radio leakage is smaller than people think. Ordinary broadcast television and FM radio are weak, spread across a wide band, and drop below the galactic background within a few tens of light years — effectively undetectable at any real distance without an implausibly large collecting area. What *is* detectable is narrow, powerful and directed: planetary radar, notably the Arecibo and Goldstone systems, which produce beams that would be visible across substantial fractions of the galaxy to a receiver in the beam. Also detectable: our atmosphere, via transmission spectroscopy, showing oxygen, methane, water and — diagnostically — chlorofluorocarbons and nitrogen dioxide, which have no natural source and constitute genuine technosignatures. Also city lights, waste heat, and the nuclear detonation signature.

On timing: the first nuclear tests were 80 years ago, so the light cone containing that information extends 80 light years. That volume holds a few thousand stars. Any response requires them to be within 40 light years for a round trip, which cuts it to a few hundred, and requires them to have been watching. It is not impossible. It is a very small window.

The second meaning is the unserious one, and since some readers will have arrived at this section expecting it: the “Law of Attraction” in the popular sense — that thoughts materialise outcomes — has no mechanism, no evidence, and a documented history running from the New Thought movement through *The Secret*. It appears in ufology through the channelling and CE-5 literature, where meditation is claimed to summon craft. The

reason I mention it at all is that CE-5 practice provides a clean natural experiment: people go to remote dark sites at night, intending to see something, and then look at the sky for hours. They see satellites, meteors, Iridium flares, aircraft, and Starlink trains, and they report success rates near 100%. That is not evidence about aliens; it is an excellent demonstration of Section 1.5.

15.3 BREAK THE UNIVERSE

The hard limit is c , and I want to explain precisely what breaks if you exceed it, because the usual objection — “you cannot reach the speed of light” — understates the problem dramatically. The problem is not that faster-than-light travel is difficult. It is that it is equivalent to time travel, and therefore to causality violation.

Special relativity’s core result is that simultaneity is relative: two events that are simultaneous in my frame are not simultaneous in yours if you are moving relative to me. For events separated by a spacelike interval — that is, further apart in space than light could cross in the time between them — there is no frame-independent ordering. Which came first depends on who is asking.

Now suppose you can send a signal faster than light. In my frame it arrives after it was sent. But there exists another valid inertial frame in which it arrives *before* it was sent, and relativity gives me no grounds to prefer my frame over that one. Chain two such signals between two relatively moving observers and you can send a message into your own past — the standard tachyonic antitelephone. You can then instruct someone to prevent you from sending it.

This is why physicists’ resistance to FTL is not conservatism. **FTL, time travel, and causality violation are the same statement in three vocabularies.** You cannot have one without the others.

What the theory does permit, formally:

The Alcubierre metric (1994) contracts space ahead of a bubble and expands it behind, moving the bubble arbitrarily fast without local FTL motion. The requirements: negative energy density — exotic matter — in quantities originally estimated at more than the mass-energy of the visible universe, later reduced by optimisation to something merely astronomical.

Negative energy density exists, in tiny amounts, via the Casimir effect and squeezed vacuum states, and is constrained by quantum inequalities that appear to forbid the amounts needed. Additional problems: the bubble interior is causally disconnected so you cannot steer, the leading edge accumulates blueshifted radiation that would sterilise the destination, and the whole construction is a global solution requiring the geometry to be pre-arranged.

Traversable wormholes require the same exotic matter to hold the throat open, and the Morris–Thorne analysis showed that any traversable wormhole can be converted into a time machine by moving one mouth, at which point Hawking’s chronology protection conjecture proposes that vacuum fluctuations pile up and destroy it.

The honest summary: general relativity permits these geometries as solutions, and quantum field theory appears to forbid the matter needed to build them. Whether that is a final answer depends on a theory of quantum gravity we do not have. I would not bet against the universe having a loophole. I would bet heavily against the loophole being cheap enough for routine cattle surveys in Nebraska.

15.4 A WRINKLE IN TIME

Time dilation is not speculative, and it is the reason interstellar travel is possible in principle without FTL at all.

The effect is measured constantly. Muons produced in the upper atmosphere reach the ground in numbers that make no sense unless their clocks run slow. Particle accelerators extend the lifetimes of unstable particles by exactly the predicted factor. Atomic clocks flown on aircraft — the Hafele–Keating experiment in 1971 — disagreed with ground clocks as predicted. GPS satellites experience about 45 microseconds per day of gravitational speeding-up and 7 microseconds of velocity slowing, for a net 38 microseconds, and the constellation would accumulate roughly 10 kilometres of position error per day if this were not corrected. Your phone’s navigation works because relativity is engineered into it.

For travel, the relevant number is the Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$. At $0.5c$, γ is about 1.15 — barely worth mentioning. At $0.99c$, about 7. At $0.999c$, about 22. And under constant 1 g proper acceleration — which is

comfortable and also solves your artificial gravity problem — the ship's clock scales logarithmically with distance: about 3.6 years of ship time to Alpha Centauri, roughly 12 years to reach 1000 light years, about 20 years to cross the galaxy, and something near 25 years to reach the Andromeda galaxy. Two and a half million years pass outside.

So a civilisation with sufficient energy can cross the galaxy within a crew's career. There is no physics problem here at all. The problem is entirely the energy and the debris.

The energy: accelerating a 1000-tonne vessel to $0.9c$ requires on the order of 10^{23} joules even with perfect conversion, which is roughly a century of total current human energy consumption, per ship, and you need the same again to stop. With a realistic drive efficiency the number becomes absurd. Antimatter would work and costs, at current production rates, something like \$60 trillion per gram.

The debris: at $0.9c$, a single interstellar dust grain of one milligram delivers kinetic energy comparable to a substantial explosive charge, and the interstellar medium is not empty. Shielding against that at relativistic speed is the real engineering barrier, and it is why the serious literature — Project Orion, Daedalus, Icarus, Breakthrough Starshot — tends to settle on velocities around $0.1c$ to $0.2c$, which makes the journey to the nearest star 20 to 40 years and reintroduces the generation-ship problem.

FUN FACT

Time dilation cuts against the visitation hypothesis in a way rarely noticed. A crew that crosses 1000 light years experiences 12 years, but 1000 years pass at home. Every mission is one-way in every sense that matters: no reporting back within a civilisational lifetime, no reinforcement, no rescue, and the sponsoring institution is very likely gone before arrival. That is a poor basis for a sustained programme of hovering over Nebraska and abducting people for decades. Interstellar travel is not exploration. It is emigration, and it is permanent.

I S . S C A S H - L A N D R U M

On the evening of 29 December 1980, on FM 1485 near Dayton, Texas, Betty Cash, Vickie

FIGURE 15.2

The witnesses' diagram of the diamond-shaped Cash–Landrum object and its escort of helicopters.

Landrum and her seven-year-old grandson Colby encountered a large diamond-shaped object above the road, emitting flame downward from its base, intensely hot and bright. They stopped the car. Betty Cash got out. The object's periodic flame bursts, they said, forced them back; the car's dashboard was hot enough that Vickie Landrum's handprint was later reported in the softened vinyl. As the object moved off they reported a formation of helicopters — they counted around 23, described as twin-rotor CH-47 Chinooks — pursuing or escorting it.

This is the strongest medical case in the record and it is the reason I include it. All three developed symptoms within hours (see Figure 15.2): nausea, vomiting, diarrhoea, severe headaches, and skin lesions. Betty Cash, who had been outside the car longest, was worst affected — she was hospitalised for two weeks, lost patches of hair, had extensive skin blistering, and suffered permanent eye damage and ongoing ill health until her death in 1998, on the anniversary of the encounter. The symptom pattern was reviewed by physicians and by Peter Gersten and by John Schuessler, an aerospace engineer who documented the case in detail.

The symptom set is consistent with acute radiation syndrome at a moderate dose combined with intense infrared and ultraviolet exposure. That is a medically distinctive presentation and it is not something the three of them could have imagined into existence.

The investigative history is where it gets bad. Cash and Landrum sued the United States government for \$20 million, alleging injury by a military craft. The case was dismissed in 1986 on the grounds that no government agency could be shown to have operated such a craft — the Army, Air Force, Navy and NASA all denied any such operation, and denied having 23 Chinooks in the air over southeast Texas that night. Chinooks are Army aircraft, and 23 of them constitutes a substantial movement that leaves paperwork.

So the case sits in a peculiar place. There is real, documented, severe injury. There is a description involving a huge, flame-emitting craft escorted by a large military helicopter formation. And there is no identified craft,

no identified radiation source, and no acknowledged operation. Proposed explanations have included a classified nuclear-thermal propulsion test, a burning gas well flare seen through unusual conditions, an aircraft mishap, and misidentification compounded by an unrelated illness. None of them accounts for all three of: the helicopters, the radiation-like injuries, and the object.

I regard Cash–Landrum as genuinely unexplained, and I want to be precise about what that means. It is unexplained in a way that points at something physical — something that burned people. The helicopter escort points hard at a human operation. That is not a small conclusion: it means either the US military irradiated three civilians and then denied it in federal court, or something else entirely happened. I do not know which. But it is the case I would put in front of anybody who thinks this field contains nothing but balloons and Venus.

15.6 UFOs AND NUCLEAR FACILITIES

The nuclear-site incidents are the best-documented category in ufology, because military bases generate paperwork and because that paperwork has been released. Robert Hastings assembled the primary collection in *UFOs and Nukes*, with over 160 military witnesses; the documents themselves come from Air Force records obtained under FOIA.

The cases worth knowing:

FIGURE 15.3

A Minuteman launch control capsule of the Echo Flight type, with the logic couplers implicated in the shutdown.

Figure 15.3. A Minuteman launch control capsule of the Echo Flight type, with the logic couplers implicated in the shutdown.

Image sourced from: https://commons.wikimedia.org/wiki/File:Minuteman_launch_control_center.jpg

Malmstrom AFB, Montana, 16 March 1967. At Echo Flight, ten Minuteman I ICBMs went from alert to “no-go” within seconds (see Figure 15.3) of each other. Robert Salas, then a deputy missile combat crew commander at Oscar Flight, reported that security police above ground described a glowing red object hovering at the gate, and that his flight also lost missiles. Boeing engineers ran an extensive fault investigation and could not reproduce a single fault mode that would drop ten separated silos simultaneously; the

eventual working hypothesis involved a noise pulse on the logic coupler. The Echo Flight shutdown is documented in the unit history. The UFO element rests on testimony.

Loring AFB, Maine, October 1975, and Wurtsmith, Malmstrom, Minot, and Falconbridge in the same period. This is a documented sequence of intrusions over Strategic Air Command weapons storage areas across the northern tier and into Canada, generating NORAD messages, security alerts, and helicopter and fighter responses. The USAF-wide message of 11 November 1975 noting these events over SAC bases is a real document. Reports described objects at low altitude over the weapons storage area, sometimes described as helicopter-like, sometimes not.

RAF Bentwaters/Woodbridge, December 1980 — the Rendlesham Forest case — involves a US Air Force base storing nuclear weapons, a memorandum from deputy base commander Lieutenant Colonel Charles Halt describing lights in the forest, indentations, elevated radiation readings, and a tape recording made in the field. Explanations include the Orfordness lighthouse, a reentering booster, and a bright star, and the radiation readings were within the range of background for the instrument used. Halt maintains his account. Section 5.2 covers the case in full.

Vandenberg, 1964, where Robert Jacobs reported that telescope footage of an Atlas missile test showed an object approaching the warhead and firing beams of light at it, after which the warhead failed. The film has never been produced.

The Soviet side matters enormously and is under-discussed. Documents released after 1991, and the KGB “Blue Folder” material, include an incident at a Ukrainian ICBM base in October 1982 in which launch systems reportedly activated spontaneously for around 15 seconds during an observed overflight. The Soviet military ran a systematic collection programme from 1978. The significance is that the phenomenon was not confined to one side’s hardware or one side’s culture, which rules out a purely American institutional explanation.

My assessment: the *correlation* is solid and the documents are real. The mechanism is not established. The strongest mundane account combines mandatory reporting, extreme observation density, genuine classified aircraft activity, electromagnetic interference from a very electrically noisy environment, and — for the Cold War intrusions — the real possibility of

Soviet reconnaissance, which would explain the secrecy better than aliens would. What that account handles least well is the simultaneity of the Malmstrom shutdowns and the Soviet mirror-image incidents.

15.7 IT AINT THE RUSSIANS!

The title is a challenge to my own argument, so let me take it seriously and then answer it.

The case that these events were Soviet: the USSR had strong motive to probe SAC readiness, ran aggressive reconnaissance including Bear-D flights and satellite coverage, and would have been very interested in whether an intrusion could degrade launch capability. American secrecy about the intrusions would then be motivated by the last thing SAC wanted publicised — that the deterrent could be tampered with. And the 1975 overflights included Canadian airspace, consistent with a polar approach.

The case against: the Soviets had their own incidents, at their own missile bases, which they also could not explain and also investigated. Both superpowers separately suspected each other and both separately found nothing. Neither found evidence of the other's involvement in archives that have since been substantially opened — and the Soviet archives that opened in the 1990s were remarkably leaky about far more sensitive matters. The performance described in the reports does not match known Soviet airframes of the period. And the reported behaviour — hovering visibly over a weapons storage area, illuminated, for extended periods — is the opposite of what a reconnaissance platform would do.

So where does that leave us? With a pattern that both nuclear superpowers documented, neither explained, and neither could attribute to the other.

I want to close the chapter on the honest version of the correlation, because this is the strongest chapter in the book for the visitation hypothesis and I am not going to undersell it. Modern sightings begin with Trinity. They concentrate around weapons facilities. The concentration persists across two adversarial systems with different cultures, different hardware and different reporting incentives. Incidents cluster around delivery and storage rather than around research. And the pattern continues: the 2004 Nimitz encounters of Section 12.5 occurred during a carrier group workup,

and a great many of the recent Navy reports come from the same kind of high-value military environment.

Against which: that is exactly where the sensors are. Every one of those environments has more radar, more trained observers, more mandatory reporting and more classified hardware than anywhere else on Earth. The correlation may be a correlation with *looking*.

I have gone back and forth on this chapter more than any other, and I will tell you where I land. The nuclear correlation is the one piece of the puzzle I cannot make disappear. It is not proof of anything. It is the reason I am still in this field.

NASA'S UNEXPLAINED FILES

"I believe that these extra-terrestrial vehicles and their crews are visiting this planet from other planets. . . Most astronauts were reluctant to discuss UFOs."

— GORDON COOPER

16.1 PROJECT PAPERCLIP

Operation Paperclip is not a conspiracy theory. It is documented American policy, and it is the single best illustration in this book of why people distrust official statements — because on this one, the paranoid version was closer to the truth than the official version, for decades.

Between 1945 and 1959, the United States brought roughly 1,600 German scientists, engineers and technicians to America, along with their families, sanitising their records to permit entry. The most prominent was Wernher von Braun, technical director of the V-2 programme at Peenemünde, who surrendered with his team in May 1945 and became, by 1960, director of NASA's Marshall Space Flight Center and the chief architect of the Saturn V.

The rest of the record: V-2 production at the Mittelwerk underground facility (see Figure 16.1) used slave labour from the Mittelbau-Dora concentration camp, where

FIGURE 16.1

V-2 assembly in the Mittelwerk tunnels, where more people died building the rockets than were killed by them.

more people died building the rockets — an estimated 20,000 — than were killed by them. Von Braun visited Mittelwerk and held SS rank. Arthur Rudolph, production manager at Mittelwerk and later project manager of the Saturn V, voluntarily left the United States in 1984 and renounced his citizenship after the Justice Department's Office of Special Investigations opened a case. Hubertus Strughold, who became the "father of space medicine" at the Air Force School of Aviation Medicine, had directed an institute associated with human experiments at Dachau. Walter Schreiber, Kurt Blome and others with medical war-crime associations were considered or employed.

The rationale was the Soviet Union, which ran its own version — Operation Osoaviakhim — removing thousands of German specialists by force in October 1946. Both superpowers built their space programmes on the same captured technology and the same people.

So what does Paperclip mean for this book?

It establishes, definitively, three things. First, the American government will conceal morally catastrophic information for strategic advantage, at scale, for decades, and lie about it in writing. Second, an institutional capacity for that kind of secrecy existed exactly when and where the UFO era began. Third — and this is the part that cuts against the conspiracy reading — it *came out*. Historians, journalists, the OSI and eventually declassification produced the full picture. Linda Hunt's *Secret Agenda* (1991) and Annie Jacobsen's *Operation Paperclip* (2014) are built from released documents.

The usual ufological deployment of Paperclip is the claim that German engineers arrived with advanced antigravity or saucer technology — the Haunebu and Vril claims of Section 19.7 — and that American aerospace performance after 1947 derives from it. The reality is that what they arrived with was liquid-fuelled rocketry, inertial guidance and systems engineering, all of it documented, all of it traceable in the Saturn V, and all of it quite enough to explain the change. The V-2's descendants went to the Moon. That is a large enough claim without adding saucers.

16.2 THE RACE TO SPACE

The sequence, because the dates carry the argument.

FIGURE 16.2

Sputnik 1: four whip antennas, a radio beacon, and a political earthquake.

Figure 16.2. Sputnik 1: four whip antennas, a radio beacon, and a political earthquake.

Image sourced from: https://commons.wikimedia.org/wiki/File:Sputnik_asm.jpg

Sputnik 1, 4 October 1957 — a 58-centimetre aluminium sphere with a radio beacon (see Figure 16.2), and a political earthquake. Sputnik 2 carried Laika a month later. Explorer 1, the first American satellite, reached orbit on 31 January 1958 after the public failure of Vanguard, and NASA was created in July 1958. Luna 2 struck the Moon in 1959; Luna 3 photographed its far side. Yuri Gagarin orbited on 12 April 1961. Alan Shepard flew suborbital on 5 May 1961, and Kennedy announced the Moon commitment on 25 May, twenty days later — which tells you what the programme was for. John Glenn orbited in February 1962. Valentina Tereshkova flew in 1963. Alexei Leonov walked in space in 1965. The Soviets lost Komarov in 1967 and the US lost the Apollo 1 crew the same year. Zond 5 sent tortoises around the Moon in 1968; Apollo 8 sent humans around it that December; Apollo 11 landed on 20 July 1969.

The astronaut sighting reports are the part relevant here, and they need sorting.

Gordon Cooper is invariably cited. He did describe, in later years, filming a saucer-shaped object at Edwards in 1957 while a test pilot — a claim that predates his spaceflight and for which no film has been produced. He also repeatedly denied seeing anything anomalous on his Mercury flight, contrary to persistent claims.

James McDivitt, Gemini 4, 1965, reported a cylindrical object with an arm. He later stated he believed it was the Titan second stage or another piece of his own launch debris, and was explicit that he did not consider it anomalous. **The photographs frequently attributed to the sighting are not the object he saw.**

Ed White, same flight, reported the same debris.

Apollo 11's “something out there” was tracked, and Aldrin has stated clearly and often that it was one of the four SLA panels from the Saturn

V third stage, travelling on a similar trajectory, and that he is frustrated at being quoted otherwise.

The Apollo 11 “we’ve been warned off the Moon” transmission is an invention. It appears in no NASA transcript, no recording, and no contemporaneous source, and traces to a 1979 book by Maurice Chatelain.

The genuine pattern is that spaceflight puts human beings in an environment full of ice crystals, vented water and urine, thermal blanket fragments, paint flecks, launch debris and other satellites, all in vacuum, all brilliantly sunlit against black, with no atmospheric cues for range or size. Astronauts see unidentified objects constantly and identify almost all of them, and the ones they cannot identify they describe as debris because they know what the environment contains.

FUN FACT

The most persistent astronaut UFO image — the “tether incident” from STS-75 in 1996, showing numerous notched discs moving near a broken satellite tether — was resolved by geometry. The objects are out-of-focus debris and ice close to the camera; the notch is the camera’s aperture diffraction pattern, visible on defocused point sources, which is why every single object has an identical notch in an identical orientation. The same signature appears in unrelated footage. It is not a fleet. It is a lens.

15.3 THE MOON LANDING IS NOT A MYTH

I include this section because if you cannot evaluate the Moon hoax claim, you cannot evaluate anything else in this book, and because the Moon hoax is the purest available example of a claim that is refuted by physical evidence anyone can check.

The evidence, in descending order of how hard it is to argue with.

Retroreflectors. Apollo 11, 14 and 15 placed corner-cube laser retroreflector arrays on the lunar surface. Observatories — McDonald, Apache Point, Côte d’Azur, Matera — have been bouncing lasers off them continuously for over fifty years, measuring the Earth–Moon distance to millimetre precision and detecting the Moon’s 3.8-centimetre annual recession. Anybody with access to a suitable telescope can do it. The Soviet Lunokhod reflectors are

also there, and the Apollo ones are in the places the missions said they were.

382 kilograms of lunar sample, distributed to laboratories worldwide including in the USSR, showing zero water content, high-temperature mineralogy, cosmic-ray exposure ages, solar wind implantation, micrometeorite impact glass and zap pits, and isotope ratios matching lunar meteorites found in Antarctica but no terrestrial rock. Fifty years of independent geochemistry by thousands of scientists across every political bloc. Nobody has produced a plausible account of faking it, because the material would have to be manufactured in an environment that did not exist.

Independent orbital imaging. The Lunar Reconnaissance Orbiter has photographed all six landing sites, showing descent stages, rover tracks, experiment packages and disturbed regolith, with shadows tracking the known sun angles. Japan's Kaguya, India's Chandrayaan, and China's Chang'e have also imaged the sites — and it is worth pausing on the fact that the Chinese space programme has independently confirmed American landing sites.

The Soviets tracked the missions, had every incentive to expose a fraud, had their own deep space network, and congratulated the crew.

400,000 participants across NASA, contractors and universities, and no deathbed confession in fifty-five years, alongside terabytes of telemetry, thousands of hours of transcripts, engineering drawings, and hardware in museums.

On the standard objections: the flag appears to wave because it has a horizontal rod along the top and the astronauts twisted the pole. Shadows appear non-parallel because of terrain and wide-angle lenses. **There are no stars because the exposure was set for a sunlit surface.** The Van Allen belts were transited quickly on a trajectory chosen to skirt the densest region, and the measured dose was about that of a chest CT. The lack of a blast crater is because the descent engine was throttled to about 3,000 pounds of thrust over a broad area in vacuum, which is nothing like a jet on concrete.

FUN FACT

My favourite argument is the one from difficulty. Faking the landings in 1969 would have required film technology that did not exist — sustained low-gravity dust behaviour, which is visible in the footage and which nobody could simulate until CGI — plus faking the sample geology, the laser

retroreflectors, the telemetry, and the Soviets. Actually going was cheaper. It usually is.

16.4 TRANSIENT LUNAR PHENOMENA

Here is the genuine lunar anomaly, and it has a NASA catalogue number.

Transient lunar phenomena are short-lived localised brightenings, obscurations, reddish or bluish glows, and star-like flashes on the lunar surface, reported by competent observers for centuries. William Herschel reported them. The Royal Greenwich Observatory recorded them. In 1968 NASA published Technical Report R-277, *Chronological Catalog of Reported Lunar Events*, compiled by Barbara Middlehurst and Patrick Moore, listing 579 events from 1540 to 1967 — prepared specifically because Apollo needed to know whether the Moon was geologically active.

The events cluster remarkably: Aristarchus accounts for roughly a third, with Plato, Alphonsus, Gassendi, Grimaldi and the Mare Crisium region making up much of the rest. That clustering is the strongest argument that something real is happening, since observer bias would not select those specific craters so consistently.

The best-documented case is 3 November 1958, when the Soviet astronomer Nikolai Kozyrev, using the Crimean Observatory's 50-inch telescope, obtained spectra of Alphonsus showing emission bands he attributed to C₂ molecules — a gas emission spectrum from the lunar surface. The plates exist. Interpretation has been disputed for sixty-five years.

Apollo carried instruments for this. Apollo 15 and 16 flew the mass spectrometer and alpha-particle spectrometer that detected radon-222 and polonium-210 emanation, with elevated readings over Aristarchus and around Mare Crisium's edges — direct evidence of gas release from the interior. Lunar Prospector confirmed radon events in 1998–99.

The leading explanations are all geological. Outgassing of radon and other volatiles through fracture systems, lofting dust that scatters light. Electrostatic dust levitation driven by solar-wind charging at the terminator. Impact flashes, which are now routinely recorded — NASA's lunar impact monitoring programme has logged hundreds, and the LCROSS and LADEE

missions studied the exosphere. Piezoelectric or triboluminescent effects from tidal stress. And the seismometers left by Apollo detected genuine shallow moonquakes for years.

So: real phenomenon, geological mechanism, still incompletely characterised. Which is exactly the status of Hessdalen in Section 13.5, and exactly what a legitimate open anomaly looks like — catalogued, instrumented, correlated with independent measurements, and nobody claiming it is a base.

IS THE HOLLOW MOON

The hollow Moon claim — and its spaceship variant, proposed by Mikhail Vasin and Alexander Shcherbakov in 1970 as the “Spaceship Moon” hypothesis — holds that the Moon is artificial or hollow, and it is a useful exercise because we have enough data to close it completely.

The numbers. Lunar mass is 7.35×10^{22} kg, determined from orbital dynamics of dozens of spacecraft and confirmed to extraordinary precision by the GRAIL mission’s twin satellites in 2012, which mapped the gravitational field to a resolution of a few kilometres. Radius is 1737 km, so mean density is 3.34 g/cm^3 — lower than Earth’s 5.51, and that difference is the actual interesting fact, because it reflects a small iron core, which is itself evidence for the giant impact origin.

A hollow shell is ruled out several ways. GRAIL’s gravity field is inconsistent with a shell: mass is distributed as expected for a differentiated rocky body, with a crust averaging around 34–43 km, a mantle, a partially molten layer, a small fluid outer core and a solid inner core of about 240 km radius. Apollo seismometers recorded shear waves propagating through the interior; shear waves do not cross a vacuum, and their travel times and shadow zones are how the core was found in the first place. Laser ranging measures the Moon’s physical libration and its moment of inertia, which fixes the internal mass distribution. And a shell of the required thinness could not support itself against self-gravity in any known material.

The “ringing like a bell” claim comes from Apollo 12, when the ascent stage was deliberately impacted and the seismic signal persisted for around 55 minutes. This is real and it has a real explanation: the lunar regolith and heavily fractured megaregolith is dry, porous, and has almost no attenuation,

so seismic energy scatters rather than damping. On Earth, water-saturated rock absorbs the energy within minutes. Long reverberation indicates a dry fractured solid, not a hollow one — the opposite of the claim.

On origin: the giant impact hypothesis is supported by the Moon's iron depletion, the identical oxygen isotope ratios between lunar and terrestrial samples — which is decisive and which no capture model explains — the Earth–Moon angular momentum budget, the evidence for a lunar magma ocean, and dynamical simulations that reproduce the system.

I E . E YOU HAVE GOT TO BE JOKING !

A short catalogue of things people have found in NASA imagery, and what they actually are, because the pattern is the lesson.

The Face on Mars. Viking 1 photographed a mesa in Cydonia in 1976 at about 50 metres per pixel, with low sun angle and a data dropout forming a nostril. Richard Hoagland built a career on it. Mars Global Surveyor imaged it at 1.5 metres per pixel in 1998 and 2001, and Mars Express in 3D: it is an eroded mesa, and the resemblance vanishes entirely.

The Mars “rat”, “spoon”, “lizard”, “Bigfoot”, “crab”, “coffin”, “doorway” — all rocks, all pareidolia, several of them centimetres across. The Curiosity “doorway” is a fracture gap about 30 cm high.

Lunar “towers”, “obelisks” and “the Shard” in Lunar Orbiter frames are film artefacts, emulsion defects and long shadows from small rocks near the terminator.

The “alien on Mars” and blue-light objects in rover images are compression artefacts, dust on the optics, and cosmic ray hits on the CCD — which produce bright single-pixel streaks that appear in one frame of a stereo pair and not the other, which is exactly how you identify them.

ISS live-feed cutoffs are loss of signal during satellite handover, and the ISS feed cuts out constantly for reasons documented in the operations schedule.

Nibiru and Planet X arrival dates have now failed continuously since 2003.

The underlying issue is that we now have petabytes of planetary imagery, publicly released — and NASA's open data policy is the reason all of this is

possible, which is worth remembering when someone tells you NASA hides things. Search a large enough noisy dataset for any shape, and you will find it. Human visual processing is aggressively tuned for faces and animals in low-information scenes, because a false positive on a predator costs nothing and a false negative costs everything. **Every one of these “discoveries” is that system working exactly as evolved.**

FUN FACT

The genuinely funny part is that NASA imagery *does* contain a real, verified alien discovery, and nobody who hunts for faces on Mars ever mentions it. Rosetta found glycine on comet 67P. Stardust returned samples with amino acids. Cassini flew through plumes of liquid water, molecular hydrogen, silica and organics erupting from Enceladus, indicating hydrothermal activity in a subsurface ocean. Titan has methane rivers and complex organic chemistry. Perseverance has found organic molecules in Jezero crater. The most extraordinary things in the archive are in the spectra, not the shadows — and reading spectra requires learning something, which is the whole reason this book exists.

16.7 TESTIMONIALS FROM ASTRONAUTS

Astronaut testimony is the strongest-sounding evidence in ufology and one of the weakest on inspection, and the reason is worth understanding, because it is not that astronauts lie.

Start with the strongest card in the deck, because everything else in this section is a variation on it. **Gordon Cooper is the best witness ufology has ever had, and he is still not good enough.** That is not a slight. Cooper was one of the original Mercury Seven, a fighter pilot in West Germany at twenty-three, an Edwards test pilot with two thousand hours in century-series jets, the man who slept in the capsule on the pad, flew twenty-two orbits alone in *Faith 7* and brought her home on manual control when the automatic system failed. If trained observation under pressure counted as proof, he would have settled the question by himself. He believed, said so publicly for forty years, and said it to the United Nations. And when you separate what he personally saw from what he was told and from what he

inferred, the residue is thinner than the citation implies — for reasons that have nothing to do with his honesty and everything to do with what any single observer, however good, can actually establish. His full case gets its own page overleaf.

The others are weaker versions of the same shape. **Edgar Mitchell**, sixth man on the Moon, was an unambiguous believer in a Roswell cover-up; he was also explicit that he had seen nothing himself and was repeating what old-timers told him after the fact. **Buzz Aldrin** described an object shadowing Apollo 11 outward, and has said repeatedly and with visible irritation that it was one of the four SLA panels jettisoned from the S-IVB (see Figure 16.3), which is exactly what the trajectory requires. **Story Musgrave** has spoken about seeing a snake-like object near the shuttle,

and has said in the same breath that it was debris. **James McDivitt** saw a cylinder on Gemini 4, said so, and spent years correcting people who inflated it. **Brian O’Leary** and **Ed White’s** reported remarks come to us third-hand. And the famous “Santa Claus” transmission from Apollo 8 was a coded announcement that Lovell’s family should expect a Christmas present, which the crew found funny at the time.

Now the pattern. **Almost every case divides into two kinds: an astronaut describing an object they themselves identified, or an astronaut expressing a private belief about events they did not witness.** Neither is what the citation implies. The quotations circulate stripped of the sentence that follows them, which is usually the explanation.

There is also a structural reason astronauts are unreliable narrators of orbital anomalies, and it has nothing to do with their competence. Low Earth orbit is full of objects. Ice crystals vent from the vehicle constantly and hang alongside it at matched velocity. Paint flecks, insulation, bolts, and the vehicle’s own jettisoned hardware travel in nearby orbits, meaning relative motion looks slow, close and controlled when the object is small and centimetres-to-metres away rather than large and distant. There is no atmosphere to give scale, no horizon for reference, and no familiar-sized

FIGURE 16.3

*An SLA panel drifting after separation
— the object Aldrin has repeatedly
identified as what Apollo 11 saw.*

Figure 16.3. An SLA panel drifting after separation — the object Aldrin has repeatedly identified as what Apollo 11 saw.

Image sourced from: https://commons.wikimedia.org/wiki/File:Apollo_SLA_panel_jettison.jpg

object in frame. A trained observer in that environment cannot judge range, size or speed, and the good ones say so.

The argument from authority fails here in a specific and instructive way. Being an expert in operating a spacecraft confers no expertise in identifying unfamiliar objects outside it. That is a different skill — optics, photogrammetry, orbital debris analysis — and it is the skill the people who examine the footage afterwards actually have. Test pilots are excellent at flying and demonstrably ordinary at estimating the distance of unidentified lights, which is the same conclusion Section 12.4 reached about pilots in the atmosphere.

None of this makes them fools. Several of these men flew machines with less computing power than a washing machine to another world, and Mitchell in particular was a serious, curious person who thought hard about consciousness for the rest of his life. Believing something on insufficient evidence is not a character flaw; it is the default human condition, and the whole point of method is that it protects people who are otherwise formidable.

FUN FACT

The most persistent quotation in this entire genre — Neil Armstrong reporting that “they” were on the Moon and warning us off — was invented in 1979 by a Chinese-language tabloid, translated back, and attributed to a fictional NASA source. Armstrong, who gave almost no interviews, publicly denied it in one of the few he did give. It remains in circulation because nobody who repeats it wanted to check, and because a man who avoids the press is a very convenient witness.

Because this section gets quoted at me more than any other, here is the whole population, one page each — Cooper first, as the strongest case and the one that deserves the most care, then the twelve. Twelve men have walked on the surface of another world, and every one of them has been conscripted at some point into an argument he did not make. They are Neil Armstrong and Buzz Aldrin (Apollo 11), Pete Conrad and Alan Bean (Apollo 12), Alan Shepard and Edgar Mitchell (Apollo 14), David Scott and James Irwin (Apollo 15), John Young and Charles Duke (Apollo 16), and Eugene Cernan and Harrison Schmitt (Apollo 17). Four are still living as this is written: Aldrin, Scott, Duke and Schmitt. What follows is what

each man actually said on the subject, which in most cases is either nothing at all or the opposite of what the internet reports.

MERCURY SEVEN • *FAITH 7*, 1963 • GEMINI 5, 1965 • COL. USAF

L . G O R D O N " G O R D O " C O O P E R J R .



Photograph: NASA (1962 official portrait). Public domain (work of the U.S. federal government).

Image sourced from: https://commons.wikimedia.org/wiki/File:Gordon_Cooper_1962_portrait.jpg

Cooper is the case to beat. Born in Shawnee, Oklahoma in 1927, he flew F-84s and F-86s with the 86th Fighter-Bomber Wing in West Germany, came home to Edwards as a test pilot on the F-102A and F-106B, was picked for the Mercury Seven in 1959, and in May 1963 flew twenty-two orbits over thirty-four hours in *Faith 7* — the last American to go to orbit alone — taking manual control for re-entry when the automatic system quit and splashing down close enough to the carrier to be photographed from her deck. He flew again on Gemini 5 with Pete Conrad, backed up Apollo 10, and retired a colonel in 1970.

He made two claims, and they are not the same kind of claim. The first: that in 1951, flying over West Germany, his squadron repeatedly saw metallic disc-shaped objects at altitudes their aircraft could not reach. The second, and the famous one: that in 1957 at Edwards a camera crew under his supervision filmed a saucer-like craft descending, extending three legs, settling on the dry lake, then lifting away — and that he sent the film up the chain to the Pentagon as ordered and never saw it again. He told that story

for the rest of his life, in a 1978 address at the United Nations, in his memoir *Leap of Faith*, and on camera shortly before his death in 2004.

Now weigh it properly, which means neither dismissing it nor rounding it up. The 1951 sightings are a trained pilot's estimate of altitude and size for distant objects with no reference scale — exactly the judgement Section 12.4 showed pilots to be poor at, and Cooper's own profession made him confident rather than accurate. The 1957 film is the crux, and **Cooper's position on it was that of a supervisor, not a witness: by his account he handled the camera crew and the paperwork, not the eyepiece.** What he could testify to first-hand was that film existed, that he forwarded it, and that it did not come back. That is a documented-sounding claim with no document at the end of it — no frame of footage, no crew member on the record corroborating the landing, no film library number. A lost film is not evidence of a saucer; it is evidence of a lost film, and the Air Force in 1957 mislaid a great deal.

So here is the honest verdict, and it is the reason he opens this section. Cooper was not a fantasist, not a self-promoter, and not confused about what he had seen. He was a serious man reporting sincerely, and after sixty years his testimony has produced no object, no image and no second witness to the central event. **That is the whole lesson of this chapter in one biography: the quality of a witness sets a ceiling on how seriously a claim should be taken, and it never substitutes for the thing itself.** If the best witness available cannot get us past hearsay about a missing reel, the problem is not the witnesses.

APOLLO 11 • COMMANDER • FIRST STEPS, 20 JULY 1969

NEIL A. ARMSTRONG



Photograph: NASA. Public domain (work of the U.S. federal government). Author not individually credited.

Image sourced from: https://commons.wikimedia.org/wiki/File:Neil_Armstrong_pose.jpg

The first man on the Moon is the most-quoted UFO witness in history and never said any of it. There is no interview, transcript, air-to-ground loop, debriefing or memoir in which Armstrong reports a lunar encounter. The famous warning — that “they” were parked on the far rim of the crater and that we were ordered off — was manufactured in 1979 and attributed to an anonymous NASA source who does not exist. Armstrong denied it directly. His reticence is precisely why he was chosen as the mouthpiece: a man who gave perhaps a dozen substantial interviews in forty-three years cannot flood the zone with corrections. When he did speak about the unexplained, it was about ice crystals, propellant venting and the difficulty of judging scale without an atmosphere — the mundane, correct answer.

APOLLO 11 • LUNAR MODULE PILOT • SECOND MAN ON THE MOON

EDWIN E. "BUZZ" ALDRIN JR.



Photograph: NASA (S69-31743). Public domain (work of the U.S. federal government).

Image sourced from: [https://commons.wikimedia.org/wiki/File:Buzz_Aldrin_\(S69-31743\).jpg](https://commons.wikimedia.org/wiki/File:Buzz_Aldrin_(S69-31743).jpg)

Aldrin is the source of the single most abused sighting in the literature, and he has spent decades trying to take it back. On the way out to the Moon the crew watched an object holding station alongside them. Aldrin described it honestly — an L-shape, then a cylinder, apparently keeping pace — and then explained, over and over, in print, on television and to anyone who would listen, that it was one of the four adapter panels jettisoned from the spent S-IVB stage, tumbling on the same trajectory because it had been let go on the same trajectory. He is on record calling the alien interpretation a misrepresentation of his own words, and has been visibly angry about it. That has not slowed the citation down by a single revolution, which tells you something about how the claim is actually used.

APOLLO 12 • COMMANDER • THIRD MAN ON THE MOON, NOVEMBER 1969

C H A R L E S " P E T E " C O N R A D J R .



Photograph: NASA. Public domain (work of the U.S. federal government).

Image sourced from: https://commons.wikimedia.org/wiki/File:Astronaut_Charles_Conrad_Jr.jpg

Conrad stepped off the ladder, noted that it might have been a small step for Neil but it was a long one for him, and proceeded to run one of the most technically productive surface missions of the programme — a pinpoint landing beside the derelict Surveyor 3 probe, whose sun-cooked hardware they cut up and brought home. He reported no anomalies of any kind. He is nonetheless a fixture on quotation lists, generally by way of remarks about the strangeness of the lunar light or the appearance of the zodiacal glow, stripped of the sentence that identifies what he was describing. Conrad died in 1999 following a motorcycle accident. **An astronaut who never made a claim cannot retract one, which makes the dead the most useful witnesses in ufology.**

APOLLO 12 • LUNAR MODULE PILOT • FOURTH MAN ON THE MOON

A L A N L . B E A N



Photograph: NASA / Johnson Space Center (S69-38859). Public domain (work of the U.S. federal government).

Image sourced from: [https://commons.wikimedia.org/wiki/File:Alan_Bean_NASA_portrait_\(S69-38859\).jpg](https://commons.wikimedia.org/wiki/File:Alan_Bean_NASA_portrait_(S69-38859).jpg)

Bean is the clearest case in the set, because he was asked the question directly in 2017 and answered it without hedging: **"I do not believe that anyone from outer space has ever visited the Earth."** His reasoning was an engineer's. Interstellar distances are not a detail to be waved away; a civilisation capable of crossing them would not need to hide, and the absence of any physical trace after decades of looking is itself the evidence. Bean spent his later life as a painter, working lunar dust from his mission patches into the pigment of canvases depicting the surface he had walked on — a man with an unusually strong claim to romance about the Moon, and no patience at all for this particular story. He died in 2018.

APOLLO 14 • COMMANDER • FIRST AMERICAN IN SPACE, FIFTH ON THE MOON

A L A N B . S H E P A R D J R .



Photograph: NASA. Public domain (work of the U.S. federal government). Cropped version of a NASA original.

Image sourced from: [https://commons.wikimedia.org/wiki/File:Alan_Shepard_Apollo_14_\(cropped\).jpg](https://commons.wikimedia.org/wiki/File:Alan_Shepard_Apollo_14_(cropped).jpg)

Shepard flew Mercury-Redstone 3 in 1961, was grounded for most of a decade by an inner-ear disorder, fought his way back onto a flight crew and landed in the Fra Mauro highlands in 1971, where he hit two golf balls with a contingency sample tool. He was famously unsentimental, famously impatient, and said nothing whatsoever about extraterrestrials. Where his name appears on sighting lists it is usually attached to a garbled version of the Mercury-era “fireflies” — which were John Glenn’s, not his, and which were ice crystals shedding from the spacecraft skin. The misattribution survives because the two men are filed together in memory as the first Americans up. Shepard died in 1998.

APOLLO 14 • LUNAR MODULE PILOT • SIXTH MAN ON THE MOON

EDGAR D. MITCHELL



Photograph: NASA (1971 official portrait). Public domain (work of the U.S. federal government).

Image sourced from: [https://commons.wikimedia.org/wiki/File:Edgar_Mitchell_\(1971_portrait\).jpg](https://commons.wikimedia.org/wiki/File:Edgar_Mitchell_(1971_portrait).jpg)

Mitchell is the one genuine believer among the twelve, and the most instructive. He stated repeatedly and publicly that he was satisfied a crash-retrieval programme existed and had been concealed. He also stated, with a scrupulousness his admirers tend to omit, that **he had seen nothing himself** — that his conviction rested on what retired officers and old-timers from the Roswell area had told him, years after the fact, in private. That is hearsay held sincerely by a remarkable man; it is not observation. Mitchell experienced something profound on the way home in 1971 and spent the rest of his life studying consciousness seriously. Believing a thing on insufficient evidence is not stupidity. It is the default setting for human beings, which is exactly why method exists. He died in 2016.

APOLLO 15 • COMMANDER • SEVENTH MAN ON THE MOON, JULY 1971

DAVID R. SCOTT



Photograph: NASA Johnson Space Center (S71-39483). Public domain (work of the U.S. federal government).

Image sourced from: [https://commons.wikimedia.org/wiki/File:PORTRAIT_-_ASTRONAUT_DAVID_SCOTT_-_COMMANDER_\(APOLLO_15\)_-_CAPE_\(S71-39483\).jpg](https://commons.wikimedia.org/wiki/File:PORTRAIT_-_ASTRONAUT_DAVID_SCOTT_-_COMMANDER_(APOLLO_15)_-_CAPE_(S71-39483).jpg)

Scott commanded the first of the extended scientific missions, drove the first lunar rover, and closed the final EVA by holding a geological hammer in one hand and a falcon feather in the other and dropping them for the television camera. They hit the dust together. That is what an astronaut looks like when he is actually testing a hypothesis in front of an audience: falsifiable, repeatable, and over in two seconds. Scott has never reported an unidentified object, and when the subject arises he redirects it to what the surface work actually found — the Genesis Rock, the mare basalts, the ages. He is one of the four moonwalkers still living.

APOLLO 15 • LUNAR MODULE PILOT • EIGHTH MAN ON THE MOON

JAMES B. IRWIN



Photograph: NASA Johnson Space Center (S71-39485). Public domain (work of the U.S. federal government).

Image sourced from: [https://commons.wikimedia.org/wiki/File:PORTRAIT_-_ASTRONAUT_JAMES_B_IRWIN_-_LUNAR_MODULE_PILOT_\(LMP\)_\(APOLLO_15\)_-_CAPE_\(S71-39485\).jpg](https://commons.wikimedia.org/wiki/File:PORTRAIT_-_ASTRONAUT_JAMES_B_IRWIN_-_LUNAR_MODULE_PILOT_(LMP)_(APOLLO_15)_-_CAPE_(S71-39485).jpg)

Irwin came home changed, left NASA within a year and founded a religious ministry, describing his time on the surface as an encounter with God rather than with anybody else. He then mounted several expeditions to Mount Ararat in search of the remains of Noah's Ark, and did not find them. This is the useful case, because Irwin is proof that a moonwalker can hold an extraordinary conviction and still produce no evidence for it. His sincerity was total; his sample return was empty. Nobody cites Irwin in support of the Ark, because the people who cite astronauts are not actually interested in what astronauts believe — only in whether the belief is the one they already hold. Irwin died in 1991, the first of the twelve to go.

APOLLO 16 • COMMANDER • NINTH MAN ON THE MOON, APRIL 1972

JOHN W. YOUNG



Photograph: NASA (1972). Public domain (work of the U.S. federal government).

Image sourced from: [https://commons.wikimedia.org/wiki/File:Astronaut_John_W._Young_\(1972\).jpg](https://commons.wikimedia.org/wiki/File:Astronaut_John_W._Young_(1972).jpg)

Young flew Gemini, flew Apollo twice, walked in the Descartes highlands, and then commanded the first flight of the Space Shuttle in 1981 — six missions across three programmes, more time in more vehicles than almost anyone. If sustained exposure to spaceflight produced encounters, Young was the maximum-likelihood candidate, and he reported none. What he did spend his later decades doing was writing internal memoranda warning that the agency was underestimating its own risks, on asteroid impacts, on supervolcanoes, on the Shuttle itself. Young was the opposite of a credulous man: he was a professional worrier who insisted on quantifying the thing he was worried about. He died in 2018.

APOLLO 16 • LUNAR MODULE PILOT • TENTH AND YOUNGEST MAN ON THE MOON

CHARLES M. DUKE JR.



Photograph: NASA (official portrait). Public domain (work of the U.S. federal government).

Image sourced from: https://commons.wikimedia.org/wiki/File:Official_NASA_portrait_Charles_Moss_Duke_Jr.jpg

Duke was thirty-six when he landed, the youngest of the twelve, and he left a family photograph in the dust with all four of them on the front and a note on the back. He later became an outspoken evangelical Christian and has spent fifty years on stages talking about the flight and his faith. He is therefore asked about aliens constantly, by audiences primed to hear yes, and he says no — that he saw nothing on the way out, nothing on the surface, and nothing on the way home. **A man who will testify publicly to a religious conversion is not a man withholding testimony out of embarrassment.** He is simply reporting what was there, which was geology. Duke is still living.

APOLLO 17 • COMMANDER • LAST MAN ON THE MOON, DECEMBER 1972

EUGENE A. CERNAN



Photograph: NASA (S71-51308). Public domain (work of the U.S. federal government).

Image sourced from: https://commons.wikimedia.org/wiki/File:Cernan_s71-51308.jpg

Cernan is the strongest card the astronaut argument holds, and it is worth playing it fairly. Asked about unidentified objects in 1973, he said he believed some of them were real and that they were "somebody else, some other civilisation." That is a moonwalker, on the record, endorsing the hypothesis. Note carefully what it is: a statement of opinion about reports made by other people. Cernan did not claim to have seen anything himself on Gemini 9, Apollo 10 or Apollo 17, and in the same era he was scathing about the notion that the landings were faked. His view was that the universe is too large for us to be alone — which is a position about probability, not a sighting. Read that way it is respectable and quite common. Read as eyewitness evidence, which is how it is always presented, it is nothing at all. Cernan died in 2017.

APOLLO 17 • LUNAR MODULE PILOT • TWELFTH AND LAST MAN TO STEP ONTO THE MOON

HARRISON H. "JACK" SCHMITT



Photograph: NASA (S71-52260). Public domain (work of the U.S. federal government).

Image sourced from: https://commons.wikimedia.org/wiki/File:Harrison_H._Schmitt.jpg

Schmitt is the only professional scientist to have worked on the lunar surface — a field geologist with a Harvard doctorate, dropped into the Taurus-Littrow valley, where he found the orange soil at Shorty crater and correctly read it as volcanic glass rather than the rust of an oxidising process. He later served a term in the United States Senate and has argued for decades that the lunar regolith's helium-3 is a strategic resource. Asked directly whether he believes aliens exist, he answers as a scientist should: the statistical odds of life somewhere else, probably carbon-based like ours, are very good — and the odds of it having come here are a different question with no evidence behind it. That distinction, between life elsewhere and visitors here, is the whole discipline compressed into one sentence, and it is fitting that the last man to leave is the one who states it most cleanly. Schmitt is still living.

ALIENS AND UNDERGROUND BASES

"I'm kinda sad, though. All these underground tunnels and lasers, but instead of having a rave we're using them for evil."

— GORDON FREEMAN

17.1 THE GREENBRIER RESORT

I want to begin this chapter with a real underground government facility, because the existence of real ones is what makes the imaginary ones plausible, and because the Greenbrier is the perfect case: it was secret, it was enormous, it was hidden under a hotel full of guests, and it stayed secret for thirty years.

The Greenbrier is a luxury resort in White Sulphur Springs, West Virginia, about 250 miles from Washington. Beginning in 1958 under the codename Project Greek Island, the Army Corps of Engineers excavated a 112,000-square-foot bunker beneath the resort's West Virginia Wing, designed to house the entire United States Congress — both chambers, plus staff, roughly 1,100 people — in the event of nuclear war. It was completed in 1962, the year of the Cuban Missile Crisis, and remained on 24-hour readiness until 1992.

FIGURE 17.1

The Greenbrier bunker blast door at White Sulphur Springs, concealed behind a wallpapered partition until 1992.

Figure 17.1. The Greenbrier bunker blast door at White Sulphur Springs, concealed behind a wallpapered partition until 1992.

Image sourced from: https://commons.wikimedia.org/wiki/File:Greenbrier_bunker_blast_door.jpg

The details are marvellous. Blast doors weighing 25 and 28 tonnes (see Figure 17.1), set in concrete walls up to five feet thick, disguised behind ordinary wallpapered partitions. A dormitory with 18 bunkrooms. A clinic with an operating theatre and a 12-bed intensive care unit. A crematorium. A decontamination chamber. Water storage, diesel generators and three 25000-gallon fuel tanks. A television studio with a photographic backdrop of the Capitol, so broadcasts could appear to originate in Washington. And the two largest halls — the Governor’s Hall and

the Mountaineer Room — were used continuously as the resort’s exhibition and conference space, with hotel guests holding trade shows inside the House and Senate chambers for three decades without knowing it.

The cover was a government-contracted data facility, staffed by employees of a fictitious company, “Forsythe Associates”, who maintained the bunker and serviced the hotel’s televisions as a day job. Supplies were rotated. Congressional leadership was briefed. And nobody talked, until Ted Gup published an investigation in the *Washington Post Magazine* on 31 May 1992, after which the facility was decommissioned. You can now take the tour, which I recommend.

Why start here? Because the Greenbrier proves several things that matter for everything that follows. Large secret facilities are genuinely built. They can be hidden in plain sight under civilian cover. Secrecy can hold for thirty years across hundreds of participants. And — critically — it ended with a journalist, a document trail, and a public tour. The secret was kept, and then it was not, and the mechanism of its ending was ordinary reporting. Hold that thought for Section 17.3.

17.2 CHEYENNE MOUNTAIN

The other real one, and the model for every fictional underground base since.

The Cheyenne Mountain Complex, near Colorado Springs, was excavated between 1961 and 1966 by removing around 700,000 tonnes of granite, creating a grid of 15 three-storey buildings inside a network of chambers roughly 600 metres beneath the summit. The buildings sit on more than a thousand steel coil springs (see Figure 17.2), each weighing about half a tonne, isolating them from ground shock — the complex was designed to survive a near miss by a multi-megaton weapon and continue operating. Access is through a tunnel with two 25-tonne blast doors, and the tunnel runs through the mountain rather than terminating, so that overpressure passes out the other side.

FIGURE 17.2

*A spring-mounted building
inside the Cheyenne Mountain
Complex, isolated from ground shock.*

Figure 17.2. A spring-mounted building inside the Cheyenne Mountain Complex, isolated from ground shock. Image sourced from: https://commons.wikimedia.org/wiki/File:Cheyenne_Mountain_building_springs.jpg

Inside: independent power generation, a six-million-gallon water reservoir in excavated rock, NBC filtration, its own fire department and medical facility, and enough supplies for a sealed crew. It was NORAD's command centre for missile warning and air sovereignty from 1966, and it is the reason a generation grew up with a mental image of officers watching a big map.

In 2006 NORAD and USNORTHCOM operations moved to Peterson Space Force Base, and Cheyenne Mountain was placed on warm standby as an alternate command centre. It remains active in that role, hosts the Cheyenne Mountain Space Force Station, and its existence, dimensions, purpose and history are completely public — there are official photographs and published construction records.

Also real, also public: the Raven Rock Mountain Complex in Pennsylvania, "Site R", the Pentagon's alternate; Mount Weather in Virginia, the FEMA special facility; the Kirtland Underground Munitions and Maintenance Storage Complex; the Utah Data Center; and the Air Force Underground Test Area facilities. The United States has built a great many holes in mountains and does not particularly hide the fact.

Which brings me to the point of these two sections. When someone tells you the government has vast underground facilities, the correct answer is yes, obviously, here are their names and construction dates. The claim in Section 17.3 is not that underground bases exist. It is a specific, much stronger claim about what is in them.

17.3 DULCE, NM

And now the actual claim, which is the Dulce base, and I will lay it out fully and then explain why it fails.

The story holds that beneath Archuleta Mesa near Dulce, New Mexico — on Jicarilla Apache land — lies a joint human–alien facility of seven levels. Level 1 security and garages. Level 2 tube shuttle station and offices. Level 3 government offices and laboratories. Level 4 mind-control and aura research. Level 5 alien housing. Level 6, called “Nightmare Hall”, containing genetic experiments and multi-species hybrids. Level 7 holding thousands of humans in cold storage vats, labelled and catalogued. The 1979 “Dulce Wars” are said to have been a firefight in which sixty-six special forces personnel died.

The provenance is fully traceable, which is what makes this case instructive rather than merely lurid.

It originates with **Paul Bennewitz**, an Albuquerque electronics businessman who in 1979–80 became convinced he was intercepting alien communications from beneath Dulce, based on radio noise and on lights over the mesa. He was located next to Kirtland Air Force Base. Air Force Office of Special Investigations agent **Richard Doty** was assigned to him, and — this is the documented part, established through Greg Bishop’s *Project Beta* and by Doty’s own later statements — Doty deliberately fed him fabricated material, including forged documents, in order to discredit and misdirect him, and to steer him away from real classified activity at Kirtland. Bennewitz’s mental health deteriorated badly; he was eventually hospitalised, and he died in 2003. The story continued without him.

The second source is **Thomas Castello**, allegedly a Dulce security officer, whose testimony supplies the seven levels and Nightmare Hall. No such person has ever been documented to exist. No employment record, no birth or death record matching the claimed details, no photographs of the man — and the “Dulce Papers” attributed to him are of unknown origin and appeared through third parties.

The third source is **John Lear** and the 1987 *Krill Papers* / Lear letter, which assembled Dulce with MJ-12, the alien treaty and the abduction programme into the shape it holds today, and which drew on Doty’s fabrications. William Cooper then incorporated it into *Behold a Pale Horse* (1991), from which it

entered general circulation.

So: the foundational documents are established forgeries produced by a US counterintelligence officer to manipulate one distressed man, and the eyewitness cannot be shown to have existed.

And then the physical problems, which are decisive on their own. A seven-level facility of the described scale requires excavation on the order of millions of cubic metres, which produces spoil that must go somewhere visible; it requires power on the order of a small city, which requires transmission lines or a generating plant; it requires ventilation shafts, cooling, water supply and sewage; and it requires a workforce that must be housed, fed, paid and rotated. Dulce is a community of about 2600 people. There is no rail spur, no transmission corridor, no heavy road traffic, and no visible spoil. Satellite imagery of Archuleta Mesa is publicly available at sub-metre resolution and shows a mesa. The Jicarilla Apache Nation, which owns the land, has repeatedly and flatly denied any such facility, and it is their land — a point that gets remarkably little attention from people who claim to be exposing the exploitation of the powerless.

Compare it to Cheyenne Mountain in Section 17.2, which is a real facility of comparable ambition, and which required 700000 tonnes of removed rock, a visible access road, published construction contracts and five years of work by identifiable contractors. That is what building a hole in a mountain looks like. It is not invisible. It cannot be.

FUN FACT

The Bennewitz affair is, to me, the darkest story in this book, and it has nothing to do with aliens. A government agency identified a vulnerable civilian, exploited his beliefs with deliberately manufactured evidence, and watched him disintegrate — for operational convenience. Doty has discussed it publicly. Nobody was disciplined. And the fabrications outlived the man by decades and are still being repeated as testimony. If you want a reason to distrust official sources, you do not need a saucer. You need a filing cabinet and a man with a badge and an objective.

17.4 JOURNEY TO THE CENTER OF THE EARTH

The hollow Earth is worth a section because, like Section 16.5, we can close it with measurements, and because the measurements are genuinely interesting.

We know the interior structure of the Earth in detail, from seismology. Every large earthquake illuminates the planet, and a global network of thousands of seismometers records the arrival times of P-waves (compressional) and S-waves (shear) at every azimuth. From that you reconstruct the interior the way a CT scanner reconstructs a body.

FIGURE 17.3

*The Earth's interior as resolved
by seismology: crust, mantle,
liquid outer core, solid inner core.*

Figure 17.3. The Earth's interior as resolved by seismology: crust, mantle, liquid outer core, solid inner core.
Image sourced from: https://commons.wikimedia.org/wiki/File:Earth_poster.svg

The results, established over a century: a crust averaging 35 km on continents and 7 km under oceans, bounded by the Mohorovičić discontinuity found in 1909 (see Figure 17.3). A mantle to 2890 km, with transition-zone discontinuities at 410 and 660 km caused by known mineral phase changes — olivine to wadsleyite to ringwoodite to perovskite — reproduced in diamond anvil cell experiments at the predicted pressures. A liquid outer core from 2890 to 5,150 km, identified by Beno Gutenberg in 1913 from the S-wave shadow zone, because shear waves cannot propagate

through liquid. And a solid inner core, discovered by Inge Lehmann in 1936 from a faint reflected phase — one of the finest pieces of inference in the history of geophysics.

Independent confirmations: the planet's mean density of 5.51 g/cm^3 against surface rock at around 2.7 requires a dense interior. The moment of inertia constrains the mass distribution. The geodynamo requires a convecting conductive liquid layer. Free oscillation modes — the whole planet ringing after great earthquakes — match the model. And the Kola Superdeep Borehole, drilled to 12.26 km over twenty years, was stopped because at 180°C the rock behaved plastically and the hole closed on the bit. That is 0.2% of the way to the centre, and it was already too hot.

The pressure at the centre is about 360 gigapascals and the temperature is comparable to the surface of the Sun. **There is no cavity, and there could not be: rock has finite strength, and a void of any scale collapses.**

FUN FACT

The hollow Earth has a distinguished pedigree, which is why I treat it gently. Edmond Halley proposed concentric shells in 1692 — a serious attempt to explain geomagnetic secular variation, using the best available physics. Leonhard Euler is often said to have supported it, probably wrongly. John Cleves Symmes lobbied Congress in 1818 for an expedition to the polar opening, and actually got a vote. Then Jules Verne in 1864 and Edgar Rice Burroughs's *Pellucidar* made it literature, and the Nazi variant made it politics. Halley's version was good science that turned out wrong. That is what good science looks like most of the time, and it is not an embarrassment.

17.5 THE NEW WORLD ORDER

I want to end this chapter by separating three things that are constantly conflated, because the conflation is how a legitimate concern becomes an unfalsifiable belief and then, sometimes, something worse.

First: real coordination among elites is real, documented, and a legitimate subject of study. The Bilderberg Meeting has met annually since 1954 with invitation-only attendance and no published minutes — its participant lists are now released, and it is a genuine off-the-record forum where influential people talk. The World Economic Forum at Davos is exactly what it appears to be. The Trilateral Commission and the Council on Foreign Relations publish extensively. Central bank coordination through the Bank for International Settlements is institutional. Lobbying, regulatory capture, revolving doors between industry and agencies, and concentrated media ownership are all measurable, and political scientists measure them. None of this requires secrecy to be harmful, and the harms are documentable in the open record. This is the version worth being angry about.

Second: the “New World Order” as an unfalsifiable master narrative — a single coordinating cabal directing all major events toward global authoritarian government, in league with non-human intelligences. This fails for the reason every grand conspiracy fails, which is that it misunderstands

institutions. **Institutions leak, compete, and are bad at coordination.** The Greenbrier held for thirty years with a few hundred people and a strong shared motive, and one journalist ended it. Watergate involved a handful of men and collapsed in two years. The Pentagon Papers, COINTELPRO, MKULTRA, Iran–Contra, the Church Committee revelations, the Tuskegee study, Paperclip, the Snowden disclosures — every one of these was a real conspiracy, and every one came out, usually through a participant with a conscience or a document that survived. The probability of a leak scales with the number of participants and the duration. A conspiracy of millions across all governments over generations has a leak probability indistinguishable from one.

Third, and this is the part I will not soften: the antisemitic core. A large portion of New World Order literature is a direct descendant of *The Protocols of the Elders of Zion*, a forgery produced by the Russian secret police around 1903, plagiarised in part from a French satire by Maurice Joly, exposed as fabricated by *The Times* of London in 1921, and used to justify mass murder. When you encounter the “international bankers”, the “Rothschild” framing, the “globalists” who control media and finance, the “cultural” subversion narrative — you are reading the Protocols with the names filed off. This is not an accusation I make loosely. It is a textual lineage, and it is traceable sentence by sentence. Section 19.7 covers the same lineage in its occult form. If you take up these ideas without knowing where they come from, you are carrying somebody else’s weapon.

So here is the discipline I would ask of you. Be suspicious of power — I am, and Chapter 8 gives you grounds. But make your suspicion *specific*. Name the agency, name the programme, name the document, name the date. Specific suspicion produced the Church Committee, the Rockefeller Commission, and the declassification of MKULTRA. Unfalsifiable suspicion produces nothing except an audience, and it has a body count.

CONTACT

“If we ever establish contact with intelligent aliens living on a planet around a distant star . . . they would be made of similar atoms to us. They could trace their origins back to the big bang 13.7 billion years ago, and they would share with us the universe’s future. However, the surest common culture would be mathematics.”

— MARTIN REES

This chapter is dedicated to the life, legacy, and scientific work of the late Professor Stephen Hawking (1942–2018).

I want to explain the dedication before anything else, because it is not decorative.

Stephen Hawking spent his career on questions about the universe that could not be answered by going and looking — the interiors of black holes, the first instant of the Big Bang, the boundary conditions of spacetime — and he answered them, or advanced them, using mathematics and rigour alone. That is the single most relevant skill set in this book. He also said two things about extraterrestrial life that are usually reported as a contradiction and are in fact a single coherent position. He said the numbers make alien life plausible. And he said that if it exists, we should probably not advertise ourselves, because “if aliens ever visit us, I think the outcome would be much as when Christopher Columbus first landed in America, which didn’t turn out very well for the Native Americans.”

That is not pessimism. It is the application of the only data point we have

— what happens on this planet when a technologically advanced culture encounters a less advanced one — to a question where we have no other data at all. Hawking was doing what I have been asking you to do for seventeen chapters: reasoning carefully from what is actually known, refusing to fill gaps with what he preferred to be true, and being willing to say something unwelcome. He was also, and this matters for a man who lost the use of his voice, obsessed with the problem of communication. A chapter about whether we could ever talk to something not from here belongs to him.

18.1 BARRIERS OF COMMUNICATION

Chapter 1's opening argument was that we cannot reliably communicate about this subject *with each other*. Section 1.1 was about the fallacy of language: that "UFO", "alien", "sighting" and "proof" mean different things to different people in the same conversation, and that a large fraction of the disagreement in this field is two people using one word for two concepts. If that is true among humans who share a species, a planet, a sensory apparatus and a hundred thousand years of common cognitive history, then the standard optimistic assumption of contact literature — that a sufficiently intelligent species will of course be able to make itself understood — deserves to be taken apart.

So let us stack up the barriers properly, from the tractable to the possibly fatal.

The physical barrier. Interstellar distance is the first and most underrated. The nearest star system is 4.24 light years away, which makes the fastest possible exchange — question, answer — an eight-and-a-half-year conversation, and that is the best case in the entire galaxy. A dialogue with something 500 light years out is not a dialogue; it is two monologues separated by a millennium, conducted between civilisations neither of which can be certain the other still exists. Every scenario in this chapter has to be read with that constraint in place. Section 18.3 is partly about the consequences.

The sensory barrier. We assume vision and hearing because we have them. There is no reason to. A species that navigates by magnetic field, electrical sense, chemical gradient or echolocation would construct a description of the world with different primitives in it. Consider that on this

planet, right now, the mantis shrimp has somewhere between twelve and sixteen photoreceptor types against our three, and can see circular polarised light — a property of light that no human being has ever perceived and never will. We share four billion years of evolution with that animal and we cannot fully imagine its visual field. Now remove the shared evolution.

The cognitive barrier. Human languages are all built on a small set of shared foundations: discrete symbols, recursive grammar, roughly linear time, agent-and-object structure, and a theory of other minds. **These are not logical necessities; they are the design features of one particular brain.** A distributed or colonial intelligence might have no concept of a singular first person. Something with no mortality might have no tense system worth the name. And any species without a theory of mind — without the ability to model what another entity knows and does not know — cannot construct a message aimed at a stranger at all, because the entire act of explaining requires imagining ignorance.

The referential barrier. This is the deep one, and it is the reason linguists are usually more pessimistic about contact than physicists are. Translation between human languages works because we share a world: I can point at water. Quine’s famous problem — he called it the indeterminacy of radical translation — is that if a stranger says “gavagai” when a rabbit runs past, you cannot determine from any amount of observation whether it means *rabbit*, *dinner*, *undetached rabbit part*, *fleeting rabbit-stage*, or *look out*. On Earth you resolve this by living alongside the speakers for a year. Across 500 light years, with no shared referents and no possibility of pointing, there is no obvious resolution procedure at all.

FUN FACT

We have a live control experiment for this and it is not going well. The Linear A script of Minoan Crete, the Indus Valley script, and Rongorongo from Easter Island remain undeciphered. These were written by *human beings*, on this planet, with human bodies, human needs and human concerns, and in the Indus case across a city civilisation of over a million people — and we cannot read a word. The failure is not one of intelligence or effort. It is the absence of a bilingual key. The Rosetta Stone is the reason we can read Egyptian; nobody has ever found the interstellar equivalent, and there is no reason to expect one to exist.

The intentional barrier. Finally, and least discussed: successful com-

munication requires both parties to want it. A species could be aware of us and uninterested, in the way that you are aware of the ant colony in the garden and have never attempted to negotiate with it. Every barrier above assumes good faith and mutual effort. That assumption is doing an enormous amount of unnoticed work in almost everything ever written about first contact.

18.2 THE UNIVERSAL LANGUAGE OF MATHEMATICS

The standard answer to everything in the previous section is mathematics, and it is a good answer. It is just not as good as it is usually claimed to be.

The argument runs: physical law is the same everywhere, therefore any technological species must have discovered the same physics, therefore we share a body of content even if we share nothing else. Any culture capable of building a radio telescope knows that hydrogen is the most abundant element, knows the prime numbers, knows the value of π , and knows the 21-centimetre hyperfine transition line of neutral hydrogen at 1420 megahertz. That last one is why SETI listens where it listens: it is a frequency any radio astronomer anywhere would already be pointing an instrument at.

This is the reasoning behind every message humanity has actually sent. The **Arecibo message** of 1974, drafted principally by Frank Drake with Carl Sagan, was 1,679 bits transmitted toward the globular cluster M13 (see Figure 18.1). The number is the key: 1,679 is the product of two primes, 23 and 73, so it can be arranged into a rectangular grid in only one sensible way, and doing so produces a bitmap showing the numbers one to ten in binary, the atomic numbers of the elements in DNA, the structure of the double helix, a human figure with its height, the population of Earth, a diagram of the solar system, and a sketch of the transmitting dish.

FIGURE 18.1

The Arecibo message arranged as 73 rows of 23 columns — the only sensible factorisation of 1,679.

Figure 18.1. The Arecibo message arranged as 73 rows of 23 columns — the only sensible factorisation of 1,679.

Image sourced from: https://commons.wikimedia.org/wiki/File:Arecibo_message.svg

The **Pioneer plaques** of 1972 and 1973 used a different key — the hyperfine transition of hydrogen as a universal ruler and clock, and a map of the Sun’s position relative to fourteen pulsars, each pulsar’s period written in binary. That map is simultaneously a location and a timestamp, since pulsar periods decay predictably, so it will tell a finder both where and *when* it was made.

Those are elegant pieces of engineering. Now the objections.

First, mathematics gives you a shared vocabulary of nouns, not a language. **Primes and hydrogen let you establish that you are both technological.** They do not let you express *we came in peace, do not do that, or we would like to negotiate*. There is no prime number for regret. Everything humans actually need to say to a stranger — intention, permission, warning, obligation — lives in exactly the part of language that physics does not underwrite.

Second, the notation is not universal even where the content is. Base ten is a consequence of having ten fingers. Left-to-right reading, our convention that a bitmap’s origin sits top-left, the very idea of encoding a picture as a raster of pixels — all of these are cultural. When Drake circulated a test version of the Arecibo-style message to distinguished scientists before transmission, including some Nobel laureates, essentially none of them decoded it correctly. Human experts, told in advance that a message was present and that it was decodable, mostly failed. That result is treated as an amusing anecdote and I think it is the most important datum in the entire discipline.

Third, our own recent history has quietly complicated the picture. When we built machines to find structure in language without being told the rules, they found it — but the representations they use internally are not human-readable, and we cannot fully explain what they encode. We have manufactured systems that manipulate our own language competently and whose internal states we cannot translate. If mutual intelligibility were as automatic as the optimistic case suggests, that should not be possible.

“Mathematics is the language in which God has written the universe.”

— Attributed to Galileo Galilei

Where does that leave the idea? I think mathematics is necessary and insufficient: the only plausible foundation for a first exchange, and nowhere near enough for a conversation. It gets you to “we are both here and we both count”. Everything past that is unsolved.

18.3 THE RIO SCALE

Suppose tomorrow somebody detects something. How excited should the rest of us be?

This is a real operational problem, not a rhetorical one, and the field's answer to it is the best piece of intellectual hygiene anywhere in this subject. The **Rio Scale** was proposed in 2000 by Iván Almár and Jill Tarter, deliberately modelled on the Torino scale for asteroid impact risk and the Richter scale for earthquakes, to give any claimed detection of extraterrestrial intelligence a single number from 0 to 10 indicating its actual significance. A revision, Rio 2.0, was published by Duncan Forgan and colleagues in 2018.

The score is built from a few independent judgements multiplied together. How consequential would the signal be if genuine — a beacon aimed at us, an omnidirectional leakage signal, an artifact in the solar system, an unambiguous physical trace? How far away is the source — within the solar system, nearby stars, across the galaxy, extragalactic? And, crucially, how credible is the detection — has it been independently confirmed by another instrument, is it a single unrepeated event, is the instrumentation understood, has interference been excluded?

That last term is what makes the scale useful, because it forces the two questions that this book keeps insisting on — *how important would this be* and *how sure are we* — to be scored *separately* and then combined. Almost every failure of reasoning in Chapter 1 is a case of importance leaking into confidence: because a claim would matter enormously if true, people treat it as better supported than it is.

FIGURE 18.2

Ehman's printout of 15 August 1977, with the sequence 6EQUJ5 circled and annotated.

Figure 18.2. Ehman's printout of 15 August 1977, with the sequence 6EQUJ5 circled and annotated.

Image sourced from: https://commons.wikimedia.org/wiki/File:Wow_signal_printout.jpg

Some worked examples. The **Wow! signal** of 15 August 1977 — Jerry Ehman's 72-second narrowband burst at the Ohio State Big Ear observatory, close to the hydrogen line, which he circled in red pen and annotated "Wow!" (see Figure 18.2) — is the most famous candidate in history and scores low, because despite decades of follow-up it has never recurred and never been independently confirmed. Repetition is not a nice-to-have; it is the whole thing. **BLC-1**, a candidate found by Breakthrough Listen in 2020 apparently

coming from Proxima Centauri, would have been extraordinary given the distance, and turned out to be terrestrial radio interference. **Oumuamua**, the interstellar object of 2017 whose unusual acceleration prompted Avi Loeb to publicly suggest an artificial origin, would score appreciably on consequence and low on credibility, because outgassing of volatiles remains an adequate explanation and the object is gone. And **Tabby's Star** — KIC 8462852, whose bizarre irregular dimming produced serious published discussion of a Dyson-sphere-like structure — is now best explained by an uneven dust cloud, because the dimming turned out to be wavelength-dependent and solid objects block all colours of light equally. That last one is a genuinely beautiful piece of reasoning and it is worth carrying with you: a single well-chosen measurement retired the exotic hypothesis, and no argument about plausibility could have done it.

FUN FACT

There is also a formal etiquette for what to do afterwards. The International Academy of Astronautics maintains a “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence” — the SETI post-detection protocols — which say, in essence: verify before announcing, tell the astronomical community and the United Nations, publish the data openly, and *do not reply* on behalf of humanity without international consultation. It has no legal force whatsoever. It is a gentlemen's agreement about the most consequential press release in history, and the honest assessment is that in the actual event it would last about forty minutes before somebody leaked it.

18.4 FIRST CONTACT. THE DEBATE

The substantive argument in contact studies is not whether they exist. It is whether we should be transmitting.

Passive SETI — listening — is uncontroversial. **METI**, Messaging to Extraterrestrial Intelligence, deliberately broadcasting to selected targets, is not, and the dispute has produced one of the few genuinely well-conducted debates in this whole field, largely because both sides are arguing under total uncertainty and both know it.

The case against transmitting rests on asymmetry. Any species that receives our signal and is capable of reaching us is, necessarily, far ahead of

us; contact between parties of grossly unequal capability has a poor historical record; and the decision is irreversible, unilateral, and taken on behalf of everyone alive and everyone not yet born. This is Hawking's position, and David Brin's, and it is essentially an argument about decision-making under uncertainty rather than about aliens: when the downside is unbounded and the upside is speculative, and when no one has the standing to consent for the species, the default is to wait.

The case for transmitting is also strong. We have been leaking radio, radar and television for a century and cannot recall it, so the horse has left; military early-warning radar is by far our loudest emission and is orders of magnitude more detectable than any deliberate message; a civilisation able to cross interstellar distance could detect our atmosphere's oxygen and methane spectroscopically without our help at all; and a galaxy in which everyone reasons like Brin is a galaxy of permanent mutual silence, which may be a sufficient explanation for the Fermi paradox all by itself. Alexander Zaitsev, who actually transmitted several messages from the Yevpatoria dish in Ukraine, made this argument and then acted on it unilaterally — which rather demonstrates the procedural objection.

My own view, offered as a view: the asymmetry argument is correct in structure and probably irrelevant in practice. We are already visible to anyone with better instruments than ours. The METI debate is therefore less about safety than about consent, and on consent the objectors are plainly right — one radio astronomer with dish time should not be able to introduce the species.

FUN FACT

The most reckless transmission ever sent was not a scientific message. In 2008 NASA broadcast the Beatles song "Across the Universe" toward Polaris to mark an anniversary. In 2005 a snack food company transmitted an advertisement toward a star system. Whatever the outcome of the METI debate, the historical record will show that humanity's considered position was overruled by a marketing department.

I B . S G U E R R I L L A W A R F A R E

Now the negative case, taken seriously rather than as entertainment. If contact went badly, what would badly actually look like?

The first thing to discard is the cinema. Every invasion narrative from *The War of the Worlds* onward — and Section 20.1 is about why that particular story has such a grip — imagines something like a human war: landings, resistance, a decisive engagement, a discovered weakness. This is a category error. A conflict between civilisations separated by even a few thousand years of technological development is not a war, in the way that a fight between an aircraft carrier and a bronze-age fleet is not a war. Where the capability gap is total, there is no engagement to speak of.

Which brings us to the section title, because guerrilla warfare is the only form of resistance that is even structurally available to the weaker party, and it is worth understanding why. Asymmetric warfare works by refusing the contest the stronger side is optimised for: dispersing so there is no target worth striking, blending into a population the aggressor cannot distinguish, imposing cost over time rather than seeking a decision, and winning by making occupation more expensive than withdrawal. Vietnam and Afghanistan are the standard illustrations, and both worked for the same reason — the stronger party wanted something that required holding ground and dealing with people.

And that is precisely the assumption that fails here. Guerrilla resistance depends on the adversary needing the territory intact and the population present. Remove that need and the strategy has nothing to grip. Worse, the classic motives for invasion mostly evaporate on inspection: water is abundant in the outer solar system and easier to collect from a comet than from the bottom of a gravity well defended by a biosphere; metals are more accessible in asteroids; and anything capable of crossing interstellar space has energy sources that make ours irrelevant. Earth's only genuinely rare feature is its biology — and that is also the one thing that would make it hazardous, since four billion years of local pathogens and a fully occupied biochemistry is a hostile environment, not a resource. If there is a real risk, it is much more likely indifference than malice: the road is not built through the anthill out of hatred.

"If aliens visit us, the outcome would be much as when Columbus landed in America, which didn't turn out well for the Native Americans."

— Stephen Hawking

There is one final scenario in this direction that I think is underrated, and it is not military at all. The most destructive contact event in human history did not involve weapons; it involved microorganisms and information. Contact between human cultures has repeatedly destroyed the weaker party through disease it had no immunity to and through the collapse of its own explanatory framework — the discovery that the world was not as it had been described. Both of those mechanisms are available to a contact event that consists of nothing but a received message.

I B . E W E C O M E I N P E A C E

The optimistic case deserves the same seriousness, and it has better arguments than its detractors admit.

Start with the selection effect. Any species that survives long enough to cross interstellar distances has, by definition, solved the problems that would otherwise have killed it: it has controlled the energies required for starflight without destroying itself, which is a filter humanity has not yet passed. Carl Sagan's version of this argument is that longevity and aggression are probably anti-correlated at the civilisational scale, because the technologies of interstellar travel are the technologies of self-annihilation. A galaxy full of aggressive young civilisations is a galaxy full of very short-lived ones. Whoever is still out there after a million years likely got that way by being stable.

Second, motive. A civilisation with no material need of us has no reason for hostility, and the same argument that undercut the invasion scenario in the last section cuts here too — but in the opposite direction it leaves curiosity standing. Information is the one thing Earth has that cannot be obtained anywhere else: an independent four-billion-year biological experiment, a second data point on how life arises and what it does. That is genuinely valuable, non-consumable, and best obtained by leaving the specimen intact and talking to it.

Third, the practice already exists. There is a serious literature on how

a benign contact might actually proceed, and its most attractive proposal is that a sufficiently advanced civilisation would not announce itself at all but leave material accessible on our terms — what the field calls the zoo hypothesis in its passive form, or in Iain Banks’s phrasing, contact conducted at the pace of the less capable party. The interesting feature of that model is that it is indistinguishable from absence, which is intellectually inconvenient and worth admitting.

And fourth, the human institutional response might be better than we fear. When COSPAR wrote planetary protection protocols, when the Antarctic Treaty set aside a continent for science, when the Outer Space Treaty of 1967 declared that space is not subject to national appropriation, the species managed — under no pressure and with no enforcement — to behave with a foresight that its general reputation does not predict. Section 18.3’s post-detection protocols are in that tradition. They are unenforceable and they were still written down, by people trying to get it right in advance. I find that more encouraging than most of what is in this book.

18.7 THE PROBLEM WITH SETI

And here is where I part company with the mainstream, because the standard summary of SETI’s sixty-year record is that we have looked and found nothing, and that summary is close to worthless.

The problem is radio, and specifically the fraction of the search space that radio SETI has actually examined. Consider what a complete search would require: every direction on the sky, every frequency from the low radio through the optical, every polarisation, every possible modulation scheme, and — this is the part that gets forgotten — every *moment in time*, because a signal you are not pointed at when it arrives is a signal that did not happen as far as your data are concerned. Now consider what has been done. A few thousand nearby stars observed attentively, mostly in a narrow band around the hydrogen line, mostly for minutes at a time, at sensitivities that would generally require a deliberate high-power beacon aimed at us to register.

Jill Tarter’s own analogy, and she ran the search for decades, is that the totality of SETI’s coverage is comparable to dipping a drinking glass into the ocean, examining it, and concluding there are no fish. The published

estimates of the fraction of the plausible search space covered come out at something like one part in 10^{18} — a number so small that “we have found nothing” carries almost no information about whether anything is there. Rounded to any practical precision, we have searched zero.

Then add the assumption problem on top, because coverage is not the only issue. Radio SETI assumes that an advanced civilisation broadcasts, deliberately, at high power, continuously, omnidirectionally or toward us specifically, using amplitude or frequency modulation on a narrow carrier, for a duration overlapping our observation, from a distance where inverse-square losses still leave a detectable signal. Every one of those is a guess derived from mid-twentieth-century human engineering. Our own radio footprint has been shrinking for thirty years: we abandoned high-power analogue broadcast for low-power digital, cable, fibre and tightly beamed satellite links, and a well-optimised digital transmission is statistically indistinguishable from noise unless you know the encoding. A civilisation like ours in 2026 would be substantially harder for SETI to detect than a civilisation like ours in 1965. The window during which a technological species is loud in radio may be a century out of a lifespan of millions of years — and if so, radio SETI is looking for a signature that barely exists.

FUN FACT

There is also the sheer arrogance of the frequency choice, which nobody likes to dwell on. We listen at the hydrogen line because *we* think it is the obvious universal channel. This is the cosmic-scale equivalent of standing in a field shouting in English on the grounds that English is widely spoken. The assumption may well be right. It is still an assumption, and it has consumed most of the field’s total observing time.

None of this is an argument that SETI is a waste. It is the most intellectually honest programme in this entire book — it states its hypothesis, it publishes null results, it does not inflate its claims, and it has been doing rigorous science on almost no money for sixty years while the rest of this subject was selling paperbacks. My complaint is not about SETI. It is about how its results get reported. Absence of evidence is evidence of absence only when the search was adequate to find the thing, and Section 1.3’s distinction between evidence and proof cuts both ways: a null result from an instrument that covers a vanishing fraction of the possibility space is not a

finding about the universe. It is a finding about the instrument. Meanwhile the interesting work has moved — optical and infrared SETI looking for nanosecond laser pulses, technosignature searches for industrial pollutants and waste heat in exoplanet atmospheres, and artifact searches within our own solar system, where anything left behind would still be sitting there regardless of whether its makers are.

18.8 MESSAGES FOR THE FUTURE

Which brings the chapter, and in a sense the argument, to the Voyager Golden Record.

In 1977 two spacecraft left Earth carrying identical gold-plated copper phonograph records (see Figure 18.3), compiled by a committee chaired by Carl Sagan, with Ann Druyan as creative director and Timothy Ferris producing. Each carries 115 analogue-encoded images, greetings in 55 languages, a twelve-minute sequence of the sounds of Earth — surf, wind, thunder, birds, whales, a kiss, a mother and child, a heartbeat, a Saturn V launch — and ninety minutes of music: Bach, Beethoven, Mozart, Stravinsky, Javanese gamelan, Senegalese percussion, Navajo

night chant, Peruvian panpipes, Azerbaijani balaban, Blind Willie Johnson's *Dark Was the Night, Cold Was the Ground*, and Chuck Berry's *Johnny B. Goode*. The cover is engraved with instructions: how to play the record, the rotational speed given in units of the hydrogen hyperfine transition, the pulsar map from the Pioneer plaques, and a diagram of hydrogen itself as a key to the whole notation.

Both spacecraft are now in interstellar space. Voyager 1 crossed the heliopause in 2012, Voyager 2 in 2018. Their power sources will fail in the 2020s or early 2030s and after that they will drift, silent, for a very long time — the records were engineered to survive on the order of a billion years, which is longer than the expected remaining habitability of the Earth's

FIGURE 18.3

The Voyager Golden Record cover: pictographic playback instructions and a pulsar map giving Earth's position.

Figure 18.3. The Voyager Golden Record cover: pictographic playback instructions and a pulsar map giving Earth's position. Image sourced from: https://commons.wikimedia.org/wiki/File:Voyager_Golden_Record_fx.png

surface.

And here is the thing I want to leave you with. The probability that either record is ever found is, realistically, indistinguishable from zero: they are moving slowly, they are not aimed at anything, and interstellar space is almost entirely empty. Sagan knew that perfectly well. He said the record would only be played if there were advanced spacefaring civilisations in interstellar space, and that launching this bottle into the cosmic ocean said something very hopeful about life on this planet.

Which means the Golden Record was never really addressed outward. Consider what it contains: not our weapons, our borders, our wars or our politics, but whale song, a mother greeting her child, a Brandenburg concerto, and a Delta blues recording made by a blind street singer in 1927. That is a species deciding what it would want to be judged by, and then committing the answer to gold and putting it somewhere it can never be revised. The message that matters is not the one a stranger might receive in a billion years. It is the one we sent ourselves in 1977, and the fact that when we seriously imagined being seen from outside, this is what we chose to show.

Section 18.2 argued that mathematics gets you to “we are both here and we both count” and no further. The Golden Record is the attempt to go further anyway — to transmit not information but *what it is like to be us* — knowing full well that nobody is listening. I think that is the most human object ever built, and it is a fitting place to end a chapter dedicated to a man who spent thirty years unable to speak and never once stopped trying to explain the universe to anybody who would listen.

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THE LOST CITY OF ATLANTIS

“And in a single day and night of misfortune all your warlike men in a body sank into the earth, and the island of Atlantis in like manner disappeared in the depths of the sea.”

— PLATO

What on earth does the Lost City of Atlantis have to do with UFOs? Well, likely more than the average individual would otherwise realize.

Here is the connection, stated plainly so you can hold me to it for the rest of the chapter. Atlantis is the master template. Almost every structural feature of modern Ufology appears first, and in fully developed form, in the Atlantis literature: a lost advanced technology, a suppressed true history, physical monuments held to be inexplicable by conventional means, a catastrophe that erased the evidence, a lineage of superior beings, and a body of channelled testimony standing in for documentation. When Erich von Däniken wrote *Chariots of the Gods* in 1968 he was not inventing a genre; he was performing a substitution, replacing the Atlanteans of the nineteenth-century occult revival with extraterrestrials and leaving the rest of the argument almost untouched. Chapter 9 covered the ancient-astronaut claims; this chapter is about where those claims actually came from.

There is a second reason, and it is the more useful one. Atlantis is a closed case. We know exactly where the story originated, we can read the original document, we can trace every subsequent embellishment to a

named person and a publication date, and we can watch a philosophical parable turn into a lost continent, then into a race of super-beings, then into a political ideology, then into a tourist industry — across twenty-three centuries, with the paper trail intact. You will never get that with Roswell. So this chapter is a laboratory: the anatomy of a myth, dissected while it is still moving.

19.1 ORIGIN OF A LEGEND

Atlantis has exactly one source, and everything else is commentary. It appears in two dialogues by Plato, the *Timaeus* and the unfinished *Critias*, written around 360 BC. There is no earlier mention anywhere — not in Homer, not in Herodotus, who was a compulsive collector of exactly this sort of story and would have loved it, not in any Egyptian record, not in any archaeological or geological trace. The tradition begins with one author and one text.

In Plato's telling, the story reaches him at fourth hand: an Egyptian priest at Sais told it to the Athenian statesman Solon, who told Dropides, who told the elder Critias, who told the younger Critias, who is relating it in the dialogue. That chain is a full nine thousand years long by the story's own accounting, and the modern reader should recognise the structure instantly, because it is the standard architecture of an unverifiable claim: a distant authority, no surviving document, and three intermediaries who are all dead.

The Atlantis described is a naval empire beyond the Pillars of Heracles, larger than Libya and Asia combined, laid out in concentric rings of land and water, rich in a metal Plato calls orichalcum, governed by descendants of Poseidon, and eventually corrupted by its own power into attacking Athens. It is defeated by the Athenians and then destroyed by earthquake and flood in a single day and night.

And that ending is the tell. The *Timaeus* and *Critias* are companion pieces to the *Republic*, and Plato says so explicitly at the start of the *Timaeus*: he wants to see the ideal state described in the *Republic* set in motion, in a war. Atlantis is the antagonist he built for that purpose — the wealthy, technologically sophisticated maritime power that overreaches and is annihilated, opposed by a virtuous, austere, land-based Athens. Plato was writing shortly after

Athens had lost the Peloponnesian War to a naval rival and had watched its own imperial overreach destroy it. He was also a man who invented illustrative stories constantly — the Cave, the Ring of Gyges, the Myth of Er — and who never once claimed any of them was history.

Aristotle, who was Plato's student for twenty years and knew perfectly well how his teacher worked, regarded the story as invented. That is about as close to a primary-source verdict as antiquity ever gives us.

FUN FACT

There *is* a real candidate for the kernel of the story and it is genuinely interesting: the Minoan civilisation of Crete, a sophisticated maritime power that was severely disrupted in the second millennium BC, and the eruption of Thera — modern Santorini — around 1600 BC, one of the largest volcanic events in human history, which destroyed the town of Akrotiri under ash and left behind a caldera whose remaining island is, as it happens, roughly ring-shaped. The dates and the scale are wrong for Plato's account by a factor of about six, and Thera is inside the Mediterranean rather than beyond Gibraltar. But a folk memory of a great island civilisation drowned in a day is not an unreasonable thing for a Greek to have inherited. This is what a responsible version of the Atlantis hypothesis looks like: smaller, later, closer, and archaeologically real.

So how did a parable become a continent? Through one book. In 1882 the American congressman Ignatius Donnelly published *Atlantis: The Antediluvian World*, and essentially every claim in modern Atlantis literature is traceable to it. Donnelly proposed that Atlantis was real, that it was the source civilisation from which Egypt, Mesopotamia, India and Mesoamerica all descended, that it explained the parallel development of pyramids and flood myths worldwide, and that survivors carried its knowledge to both hemispheres. His method was to collect superficial similarities between distant cultures and treat each one as evidence of common origin. Section 1.5 on correlation and causation is the whole refutation, and Donnelly is the best available worked example of what that fallacy produces when applied with energy across six hundred pages.

Then Helena Blavatsky, founder of Theosophy, absorbed Donnelly into an occult cosmology in *The Secret Doctrine* of 1888, adding root races, psychic powers, and a spiritual hierarchy; and Edgar Cayce, the American "sleeping prophet", supplied thousands of trance readings between 1923 and 1945

describing Atlantean technology in detail — aircraft, energy crystals, a great central power source. Every subsequent variant of the story descends from those three. Not one of them was an archaeologist. Not one produced a physical object.

19.2 THE ATLANTEAN ERA

Suppose you wanted to take the strong claim seriously — a technological civilisation predating the conventional record. What would the evidence look like, and does any of it exist?

This is a fair question and it deserves better than mockery, because the honest answer is that the claim is not *logically* absurd. Plenty of prehistory is genuinely unaccounted for, and the last fifty years have repeatedly pushed the beginnings of complex human activity earlier than expected. Göbekli Tepe in southeastern Turkey is the standout: monumental carved stone enclosures dated to around 9500 BC, some seven thousand years before Stonehenge, built by people who were supposed to be pre-agricultural hunter-gatherers incapable of that kind of organisation. It genuinely overturned a textbook consensus. Sea levels at the end of the last glacial period were roughly 120 metres lower than today, so a great deal of the most desirable coastal land occupied by late Paleolithic humans is now underwater — Doggerland between Britain and Denmark, Sundaland in Southeast Asia, the Persian Gulf basin. There is real drowned prehistory down there, and very little of it has been surveyed.

So the modest claim — that humans were doing more, earlier, than we credited, and that a lot of the evidence is submerged — is not only defensible, it is being confirmed piece by piece by ordinary archaeology. Graham Hancock's popular case rests on this ground and on the Younger Dryas impact hypothesis, the proposal that a cometary event around 12900 years ago triggered abrupt climatic reversal and catastrophic flooding. That hypothesis is scientifically live and contested; some genuinely serious people support it, and the evidence for it is under active dispute rather than dismissed.

But the strong claim — an advanced, technological, metal-working, energy-generating civilisation — fails on a specific and I think unanswerable point, which is worth learning because it applies far beyond this chapter.

Technology is not a single artifact. It is an infrastructure, and infrastructure leaves a fingerprint that cannot be avoided or erased.

Metallurgy requires ore extraction, and mining leaves spoil heaps and altered landscapes that persist for tens of thousands of years; we can still find Bronze Age copper workings. Smelting releases lead and copper into the atmosphere, and that signal is recorded in Greenland ice cores and in lake sediments — we can read Roman smelting activity in the ice, year by year, and see it collapse with the Empire. Agriculture at scale alters pollen profiles in bog and lake cores, and those profiles have been read continuously back through the Pleistocene. A large population leaves skeletal remains, middens, and domesticated animal and plant genomes; domestication is a genetic event with a reconstructible date. Any energy technology leaves isotopic or chemical traces. And a global civilisation would have distributed its material culture along its trade routes, which is exactly how we detect every other ancient network.

None of these signatures exists before the conventional record. Not one. The ice cores are clean, the pollen is clean, the sediment metals are clean. And here is the part that matters: these datasets were not assembled to refute Atlantis. They were built by climate scientists, palaeobotanists and geochemists pursuing entirely unrelated questions, which makes them about as close to an unbiased test as this subject will ever get. A civilisation can hide its cities under the sea. It cannot hide its smelters from the Greenland ice sheet.

"A lost city can be buried. A lost industry cannot — the atmosphere keeps the receipts."

19.3 CRYSTAL SKULLS

The crystal skulls are the bridge that supposedly ties Atlantis to Egypt, and they are also the cleanest debunking in this entire book, which is why I am giving them a full section rather than a footnote. If you only remember one case study from this chapter, remember this one, because it shows you what a definitive answer actually looks like.

The claim: a set of skulls carved from single blocks of quartz crystal, of unknown antiquity, attributed to Aztec, Maya or Atlantean craftsmen,

exhibiting workmanship allegedly impossible without modern tools, and — in the esoteric version — constituting a set of thirteen which, if reunited, would release ancient knowledge. Cayce's readings supplied the Atlantean provenance; the Mitchell-Hedges skull supplied the drama; *Indiana Jones and the Kingdom of the Crystal Skull* supplied the audience.

The answer came from a microscope. Researchers at the British Museum and the Smithsonian examined the surfaces of the major skulls under scanning electron microscopy, looking at the tool marks. **This is the crucial methodological point: worked stone records the tool that worked it.** Hand-lapping with abrasive slurry, which is what pre-Columbian lapidaries used, leaves irregular, variably-oriented scratches. A rotary wheel leaves parallel striations of uniform depth, and the marks change character where the wheel enters a curve. Diamond-loaded rotary tools leave a signature different again.

What the examinations found, in the British Museum skull and the Smithsonian's, was regular parallel rotary striations — unambiguous evidence of a jeweller's lapidary wheel, a technology not present in the Americas before European contact. The Smithsonian skull additionally showed traces of silicon carbide, an abrasive not synthesised until the late nineteenth century, which dates the object to after 1890 outright. Isotopic and inclusion analysis of the quartz pointed to sources in Brazil and Madagascar rather than Mexico. And provenance research closed the circle: essentially every known skull can be traced to the Paris antiquities market of the 1860s–1880s, and specifically to the dealer Eugène Boban, who sold to museums during exactly the period when Mexican antiquities were fashionable and unregulated. The British Museum's came through Tiffany & Co. in 1897.

As for the Mitchell-Hedges skull, said to have been discovered by Anna Mitchell-Hedges at the Maya site of Lubaantun in Belize in 1924 on her seventeenth birthday: F. A. Mitchell-Hedges never mentioned it in his accounts of the expedition, his daughter was not documented as present, and auction records show he purchased it from Sydney Burney at Sotheby's in 1943. It was examined in 2008 and shows the same rotary tooling.

No crystal skull has ever been recovered from a documented archaeological excavation. Not one, in a century and a half, out of thousands of professionally excavated Mesoamerican sites — and Mesoamerican archaeology is not short of quartz artifacts, so the absence is meaningful.

FIGURE 19.1

The British Museum crystal skull, acquired in 1897 — microscopy revealed rotary tool marks inconsistent with pre-Columbian lapidary technique.

Figure 19.1. The British Museum crystal skull, acquired in 1897 — microscopy revealed rotary tool marks inconsistent with pre-Columbian lapidary technique.

Image sourced from: <https://example.org/placeholder-skull.jpg>

I dwell on this because of what it demonstrates. A physical object was available. A specific, testable, falsifiable prediction followed from the claim — that the tool marks would be consistent with hand-lapping. The test was performed. The prediction failed. The case closed. This is precisely what Ufology cannot do, and the reason is not that skeptics are closed-minded. It is that Ufology has no object. Give me a piece of a craft and I will give you the isotope ratios inside a week, and Section 1.3 explains why that single measurement would settle in one afternoon a question that testimony has failed to settle in eighty years.

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19.4 PYRAMIDS

The other bridge to Egypt is the pyramid argument, and it comes in two parts: that pyramids appear worldwide and therefore share a source, and

that the Great Pyramid could not have been built with the tools available. Both are wrong, and they are wrong in instructive ways.

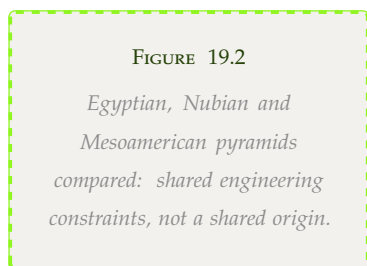


Figure 19.2. Egyptian, Nubian and Mesoamerican pyramids compared: shared engineering constraints, not a shared origin.

Image sourced from: https://commons.wikimedia.org/wiki/File:Pyramid_size_comparison.svg

On the worldwide distribution: pyramids appear independently in Egypt, Mesoamerica, Mesopotamia, Nubia, China, and Java, and the inference to a common Atlantean origin ignores the far simpler explanation, which is structural (see Figure 19.2). If you want to build something very tall out of stone or mud brick and you have no steel, no concrete and no arch, your options are severely constrained: the load must be carried in compression to the ground, so each course has to be smaller than the one below. Everything you can build under those constraints is a pyramid, a cone, or a heap. The shape is not

a cultural signature, it is a consequence of the material. And on inspection the resemblance dissolves: Egyptian pyramids are smooth-sided tombs built over a few centuries in the third millennium BC; Mesoamerican pyramids are stepped platforms with temples on top, used for ritual performance, built two to three thousand years later, frequently constructed in layers over earlier versions; ziggurats are terraced administrative-religious structures of mud brick. Different shapes, different construction methods, different purposes, separated by millennia. The only thing they share is that they get narrower going up, which is gravity's requirement, not Atlantis's.

On the construction of the Great Pyramid, the honest position is that we know a great deal more than the mystery literature admits. We have the quarries — the bulk limestone came from the Giza plateau itself, immediately adjacent, and you can see the quarry scars. We have the tools: copper chisels, wooden mallets, dolerite pounders, and abrasive sand, all recovered. We have the ramps, or at least several of them, including a well-preserved inclined system with post holes at the Hatnub alabaster quarry. We have the workers' town at Giza excavated by Mark Lehner and Zahi Hawass, complete with bakeries, breweries, dormitories and the cattle bones to feed several thousand people. We have the workers' own graffiti inside relieving chambers naming their gangs. And since 2013 we have the Merer diary — papyri found at Wadi al-Jarf, the oldest inscribed papyri ever recovered,

which are the logbook of a boat crew transporting Tura limestone casing blocks to Giza during the reign of Khufu. It is an administrative record of the pyramid being built, written by a man who was doing it. We know his name.

FUN FACT

The “we could not build it today” claim is worth pausing on, because it is false in an interesting way. We could build it easily; we simply would not, because there is no reason to spend twenty years and a national labour surplus stacking limestone. The real achievement at Giza was never technological — it was *organisational*. Feeding, housing, rotating and coordinating tens of thousands of workers for two decades, with logistics, rations and quarry schedules, is a genuinely staggering administrative accomplishment by a state that had existed for only a few centuries. The mystery literature’s insistence that aliens must have done it is, when you look at it squarely, a refusal to credit ancient Egyptians with competence. That is the part I object to. Not the aliens — the condescension.

19.5 THE ISLES OF LEMURIA

Lemuria is my favourite object lesson in this chapter, because unlike Atlantis it did not begin as a parable. It began as a legitimate scientific hypothesis, and watching it decay is genuinely instructive.

In 1864 the English zoologist Philip Sclater noticed a real biogeographical puzzle: lemur fossils and related primates were distributed across Madagascar and India but were absent from Africa and the Middle East in between. To explain it, he proposed a submerged landmass in the Indian Ocean connecting the two, and named it Lemuria. This was reasonable science for 1864. Continental drift had not been formulated, sunken land bridges were the available explanatory tool, and Ernst Haeckel and others took the idea seriously enough to suggest Lemuria as a possible cradle of humanity.

Then plate tectonics arrived in the 1960s and solved the problem properly: Madagascar and India were physically joined as part of Gondwana and subsequently separated by seafloor spreading. The distribution needed no sunken continent, and Lemuria was retired — correctly, on evidence, by the scientific community that had proposed it. That is how the process is

supposed to work, and it did.

Except the name had already escaped. Between 1864 and 1960, Helena Blavatsky had incorporated Lemuria into Theosophy as the home of the Third Root Race; the Anglo-Indian writer James Churchward had relabelled a similar Pacific continent “Mu” and produced a series of books about it beginning in 1926, based on tablets he claimed to have seen in an Indian temple and which no one else ever saw; and Tamil nationalist writers had adopted Lemuria as Kumari Kandam, an ancestral Tamil homeland. By the time the geologists had finished, Lemuria had a devoted following that had no interest in geology.

It survives today as a spiritual homeland, with Mount Shasta in California as a claimed refuge for surviving Lemurians. There is also a genuine and lovely twist: in 1999 an actual submerged continental fragment, the Mascarene Plateau, was investigated in the Indian Ocean, and in 2013 zircon evidence supported a landmass now called Mauritia beneath Mauritius. There really was drowned continental crust there. It sank tens of millions of years before any primate existed, and no lemur ever walked on it.

The lesson: a hypothesis can be retracted by the scientific community and remain in circulation forever, because the popular version was never held for the reason the scientific version was proposed. Once a claim acquires an emotional constituency, evidence loses jurisdiction over it.

19.6 THE SCIENCE OF POLE

SHIFTING

Catastrophic pole shift is the mechanism that all of this needs, so let us be precise about it, because there are three completely different phenomena sharing the phrase and conflating them is where the whole argument lives.

Geomagnetic reversal is real, common, and well documented. The Earth’s magnetic field has switched polarity hundreds of times; we read the record directly in the symmetrical magnetic striping of oceanic crust either side of the mid-ocean ridges, which is one of the most beautiful datasets in science. The most recent full reversal, the Brunhes-Matuyama, was around 780,000 years ago. The process takes somewhere between a thousand and ten thousand years, during which field strength drops and the field becomes

chaotic. The mass extinction record shows no correlation with reversals, which is the decisive point. The magnetic field flipping is a magnetometer event, not a geological catastrophe.

True polar wander is also real, also documented, and also slow: the entire solid Earth reorients relative to its spin axis, at rates on the order of one degree per million years. Over geological time this has moved continents through climate zones, which matters enormously for palaeoclimate and not at all for anybody's afternoon.

Crustal displacement — the claim that the entire lithosphere can slip catastrophically over the mantle in a matter of hours or days, relocating the poles and destroying civilisations — is the one the Atlantis literature requires, and it is not a real phenomenon. It was proposed by Charles Hapgood in *Earth's Shifting Crust* in 1958, and it acquired lasting credibility from a supportive foreword by Albert Einstein, who was not a geologist and who was writing before plate tectonics existed. That foreword has been doing rhetorical work for seventy years, and Section 1.8's point about credentials not being a method is nowhere better illustrated: the greatest physicist of the century wrote an approving note about a geological mechanism outside his field, in advance of the theory that would explain the actual behaviour of the crust, and the note has outlived the hypothesis by five decades.

Why it fails: the lithosphere is not a shell sitting loosely on a lubricated sphere. It is broken into plates coupled to mantle convection, and it moves at centimetres per year because that is the rate at which mantle rock deforms. Moving it thousands of kilometres in days would require energy input comparable to a major impact and would produce heating, deformation, and a stratigraphic signature that is simply not present. And the empirical test is decisive and simple: continuous ice core records from Greenland and Antarctica extend back 120,000 and 800,000 years respectively, with annual layering. If the poles had relocated, the ice would have melted. The ice is still there, layer by layer, all the way down. That single observation closes the question.

FUN FACT

Hapgood does deserve credit for one thing. He wrote extensively on the Piri Reis map of 1513, arguing it depicted an ice-free Antarctic coastline. It does not — the southern portion is a distorted rendering of the South American

coast, and the map's own annotations cite Columbus-era Portuguese sources — but the analysis prompted real cartographic scholarship on early modern maps. Even a wrong hypothesis, pursued carefully enough to be checkable, produces something. That is more than most of the material in this chapter can claim.

19.7 THE ARYAN RACE

I need to be extremely clear about what this section is and is not, and I am going to state it in plain terms before I write another word.

This section is not an argument that Nazi ideology contained anything true, valuable, or worth recovering. It contained nothing of the sort. Its racial doctrine was false as science and monstrous as ethics, and it was used to justify the industrialised murder of eleven million people. Nothing below qualifies that.

But you cannot tell the story of UFOs without asking why a political party was so deeply invested in the occult, and answering that question properly is one of the most important things in this book — because the Nazi case is the definitive demonstration that pseudo-history is not harmless. Everywhere else in these twenty chapters, being wrong about ancient civilisations costs somebody the price of a paperback. Here it cost eleven million lives. That is why the chapter cannot end at crystal skulls.

19.7.1 WHERE THE WORD CAME FROM

“Aryan” began as a linguistic term and was corrupted into a racial one, and the corruption is traceable. In the late eighteenth century, William Jones observed that Sanskrit, Greek, Latin, Persian and the Germanic languages share systematic structural correspondences implying descent from a common ancestral language. This was a real and important discovery: Proto-Indo-European is a genuine reconstruction, and the language family is not in dispute. *Ārya* was a self-designation used in Sanskrit and Old Persian texts, meaning something like “noble”.

The fallacy was to convert a language family into a biological race, and then into a hierarchy. Once that move was made — principally by

nineteenth-century writers such as Arthur de Gobineau and later Houston Stewart Chamberlain — a technical observation about verb endings became a claim about superior blood. And note the structure, because it is the same structure as Donnelly's: observe a real pattern of similarity, infer common origin, then infer a hierarchy that the evidence never contained.

19.7.2 WHY THE OCCULT

Here is the actual answer to the question, and it is not mystical. It is that the ideology required a history it did not have.

If your political programme asserts that a particular people are the original and superior human stock, you need historical evidence of their primacy. There isn't any. So it must be supplied, and the only available suppliers were the occult and pseudo-archaeological traditions this chapter has been describing — Blavatsky's root races, Atlantis and Hyperborea as ancestral northern homelands, and the German Völkisch mystical movements of the early twentieth century. The Thule Society, named for a legendary northern land, was one such group and several early Nazi figures passed through its orbit. The swastika itself was appropriated from Indian iconography precisely because of the Indo-European language connection.

This produced state-funded pseudo-archaeology. Heinrich Himmler founded the Ahnenerbe in 1935 as a research institute for ancestral heritage, and it mounted real expeditions — Tibet in 1938–39 under Ernst Schäfer, Scandinavia, the Middle East — looking for evidence of ancient Aryan presence. It found none, because none exists. What it produced instead was fabricated interpretation, and later, under Himmler's direction, some of its personnel conducted lethal human experiments at Dachau and assembled a collection of murdered Jewish prisoners' skeletons at Strasbourg. That is where pseudo-archaeology ended up: not in a bad book, in an anatomy department.

So the answer to “why was a political party into the occult?” is that the occult was the only place that would give them the past they needed. A regime that had decided on its conclusion in advance went shopping for evidence, and mysticism was the one vendor with the product in stock. That is the deepest lesson in this book about what happens when you decide what is true before you look.

19.7.3 THE NAZI UFO CLAIMS

The postwar mythology — Haunebu and Vril flying discs, an Antarctic base at Neuschwabenland, a surviving Reich in a hollow Earth, and Nazi technology as the true origin of flying saucers — is fabricated, and unusually well documented as fabrication. The Haunebu “blueprints” first appeared in the 1970s, primarily via the Austrian occult writer Ernst Zündel, a Holocaust denier who sold them by mail order, and are traceable to that source. No such craft appears in the captured German technical archives, which are extensive, exhaustively examined, and include genuinely exotic programmes the Allies were very interested in.

What was real: the Germans built the world’s first operational jet fighter, the first ballistic missile, the first cruise missile, the first guided anti-ship weapon, and several genuinely unusual airframes including the Horten flying wings (see Figure 19.3), which look strikingly like a modern stealth aircraft in planform. Chapter 12 and Section 3.2 cover why that matters: captured German engineers and designs went into American and Soviet programmes through Operation Paperclip, and postwar flying-wing and delta prototypes are a genuine part of the flying-saucer story. The real history is remarkable and requires no discs.

As for Antarctica: the German Neuschwabenland expedition of 1938–39 was a real whaling-and-survey reconnaissance that mapped coastline and dropped territorial markers. It built no base. Operation Highjump, which sits alongside the claims in Chapter 17, was a large 1946–47 U.S. Navy Antarctic exercise that has been retrofitted into a secret war against a Nazi redoubt; its actual purpose was cold-weather operational training and establishing a presence, and its records are declassified and dull.

I want to name the mechanism at work in this particular myth, because it is the ugliest thing in this subject. Nazi UFO lore functions as glamour. It recasts a genocidal bureaucracy as a mysterious technological elite, and that recasting has been used deliberately by neo-Nazi publishers as a recruitment aesthetic — which is precisely why several of its principal sources were

FIGURE 19.3

*The Horten Ho 229 in planform
— a genuine wartime prototype
that looks startlingly modern.*

Figure 19.3. The Horten Ho 229 in planform — a genuine wartime prototype that looks startlingly modern.

Image sourced from: https://commons.wikimedia.org/wiki/File:Horten_Ho_229_planform.jpg

Holocaust deniers rather than aviation historians. When you encounter this material, that is what it is for.

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 erbe and Nazi racial ideology.

19.8 USO'S

One last topic, and it returns us to water, which is where the chapter began.

USO stands for Unidentified Submerged Object, and the category is the reason the Pentagon broadened UAP from “aerial” to “anomalous” in 2022. The claims: objects entering and leaving the sea without deceleration, sonar contacts moving at implausible speed, and transmedium travel — the same craft operating in air and water. Section 12.5's Nimitz case is the flagship, since Fravor reported a disturbance in the water beneath the object.

The case that most deserves attention here is not American. Between 1980 and 1986 the Swedish Navy repeatedly hunted submarine contacts in its territorial waters, depth-charged them, and never once recovered or identified a vessel. This is the best-documented case of a technologically advanced navy repeatedly failing to identify contacts in its own coastal waters, over six years, with full political backing. It ended when a 1995 review concluded that a substantial proportion of the contacts had been biological — and specifically, that a class of characteristic acoustic signature had been produced by herring, releasing gas. That finding remains contested in Sweden, and I present it as the honest state of the argument rather than as a settled joke.

And that is precisely why I include it. The underwater environment is the most hostile sensing environment on the planet. Sound speed varies with temperature, salinity and pressure, producing refraction, shadow zones and convergence zones where a contact can appear at the wrong range or bearing entirely, or vanish; the thermocline reflects sound; biologics produce signatures that fooled a NATO navy for six years; and the deep

scattering layer, a dense band of marine organisms that rises and falls daily, was originally mistaken for a false sea floor. If Section 12.6's atmospheric optics explain a large fraction of aerial sightings, underwater acoustics are worse by a considerable margin, and the practitioners know it.

Which makes the honest summary of USOs uncomfortable in both directions. Sonar operators of the world's most capable navies report contacts they cannot identify — and their instrument is the least reliable one in the whole of this book. There is also the mundane inventory: unmanned underwater vehicles have been operational for decades, the seabed is littered with wrecks and unexploded ordnance, and the deep ocean remains genuinely, overwhelmingly unexplored — something like twenty-five per cent of the ocean floor is mapped at modern resolution, and new large animals are still being described.

So Atlantis, in the end, hands us the right closing thought by accident. There really is a vast unexplored region of this planet that we cannot see into and where our instruments deceive us. It is not hiding a lost civilisation — the ice cores rule that out — but it is hiding a great deal, and the distinction between “we do not know what is down there” and “we know what is down there and it is extraordinary” is the same distinction this book has been drawing since page one. The ocean is a genuine mystery. That is different, and better, than a solved one dressed up as a mystery.

THE MEDIA AND DISCLOSURE

“The media’s the most powerful entity on earth. They have the power to make the innocent guilty and to make the guilty innocent, and that’s power. Because they control the minds of the masses.”

— MALCOLM X

This is where the book closes the loop.

We opened by dedicating this text to Arthur William Bell, and we opened Chapter 1 by arguing that the central problem in Ufology is not a shortage of sightings but a failure of language — that we cannot investigate a subject we cannot define. This final chapter is about the machine that did the defining. Not the Air Force, not the scientists, not even the witnesses. The media.

Every term in this book reached you through a broadcast, a film, a newspaper column or a podcast. “Flying saucer” was a reporter’s phrase. “Little green men” came from pulp fiction. The image in your head when you read the word “alien” — grey skin, black eyes, oversized cranium — is a specific visual design that spread through television and film in the late 1970s and 1980s, and Section 7.3’s discussion of abduction narratives showed what happens to testimony once a template is in circulation. The subject we have spent twenty chapters examining is, to a degree that should bother us, a media artifact.

So the last thing a skeptic needs is not another case file. It is an understanding of the delivery system.

20.1 WAR OF THE WORLDS

On the evening of 30 October 1938, the Columbia Broadcasting System aired an adaptation of H. G. Wells's *The War of the Worlds*, directed by and starring the twenty-three-year-old Orson Welles. The innovation was structural: rather than narrating a story, the first two-thirds of the broadcast imitated the form of a live radio evening — dance music from a hotel ballroom, interrupted by increasingly urgent news bulletins, remote reports from Grover's Mill, New Jersey, and an on-air transmission that cut to dead silence.

What happened next is the most repeated media anecdote of the twentieth century, and most of what you have heard about it is wrong. This matters more than the story itself, so let us do it properly.

FIGURE 20.1

Orson Welles at the microphone in 1938, and the next morning's front pages that manufactured the panic.

Figure 20.1. Orson Welles at the microphone in 1938, and the next morning's front pages that manufactured the panic. Image sourced from: https://commons.wikimedia.org/wiki/File:Orson_Welles_1938.jpg

The legend says a nationwide panic ensued (see Figure 20.1): millions fleeing, highways jammed, suicides, mobs. Subsequent scholarship — particularly Jefferson Pooley and Michael Socolow's work, and the media historian W. Joseph Campbell's — has established that the audience was small. The broadcast ran against *The Chase and Sanborn Hour* with Edgar Bergen, which was the most popular programme in America, and the C. E. Hooper ratings service found roughly two per cent of surveyed households listening to Welles.

There was no confirmed death, no verified suicide, and the “jammed highways” appear in almost no contemporaneous local reporting. Some listeners were genuinely frightened, some called their local stations, and that is approximately the extent of it.

So where did the panic come from? Overwhelmingly from newspapers. The next morning's press ran the story hard and kept running it, and there is a straightforward commercial explanation: radio was five years into eviscerating newspaper advertising revenue, and a story establishing that the new medium was dangerous and irresponsible was extremely useful to the old one. The panic was not an event that the press covered. It was a product the press manufactured, out of an incentive it had.

FUN FACT

There is a delicious recursion here. The single most cited example of “the media can make people believe anything” is itself a thing the media made people believe. If you want one case study to keep in your pocket for the rest of your life, it is this one: a story that is famous for demonstrating credulity, which almost everyone who repeats it has accepted uncritically, and which survives because it is too good a parable to check.

The genuinely important consequence is that the broadcast established a *format*. Welles proved that the conventions of authoritative news — the calm anchor, the remote correspondent, the interruption — carry credibility independent of content. Nine years later, when the phrase “flying saucer” entered the wire services after Kenneth Arnold, the machinery for turning an ambiguous sighting into a national phenomenon overnight was already built and tested.

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20.2 POWER OF PROPAGANDA

I want to use this section narrowly and technically, because “propaganda” has become a word people throw at coverage they dislike. What I mean by it is specific: the deliberate shaping of belief through control of framing and repetition rather than through falsehood. The most effective propaganda contains very few lies. It works by selection.

Four mechanisms are worth naming, because you will find all four in the UFO literature and in the official response to it.

Framing. The same facts produce different conclusions depending on the container. “Air Force releases report on Roswell debris” and “Air Force issues second explanation for Roswell” describe an identical event. Neither is false. Chapter 14 is a case study in framing: the 1994 and 1997 reports were reasonable documents that arrived so late, and after so much prior

official vagueness, that their framing was decided before they were read.

The illusory truth effect. Repeated statements are rated more likely to be true, and this occurs independently of the source's credibility and independently of the listener's prior knowledge. This is well replicated experimental psychology, not commentary, and it is the single most important finding in this chapter, because it means that the sheer *volume* of UFO content generates belief mechanically. No one has to lie. Something merely has to be said often.

Availability and the vividness advantage. We estimate frequency by how easily examples come to mind. A dramatic televised abduction account is retrieved effortlessly; a statistical base rate is not retrieved at all. **This is why one good story outperforms a hundred good studies.**

Selective silence. Section 3.2 covered what military secrecy actually requires: a programme must not merely be hidden, its absence must be explained. When an agency refuses to comment, it is not producing an absence of information — it is producing information, and the audience will fill the gap with the most interesting available hypothesis. The Air Force's handling of the U-2 and the Blackbird did more to build the flying-saucer mythology than any believer, and it did it by saying nothing at all. Secrecy is not the opposite of publicity. It is a form of it.

"Propaganda rarely needs to tell you something false. It only needs to decide which true things you hear, and how often."

And because I have been asking you all book to apply these tools to believers, apply them here to me. This book selects. I chose which cases to include, I put the debunked crystal skulls in a full section and compressed six decades of sonar contacts into a few paragraphs, and I framed each chapter with a quotation I picked. I have tried to flag my uncertainty where it exists, but the structure of a textbook is itself a framing device, and you should read this one the way I have taught you to read everything else in the field.

20.3 THE INFLUENCE OF HOLLYWOOD

The traffic between screen and sighting runs in both directions, and the chronology is the argument.

The pattern begins immediately. *The Day the Earth Stood Still* and *The Thing from Another World* both appeared in 1951, four years after Arnold, and between them established the two enduring templates: the benevolent emissary with a warning, and the infiltrating menace. Cold War anxiety wore a flying saucer for a decade after that — *Invasion of the Body Snatchers* in 1956 is a film about aliens in the same sense that *Moby-Dick* is a book about whaling.

Then come the two that changed the field. *Close Encounters of the Third Kind* in 1977 took its title and its classification scheme directly from J. Allen Hynek, who consulted on the film and appears in it, and it did something no previous film had: it made contact benign, awe-inspiring, and quasi-religious. Chapter 18's discussion of what contact would actually involve is in large part a conversation with this film, because Spielberg's version — lights, music, communication as a five-note motif — is the version in most people's imagination.

Fire in the Sky in 1993 dramatised the Travis Walton case, and *The X-Files*, running from 1993 to 2002, did something more consequential than any single film. It made “the government is hiding it” the default frame rather than a fringe position, gave the culture the phrase “trust no one”, and coincided almost exactly with the peak years of *Coast to Coast AM*. The most influential UFO text of the 1990s was a network drama, and the second most influential was a radio show.

Here is the part that should trouble a serious investigator. *Communion*, Whitley Strieber's account, was published in 1987 and its cover carried a specific rendering (see Figure 20.2) of a grey-skinned, large-eyed being. Abduction reports from before that period describe a much wider range of entities: hairy dwarfs in the South American reports, tall Nordic figures, robotic forms, one-off oddities. The reports converge on the **Grey** after the image saturates the culture, and they con-

FIGURE 20.2

The 1987 cover art of Communion — the point after which reported entities converge on a single face.

Figure 20.2. The 1987 cover art of *Communion* — the point after which reported entities converge on a single face.

Image sourced from: https://commons.wikimedia.org/wiki/File:Communion_book_cover_art.jpg

verge further after *The X-Files*. Section 7.1 laid out the mechanism — suggestibility under hypnosis, source-monitoring error, the reconstructive nature of memory — and this is the epidemiological evidence for it. Testimony tracks the movies.

FUN FACT

Before *The Day the Earth Stood Still*, “the aliens” in American popular culture were frequently invertebrate, gaseous, or unrepresentable. The reason your mental image is humanoid, bipedal, and roughly your own height is production economics: it is far cheaper to put an actor in a suit. The dominant visual hypothesis about extraterrestrial biology in the twentieth century was determined by studio budgets.

None of this proves that no one saw anything. It proves that the *description* of what they saw is not independent evidence, and independence is the entire value of corroboration. A thousand witnesses who all watched the same film are, evidentially, closer to one witness than to a thousand.

20.4 ART BELL

Arthur William Bell III was born on 17 June 1945 in Jacksonville, North Carolina, the son of a Marine Corps captain and a Marine drill instructor, and he was a radio obsessive essentially from childhood — a licensed amateur operator as a boy, callsign eventually W6OBB. He served in the United States Air Force during the Vietnam War as a medic, and he ran a pirate station from his barracks in Amarillo before being stationed overseas.

And then there is the record. While working as a disc jockey at KSBK in Okinawa, Japan — at the time the only English-language commercial station in the country — Art Bell set a Guinness world record by broadcasting solo for 116 straight hours. One man, on air, alone, for very nearly five days.

I want to sit with that for a moment, because it is not a piece of trivia. It is the single most explanatory fact about him. Nobody talks continuously for 116 hours unless the act of talking to strangers over a transmitter is the thing they are actually for. Every characteristic of the show he would later build — the willingness to let a caller run, the tolerance for silence, the absence of

a screening panel and a producer's clock, the ability to hold five hours of unstructured live conversation together night after night — is visible in that record decades before *Coast to Coast AM* existed. The record was not a stunt he pulled. It was a description of him.

While in Okinawa he also, characteristically, involved himself in something with no professional upside: he chartered aircraft and helped move Vietnamese orphans out of the country during the war's collapse. He later worked in cable television management in Nevada, which is where the radio career restarted.

What made him extraordinary as a broadcaster was a specific and rare skill, and it is one I have been teaching in a different vocabulary for twenty chapters. Art Bell took guests seriously without endorsing them. He would let a man explain, at length and without interruption or ridicule, that he had been taken aboard a craft — and he would ask precise follow-up questions, and he would not say he believed it. He described himself as a skeptic who found the subject fascinating. That posture — genuine curiosity plus withheld assent — is exactly the “doubt-driven skepticism” of Section 1.7, and Bell practised it live, unscripted, for hours, in front of millions, better than most people manage in writing with time to revise.

He also had the instrument for it. The voice, the desert studio in Pahrump with the transmitter on his own property, the sign-on from “the high desert and the great American Southwest”, the deliberate intimacy of talking to one person at three in the morning. Radio at that hour is a peculiar technology: the listener is alone, usually driving or unable to sleep, and the voice is inside the car with them. Bell understood that better than anyone who has done the job.

He retired, unretired, retired again, and retired again — repeatedly, and for reasons that were variously personal, family-related, and never fully explained — which frustrated his audience for twenty years. He died at his home in Pahrump on 13 April 2018, on Friday the 13th, which he would have enjoyed enormously.

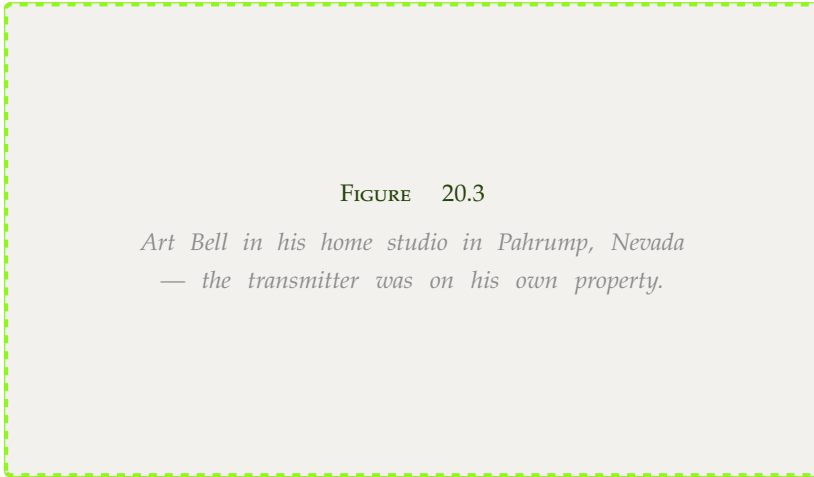


Figure 20.3. Art Bell in his home studio in Pahrump, Nevada — the transmitter was on his own property.

Image sourced from: <https://example.org/placeholder-artbell.jpg>

20.5 COAST TO COAST AM

Radio has proven to be one of the most successful and resilient technologies ever contrived. Having survived the development of not one, but two advances in technology that were otherwise poised to render radio obsolete. First with the coming of television, and next with the inception and later explosion of the internet nearing the end of the twentieth century. Yet even today, more than twenty years later, radio is at the forefront of media consumption — not drowned out by, but rather alongside of the internet and TV. Radio may even be attributed as the inspiration for the widely popular spread of podcasts as a form of entertainment.

Podcasts have exploded in popularity recently, with many new additions showing up daily, trying to emulate — reproduce, if you will — the form of other popular shows such as *The Joe Rogan Experience*, *Freakonomics*, or *Serial*. Podcasts have something cool, something unique about them that separates and distinguishes them from the rest of radio. They exist to serve a niche, fill a void, occupy some curiosity of the human condition that otherwise isn't served through traditional channels. Podcasts are NOT — well, some may be — simply nothing but news, weather or sports. They are a breath

of fresh air, and for many the discovery that such a media exists is like an escape from the monotonous day to day they have known their whole life.

Face it: news is boring. Weather and sports are not much more appealing. At least in the form of radio, let me be clear. If you are going to listen to the announcer of your favorite sports match, would it not be more entertaining, or just as easy, to pull up the stream on YouTube or through your TV and receive visuals as well? If you are going to hear how Trump is single-handedly running the nation while saving all the starving kids of Africa and finding a cure to cancer while he sleeps, then why not just read that online? While things of this nature have always existed as a part of radio, and been consumed through channels on the FM dial, AM radio is different. It's like a wildcard, a stab into the darkness, and an alternative world in the space of NPR — National Public Radio. And in the more than hundred-year history of radio stations broadcasting content into the great unknown, none other has been as provocative, as influential, and downright weird as *Coast to Coast AM*.

Beginning in 1984, at a time when nearly all non-music orientated radio stations were focused on covering the standards — that being news, politics, and sports — *Coast to Coast* was different. It began in the desert of Las Vegas when a man by the name of Art Bell was approached with an opportunity to move his daytime talk show to the wee hours of the night. Thrilled with the chance, Art gleefully accepted, and while currently working as the manager of a nearby cable company, he quickly found himself pulled back into the world of radio he grew up on. While initially running as another typical conservative-orientated talk show, Art would ever so often inject things of his own curiosity into the program. This proved extremely successful, in addition to improving the ratings of the show, and so in 1992 Art Bell's all-night show was rebranded as *Coast to Coast AM* and taken nationwide.¹ While *Coast to Coast* was in fact an FM broadcast station, it mirrored much of the content style of things found on the AM dial. Instead of music, it was people talking for hours upon hours on whatever the subject at hand may be, all under the guidance of the show's host.

And the subject of choice, you ask? Well, anything and everything — from the occult, to ghosts and demons, aliens and UFOs, crop circles and the Bermuda Triangle, ley lines and ancient history, people who claim to be

¹See https://rationalwiki.org/wiki/Coast_to_Coast_AM.

abducted, people who claim to be abducting, vampires, werewolves, men who superglue their mouth shut, the men in black, people pursued by the men in black, people pursuing the men in black, some guy who collects toenails, Roswell and alien autopsies, government secrets and Project Blue Book, space, time, the whole nine yards, and yes, everything INCLUDING the kitchen sink!!! In other words, that show was WEEEIRD!

What is worth adding, from the perspective of the twenty chapters behind us, is how much of this book would not exist without that show. Chapter 13 is largely about Linda Moulton Howe, whose reporting reached a mass audience primarily as a recurring *Coast to Coast* guest. Section 17.3 and the underground-base literature, the Mothman revival, remote viewing, the Hale-Bopp companion object of 1997 — which Bell later publicly acknowledged handling badly, and which intersected tragically with the Heaven's Gate deaths discussed in Chapter 10 — and a substantial fraction of the named figures in this text either broke their material on that programme or became known through it. Art Bell did not merely report on the field. Between 1992 and 2003 he was very nearly the field's distribution network.

The show continues today under later hosts, primarily George Noory, and syndicated on hundreds of stations. It is a different programme, and I will leave it at that.

FUN FACT

Bell's open-phones formats are an accidental monument to everything in Chapter 1. "Ghost to Ghost AM" on Halloween took unscreened listener stories on air. There was no verification, no follow-up, no possibility of investigation — and that was the point; it was theatre, and Bell said so. But those broadcasts entered the literature anyway, cited later by authors who no longer remembered that the origin was an unscreened Halloween call-in. That is how anecdote becomes source material: not by anyone lying, but by the label falling off.

20.6 12-16-17

On 16 December 2017, *The New York Times* published a story titled "Glowing Auras and Black Money: The Pentagon's Mysterious U.F.O. Program," reporting the existence of the Advanced Aerospace Threat Identification Program, a Defense Department effort funded at roughly \$22 million

between 2007 and 2012, largely at the direction of Senator Harry Reid. Alongside it the paper published gun-camera footage from Navy aircraft, including the *Nimitz* encounter described in Section 12.5.

I have given this date its own section, and the date is the section title, because it is the dividing line. Everything in this book belongs to one side of it or the other.

Before 16 December 2017, the structure of the subject was fixed and had been for seventy years. Civilian researchers gathered testimony; the government's public position was that there was nothing to investigate and had not been since Blue Book closed in 1969; the burden fell entirely on believers to produce evidence; and the only available channels were paperbacks, conferences, late-night radio and the internet. Chapters 8 through 15 describe that world.

After that date, the Department of Defense had confirmed in a newspaper of record that it had run a programme, that it possessed sensor data it could not explain, and that it considered the question a matter of flight safety and airspace security. Within four years there was a statutory reporting office, an unclassified Office of the Director of National Intelligence preliminary assessment in June 2021, congressional hearings for the first time in over fifty years, and the term **UAP** in the United States Code.

What actually changed was jurisdiction. The question moved out of the hands of enthusiasts and into the hands of institutions with sensors, budgets, subpoena power and an obligation to report. That is a smaller and more bureaucratic transformation than the excitement of that week suggested — and it is also a far more consequential one, because for the first time the subject acquired the one thing Chapter 1 said it never had: an official definition, a designated custodian, and a requirement to produce data rather than testimony.

So the old guard and the new guard differ less in what they believe than in what they consider evidence. The old guard was built on witnesses, because witnesses were all anyone had; its craft was interviewing, archiving, and the freedom-of-information request, and its heroes were researchers who spent decades on cases the state refused to acknowledge. Its besetting weakness was that testimony accumulates without ever converging, which is why eighty years produced a vast literature and no settled fact. The new guard is built on instrumentation — radar tracks, infrared imagery, calibrated sensor fusion — and its besetting weakness is the opposite: sensors have

artifacts, and Section 12.5 showed how much of the celebrated Navy footage is arguably explicable as parallax, glare, and gimbal rotation. Neither guard has the answer. But only one of them has instruments capable of producing one, and that is why the shift matters.

A personal note, since this is the last chapter and I have earned it. This happened just a few months after I started my UFO club. I had spent that autumn arguing with people who thought the subject was beneath discussion, and then in the middle of December the Pentagon confirmed it had been quietly funding exactly the thing I was being told did not exist. I would like to report that I handled this gracefully. I did not; I was insufferable for approximately a month. What I did not understand at the time, and what this whole book has been my attempt to work out, is that the story did not vindicate believers at all. It vindicated *procedure*. The reason we now have hearings and a reporting office is not that someone finally believed the witnesses. It is that somebody produced a sensor record, wrote it up, and filed it through a channel that could not simply ignore it.

That is the whole thesis of this textbook in one sentence. The question was never whether something is out there. It is whether we are willing to build the tools, the definitions and the discipline required to find out — and to accept the answer if it turns out to be boring.

Eighty years of testimony moved this subject almost nowhere. One reporting requirement moved it further in four years than the previous seventy combined. Keep looking up. Then write down what your instrument said, not what you hoped it said.

“And so, from the high desert, good night.”

— Art Bell, sign-off

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M

G L O S S A R Y

Terms are defined as they are used in this book. Where a term is contested — and a great many of them are — I have said so rather than picking a side and pretending the argument is settled.

A

Abduction (alien) A reported experience of being taken, usually involuntarily, aboard a craft or into the control of non-human entities. Discussed in Chapter 7. The reports are real; the question at issue is what produces them.

AATIP Advanced Aerospace Threat Identification Program. A U.S. Department of Defense effort, funded at roughly \$22 million between 2007 and 2012, revealed publicly on 16 December 2017.

Ahnenerbe A research institute founded by Heinrich Himmler in 1935 to seek evidence of ancient Germanic and “Aryan” presence. It produced no valid archaeology and some of its personnel committed atrocities. See Section 19.7.

Anecdotal evidence Testimony from individual experience, unaccompanied by measurement or independent record. It is evidence — it is simply the weakest class of it, and it does not become stronger by accumulating.

Argument from ignorance The fallacy of treating the absence of an explanation as support for a particular explanation. “Unidentified” means unidentified.

Atlantis An island civilisation described by Plato in the *Timaeus* and *Critias* (c. 360 BC) as an illustrative antagonist. See Chapter 19.

B

Base rate The underlying frequency of an event in a population. Ignoring it — the base rate fallacy — is the most common error in reasoning about rare events, including sightings.

Blue Book (Project) The U.S. Air Force's public UFO investigation, 1952–1969. It examined 12,618 reports and classified 701 as unidentified.

Burden of proof The obligation to supply support for a claim, which rests on the party making it. See Section 1.2.

C

Close encounter A sighting at short range, classified by J. Allen Hynek into kinds: CE-1 (observed at close range), CE-2 (with physical effects), CE-3 (with occupants visible).

Confirmation bias The tendency to seek, notice, and remember information consistent with what one already believes. It afflicts skeptics and believers identically. See Section 1.6.

Correlation A statistical relationship between two variables. It does not establish that either causes the other. See Section 1.5.

Crustal displacement The hypothesis, proposed by Charles Hapgood in 1958, that the Earth's lithosphere can slip catastrophically over the mantle. Not supported; see Section 19.6.

Cryptid An animal whose existence is claimed but not established by specimen, type description, or documented remains.

D

Disclosure In UFO discourse, the anticipated official admission of non-human intelligence. In practice what has occurred is narrower: institutional acknowledgement of unexplained sensor data.

Drake equation A 1961 framework by Frank Drake for estimating the number of communicating civilisations in the galaxy. Its value is pedagogical: it organises our ignorance into named variables. See Chapter 6.

E

EBE Extraterrestrial Biological Entity. A term of art from U.S. documents and folklore for a non-human organism of off-world origin.

ESP Extrasensory perception. An umbrella term covering telepathy, clairvoyance and precognition. See Section 7.7.

Exotic matter Hypothetical material with negative energy density, required by most faster-than-light and traversable-wormhole solutions. No such matter has been observed in bulk.

F

Falsifiability The property of a claim that some possible observation could show it false. A claim that no observation could contradict is not a scientific claim, however true it may be.

Fermi paradox The tension between the apparent likelihood of extraterrestrial civilisations and the absence of any evidence of them.

Foo fighter A term used by Allied aircrews in the Second World War for luminous objects observed accompanying aircraft. See Section 12.3.

Fravor, David U.S. Navy commander who reported the 2004 *Nimitz* “Tic Tac” encounter. See Section 12.5.

G

Ganzfeld An experimental protocol in parapsychology using uniform sensory input to reduce noise, historically the most-cited body of Psi evidence and the subject of long dispute over methodology and replication.

Grandfather paradox The self-consistency problem raised by backwards time travel: preventing the conditions of one's own journey. See Chapter 5.

Grey The dominant modern description of a reported entity: small, hairless, grey-skinned, with large black eyes. Its cultural saturation dates from the late 1980s, which matters when evaluating testimony that describes it.

H

Howe, Linda Moulton Investigative journalist and documentarian, best known for *A Strange Harvest* (1980) and *Earthfiles*. See Chapter 13.

Hynek, J. Allen Astronomer, scientific consultant to Blue Book, later founder of the Center for UFO Studies. The field's most consequential convert from skeptic to serious investigator.

I

IFO Identified Flying Object. What the overwhelming majority of UFOs become on investigation.

Illusory truth effect The experimentally robust finding that repeated statements are judged more likely to be true, irrespective of source credibility. See Section 20.2.

L

Lemuria A hypothetical submerged landmass proposed by Philip Sclater in 1864 to explain lemur distribution, superseded by plate tectonics and subsequently absorbed into occult literature. See Section 19.5.

Lenticular cloud A stationary, smooth, lens-shaped cloud formed in the lee of terrain. One of the most reliable producers of genuine “flying saucer” photographs.

M

Majestic 12 A purported secret committee described in documents that surfaced in 1984. The documents are generally regarded as fabricated, on typographic and procedural grounds.

Men in Black Figures reported to appear after sightings to intimidate witnesses. See Chapter 8.

Mutilation (cattle) Livestock deaths featuring apparently precise tissue removal and reported absence of blood. See Chapter 13.

N

Nimitz encounter A series of November 2004 radar and visual contacts by aircraft from Carrier Strike Group Eleven off Southern California, including the FLIR1 “Tic Tac” video. See Section 12.5.

Null hypothesis The default assumption that an observed effect is absent, or attributable to chance. It is not a belief; it is a starting position that evidence must move.

O

Occam's razor The principle that, among explanations accounting equally well for the evidence, the one requiring the fewest unsupported assumptions is preferred. It is a heuristic for choosing what to test first, not a proof.

Operation Paperclip The postwar U.S. programme that recruited German scientists and engineers, with direct consequences for aerospace development and, indirectly, for the flying-saucer era.

P

Paperclip See Operation Paperclip.

Parapsychology The scientific study of claimed anomalous mental phenomena. See Section 7.7.

Peer review Pre-publication evaluation of a manuscript by other specialists. It filters, it does not certify. See Section 1.8.

Pole shift Ambiguous term. Distinguish geomagnetic reversal (real, slow), true polar wander (real, very slow), and catastrophic crustal displacement (not supported). See Section 19.6.

Precognition Claimed acquisition of information about future events by non-inferential means.

Psi Collective term for claimed anomalous phenomena studied by parapsychology, covering both ESP and psychokinesis.

Psychokinesis (PK) Claimed direct mental influence on physical systems. Conventionally divided into macro-PK (visible object movement) and micro-PK (statistical deviation in random systems).

R

Radiosonde A balloon-borne instrument package. The Roswell debris was officially attributed to equipment of this general class, specifically a Project Mogul array. See Chapter 14.

Remote viewing Claimed perception of distant locations, investigated under U.S. government funding in programmes including Stargate.

Roswell The July 1947 debris recovery near Roswell, New Mexico. The founding case of American Ufology. See Chapter 14.

S

SETI Search for Extraterrestrial Intelligence. Primarily radio and optical survey work; see Sections 2.4 and 18.7 for the case that its search space is minute.

Skepticism The practice of proportioning belief to evidence, including belief in one's own conclusions. Not the same as denial, which is a conclusion held in advance.

Sonoluminescence Light emission from collapsing cavitation bubbles — included here as a reminder that genuinely bizarre physical phenomena exist and were established by measurement.

Swamp gas Methane from decaying organic matter, capable of producing ignis fatuus. Notorious as an over-applied official explanation, which damaged the credibility of subsequent correct ones.

T

Telepathy Claimed direct mind-to-mind transfer of information.

Temperature inversion An atmospheric layer in which temperature rises with altitude, capable of refracting light and radar and producing displaced or spurious returns. One of the most important mundane explanations in the field.

Tic Tac Informal name for the object in the 2004 *Nimitz* FLIR1 footage.

Trace evidence Physical residue associated with a reported event — soil alteration, radiation, material fragments. The category the field most needs and least has.

U

UAP Unidentified Anomalous Phenomena (formerly Unidentified Aerial Phenomena). The current official term, broadened in 2022 to include submerged and transmedium objects.

UFO Unidentified Flying Object. Coined by the U.S. Air Force in 1953 to replace “flying saucer”. Note what it does and does not assert: it is a statement about the observer’s knowledge, not about the object.

USO Unidentified Submerged Object. See Section 19.8.

W

Wormhole A hypothetical topological connection between separated regions of spacetime. Permitted by general relativity as a solution; not demonstrated to exist, and not demonstrated to be traversable. See Chapter 11.

REFERENCES & FURTHER READING

Chapter-level source notes appear throughout the text. What follows is the consolidated list, plus recommendations for going further. I have deliberately included work I disagree with, because you cannot evaluate an argument you have only encountered in summary — least of all in a summary written by someone with a position.

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REFERENCE AND LOOKUP TOOLS

National Archives (US), Record Group 341, and the online Blue Book collection at <https://www.archives.gov>. Start here rather than with a summary of a summary.

CIA Records Search Tool (CREST), <https://www.cia.gov/readingroom>. Declassified remote-viewing, balloon and reconnaissance files, full text.

U.S. Naval Observatory and the Astronomical Almanac; also any perpetual

calendar, for checking what day of the week a witness's date actually fell on.

Heavens-Above and Stellarium, for reconstructing what was in the sky at a given place and time. Half the cases in this book dissolve at this step.

NOAA and national meteorological archives, for weather-balloon launch schedules, temperature inversions and cloud conditions on the night in question.

ON IMAGES

Figure placeholders throughout this edition carry the source URL of the intended image in the caption. Where an image is not in the public domain or under a permissive licence, permission must be cleared before commercial publication. Nothing in this edition should be reproduced on the assumption that clearance has been obtained.

*The Science of Ufology 1.0 — first edition.
Corrections, disagreements and better sources are actively welcome.
That is, after all, the entire point.*

ABOUT THE AUTHOR

E V A N S V E N D S E N



The author at the Area 51 exhibit — *Myth or Reality*. Thumbs up for the myth; the reality is in your hands.

Evan Svendsen is a game designer and software developer from Lethbridge, Alberta. He trained at Lethbridge Polytechnic, works mostly in C# and Unity, and has spent more than a decade building and publishing independent games through Tech Skull Studios — the small Lethbridge outfit he helped start in 2013, whose titles have shipped on Android and iOS. He also writes on technology, transit and the strange at <https://evansvendsen.substack.com>.

Game development turned out to be unexpectedly good training for this book. A designer spends his working life asking what a system is actually doing rather than what it appears to be doing, reproducing a bug before believing it exists, and separating the interesting explanation from the correct one. Ufology, as the preceding twenty chapters have argued at length, suffers badly from a shortage of exactly that habit.

His interests outside software run to Alberta infrastructure — he is behind the Prairie Lanes rail and transit concept — and to the odd corner of provincial history where those interests meet this one. Alberta is, after all, the home of the world's first UFO landing pad, built at St. Paul in 1967 for Canada's centennial and still standing; the author has publicly proposed that the town lean into it with a UFO-themed skatepark. He remains open to the possibility that something is flying around up there. He is simply not willing to accept it on the strength of a blurry photograph.

Corrections, better sources and well-argued disagreement reach the author faster than any of the addresses in Chapter 20 will reach anyone else.